HW 1 Report

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https://github.com/shobhit-05/cse493s-spring26-hw1

Part 0

In [9]:
```
import json
import subprocess
from pathlib import Path

import matplotlib.pyplot as plt
import pandas as pd
```

In [10]:
```
ROOT = Path.cwd()
PART0_DIR = ROOT / "out" / "submission" / "part0"
PLOTS_DIR = PART0_DIR / "plots"
TABLES_DIR = PART0_DIR / "tables"

SANITY_FULL = PART0_DIR / "sanity_full"
SANITY_MASK3 = PART0_DIR / "sanity_mask3"

PLOTS_DIR.mkdir(parents=True, exist_ok=True)
TABLES_DIR.mkdir(parents=True, exist_ok=True)

print("root:", ROOT)
print("part0:", PART0_DIR)
```

```
root: /Users/shobhit/Documents/UW/CSE 493/cse493s-spring26-hw1
part0: /Users/shobhit/Documents/UW/CSE 493/cse493s-spring26-hw1/out/submissio
n/part0
```

In [11]:
```
def run(cmd: str):
    print(f"\n$ {cmd}")
    result = subprocess.run(cmd, shell=True)
    if result.returncode != 0:
        raise RuntimeError(f"Command failed with code {result.returncode}: {cm


def load_json(path: Path):
    with open(path, "r") as f:
        return json.load(f)
```

Train Sanity Models

In [12]:
```
run(
    "python train.py --mode sanity --steps 1200 --mask_prefix_tokens 0 "
    "--batch_size 1 --lr 1e-2 --eval_interval 50 --log_interval 100 "
    ut/submission/part0/sanity_full --n_layer 1 --n_head 2 --n_emb
)
```

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```
run(
    "python train.py --mode sanity --steps 1200 --mask_prefix_tokens 3 "
    "--batch_size 1 --lr 1e-2 --eval_interval 50 --log_interval 100 "
    "--out_dir out/submission/part0/sanity_mask3 --n_layer 1 --n_head 2 --n_eml
)
```

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```
$ python train.py --mode sanity --steps 1200 --mask_prefix_tokens 0 --batch_si
ze 1 --lr 1e-2 --eval_interval 50 --log_interval 100 --out_dir out/submission/
part0/sanity_full --n_layer 1 --n_head 2 --n_embd 32 --bias
number of parameters: 0.01M
device=mps train=1 val=1 test=1 batch_size=1
step=1 train_loss=1.630736 train_acc=nan val_loss=1.630736 val_acc=nan test_lo
ss=1.630736 test_acc=nan
step=100 train_loss=0.000360 train_acc=nan val_loss=0.000360 val_acc=nan test_
loss=0.000360 test_acc=nan
step=200 train_loss=0.000039 train_acc=nan val_loss=0.000039 val_acc=nan test_
loss=0.000039 test_acc=nan
step=300 train_loss=0.000002 train_acc=nan val_loss=0.000002 val_acc=nan test_
loss=0.000002 test_acc=nan
step=400 train_loss=0.000000 train_acc=nan val_loss=0.000000 val_acc=nan test_
loss=0.000000 test_acc=nan
step=500 train_loss=0.000000 train_acc=nan val_loss=0.000000 val_acc=nan test_
loss=0.000000 test_acc=nan
step=600 train_loss=0.000000 train_acc=nan val_loss=0.000000 val_acc=nan test_
loss=0.000000 test_acc=nan
step=700 train_loss=0.000000 train_acc=nan val_loss=0.000000 val_acc=nan test_
loss=0.000000 test_acc=nan
step=800 train_loss=0.000000 train_acc=nan val_loss=0.000000 val_acc=nan test_
loss=0.000000 test_acc=nan
step=900 train_loss=0.000000 train_acc=nan val_loss=0.000000 val_acc=nan test_
loss=0.000000 test_acc=nan
step=1000 train_loss=0.000000 train_acc=nan val_loss=0.000000 val_acc=nan test
_loss=0.000000 test_acc=nan
step=1100 train_loss=0.000000 train_acc=nan val_loss=0.000000 val_acc=nan test
_loss=0.000000 test_acc=nan
step=1200 train_loss=0.000000 train_acc=nan val_loss=0.000000 val_acc=nan test
_loss=0.000000 test_acc=nan
generated: ['<BOS>', 'I', 'love', 'machine', 'learning', '<EOS>']
best metrics: {
  "step": 450,
  "train_loss": 0.0,
  "train_acc": NaN,
  "val_loss": 0.0,
  "val_acc": NaN,
  "test_loss": 0.0,
  "test_acc": NaN,
  "elapsed_sec": 1.7997851371765137
}
checkpoint dir: out/submission/part0/sanity_full

$ python train.py --mode sanity --steps 1200 --mask_prefix_tokens 3 --batch_si
ze 1 --lr 1e-2 --eval_interval 50 --log_interval 100 --out_dir out/submission/
part0/sanity_mask3 --n_layer 1 --n_head 2 --n_embd 32 --bias
number of parameters: 0.01M
device=mps train=1 val=1 test=1 batch_size=1
step=1 train_loss=1.456505 train_acc=nan val_loss=1.456505 val_acc=nan test_lo
ss=1.456505 test_acc=nan
step=100 train_loss=0.000172 train_acc=nan val_loss=0.000172 val_acc=nan test_
loss=0.000172 test_acc=nan
step=200 train_loss=0.000028 train_acc=nan val_loss=0.000028 val_acc=nan test_
loss=0.000028 test_acc=nan
step=300 train_loss=0.000002 train_acc=nan val_loss=0.000002 val_acc=nan test_
loss=0.000002 test_acc=nan
step=400 train_loss=0.000000 train_acc=nan val_loss=0.000000 val_acc=nan test_
st_acc=nan
step=500 train_loss=0.000000 train_acc=nan val_loss=0.000000 val_acc=nan test_
```

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```
loss=0.000000 test_acc=nan
step=600 train_loss=0.000000 train_acc=nan val_loss=0.000000 val_acc=nan test_
loss=0.000000 test_acc=nan
step=700 train_loss=0.000000 train_acc=nan val_loss=0.000000 val_acc=nan test_
loss=0.000000 test_acc=nan
step=800 train_loss=0.000000 train_acc=nan val_loss=0.000000 val_acc=nan test_
loss=0.000000 test_acc=nan
step=900 train_loss=0.000000 train_acc=nan val_loss=0.000000 val_acc=nan test_
loss=0.000000 test_acc=nan
step=1000 train_loss=0.000000 train_acc=nan val_loss=0.000000 val_acc=nan test
_loss=0.000000 test_acc=nan
step=1100 train_loss=0.000014 train_acc=nan val_loss=0.000014 val_acc=nan test
_loss=0.000014 test_acc=nan
step=1200 train_loss=0.000001 train_acc=nan val_loss=0.000001 val_acc=nan test
_loss=0.000001 test_acc=nan
generated: ['<BOS>', 'learning', '<EOS>', 'learning', '<EOS>', 'learning']
best metrics: {
  "step": 400,
  "train_loss": 0.0,
  "train_acc": NaN,
  "val_loss": 0.0,
  "val_acc": NaN,
  "test_loss": 0.0,
  "test_acc": NaN,
  "elapsed_sec": 1.6190910339355469
}
checkpoint dir: out/submission/part0/sanity_mask3
```

Inference and Contract Verification

In [ ]:

```
# Memorization generation
run("python inference.py --checkpoint_dir out/submission/part0/sanity_full --ta

# Suffix generation from a fixed prefix
run(
    "python inference.py --checkpoint_dir out/submission/part0/sanity_mask3 "
    "--task custom --prompt_tokens '<BOS>,I,love,machine' --max_new_tokens 2"
)

# Contract check
run(
    "python - <<'PY'\n"
    "from part_0_1_contract import load_model_and_tokenizer, get_bos_token\n"
    "m, tok = load_model_and_tokenizer('out/submission/part0/sanity_full')\n"
    "print('BOS:', get_bos_token(tok))\n"
    "PY"
)
```

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```
$ python inference.py --checkpoint_dir out/submission/part0/sanity_full --task
sanity
device=mps
number of parameters: 0.01M
generated_ids: [1, 3, 4, 5, 6, 2]
generated_tokens: ['<BOS>', 'I', 'love', 'machine', 'learning', '<EOS>']

$ python inference.py --checkpoint_dir out/submission/part0/sanity_mask3 --tas
k custom --prompt_tokens '<BOS>,I,love,machine' --max_new_tokens 2
device=mps
number of parameters: 0.01M
prompt_tokens: ['<BOS>', 'I', 'love', 'machine']
generated_ids: [1, 3, 4, 5, 6, 2]
generated_tokens: ['<BOS>', 'I', 'love', 'machine', 'learning', '<EOS>']

$ python - <<'PY'
from part_0_1_contract import load_model_and_tokenizer, get_bos_token
m, tok = load_model_and_tokenizer('out/submission/part0/sanity_full')
print('BOS:', get_bos_token(tok))
PY
number of parameters: 0.01M
BOS: <BOS>
```

Summarize Metrics

In [14]:

```python
m_full = load_json(SANITY_FULL / "metrics.json")
m_mask = load_json(SANITY_MASK3 / "metrics.json")

summary_df = pd.DataFrame([
    {
        "run": "sanity_full",
        "mask_prefix_tokens": 0,
        "best_step": m_full.get("step"),
        "train_loss": m_full.get("train_loss"),
        "val_loss": m_full.get("val_loss"),
        "test_loss": m_full.get("test_loss"),
    },
    {
        "run": "sanity_mask3",
        "mask_prefix_tokens": 3,
        "best_step": m_mask.get("step"),
        "train_loss": m_mask.get("train_loss"),
        "val_loss": m_mask.get("val_loss"),
        "test_loss": m_mask.get("test_loss"),
    },
])

summary_df.to_csv(TABLES_DIR / "part0_summary.csv", index=False)
summary_df
```

Out[14]:

	run	mask_prefix_tokens	best_step	train_loss	val_loss	test_loss
0	sanity_full	0	450	0.0	0.0	0.0
1	sanity_mask3	3	400	0.0	0.0	0.0

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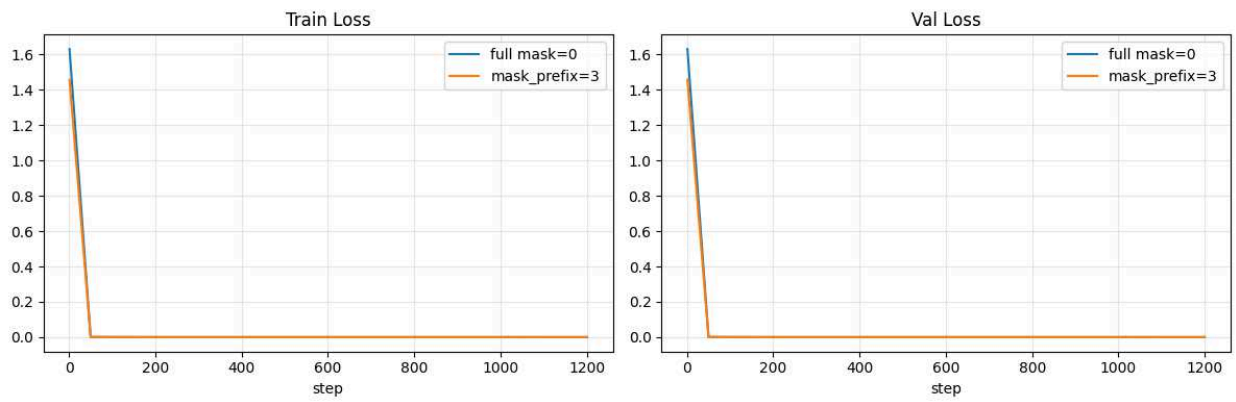
Plot Training Curves

In [15]:
```
h_full = pd.read_csv(SANITY_FULL / "history.csv")
h_mask = pd.read_csv(SANITY_MASK3 / "history.csv")

fig, axes = plt.subplots(1, 2, figsize=(12, 4), sharex=True)
axes[0].plot(h_full["step"], h_full["train_loss"], label="full mask=0")
axes[0].plot(h_mask["step"], h_mask["train_loss"], label="mask_prefix=3")
axes[0].set_title("Train Loss")
axes[0].set_xlabel("step")
axes[0].grid(alpha=0.3)
axes[0].legend()

axes[1].plot(h_full["step"], h_full["val_loss"], label="full mask=0")
axes[1].plot(h_mask["step"], h_mask["val_loss"], label="mask_prefix=3")
axes[1].set_title("Val Loss")
axes[1].set_xlabel("step")
axes[1].grid(alpha=0.3)
axes[1].legend()

fig.tight_layout()
fig.savefig(PLOTS_DIR / "part0_loss_curves.png", dpi=180)
plt.show()
```



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Part 1

In [12]:
```
import json
import os
import subprocess
from pathlib import Path

import matplotlib.pyplot as plt
import pandas as pd

from train import build_modular_dataset
```

In [13]:
```
ROOT = Path.cwd()
PART1_DIR = ROOT / "out" / "submission" / "part1"
PLOTS_DIR = PART1_DIR / "plots"
TABLES_DIR = PART1_DIR / "tables"

PART1_DIR.mkdir(parents=True, exist_ok=True)
PLOTS_DIR.mkdir(parents=True, exist_ok=True)
TABLES_DIR.mkdir(parents=True, exist_ok=True)

print("root:", ROOT)
print("part1:", PART1_DIR)
```

```
root: /Users/shobhit/Documents/UW/CSE 493/cse493s-spring26-hw1
part1: /Users/shobhit/Documents/UW/CSE 493/cse493s-spring26-hw1/out/submissio
n/part1
```

In [ ]:
```
FAST_DEBUG = False

WARMUP_STEPS = 1500 if FAST_DEBUG else 100000
GROKKING_STEPS = 5000 if FAST_DEBUG else 500000
ABLATION_STEPS = 5000 if FAST_DEBUG else GROKKING_STEPS

EVAL_INTERVAL = 100 if FAST_DEBUG else 500
LOG_INTERVAL = 100 if FAST_DEBUG else 500
GROKKING_EVAL_INTERVAL = 200 if FAST_DEBUG else 2000
GROKKING_LOG_INTERVAL = 200 if FAST_DEBUG else 2000

BASE_OPTIMIZER = "adam"

WARMUP_WEIGHT_DECAY = 0.01
WARMUP_LR = 1e-3
WARMUP_MIN_LR = 1e-4
WARMUP_LR_SCHEDULE = "cosine"
WARMUP_WARMUP_STEPS = 1000 if not FAST_DEBUG else 50
WARMUP_GRAD_CLIP = 1.0

GROKKING_WEIGHT_DECAY = 0.01
GROKKING_LR = 3e-3
GROKKING_MIN_LR = 1e-4
GROKKING_LR_SCHEDULE = "cosine"
GROKKING_WARMUP_STEPS = 1000 if not FAST_DEBUG else 50
GROKKING_GRAD_CLIP = 1.0

print("FAST_DEBUG:", FAST_DEBUG)
```

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```
print("WARMUP_STEPS:", WARMUP_STEPS)
print("GROKKING_STEPS:", GROKKING_STEPS)
print("ABLATION_STEPS:", ABLATION_STEPS)
print("BASE_OPTIMIZER:", BASE_OPTIMIZER)
print("WARMUP_LR_SCHEDULE:", WARMUP_LR_SCHEDULE)
print("GROKKING_LR:", GROKKING_LR)
print("GROKKING_EVAL_INTERVAL:", GROKKING_EVAL_INTERVAL)
```

```
FAST_DEBUG: False
WARMUP_STEPS: 100000
GROKKING_STEPS: 500000
ABLATION_STEPS: 500000
BASE_OPTIMIZER: adam
WARMUP_LR_SCHEDULE: cosine
GROKKING_LR: 0.003
GROKKING_EVAL_INTERVAL: 2000
```

In [15]:

```
def run(cmd: str):
    print(f"\n$ {cmd}")
    result = subprocess.run(cmd, shell=True)
    if result.returncode != 0:
        raise RuntimeError(f"Command failed with code {result.returncode}: {cm


def load_history(run_dir: Path) -> pd.DataFrame:
    path = run_dir / "history.csv"
    if not path.exists():
        raise FileNotFoundError(path)
    return pd.read_csv(path)


def load_metrics(run_dir: Path) -> dict:
    path = run_dir / "metrics.json"
    if not path.exists():
        raise FileNotFoundError(path)
    with open(path, "r") as f:
        return json.load(f)
```

1.1 Data Generation

In [5]:

```
rows = []
sample_rows = []

for p in [97, 113]:
    for op in ["+", "-", "/"]:
        split_dir = PART1_DIR / "splits" / f"p{p}_{'div' if op=='/' else ('add
        tokenizer, train_data, val_data, test_data = build_modular_dataset(
            p=p,
            op=op,
            train_frac=0.3,
            val_frac=0.1,
            seed=42,
            save_split_dir=str(split_dir),
        )

        total = len(train_data) + len(val_data) + len(test_data)
        expected = p * (p - 1) if op == "/" else p * p
```

Loading [MathJax]/extensions/Safe.js

```
        rows.append(
            {
                "p": p,
                "op": op,
                "train": len(train_data),
                "val": len(val_data),
                "test": len(test_data),
                "total": total,
                "expected_total": expected,
                "matches_expected": total == expected,
            }
        )

        for split_name, split_data in [("train", train_data), ("val", val_data
            s = split_data[0]
            full_ids = [s["x"][0]] + s["y"]
            sample_rows.append(
                {
                    "p": p,
                    "op": op,
                    "split": split_name,
                    "tokens": tokenizer.decode_ids(full_ids),
                    "target_pos": s["target_pos"],
                    "loss_mask": s["loss_mask"],
                }
            )

counts_df = pd.DataFrame(rows).sort_values(["p", "op"]).reset_index(drop=True)
samples_df = pd.DataFrame(sample_rows)

counts_df.to_csv(TABLES_DIR / "1_1_split_counts.csv", index=False)
samples_df.to_json(TABLES_DIR / "1_1_samples.json", orient="records", indent=2

counts_df
```

Out[5]:

	p	op	train	val	test	total	expected_total	matches_expected
0	97	+	2822	940	5647	9409	9409	True
1	97	-	2822	940	5647	9409	9409	True
2	97	/	2793	931	5588	9312	9312	True
3	113	+	3830	1276	7663	12769	12769	True
4	113	-	3830	1276	7663	12769	12769	True
5	113	/	3796	1265	7595	12656	12656	True

In [6]:

```
samples_df.head(18)
```

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Out[6]:

	p	op	split	tokens	target_pos	loss_mask
0	97	+	train	[<BOS>, 73, +, 93, =, 69, <EOS>]	4	[0.0, 0.0, 0.0, 0.0, 1.0, 0.0]
1	97	+	val	[<BOS>, 10, +, 14, =, 24, <EOS>]	4	[0.0, 0.0, 0.0, 0.0, 1.0, 0.0]
2	97	+	test	[<BOS>, 16, +, 2, =, 18, <EOS>]	4	[0.0, 0.0, 0.0, 0.0, 1.0, 0.0]
3	97	-	train	[<BOS>, 73, -, 93, =, 77, <EOS>]	4	[0.0, 0.0, 0.0, 0.0, 1.0, 0.0]
4	97	-	val	[<BOS>, 10, -, 14, =, 93, <EOS>]	4	[0.0, 0.0, 0.0, 0.0, 1.0, 0.0]
5	97	-	test	[<BOS>, 16, -, 2, =, 14, <EOS>]	4	[0.0, 0.0, 0.0, 0.0, 1.0, 0.0]
6	97	/	train	[<BOS>, 24, /, 60, =, 78, <EOS>]	4	[0.0, 0.0, 0.0, 0.0, 1.0, 0.0]
7	97	/	val	[<BOS>, 13, /, 2, =, 55, <EOS>]	4	[0.0, 0.0, 0.0, 0.0, 1.0, 0.0]
8	97	/	test	[<BOS>, 77, /, 13, =, 88, <EOS>]	4	[0.0, 0.0, 0.0, 0.0, 1.0, 0.0]
9	113	+	train	[<BOS>, 71, +, 4, =, 75, <EOS>]	4	[0.0, 0.0, 0.0, 0.0, 1.0, 0.0]
10	113	+	val	[<BOS>, 18, +, 10, =, 28, <EOS>]	4	[0.0, 0.0, 0.0, 0.0, 1.0, 0.0]
11	113	+	test	[<BOS>, 59, +, 81, =, 27, <EOS>]	4	[0.0, 0.0, 0.0, 0.0, 1.0, 0.0]
12	113	-	train	[<BOS>, 71, -, 4, =, 67, <EOS>]	4	[0.0, 0.0, 0.0, 0.0, 1.0, 0.0]
13	113	-	val	[<BOS>, 18, -, 10, =, 8, <EOS>]	4	[0.0, 0.0, 0.0, 0.0, 1.0, 0.0]
14	113	-	test	[<BOS>, 59, -, 81, =, 91, <EOS>]	4	[0.0, 0.0, 0.0, 0.0, 1.0, 0.0]
15	113	/	train	[<BOS>, 0, /, 64, =, 0, <EOS>]	4	[0.0, 0.0, 0.0, 0.0, 1.0, 0.0]
16	113	/	val	[<BOS>, 57, /, 82, =, 82, <EOS>]	4	[0.0, 0.0, 0.0, 0.0, 1.0, 0.0]
17	113	/	test	[<BOS>, 35, /, 100, =, 6, <EOS>]	4	[0.0, 0.0, 0.0, 0.0, 1.0, 0.0]

1.1 Write-Up

We generated modular equation datasets in the form `<BOS> a op b = c <EOS>` for `op in {+, −, /}` and `p in {97, 113}`. For division, we excluded `b=0`, so division has fewer examples than addition/subtraction.

Key dataset sizes (train/val/test):

- `p=97, +`: `2822 / 940 / 5647` (total `9409 = 97*97`)
- `p=97, −`: `2822 / 940 / 5647` (total `9409 = 97*97`)
- `p=97, /`: `2793 / 931 / 5588` (total `9312 = 97*96`)
- `p=113, +`: `3830 / 1276 / 7663` (total `12769 = 113*113`)
- `p=113, −`: `3830 / 1276 / 7663` (total `12769 = 113*113`)
- `p=113, /`: `3796 / 1265 / 7595` (total `12656 = 113*112`)

All totals match expected counts (see `tables/1_1_split_counts.csv`), so data generation and split construction are correct.

1.2 Warmup - Addition/Subtraction Experiments

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In [7]:
```
import json
import shlex
import sys

SKIP_EXISTING_RUNS = True

def _latest_logged_step(run_dir: Path):
    history_path = run_dir / "history.json"

    if not history_path.exists():
        return None
    try:
        hist = json.loads(history_path.read_text())
    except Exception:
        return None
    if not hist:
        return None

    return hist[-1].get("step")

def _is_completed_run(run_dir: Path, target_steps: int):
    latest = _latest_logged_step(run_dir)
    ckpt_exists = (run_dir / "model_last.pt").exists() or (run_dir / "model.pt'

    if isinstance(latest, int) and latest >= target_steps and ckpt_exists:
        return True

    return False

def run_train(run_dir: Path, op: str, p: int, n_layer: int, seed: int, steps: i
    weight_decay: float = WARMUP_WEIGHT_DECAY,
    optimizer: str = BASE_OPTIMIZER, lr: float = WARMUP_LR, min_lr: float = WAI
    lr_schedule: str = WARMUP_LR_SCHEDULE, warmup_steps: int = WARMUP_WARMUP_ST
    grad_clip: float = WARMUP_GRAD_CLIP, skip_existing: bool = SKIP_EXISTING_RU
):
    if skip_existing and _is_completed_run(run_dir, steps):
        print(f"[skip] already complete: {run_dir}")
        return False

    out_dir_q = shlex.quote(str(run_dir))
    op_q = shlex.quote(op)

    cmd = (
        f"{shlex.quote(sys.executable)} train.py "
        f"--mode modular --op {op_q} --p {p} "
        f"--n_layer {n_layer} --n_head 4 --n_embd 128 "
        f"--steps {steps} --seed {seed} "
        f"--batch_size 128 --lr {lr} --min_lr {min_lr} --lr_schedule {lr_sched
        f"--warmup_steps {warmup_steps} --grad_clip {grad_clip} "
        f"--beta1 0.9 --beta2 0.95 --weight_decay {weight_decay} --optimizer {
        f"--eval_interval {EVAL_INTERVAL} --log_interval {LOG_INTERVAL} "
        f"--out_dir {out_dir_q} --bias --resume"
    )
    run(cmd)
    return True


warmup_runs = []
executed_warmup = 0
```

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```python
for op in ["+", "-"]:
    op_name = "add" if op == "+" else "sub"
    for p in [97, 113]:
        for n_layer in [1, 2]:
            run_dir = PART1_DIR / "warmup" / f"{op_name}_p{p}_l{n_layer}_s42"
            warmup_runs.append({
                "name": run_dir.name,
                "run_dir": run_dir,
                "op": op,
                "p": p,
                "n_layer": n_layer,
                "seed": 42,
            })
            executed_warmup += int(run_train(
                run_dir=run_dir,
                op=op,
                p=p,
                n_layer=n_layer,
                seed=42,
                steps=WARMUP_STEPS,
                weight_decay=WARMUP_WEIGHT_DECAY,
                optimizer=BASE_OPTIMIZER,
                lr=WARMUP_LR,
                min_lr=WARMUP_MIN_LR,
                lr_schedule=WARMUP_LR_SCHEDULE,
                warmup_steps=WARMUP_WARMUP_STEPS,
                grad_clip=WARMUP_GRAD_CLIP,
            ))

# Required 3 random restarts for (+, p=97, 1-layer)
restart_runs = []
executed_restarts = 0
for seed in [101, 202, 303]:
    run_dir = PART1_DIR / "warmup_restarts" / f"add_p97_l1_s{seed}"
    restart_runs.append({
        "name": run_dir.name,
        "run_dir": run_dir,
        "op": "+",
        "p": 97,
        "n_layer": 1,
        "seed": seed,
    })
    executed_restarts += int(run_train(
        run_dir=run_dir,
        op="+",
        p=97,
        n_layer=1,
        seed=seed,
        steps=WARMUP_STEPS,
        weight_decay=WARMUP_WEIGHT_DECAY,
        optimizer=BASE_OPTIMIZER,
        lr=WARMUP_LR,
        min_lr=WARMUP_MIN_LR,
        lr_schedule=WARMUP_LR_SCHEDULE,
        warmup_steps=WARMUP_WARMUP_STEPS,
        grad_clip=WARMUP_GRAD_CLIP,
    ))

print("warmup runs configured:", len(warmup_runs), "executed now:", executed_wa
print("restart runs configured:", len(restart_runs), "executed now:", executed_
```

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```
$ /opt/homebrew/Caskroom/miniconda/base/envs/cse493/bin/python train.py --mode
modular --op + --p 97 --n_layer 1 --n_head 4 --n_embd 128 --steps 100000 --see
d 42 --batch_size 128 --lr 0.001 --min_lr 0.0001 --lr_schedule cosine --warmup
_steps 1000 --grad_clip 1.0 --beta1 0.9 --beta2 0.95 --weight_decay 0.01 --opt
imizer adam --eval_interval 500 --log_interval 500 --out_dir '/Users/shobhit/D
ocuments/UW/CSE 493/cse493s-spring26-hw1/out/submission/part1/warmup/add_p97_l
1_s42' --bias --resume
number of parameters: 0.21M
[resume] model_last.pt not found in /Users/shobhit/Documents/UW/CSE 493/cse493
s-spring26-hw1/out/submission/part1/warmup/add_p97_l1_s42; starting fresh
device=mps train=2822 val=940 test=5647 batch_size=128
step=1 train_loss=4.680046 train_acc=0.0000 val_loss=4.690321 val_acc=0.0000 t
est_loss=4.690012 test_acc=0.0000
step=500 train_loss=4.291574 train_acc=0.0804 val_loss=4.851421 val_acc=0.0021
test_loss=4.842137 test_acc=0.0012
step=1000 train_loss=4.252802 train_acc=0.0840 val_loss=4.876426 val_acc=0.000
0 test_loss=4.872337 test_acc=0.0007
step=1500 train_loss=4.366595 train_acc=0.0517 val_loss=4.794629 val_acc=0.005
3 test_loss=4.789520 test_acc=0.0016
step=2000 train_loss=4.313904 train_acc=0.0539 val_loss=4.886852 val_acc=0.004
3 test_loss=4.876851 test_acc=0.0030
step=2500 train_loss=4.262407 train_acc=0.0581 val_loss=4.925606 val_acc=0.004
3 test_loss=4.934446 test_acc=0.0018
step=3000 train_loss=4.239371 train_acc=0.0620 val_loss=4.971265 val_acc=0.002
1 test_loss=4.976235 test_acc=0.0019
step=3500 train_loss=4.228320 train_acc=0.0670 val_loss=4.963817 val_acc=0.001
1 test_loss=4.970831 test_acc=0.0018
step=4000 train_loss=4.202864 train_acc=0.0719 val_loss=4.955878 val_acc=0.004
3 test_loss=4.984765 test_acc=0.0014
step=4500 train_loss=4.201541 train_acc=0.0687 val_loss=4.952686 val_acc=0.003
2 test_loss=4.975622 test_acc=0.0016
step=5000 train_loss=4.085493 train_acc=0.0836 val_loss=4.685148 val_acc=0.008
5 test_loss=4.716004 test_acc=0.0089
step=5500 train_loss=2.975625 train_acc=0.2052 val_loss=3.190342 val_acc=0.091
5 test_loss=3.198767 test_acc=0.0838
step=6000 train_loss=2.826987 train_acc=0.2155 val_loss=3.064893 val_acc=0.089
4 test_loss=3.080225 test_acc=0.0843
step=6500 train_loss=2.740622 train_acc=0.2452 val_loss=3.003129 val_acc=0.106
4 test_loss=3.018394 test_acc=0.0931
step=7000 train_loss=2.687376 train_acc=0.2477 val_loss=2.945366 val_acc=0.100
0 test_loss=2.956602 test_acc=0.1013
step=7500 train_loss=2.641785 train_acc=0.2583 val_loss=2.878980 val_acc=0.105
3 test_loss=2.875499 test_acc=0.1197
step=8000 train_loss=2.626563 train_acc=0.2438 val_loss=2.815283 val_acc=0.112
8 test_loss=2.813849 test_acc=0.1280
step=8500 train_loss=2.605451 train_acc=0.2580 val_loss=2.769430 val_acc=0.128
7 test_loss=2.777018 test_acc=0.1413
step=9000 train_loss=2.597300 train_acc=0.2473 val_loss=2.735114 val_acc=0.142
6 test_loss=2.743243 test_acc=0.1514
step=9500 train_loss=2.593339 train_acc=0.2449 val_loss=2.722503 val_acc=0.146
8 test_loss=2.729522 test_acc=0.1541
step=10000 train_loss=2.580122 train_acc=0.2197 val_loss=2.722600 val_acc=0.12
98 test_loss=2.730527 test_acc=0.1427
step=10500 train_loss=2.567456 train_acc=0.2707 val_loss=2.698981 val_acc=0.17
34 test_loss=2.710441 test_acc=0.1668
step=11000 train_loss=2.547450 train_acc=0.2792 val_loss=2.677479 val_acc=0.16
06 test_loss=2.677223 test_acc=0.1709
step=11500 train_loss=2.546251 train_acc=0.2771 val_loss=2.673181 val_acc=0.15
78417 test_acc=0.1613
step=12000 train_loss=2.520902 train_acc=0.2892 val_loss=2.639113 val_acc=0.18
```

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```
09 test_loss=2.636226 test_acc=0.1819
step=12500 train_loss=2.511727 train_acc=0.2725 val_loss=2.604005 val_acc=0.17
45 test_loss=2.605465 test_acc=0.1831
step=13000 train_loss=2.503371 train_acc=0.2757 val_loss=2.594815 val_acc=0.16
70 test_loss=2.595829 test_acc=0.1845
step=13500 train_loss=2.499043 train_acc=0.2807 val_loss=2.593399 val_acc=0.20
32 test_loss=2.593573 test_acc=0.1897
step=14000 train_loss=2.498539 train_acc=0.2459 val_loss=2.607878 val_acc=0.17
23 test_loss=2.605841 test_acc=0.1766
step=14500 train_loss=2.497939 train_acc=0.2580 val_loss=2.591759 val_acc=0.18
62 test_loss=2.591385 test_acc=0.1799
step=15000 train_loss=2.486823 train_acc=0.2771 val_loss=2.573613 val_acc=0.18
40 test_loss=2.577578 test_acc=0.1847
step=15500 train_loss=2.494568 train_acc=0.2881 val_loss=2.586677 val_acc=0.19
15 test_loss=2.580081 test_acc=0.1927
step=16000 train_loss=2.486585 train_acc=0.2771 val_loss=2.558024 val_acc=0.19
79 test_loss=2.565138 test_acc=0.1967
step=16500 train_loss=2.484625 train_acc=0.2807 val_loss=2.564706 val_acc=0.20
21 test_loss=2.567482 test_acc=0.1891
step=17000 train_loss=2.489149 train_acc=0.2934 val_loss=2.560464 val_acc=0.22
02 test_loss=2.563304 test_acc=0.2153
step=17500 train_loss=2.490251 train_acc=0.2764 val_loss=2.543699 val_acc=0.24
04 test_loss=2.545759 test_acc=0.2194
step=18000 train_loss=2.493398 train_acc=0.2647 val_loss=2.536788 val_acc=0.21
28 test_loss=2.533335 test_acc=0.2120
step=18500 train_loss=2.498817 train_acc=0.2718 val_loss=2.543392 val_acc=0.20
43 test_loss=2.538651 test_acc=0.2166
step=19000 train_loss=2.496688 train_acc=0.2931 val_loss=2.549536 val_acc=0.23
30 test_loss=2.540983 test_acc=0.2456
step=19500 train_loss=2.512744 train_acc=0.2583 val_loss=2.559118 val_acc=0.20
11 test_loss=2.556145 test_acc=0.2090
step=20000 train_loss=2.511000 train_acc=0.2707 val_loss=2.556017 val_acc=0.20
74 test_loss=2.555501 test_acc=0.2145
step=20500 train_loss=2.521388 train_acc=0.2590 val_loss=2.559144 val_acc=0.20
43 test_loss=2.560437 test_acc=0.2075
step=21000 train_loss=2.514118 train_acc=0.2771 val_loss=2.555961 val_acc=0.21
38 test_loss=2.555945 test_acc=0.2245
step=21500 train_loss=2.511430 train_acc=0.2392 val_loss=2.560234 val_acc=0.18
83 test_loss=2.554365 test_acc=0.1874
step=22000 train_loss=2.512918 train_acc=0.2768 val_loss=2.559187 val_acc=0.22
45 test_loss=2.556726 test_acc=0.2311
step=22500 train_loss=2.522642 train_acc=0.2736 val_loss=2.554632 val_acc=0.23
94 test_loss=2.560420 test_acc=0.2129
step=23000 train_loss=2.520593 train_acc=0.2732 val_loss=2.561226 val_acc=0.20
00 test_loss=2.561756 test_acc=0.2173
step=23500 train_loss=2.520580 train_acc=0.2959 val_loss=2.561782 val_acc=0.22
87 test_loss=2.559389 test_acc=0.2350
step=24000 train_loss=2.521790 train_acc=0.2491 val_loss=2.556531 val_acc=0.20
00 test_loss=2.557777 test_acc=0.1971
step=24500 train_loss=2.518021 train_acc=0.2665 val_loss=2.559878 val_acc=0.23
09 test_loss=2.563882 test_acc=0.2152
step=25000 train_loss=2.508178 train_acc=0.2661 val_loss=2.557975 val_acc=0.21
60 test_loss=2.551640 test_acc=0.2171
step=25500 train_loss=2.508416 train_acc=0.2679 val_loss=2.555358 val_acc=0.19
36 test_loss=2.548385 test_acc=0.2192
step=26000 train_loss=2.509482 train_acc=0.2640 val_loss=2.547774 val_acc=0.21
70 test_loss=2.545831 test_acc=0.2164
step=26500 train_loss=2.514992 train_acc=0.2544 val_loss=2.565622 val_acc=0.20
57198 test_acc=0.2141
step=27000 train_loss=2.517883 train_acc=0.2484 val_loss=2.557531 val_acc=0.20
```

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```
53 test_loss=2.562987 test_acc=0.1973
step=27500 train_loss=2.511964 train_acc=0.2796 val_loss=2.552083 val_acc=0.22
77 test_loss=2.555291 test_acc=0.2217
step=28000 train_loss=2.510786 train_acc=0.2732 val_loss=2.550921 val_acc=0.22
66 test_loss=2.553556 test_acc=0.2180
step=28500 train_loss=2.511601 train_acc=0.2491 val_loss=2.558044 val_acc=0.18
40 test_loss=2.556720 test_acc=0.1914
step=29000 train_loss=2.506987 train_acc=0.3005 val_loss=2.549455 val_acc=0.23
94 test_loss=2.552667 test_acc=0.2387
step=29500 train_loss=2.500973 train_acc=0.2778 val_loss=2.554263 val_acc=0.21
70 test_loss=2.547065 test_acc=0.2180
step=30000 train_loss=2.502951 train_acc=0.2498 val_loss=2.556037 val_acc=0.19
04 test_loss=2.549793 test_acc=0.1953
step=30500 train_loss=2.501319 train_acc=0.2371 val_loss=2.535905 val_acc=0.18
94 test_loss=2.539737 test_acc=0.1955
step=31000 train_loss=2.502086 train_acc=0.2778 val_loss=2.550952 val_acc=0.21
70 test_loss=2.547697 test_acc=0.2221
step=31500 train_loss=2.503458 train_acc=0.2406 val_loss=2.557065 val_acc=0.19
36 test_loss=2.548430 test_acc=0.1930
step=32000 train_loss=2.503925 train_acc=0.2775 val_loss=2.551655 val_acc=0.21
06 test_loss=2.544878 test_acc=0.2182
step=32500 train_loss=2.501864 train_acc=0.2874 val_loss=2.546367 val_acc=0.22
34 test_loss=2.541808 test_acc=0.2318
step=33000 train_loss=2.504027 train_acc=0.2509 val_loss=2.549071 val_acc=0.17
87 test_loss=2.546416 test_acc=0.1959
step=33500 train_loss=2.501125 train_acc=0.2725 val_loss=2.542060 val_acc=0.19
89 test_loss=2.539296 test_acc=0.2175
step=34000 train_loss=2.502450 train_acc=0.2615 val_loss=2.543656 val_acc=0.21
81 test_loss=2.546986 test_acc=0.2098
step=34500 train_loss=2.497976 train_acc=0.2785 val_loss=2.542978 val_acc=0.22
66 test_loss=2.541745 test_acc=0.2237
step=35000 train_loss=2.501318 train_acc=0.2814 val_loss=2.544883 val_acc=0.22
77 test_loss=2.542953 test_acc=0.2254
step=35500 train_loss=2.503520 train_acc=0.3051 val_loss=2.549046 val_acc=0.23
83 test_loss=2.544469 test_acc=0.2488
step=36000 train_loss=2.495910 train_acc=0.2463 val_loss=2.542694 val_acc=0.18
72 test_loss=2.543399 test_acc=0.1951
step=36500 train_loss=2.492253 train_acc=0.2899 val_loss=2.531820 val_acc=0.23
83 test_loss=2.537160 test_acc=0.2400
step=37000 train_loss=2.490844 train_acc=0.2980 val_loss=2.531253 val_acc=0.24
89 test_loss=2.529242 test_acc=0.2527
step=37500 train_loss=2.488593 train_acc=0.3051 val_loss=2.531035 val_acc=0.25
00 test_loss=2.530407 test_acc=0.2477
step=38000 train_loss=2.490300 train_acc=0.2810 val_loss=2.536173 val_acc=0.23
72 test_loss=2.536527 test_acc=0.2281
step=38500 train_loss=2.485962 train_acc=0.2739 val_loss=2.527419 val_acc=0.20
85 test_loss=2.522318 test_acc=0.2237
step=39000 train_loss=2.484541 train_acc=0.2647 val_loss=2.527853 val_acc=0.22
55 test_loss=2.529297 test_acc=0.2127
step=39500 train_loss=2.487487 train_acc=0.3009 val_loss=2.531737 val_acc=0.24
26 test_loss=2.526871 test_acc=0.2353
step=40000 train_loss=2.489526 train_acc=0.2573 val_loss=2.538394 val_acc=0.20
96 test_loss=2.536524 test_acc=0.2022
step=40500 train_loss=2.486360 train_acc=0.2495 val_loss=2.531284 val_acc=0.20
00 test_loss=2.524054 test_acc=0.2074
step=41000 train_loss=2.489846 train_acc=0.2530 val_loss=2.533050 val_acc=0.20
64 test_loss=2.533044 test_acc=0.2005
step=41500 train_loss=2.484125 train_acc=0.2750 val_loss=2.527815 val_acc=0.22
26850 test_acc=0.2233
step=42000 train_loss=2.481917 train_acc=0.3214 val_loss=2.524859 val_acc=0.25
```

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```
43 test_loss=2.522623 test_acc=0.2568
step=42500 train_loss=2.482761 train_acc=0.2711 val_loss=2.525998 val_acc=0.20
74 test_loss=2.522852 test_acc=0.2160
step=43000 train_loss=2.480908 train_acc=0.2714 val_loss=2.521709 val_acc=0.20
32 test_loss=2.517132 test_acc=0.2187
step=43500 train_loss=2.483667 train_acc=0.2619 val_loss=2.535961 val_acc=0.19
26 test_loss=2.531928 test_acc=0.2150
step=44000 train_loss=2.478145 train_acc=0.2502 val_loss=2.511646 val_acc=0.20
53 test_loss=2.518571 test_acc=0.1962
step=44500 train_loss=2.481050 train_acc=0.3179 val_loss=2.529216 val_acc=0.26
49 test_loss=2.524625 test_acc=0.2570
step=45000 train_loss=2.472929 train_acc=0.2856 val_loss=2.518191 val_acc=0.22
87 test_loss=2.515878 test_acc=0.2191
step=45500 train_loss=2.473333 train_acc=0.2874 val_loss=2.520364 val_acc=0.22
55 test_loss=2.514816 test_acc=0.2384
step=46000 train_loss=2.472397 train_acc=0.2693 val_loss=2.522203 val_acc=0.20
32 test_loss=2.518185 test_acc=0.2150
step=46500 train_loss=2.476733 train_acc=0.3154 val_loss=2.518159 val_acc=0.27
23 test_loss=2.518863 test_acc=0.2467
step=47000 train_loss=2.477690 train_acc=0.2605 val_loss=2.519924 val_acc=0.18
83 test_loss=2.516710 test_acc=0.2191
step=47500 train_loss=2.471880 train_acc=0.3207 val_loss=2.517470 val_acc=0.24
57 test_loss=2.510860 test_acc=0.2522
step=48000 train_loss=2.472548 train_acc=0.2799 val_loss=2.513973 val_acc=0.21
91 test_loss=2.512431 test_acc=0.2233
step=48500 train_loss=2.466011 train_acc=0.2605 val_loss=2.511992 val_acc=0.21
28 test_loss=2.507069 test_acc=0.2047
step=49000 train_loss=2.460642 train_acc=0.3140 val_loss=2.505070 val_acc=0.25
74 test_loss=2.507584 test_acc=0.2534
step=49500 train_loss=2.465867 train_acc=0.2959 val_loss=2.514051 val_acc=0.24
57 test_loss=2.513805 test_acc=0.2391
step=50000 train_loss=2.469513 train_acc=0.2803 val_loss=2.517087 val_acc=0.22
02 test_loss=2.513032 test_acc=0.2231
step=50500 train_loss=2.469593 train_acc=0.2803 val_loss=2.518390 val_acc=0.23
09 test_loss=2.514794 test_acc=0.2261
step=51000 train_loss=2.463053 train_acc=0.3147 val_loss=2.505549 val_acc=0.25
85 test_loss=2.502486 test_acc=0.2564
step=51500 train_loss=2.463096 train_acc=0.3221 val_loss=2.504173 val_acc=0.25
53 test_loss=2.503266 test_acc=0.2617
step=52000 train_loss=2.465979 train_acc=0.2931 val_loss=2.512372 val_acc=0.23
62 test_loss=2.509662 test_acc=0.2412
step=52500 train_loss=2.463464 train_acc=0.3395 val_loss=2.505428 val_acc=0.25
32 test_loss=2.503642 test_acc=0.2732
step=53000 train_loss=2.456976 train_acc=0.2906 val_loss=2.495411 val_acc=0.24
04 test_loss=2.496974 test_acc=0.2401
step=53500 train_loss=2.456422 train_acc=0.3264 val_loss=2.501288 val_acc=0.24
89 test_loss=2.495266 test_acc=0.2603
step=54000 train_loss=2.459181 train_acc=0.2970 val_loss=2.495804 val_acc=0.21
81 test_loss=2.501411 test_acc=0.2403
step=54500 train_loss=2.454763 train_acc=0.2916 val_loss=2.499371 val_acc=0.24
89 test_loss=2.499461 test_acc=0.2357
step=55000 train_loss=2.456887 train_acc=0.2668 val_loss=2.500918 val_acc=0.20
64 test_loss=2.502184 test_acc=0.2116
step=55500 train_loss=2.456501 train_acc=0.3150 val_loss=2.507620 val_acc=0.24
04 test_loss=2.502562 test_acc=0.2469
step=56000 train_loss=2.456492 train_acc=0.2789 val_loss=2.501636 val_acc=0.21
06 test_loss=2.497350 test_acc=0.2208
step=56500 train_loss=2.454083 train_acc=0.3140 val_loss=2.495660 val_acc=0.24
95959 test_acc=0.2515
step=57000 train_loss=2.451831 train_acc=0.2718 val_loss=2.495192 val_acc=0.22
```

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```
23 test_loss=2.495176 test_acc=0.2148
step=57500 train_loss=2.449980 train_acc=0.3097 val_loss=2.498448 val_acc=0.22
66 test_loss=2.492434 test_acc=0.2479
step=58000 train_loss=2.451898 train_acc=0.3083 val_loss=2.496542 val_acc=0.23
51 test_loss=2.494370 test_acc=0.2419
step=58500 train_loss=2.451557 train_acc=0.2909 val_loss=2.494799 val_acc=0.21
17 test_loss=2.494196 test_acc=0.2251
step=59000 train_loss=2.449316 train_acc=0.2941 val_loss=2.496895 val_acc=0.21
91 test_loss=2.489702 test_acc=0.2341
step=59500 train_loss=2.449485 train_acc=0.2771 val_loss=2.494266 val_acc=0.21
91 test_loss=2.491055 test_acc=0.2159
step=60000 train_loss=2.447207 train_acc=0.2966 val_loss=2.487997 val_acc=0.24
15 test_loss=2.489079 test_acc=0.2385
step=60500 train_loss=2.442447 train_acc=0.2987 val_loss=2.484715 val_acc=0.23
30 test_loss=2.484642 test_acc=0.2376
step=61000 train_loss=2.444039 train_acc=0.2977 val_loss=2.485974 val_acc=0.24
26 test_loss=2.482370 test_acc=0.2391
step=61500 train_loss=2.444824 train_acc=0.2984 val_loss=2.486161 val_acc=0.24
36 test_loss=2.483824 test_acc=0.2424
step=62000 train_loss=2.440870 train_acc=0.3469 val_loss=2.486743 val_acc=0.27
34 test_loss=2.482951 test_acc=0.2750
step=62500 train_loss=2.439689 train_acc=0.2931 val_loss=2.483198 val_acc=0.23
83 test_loss=2.479230 test_acc=0.2389
step=63000 train_loss=2.439520 train_acc=0.3296 val_loss=2.484056 val_acc=0.27
23 test_loss=2.483006 test_acc=0.2672
step=63500 train_loss=2.443048 train_acc=0.2980 val_loss=2.484366 val_acc=0.24
36 test_loss=2.483865 test_acc=0.2389
step=64000 train_loss=2.437643 train_acc=0.3636 val_loss=2.480541 val_acc=0.29
15 test_loss=2.479764 test_acc=0.2865
step=64500 train_loss=2.438331 train_acc=0.2842 val_loss=2.487981 val_acc=0.21
91 test_loss=2.480161 test_acc=0.2336
step=65000 train_loss=2.435736 train_acc=0.3225 val_loss=2.481805 val_acc=0.25
32 test_loss=2.478987 test_acc=0.2635
step=65500 train_loss=2.435844 train_acc=0.3395 val_loss=2.477734 val_acc=0.27
13 test_loss=2.479382 test_acc=0.2633
step=66000 train_loss=2.434454 train_acc=0.2807 val_loss=2.480571 val_acc=0.21
70 test_loss=2.475410 test_acc=0.2293
step=66500 train_loss=2.432741 train_acc=0.2768 val_loss=2.476095 val_acc=0.21
60 test_loss=2.474118 test_acc=0.2164
step=67000 train_loss=2.431967 train_acc=0.3349 val_loss=2.472960 val_acc=0.25
53 test_loss=2.472614 test_acc=0.2658
step=67500 train_loss=2.432959 train_acc=0.3161 val_loss=2.474685 val_acc=0.25
53 test_loss=2.473081 test_acc=0.2515
step=68000 train_loss=2.431480 train_acc=0.2941 val_loss=2.474644 val_acc=0.22
77 test_loss=2.473389 test_acc=0.2330
step=68500 train_loss=2.431157 train_acc=0.3218 val_loss=2.473451 val_acc=0.25
11 test_loss=2.469970 test_acc=0.2571
step=69000 train_loss=2.430185 train_acc=0.3147 val_loss=2.473227 val_acc=0.24
36 test_loss=2.473052 test_acc=0.2548
step=69500 train_loss=2.429356 train_acc=0.3246 val_loss=2.473183 val_acc=0.24
68 test_loss=2.471187 test_acc=0.2639
step=70000 train_loss=2.427383 train_acc=0.3285 val_loss=2.470674 val_acc=0.25
32 test_loss=2.469804 test_acc=0.2663
step=70500 train_loss=2.426522 train_acc=0.2853 val_loss=2.470089 val_acc=0.23
40 test_loss=2.466698 test_acc=0.2307
step=71000 train_loss=2.424802 train_acc=0.2987 val_loss=2.467966 val_acc=0.22
77 test_loss=2.466130 test_acc=0.2348
step=71500 train_loss=2.423676 train_acc=0.2980 val_loss=2.467859 val_acc=0.24
64895 test_acc=0.2330
step=72000 train_loss=2.424980 train_acc=0.3168 val_loss=2.465575 val_acc=0.26
```

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```
28 test_loss=2.466925 test_acc=0.2532
step=72500 train_loss=2.421915 train_acc=0.2991 val_loss=2.467360 val_acc=0.22
55 test_loss=2.464314 test_acc=0.2304
step=73000 train_loss=2.424012 train_acc=0.3186 val_loss=2.471253 val_acc=0.25
00 test_loss=2.469499 test_acc=0.2555
step=73500 train_loss=2.423826 train_acc=0.3111 val_loss=2.471885 val_acc=0.24
26 test_loss=2.466706 test_acc=0.2481
step=74000 train_loss=2.427900 train_acc=0.3242 val_loss=2.472628 val_acc=0.25
85 test_loss=2.469504 test_acc=0.2552
step=74500 train_loss=2.422613 train_acc=0.3097 val_loss=2.466271 val_acc=0.22
77 test_loss=2.462677 test_acc=0.2554
step=75000 train_loss=2.423463 train_acc=0.3544 val_loss=2.467867 val_acc=0.27
13 test_loss=2.465999 test_acc=0.2814
step=75500 train_loss=2.422587 train_acc=0.2913 val_loss=2.472810 val_acc=0.23
72 test_loss=2.466409 test_acc=0.2260
step=76000 train_loss=2.420657 train_acc=0.3310 val_loss=2.458880 val_acc=0.26
81 test_loss=2.461822 test_acc=0.2598
step=76500 train_loss=2.418584 train_acc=0.3168 val_loss=2.462651 val_acc=0.25
53 test_loss=2.460409 test_acc=0.2559
step=77000 train_loss=2.417803 train_acc=0.3306 val_loss=2.460645 val_acc=0.27
13 test_loss=2.460571 test_acc=0.2672
step=77500 train_loss=2.418392 train_acc=0.3086 val_loss=2.462414 val_acc=0.24
57 test_loss=2.460033 test_acc=0.2485
step=78000 train_loss=2.417700 train_acc=0.3242 val_loss=2.463019 val_acc=0.25
85 test_loss=2.459364 test_acc=0.2575
step=78500 train_loss=2.416389 train_acc=0.3086 val_loss=2.460156 val_acc=0.23
40 test_loss=2.457723 test_acc=0.2481
step=79000 train_loss=2.415656 train_acc=0.3636 val_loss=2.458356 val_acc=0.30
53 test_loss=2.455186 test_acc=0.2934
step=79500 train_loss=2.414269 train_acc=0.3143 val_loss=2.455023 val_acc=0.26
06 test_loss=2.454588 test_acc=0.2469
step=80000 train_loss=2.412689 train_acc=0.3299 val_loss=2.455525 val_acc=0.26
70 test_loss=2.454366 test_acc=0.2559
step=80500 train_loss=2.413091 train_acc=0.3288 val_loss=2.459016 val_acc=0.25
21 test_loss=2.454822 test_acc=0.2628
step=81000 train_loss=2.412585 train_acc=0.3317 val_loss=2.455776 val_acc=0.27
02 test_loss=2.454577 test_acc=0.2725
step=81500 train_loss=2.411339 train_acc=0.3232 val_loss=2.456682 val_acc=0.26
49 test_loss=2.453366 test_acc=0.2658
step=82000 train_loss=2.411001 train_acc=0.3157 val_loss=2.456372 val_acc=0.24
79 test_loss=2.453366 test_acc=0.2536
step=82500 train_loss=2.410830 train_acc=0.3186 val_loss=2.453148 val_acc=0.26
17 test_loss=2.451697 test_acc=0.2536
step=83000 train_loss=2.409490 train_acc=0.3164 val_loss=2.450005 val_acc=0.24
47 test_loss=2.449935 test_acc=0.2515
step=83500 train_loss=2.409937 train_acc=0.3451 val_loss=2.452085 val_acc=0.27
98 test_loss=2.449548 test_acc=0.2763
step=84000 train_loss=2.409437 train_acc=0.3161 val_loss=2.456737 val_acc=0.24
57 test_loss=2.452909 test_acc=0.2532
step=84500 train_loss=2.408330 train_acc=0.3101 val_loss=2.454137 val_acc=0.24
15 test_loss=2.451892 test_acc=0.2449
step=85000 train_loss=2.407643 train_acc=0.3398 val_loss=2.451968 val_acc=0.27
23 test_loss=2.448628 test_acc=0.2709
step=85500 train_loss=2.407390 train_acc=0.3246 val_loss=2.449530 val_acc=0.25
74 test_loss=2.449721 test_acc=0.2545
step=86000 train_loss=2.408503 train_acc=0.3342 val_loss=2.452173 val_acc=0.25
53 test_loss=2.452347 test_acc=0.2649
step=86500 train_loss=2.406988 train_acc=0.3253 val_loss=2.451241 val_acc=0.25
50059 test_acc=0.2642
step=87000 train_loss=2.406175 train_acc=0.3342 val_loss=2.452259 val_acc=0.26
```

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```
28 test_loss=2.449194 test_acc=0.2672
step=87500 train_loss=2.405966 train_acc=0.3395 val_loss=2.449789 val_acc=0.28
40 test_loss=2.447199 test_acc=0.2747
step=88000 train_loss=2.405213 train_acc=0.3420 val_loss=2.446824 val_acc=0.27
45 test_loss=2.445544 test_acc=0.2716
step=88500 train_loss=2.405369 train_acc=0.3161 val_loss=2.448758 val_acc=0.25
32 test_loss=2.446624 test_acc=0.2539
step=89000 train_loss=2.405179 train_acc=0.3253 val_loss=2.448627 val_acc=0.27
55 test_loss=2.446867 test_acc=0.2596
step=89500 train_loss=2.404967 train_acc=0.3427 val_loss=2.448132 val_acc=0.26
06 test_loss=2.444742 test_acc=0.2706
step=90000 train_loss=2.404623 train_acc=0.2952 val_loss=2.447340 val_acc=0.24
36 test_loss=2.444964 test_acc=0.2332
step=90500 train_loss=2.404465 train_acc=0.3203 val_loss=2.444910 val_acc=0.25
32 test_loss=2.446855 test_acc=0.2577
step=91000 train_loss=2.402957 train_acc=0.3200 val_loss=2.446516 val_acc=0.25
64 test_loss=2.444667 test_acc=0.2573
step=91500 train_loss=2.402597 train_acc=0.3331 val_loss=2.445907 val_acc=0.26
60 test_loss=2.444633 test_acc=0.2646
step=92000 train_loss=2.402969 train_acc=0.3377 val_loss=2.447023 val_acc=0.25
85 test_loss=2.445548 test_acc=0.2697
step=92500 train_loss=2.403064 train_acc=0.3225 val_loss=2.446264 val_acc=0.26
60 test_loss=2.445020 test_acc=0.2543
step=93000 train_loss=2.402340 train_acc=0.3161 val_loss=2.449010 val_acc=0.26
06 test_loss=2.446767 test_acc=0.2476
step=93500 train_loss=2.401796 train_acc=0.3094 val_loss=2.448927 val_acc=0.22
02 test_loss=2.443999 test_acc=0.2456
step=94000 train_loss=2.401289 train_acc=0.3249 val_loss=2.446194 val_acc=0.25
96 test_loss=2.443526 test_acc=0.2564
step=94500 train_loss=2.401224 train_acc=0.3246 val_loss=2.446095 val_acc=0.25
96 test_loss=2.445333 test_acc=0.2582
step=95000 train_loss=2.401171 train_acc=0.3480 val_loss=2.445818 val_acc=0.27
98 test_loss=2.443060 test_acc=0.2817
step=95500 train_loss=2.400913 train_acc=0.3249 val_loss=2.444989 val_acc=0.25
64 test_loss=2.443060 test_acc=0.2555
step=96000 train_loss=2.399952 train_acc=0.3228 val_loss=2.443539 val_acc=0.26
17 test_loss=2.440347 test_acc=0.2608
step=96500 train_loss=2.399289 train_acc=0.2973 val_loss=2.443129 val_acc=0.23
83 test_loss=2.440220 test_acc=0.2415
step=97000 train_loss=2.398763 train_acc=0.3441 val_loss=2.443342 val_acc=0.28
51 test_loss=2.441491 test_acc=0.2770
step=97500 train_loss=2.398860 train_acc=0.3398 val_loss=2.442025 val_acc=0.26
60 test_loss=2.441518 test_acc=0.2711
step=98000 train_loss=2.399752 train_acc=0.3412 val_loss=2.444038 val_acc=0.26
91 test_loss=2.443902 test_acc=0.2701
step=98500 train_loss=2.399891 train_acc=0.3257 val_loss=2.442884 val_acc=0.26
49 test_loss=2.441019 test_acc=0.2651
step=99000 train_loss=2.399661 train_acc=0.3235 val_loss=2.439905 val_acc=0.25
21 test_loss=2.439464 test_acc=0.2580
step=99500 train_loss=2.399457 train_acc=0.3809 val_loss=2.443662 val_acc=0.29
47 test_loss=2.441807 test_acc=0.3060
step=100000 train_loss=2.399731 train_acc=0.3583 val_loss=2.441948 val_acc=0.2
809 test_loss=2.441063 test_acc=0.2794
best metrics: {
  "step": 99000,
  "lr": 0.00010022655592756672,
  "train_loss": 2.399661273165513,
  "train_acc": 0.3235294117647059,
  "val_loss": 2.4399045568831426,
  "val_acc": 0.2521276595744681,
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```
  "test_loss": 2.4394639558827578,
  "test_acc": 0.2580131043031698,
  "elapsed_sec": 819.7059659957886
}
checkpoint dir: /Users/shobhit/Documents/UW/CSE 493/cse493s-spring26-hw1/out/s
ubmission/part1/warmup/add_p97_l1_s42

$ /opt/homebrew/Caskroom/miniconda/base/envs/cse493/bin/python train.py --mode
modular --op + --p 97 --n_layer 2 --n_head 4 --n_embd 128 --steps 100000 --see
d 42 --batch_size 128 --lr 0.001 --min_lr 0.0001 --lr_schedule cosine --warmup
_steps 1000 --grad_clip 1.0 --beta1 0.9 --beta2 0.95 --weight_decay 0.01 --opt
imizer adam --eval_interval 500 --log_interval 500 --out_dir '/Users/shobhit/D
ocuments/UW/CSE 493/cse493s-spring26-hw1/out/submission/part1/warmup/add_p97_l
2_s42' --bias --resume
number of parameters: 0.41M
[resume] model_last.pt not found in /Users/shobhit/Documents/UW/CSE 493/cse493
s-spring26-hw1/out/submission/part1/warmup/add_p97_l2_s42; starting fresh
device=mps train=2822 val=940 test=5647 batch_size=128
step=1 train_loss=4.689393 train_acc=0.0000 val_loss=4.691188 val_acc=0.0000 t
est_loss=4.690660 test_acc=0.0000
step=500 train_loss=4.333275 train_acc=0.0634 val_loss=4.802538 val_acc=0.0053
test_loss=4.820375 test_acc=0.0025
step=1000 train_loss=4.259060 train_acc=0.0751 val_loss=4.860297 val_acc=0.000
0 test_loss=4.871773 test_acc=0.0021
step=1500 train_loss=4.385791 train_acc=0.0524 val_loss=4.767827 val_acc=0.000
0 test_loss=4.773481 test_acc=0.0012
step=2000 train_loss=4.326517 train_acc=0.0532 val_loss=4.859171 val_acc=0.007
4 test_loss=4.864645 test_acc=0.0028
step=2500 train_loss=4.285777 train_acc=0.0585 val_loss=4.883049 val_acc=0.002
1 test_loss=4.913794 test_acc=0.0018
step=3000 train_loss=4.262290 train_acc=0.0638 val_loss=4.923098 val_acc=0.002
1 test_loss=4.943103 test_acc=0.0019
step=3500 train_loss=4.251103 train_acc=0.0695 val_loss=4.910588 val_acc=0.000
0 test_loss=4.937726 test_acc=0.0019
step=4000 train_loss=4.242089 train_acc=0.0634 val_loss=4.941747 val_acc=0.000
0 test_loss=4.959186 test_acc=0.0005
step=4500 train_loss=4.229495 train_acc=0.0673 val_loss=4.943945 val_acc=0.004
3 test_loss=4.947285 test_acc=0.0021
step=5000 train_loss=4.181971 train_acc=0.0751 val_loss=4.858868 val_acc=0.006
4 test_loss=4.861491 test_acc=0.0058
step=5500 train_loss=3.720132 train_acc=0.0953 val_loss=3.951950 val_acc=0.045
7 test_loss=3.962368 test_acc=0.0352
step=6000 train_loss=2.926617 train_acc=0.2009 val_loss=3.099427 val_acc=0.085
1 test_loss=3.104523 test_acc=0.0905
step=6500 train_loss=2.742839 train_acc=0.2364 val_loss=2.930387 val_acc=0.114
9 test_loss=2.942706 test_acc=0.1096
step=7000 train_loss=2.676455 train_acc=0.2392 val_loss=2.892224 val_acc=0.104
3 test_loss=2.903222 test_acc=0.1132
step=7500 train_loss=2.648936 train_acc=0.2601 val_loss=2.848167 val_acc=0.148
9 test_loss=2.849788 test_acc=0.1374
step=8000 train_loss=2.627845 train_acc=0.2566 val_loss=2.813328 val_acc=0.140
4 test_loss=2.806661 test_acc=0.1408
step=8500 train_loss=2.598955 train_acc=0.2605 val_loss=2.765545 val_acc=0.138
3 test_loss=2.774818 test_acc=0.1369
step=9000 train_loss=2.551148 train_acc=0.2718 val_loss=2.697375 val_acc=0.155
3 test_loss=2.698288 test_acc=0.1557
step=9500 train_loss=2.538337 train_acc=0.2771 val_loss=2.646909 val_acc=0.173
4 test_loss=2.646186 test_acc=0.1852
step=10000 train_loss=2.521547 train_acc=0.2452 val_loss=2.610778 val_acc=0.16
49 test_loss=2.611599 test_acc=0.1672
```

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```
step=10500 train_loss=2.517276 train_acc=0.2877 val_loss=2.607226 val_acc=0.20
85 test_loss=2.611562 test_acc=0.1996
step=11000 train_loss=2.502353 train_acc=0.2764 val_loss=2.602182 val_acc=0.19
26 test_loss=2.600173 test_acc=0.1829
step=11500 train_loss=2.508415 train_acc=0.2622 val_loss=2.605891 val_acc=0.16
60 test_loss=2.610211 test_acc=0.1725
step=12000 train_loss=2.505392 train_acc=0.2792 val_loss=2.604056 val_acc=0.17
34 test_loss=2.600893 test_acc=0.1849
step=12500 train_loss=2.495790 train_acc=0.2782 val_loss=2.587775 val_acc=0.19
15 test_loss=2.588515 test_acc=0.1838
step=13000 train_loss=2.491762 train_acc=0.2757 val_loss=2.583105 val_acc=0.17
98 test_loss=2.584559 test_acc=0.1849
step=13500 train_loss=2.490241 train_acc=0.2934 val_loss=2.584758 val_acc=0.19
68 test_loss=2.582606 test_acc=0.1902
step=14000 train_loss=2.488643 train_acc=0.2519 val_loss=2.597054 val_acc=0.17
23 test_loss=2.597197 test_acc=0.1684
step=14500 train_loss=2.503355 train_acc=0.2654 val_loss=2.591879 val_acc=0.17
87 test_loss=2.591458 test_acc=0.1847
step=15000 train_loss=2.487953 train_acc=0.2736 val_loss=2.572584 val_acc=0.19
68 test_loss=2.576192 test_acc=0.1858
step=15500 train_loss=2.494087 train_acc=0.2934 val_loss=2.586300 val_acc=0.18
94 test_loss=2.579058 test_acc=0.2003
step=16000 train_loss=2.485361 train_acc=0.2831 val_loss=2.561736 val_acc=0.19
26 test_loss=2.567866 test_acc=0.1877
step=16500 train_loss=2.488093 train_acc=0.2920 val_loss=2.572272 val_acc=0.20
85 test_loss=2.575252 test_acc=0.1948
step=17000 train_loss=2.481635 train_acc=0.3079 val_loss=2.563243 val_acc=0.20
96 test_loss=2.568182 test_acc=0.2093
step=17500 train_loss=2.490675 train_acc=0.2732 val_loss=2.559775 val_acc=0.21
81 test_loss=2.562831 test_acc=0.2019
step=18000 train_loss=2.494086 train_acc=0.2544 val_loss=2.546457 val_acc=0.18
30 test_loss=2.543999 test_acc=0.1918
step=18500 train_loss=2.488004 train_acc=0.2644 val_loss=2.533605 val_acc=0.20
00 test_loss=2.529352 test_acc=0.2056
step=19000 train_loss=2.492209 train_acc=0.2707 val_loss=2.544653 val_acc=0.23
62 test_loss=2.536634 test_acc=0.2277
step=19500 train_loss=2.490548 train_acc=0.2916 val_loss=2.539969 val_acc=0.22
45 test_loss=2.536375 test_acc=0.2300
step=20000 train_loss=2.494300 train_acc=0.2534 val_loss=2.538055 val_acc=0.20
32 test_loss=2.539181 test_acc=0.2049
step=20500 train_loss=2.490598 train_acc=0.2760 val_loss=2.530772 val_acc=0.22
23 test_loss=2.530640 test_acc=0.2244
step=21000 train_loss=2.482827 train_acc=0.2711 val_loss=2.525886 val_acc=0.21
06 test_loss=2.524497 test_acc=0.2176
step=21500 train_loss=2.486866 train_acc=0.2640 val_loss=2.537837 val_acc=0.20
74 test_loss=2.529765 test_acc=0.2063
step=22000 train_loss=2.486758 train_acc=0.2807 val_loss=2.533324 val_acc=0.21
91 test_loss=2.530946 test_acc=0.2309
step=22500 train_loss=2.492143 train_acc=0.2597 val_loss=2.524764 val_acc=0.22
98 test_loss=2.530378 test_acc=0.2068
step=23000 train_loss=2.498216 train_acc=0.2888 val_loss=2.537130 val_acc=0.21
70 test_loss=2.537799 test_acc=0.2286
step=23500 train_loss=2.496026 train_acc=0.2860 val_loss=2.538818 val_acc=0.22
55 test_loss=2.534574 test_acc=0.2304
step=24000 train_loss=2.514669 train_acc=0.2413 val_loss=2.548702 val_acc=0.20
21 test_loss=2.549808 test_acc=0.2040
step=24500 train_loss=2.511415 train_acc=0.2704 val_loss=2.551892 val_acc=0.22
66 test_loss=2.557560 test_acc=0.2136
step=25000 train_loss=2.515238 train_acc=0.2679 val_loss=2.564115 val_acc=0.19
79 test_loss=2.559261 test_acc=0.2123
```

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```
step=25500 train_loss=2.513630 train_acc=0.2619 val_loss=2.559923 val_acc=0.20
43 test_loss=2.553805 test_acc=0.2065
step=26000 train_loss=2.515318 train_acc=0.2537 val_loss=2.553148 val_acc=0.20
74 test_loss=2.552046 test_acc=0.2083
step=26500 train_loss=2.517021 train_acc=0.2679 val_loss=2.566450 val_acc=0.19
79 test_loss=2.557933 test_acc=0.2173
step=27000 train_loss=2.516503 train_acc=0.2413 val_loss=2.555682 val_acc=0.19
79 test_loss=2.561264 test_acc=0.1934
step=27500 train_loss=2.517474 train_acc=0.2796 val_loss=2.559700 val_acc=0.21
91 test_loss=2.560119 test_acc=0.2240
step=28000 train_loss=2.507958 train_acc=0.2877 val_loss=2.549432 val_acc=0.21
70 test_loss=2.551809 test_acc=0.2206
step=28500 train_loss=2.513476 train_acc=0.2275 val_loss=2.560748 val_acc=0.17
98 test_loss=2.559111 test_acc=0.1755
step=29000 train_loss=2.511549 train_acc=0.2807 val_loss=2.555227 val_acc=0.22
77 test_loss=2.556448 test_acc=0.2253
step=29500 train_loss=2.511469 train_acc=0.2888 val_loss=2.563343 val_acc=0.22
98 test_loss=2.557160 test_acc=0.2307
step=30000 train_loss=2.519991 train_acc=0.2349 val_loss=2.573592 val_acc=0.18
62 test_loss=2.567458 test_acc=0.1803
step=30500 train_loss=2.509168 train_acc=0.2162 val_loss=2.542603 val_acc=0.16
81 test_loss=2.547526 test_acc=0.1804
step=31000 train_loss=2.502883 train_acc=0.2814 val_loss=2.552293 val_acc=0.23
30 test_loss=2.548506 test_acc=0.2276
step=31500 train_loss=2.512931 train_acc=0.2509 val_loss=2.566167 val_acc=0.19
36 test_loss=2.558047 test_acc=0.2010
step=32000 train_loss=2.510626 train_acc=0.2764 val_loss=2.557603 val_acc=0.21
49 test_loss=2.550991 test_acc=0.2256
step=32500 train_loss=2.514893 train_acc=0.2948 val_loss=2.559480 val_acc=0.22
77 test_loss=2.555700 test_acc=0.2375
step=33000 train_loss=2.503557 train_acc=0.2473 val_loss=2.549703 val_acc=0.17
98 test_loss=2.546655 test_acc=0.1874
step=33500 train_loss=2.504463 train_acc=0.2707 val_loss=2.545295 val_acc=0.20
74 test_loss=2.542917 test_acc=0.2224
step=34000 train_loss=2.503235 train_acc=0.2530 val_loss=2.544384 val_acc=0.19
68 test_loss=2.547474 test_acc=0.2075
step=34500 train_loss=2.509633 train_acc=0.2842 val_loss=2.555583 val_acc=0.23
51 test_loss=2.551581 test_acc=0.2297
step=35000 train_loss=2.505692 train_acc=0.2977 val_loss=2.551460 val_acc=0.24
47 test_loss=2.546932 test_acc=0.2433
step=35500 train_loss=2.506007 train_acc=0.3023 val_loss=2.551492 val_acc=0.24
15 test_loss=2.547313 test_acc=0.2437
step=36000 train_loss=2.500129 train_acc=0.2523 val_loss=2.546903 val_acc=0.19
26 test_loss=2.547225 test_acc=0.1992
step=36500 train_loss=2.500499 train_acc=0.2938 val_loss=2.539714 val_acc=0.23
19 test_loss=2.544474 test_acc=0.2430
step=37000 train_loss=2.498998 train_acc=0.2835 val_loss=2.541510 val_acc=0.24
04 test_loss=2.537423 test_acc=0.2334
step=37500 train_loss=2.498090 train_acc=0.2828 val_loss=2.543355 val_acc=0.22
13 test_loss=2.541053 test_acc=0.2276
step=38000 train_loss=2.499335 train_acc=0.2877 val_loss=2.545220 val_acc=0.23
72 test_loss=2.546870 test_acc=0.2357
step=38500 train_loss=2.497685 train_acc=0.2608 val_loss=2.538996 val_acc=0.20
21 test_loss=2.534007 test_acc=0.2127
step=39000 train_loss=2.502525 train_acc=0.2711 val_loss=2.547429 val_acc=0.22
13 test_loss=2.546317 test_acc=0.2219
step=39500 train_loss=2.493078 train_acc=0.2962 val_loss=2.539036 val_acc=0.23
30 test_loss=2.532192 test_acc=0.2325
step=40000 train_loss=2.493172 train_acc=0.2502 val_loss=2.543468 val_acc=0.20
43 test_loss=2.540023 test_acc=0.2070
```

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```
step=40500 train_loss=2.495045 train_acc=0.2679 val_loss=2.540827 val_acc=0.21
49 test_loss=2.533050 test_acc=0.2162
step=41000 train_loss=2.493500 train_acc=0.2406 val_loss=2.535209 val_acc=0.20
32 test_loss=2.536639 test_acc=0.1900
step=41500 train_loss=2.489933 train_acc=0.2697 val_loss=2.534096 val_acc=0.21
70 test_loss=2.533877 test_acc=0.2157
step=42000 train_loss=2.487471 train_acc=0.3101 val_loss=2.530976 val_acc=0.25
53 test_loss=2.528321 test_acc=0.2495
step=42500 train_loss=2.485993 train_acc=0.2519 val_loss=2.530431 val_acc=0.20
11 test_loss=2.526012 test_acc=0.2013
step=43000 train_loss=2.484260 train_acc=0.2775 val_loss=2.526460 val_acc=0.21
28 test_loss=2.521482 test_acc=0.2206
step=43500 train_loss=2.484859 train_acc=0.2707 val_loss=2.535218 val_acc=0.20
43 test_loss=2.532270 test_acc=0.2113
step=44000 train_loss=2.484728 train_acc=0.2509 val_loss=2.519597 val_acc=0.20
74 test_loss=2.525312 test_acc=0.1955
step=44500 train_loss=2.474928 train_acc=0.3239 val_loss=2.522465 val_acc=0.26
81 test_loss=2.517333 test_acc=0.2646
step=45000 train_loss=2.475708 train_acc=0.2977 val_loss=2.519755 val_acc=0.24
15 test_loss=2.516801 test_acc=0.2332
step=45500 train_loss=2.474786 train_acc=0.2877 val_loss=2.521850 val_acc=0.22
55 test_loss=2.515536 test_acc=0.2306
step=46000 train_loss=2.477327 train_acc=0.2608 val_loss=2.528418 val_acc=0.20
43 test_loss=2.522931 test_acc=0.2090
step=46500 train_loss=2.476104 train_acc=0.3115 val_loss=2.516976 val_acc=0.27
02 test_loss=2.517049 test_acc=0.2483
step=47000 train_loss=2.475716 train_acc=0.2686 val_loss=2.517158 val_acc=0.20
21 test_loss=2.514452 test_acc=0.2231
step=47500 train_loss=2.474397 train_acc=0.3101 val_loss=2.521356 val_acc=0.23
72 test_loss=2.512806 test_acc=0.2477
step=48000 train_loss=2.472647 train_acc=0.2757 val_loss=2.515096 val_acc=0.22
55 test_loss=2.513040 test_acc=0.2203
step=48500 train_loss=2.480106 train_acc=0.2970 val_loss=2.526780 val_acc=0.24
57 test_loss=2.520978 test_acc=0.2267
step=49000 train_loss=2.470584 train_acc=0.2807 val_loss=2.514452 val_acc=0.24
04 test_loss=2.516951 test_acc=0.2265
step=49500 train_loss=2.466044 train_acc=0.2902 val_loss=2.513974 val_acc=0.24
36 test_loss=2.513353 test_acc=0.2362
step=50000 train_loss=2.466650 train_acc=0.2860 val_loss=2.514986 val_acc=0.22
02 test_loss=2.509245 test_acc=0.2284
step=50500 train_loss=2.468192 train_acc=0.2874 val_loss=2.517077 val_acc=0.24
04 test_loss=2.512752 test_acc=0.2350
step=51000 train_loss=2.468376 train_acc=0.3111 val_loss=2.511684 val_acc=0.25
00 test_loss=2.508205 test_acc=0.2619
step=51500 train_loss=2.464472 train_acc=0.3104 val_loss=2.506215 val_acc=0.25
00 test_loss=2.504776 test_acc=0.2493
step=52000 train_loss=2.464222 train_acc=0.2973 val_loss=2.510275 val_acc=0.23
51 test_loss=2.507325 test_acc=0.2460
step=52500 train_loss=2.461438 train_acc=0.3072 val_loss=2.505313 val_acc=0.22
87 test_loss=2.500916 test_acc=0.2529
step=53000 train_loss=2.461809 train_acc=0.2796 val_loss=2.501806 val_acc=0.22
23 test_loss=2.502010 test_acc=0.2327
step=53500 train_loss=2.465061 train_acc=0.3338 val_loss=2.511886 val_acc=0.26
17 test_loss=2.504133 test_acc=0.2647
step=54000 train_loss=2.456339 train_acc=0.3097 val_loss=2.494487 val_acc=0.23
30 test_loss=2.498426 test_acc=0.2474
step=54500 train_loss=2.456397 train_acc=0.2768 val_loss=2.501161 val_acc=0.23
62 test_loss=2.500182 test_acc=0.2180
step=55000 train_loss=2.455630 train_acc=0.2828 val_loss=2.501096 val_acc=0.21
49 test_loss=2.500927 test_acc=0.2233
```

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```
step=55500 train_loss=2.455548 train_acc=0.2814 val_loss=2.507761 val_acc=0.21
60 test_loss=2.501517 test_acc=0.2230
step=56000 train_loss=2.460683 train_acc=0.2828 val_loss=2.507264 val_acc=0.21
60 test_loss=2.502767 test_acc=0.2253
step=56500 train_loss=2.459624 train_acc=0.3030 val_loss=2.501571 val_acc=0.23
40 test_loss=2.501057 test_acc=0.2442
step=57000 train_loss=2.457509 train_acc=0.2931 val_loss=2.501977 val_acc=0.23
09 test_loss=2.500478 test_acc=0.2325
step=57500 train_loss=2.455858 train_acc=0.3005 val_loss=2.505319 val_acc=0.21
70 test_loss=2.498749 test_acc=0.2412
step=58000 train_loss=2.452144 train_acc=0.3125 val_loss=2.498745 val_acc=0.24
47 test_loss=2.494683 test_acc=0.2488
step=58500 train_loss=2.450007 train_acc=0.3207 val_loss=2.495541 val_acc=0.23
40 test_loss=2.492628 test_acc=0.2495
step=59000 train_loss=2.451976 train_acc=0.3001 val_loss=2.499755 val_acc=0.21
91 test_loss=2.491780 test_acc=0.2481
step=59500 train_loss=2.447418 train_acc=0.2892 val_loss=2.493582 val_acc=0.21
81 test_loss=2.488703 test_acc=0.2270
step=60000 train_loss=2.450062 train_acc=0.2945 val_loss=2.491645 val_acc=0.24
68 test_loss=2.492734 test_acc=0.2330
step=60500 train_loss=2.448270 train_acc=0.2849 val_loss=2.491744 val_acc=0.22
45 test_loss=2.490279 test_acc=0.2286
step=61000 train_loss=2.443070 train_acc=0.2987 val_loss=2.487497 val_acc=0.23
94 test_loss=2.481692 test_acc=0.2398
step=61500 train_loss=2.444396 train_acc=0.3335 val_loss=2.485986 val_acc=0.28
30 test_loss=2.482646 test_acc=0.2801
step=62000 train_loss=2.439855 train_acc=0.3402 val_loss=2.486024 val_acc=0.27
34 test_loss=2.481301 test_acc=0.2692
step=62500 train_loss=2.441024 train_acc=0.2934 val_loss=2.485203 val_acc=0.24
36 test_loss=2.480843 test_acc=0.2346
step=63000 train_loss=2.444123 train_acc=0.3129 val_loss=2.490571 val_acc=0.25
85 test_loss=2.487733 test_acc=0.2460
step=63500 train_loss=2.441977 train_acc=0.2842 val_loss=2.485357 val_acc=0.23
09 test_loss=2.482814 test_acc=0.2327
step=64000 train_loss=2.440609 train_acc=0.3544 val_loss=2.484012 val_acc=0.28
94 test_loss=2.482760 test_acc=0.2871
step=64500 train_loss=2.440142 train_acc=0.2814 val_loss=2.491197 val_acc=0.21
70 test_loss=2.482466 test_acc=0.2283
step=65000 train_loss=2.438648 train_acc=0.3125 val_loss=2.484638 val_acc=0.24
68 test_loss=2.481344 test_acc=0.2578
step=65500 train_loss=2.437814 train_acc=0.3388 val_loss=2.482004 val_acc=0.26
60 test_loss=2.481238 test_acc=0.2639
step=66000 train_loss=2.438573 train_acc=0.3019 val_loss=2.486677 val_acc=0.22
13 test_loss=2.479520 test_acc=0.2493
step=66500 train_loss=2.435323 train_acc=0.3030 val_loss=2.480077 val_acc=0.23
51 test_loss=2.476190 test_acc=0.2355
step=67000 train_loss=2.434525 train_acc=0.3157 val_loss=2.475501 val_acc=0.24
15 test_loss=2.475182 test_acc=0.2534
step=67500 train_loss=2.430654 train_acc=0.3409 val_loss=2.473687 val_acc=0.27
34 test_loss=2.470870 test_acc=0.2690
step=68000 train_loss=2.431180 train_acc=0.3072 val_loss=2.474875 val_acc=0.24
68 test_loss=2.473263 test_acc=0.2490
step=68500 train_loss=2.432625 train_acc=0.3058 val_loss=2.477652 val_acc=0.23
40 test_loss=2.472477 test_acc=0.2490
step=69000 train_loss=2.432976 train_acc=0.3239 val_loss=2.476400 val_acc=0.26
49 test_loss=2.475773 test_acc=0.2644
step=69500 train_loss=2.430026 train_acc=0.3143 val_loss=2.474382 val_acc=0.24
79 test_loss=2.472134 test_acc=0.2534
step=70000 train_loss=2.428277 train_acc=0.3296 val_loss=2.472187 val_acc=0.25
64 test_loss=2.470603 test_acc=0.2672
```

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```
step=70500 train_loss=2.430882 train_acc=0.2991 val_loss=2.474655 val_acc=0.24
57 test_loss=2.470876 test_acc=0.2469
step=71000 train_loss=2.428003 train_acc=0.3140 val_loss=2.471881 val_acc=0.24
04 test_loss=2.469470 test_acc=0.2476
step=71500 train_loss=2.428621 train_acc=0.3033 val_loss=2.473320 val_acc=0.25
11 test_loss=2.469569 test_acc=0.2391
step=72000 train_loss=2.427158 train_acc=0.3104 val_loss=2.468886 val_acc=0.25
96 test_loss=2.468714 test_acc=0.2502
step=72500 train_loss=2.423454 train_acc=0.2970 val_loss=2.470266 val_acc=0.23
30 test_loss=2.466259 test_acc=0.2288
step=73000 train_loss=2.423929 train_acc=0.3097 val_loss=2.471556 val_acc=0.24
57 test_loss=2.468664 test_acc=0.2430
step=73500 train_loss=2.424740 train_acc=0.3111 val_loss=2.473797 val_acc=0.24
26 test_loss=2.468412 test_acc=0.2449
step=74000 train_loss=2.423230 train_acc=0.3253 val_loss=2.468976 val_acc=0.25
96 test_loss=2.464153 test_acc=0.2534
step=74500 train_loss=2.422930 train_acc=0.3285 val_loss=2.466906 val_acc=0.24
15 test_loss=2.462527 test_acc=0.2683
step=75000 train_loss=2.421850 train_acc=0.3749 val_loss=2.466829 val_acc=0.27
77 test_loss=2.464412 test_acc=0.2971
step=75500 train_loss=2.420296 train_acc=0.2923 val_loss=2.470858 val_acc=0.23
19 test_loss=2.463793 test_acc=0.2288
step=76000 train_loss=2.418354 train_acc=0.3274 val_loss=2.457632 val_acc=0.26
81 test_loss=2.459614 test_acc=0.2587
step=76500 train_loss=2.416002 train_acc=0.3125 val_loss=2.460638 val_acc=0.25
43 test_loss=2.457209 test_acc=0.2461
step=77000 train_loss=2.415586 train_acc=0.3363 val_loss=2.459833 val_acc=0.27
34 test_loss=2.458266 test_acc=0.2709
step=77500 train_loss=2.416878 train_acc=0.3193 val_loss=2.461792 val_acc=0.25
32 test_loss=2.458594 test_acc=0.2577
step=78000 train_loss=2.416958 train_acc=0.3221 val_loss=2.463049 val_acc=0.25
21 test_loss=2.458762 test_acc=0.2525
step=78500 train_loss=2.416156 train_acc=0.3203 val_loss=2.461358 val_acc=0.23
62 test_loss=2.457646 test_acc=0.2518
step=79000 train_loss=2.416347 train_acc=0.3540 val_loss=2.460336 val_acc=0.30
00 test_loss=2.455483 test_acc=0.2871
step=79500 train_loss=2.413769 train_acc=0.3005 val_loss=2.455572 val_acc=0.25
32 test_loss=2.453914 test_acc=0.2369
step=80000 train_loss=2.413443 train_acc=0.3310 val_loss=2.456512 val_acc=0.27
45 test_loss=2.454584 test_acc=0.2603
step=80500 train_loss=2.412991 train_acc=0.3249 val_loss=2.459904 val_acc=0.24
79 test_loss=2.454974 test_acc=0.2555
step=81000 train_loss=2.412021 train_acc=0.3278 val_loss=2.456431 val_acc=0.26
49 test_loss=2.453830 test_acc=0.2672
step=81500 train_loss=2.410889 train_acc=0.3225 val_loss=2.456984 val_acc=0.25
85 test_loss=2.452666 test_acc=0.2637
step=82000 train_loss=2.411891 train_acc=0.3398 val_loss=2.458584 val_acc=0.27
34 test_loss=2.454218 test_acc=0.2731
step=82500 train_loss=2.411105 train_acc=0.3249 val_loss=2.455407 val_acc=0.24
79 test_loss=2.452104 test_acc=0.2591
step=83000 train_loss=2.410000 train_acc=0.3122 val_loss=2.451131 val_acc=0.24
47 test_loss=2.450661 test_acc=0.2504
step=83500 train_loss=2.410324 train_acc=0.3579 val_loss=2.454062 val_acc=0.27
02 test_loss=2.449841 test_acc=0.2846
step=84000 train_loss=2.409887 train_acc=0.3161 val_loss=2.458255 val_acc=0.24
36 test_loss=2.453170 test_acc=0.2444
step=84500 train_loss=2.408625 train_acc=0.3037 val_loss=2.455856 val_acc=0.23
62 test_loss=2.452333 test_acc=0.2375
_loss=2.407591 train_acc=0.3459 val_loss=2.452651 val_acc=0.27
77 test_loss=2.448343 test_acc=0.2791
```

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```
step=85500 train_loss=2.406568 train_acc=0.3253 val_loss=2.449106 val_acc=0.27
02 test_loss=2.448315 test_acc=0.2598
step=86000 train_loss=2.406660 train_acc=0.3395 val_loss=2.450790 val_acc=0.26
60 test_loss=2.450327 test_acc=0.2702
step=86500 train_loss=2.405632 train_acc=0.3405 val_loss=2.451026 val_acc=0.26
60 test_loss=2.448371 test_acc=0.2743
step=87000 train_loss=2.405407 train_acc=0.3225 val_loss=2.452586 val_acc=0.24
68 test_loss=2.448033 test_acc=0.2608
step=87500 train_loss=2.405319 train_acc=0.3391 val_loss=2.450013 val_acc=0.27
98 test_loss=2.446658 test_acc=0.2764
step=88000 train_loss=2.404620 train_acc=0.3568 val_loss=2.447188 val_acc=0.28
09 test_loss=2.444745 test_acc=0.2798
step=88500 train_loss=2.403604 train_acc=0.3193 val_loss=2.447909 val_acc=0.26
60 test_loss=2.444748 test_acc=0.2580
step=89000 train_loss=2.404057 train_acc=0.3285 val_loss=2.448778 val_acc=0.27
34 test_loss=2.445874 test_acc=0.2646
step=89500 train_loss=2.403791 train_acc=0.3462 val_loss=2.448345 val_acc=0.28
30 test_loss=2.443580 test_acc=0.2738
step=90000 train_loss=2.402771 train_acc=0.3200 val_loss=2.447208 val_acc=0.24
57 test_loss=2.442855 test_acc=0.2476
step=90500 train_loss=2.401875 train_acc=0.3264 val_loss=2.443037 val_acc=0.25
32 test_loss=2.443809 test_acc=0.2614
step=91000 train_loss=2.402470 train_acc=0.3133 val_loss=2.447050 val_acc=0.24
89 test_loss=2.443947 test_acc=0.2490
step=91500 train_loss=2.402010 train_acc=0.3303 val_loss=2.446006 val_acc=0.26
28 test_loss=2.443989 test_acc=0.2608
step=92000 train_loss=2.401845 train_acc=0.3313 val_loss=2.447063 val_acc=0.25
21 test_loss=2.444300 test_acc=0.2639
step=92500 train_loss=2.401610 train_acc=0.3405 val_loss=2.445259 val_acc=0.28
09 test_loss=2.443348 test_acc=0.2672
step=93000 train_loss=2.400431 train_acc=0.3306 val_loss=2.448269 val_acc=0.27
02 test_loss=2.444656 test_acc=0.2623
step=93500 train_loss=2.400286 train_acc=0.3164 val_loss=2.448081 val_acc=0.22
98 test_loss=2.442170 test_acc=0.2502
step=94000 train_loss=2.400188 train_acc=0.3179 val_loss=2.446350 val_acc=0.25
85 test_loss=2.442013 test_acc=0.2497
step=94500 train_loss=2.400003 train_acc=0.3331 val_loss=2.445277 val_acc=0.26
17 test_loss=2.443942 test_acc=0.2647
step=95000 train_loss=2.399797 train_acc=0.3583 val_loss=2.444917 val_acc=0.28
72 test_loss=2.441327 test_acc=0.2911
step=95500 train_loss=2.399427 train_acc=0.3427 val_loss=2.444735 val_acc=0.27
55 test_loss=2.441446 test_acc=0.2748
step=96000 train_loss=2.398628 train_acc=0.3427 val_loss=2.443064 val_acc=0.26
49 test_loss=2.438924 test_acc=0.2693
step=96500 train_loss=2.398374 train_acc=0.3044 val_loss=2.443228 val_acc=0.23
94 test_loss=2.439296 test_acc=0.2391
step=97000 train_loss=2.398384 train_acc=0.3480 val_loss=2.444192 val_acc=0.27
98 test_loss=2.441099 test_acc=0.2784
step=97500 train_loss=2.398161 train_acc=0.3405 val_loss=2.442327 val_acc=0.26
49 test_loss=2.440651 test_acc=0.2685
step=98000 train_loss=2.398544 train_acc=0.3405 val_loss=2.443197 val_acc=0.27
55 test_loss=2.442240 test_acc=0.2679
step=98500 train_loss=2.399177 train_acc=0.3327 val_loss=2.442759 val_acc=0.26
49 test_loss=2.440437 test_acc=0.2683
step=99000 train_loss=2.398784 train_acc=0.3395 val_loss=2.439876 val_acc=0.26
06 test_loss=2.438467 test_acc=0.2667
step=99500 train_loss=2.398298 train_acc=0.3813 val_loss=2.442663 val_acc=0.29
89 test_loss=2.440368 test_acc=0.3088
n_loss=2.398310 train_acc=0.3561 val_loss=2.441284 val_acc=0.2
904 test_loss=2.439653 test_acc=0.2863
```

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```
best metrics: {
  "step": 99000,
  "lr": 0.00010022655592756672,
  "train_loss": 2.3987835359269214,
  "train_acc": 0.33947554925584694,
  "val_loss": 2.4398755519948105,
  "val_acc": 0.26063829787234044,
  "test_loss": 2.4384674552227095,
  "test_acc": 0.2666902780237294,
  "elapsed_sec": 2778.998263835907
}
checkpoint dir: /Users/shobhit/Documents/UW/CSE 493/cse493s-spring26-hw1/out/s
ubmission/part1/warmup/add_p97_l2_s42

$ /opt/homebrew/Caskroom/miniconda/base/envs/cse493/bin/python train.py --mode
modular --op + --p 113 --n_layer 1 --n_head 4 --n_embd 128 --steps 100000 --se
ed 42 --batch_size 128 --lr 0.001 --min_lr 0.0001 --lr_schedule cosine --warmu
p_steps 1000 --grad_clip 1.0 --beta1 0.9 --beta2 0.95 --weight_decay 0.01 --op
timizer adam --eval_interval 500 --log_interval 500 --out_dir '/Users/shobhit/
Documents/UW/CSE 493/cse493s-spring26-hw1/out/submission/part1/warmup/add_p113
_l1_s42' --bias --resume
number of parameters: 0.21M
[resume] model_last.pt not found in /Users/shobhit/Documents/UW/CSE 493/cse493
s-spring26-hw1/out/submission/part1/warmup/add_p113_l1_s42; starting fresh
device=mps train=3830 val=1276 test=7663 batch_size=128
step=1 train_loss=4.814455 train_acc=0.0005 val_loss=4.812633 val_acc=0.0000 t
est_loss=4.816957 test_acc=0.0010
step=500 train_loss=4.512485 train_acc=0.0614 val_loss=4.923081 val_acc=0.0016
test_loss=4.937055 test_acc=0.0018
step=1000 train_loss=4.485818 train_acc=0.0535 val_loss=4.956103 val_acc=0.001
6 test_loss=4.961850 test_acc=0.0017
step=1500 train_loss=4.569907 train_acc=0.0439 val_loss=4.904477 val_acc=0.003
1 test_loss=4.906747 test_acc=0.0018
step=2000 train_loss=4.513410 train_acc=0.0420 val_loss=4.977387 val_acc=0.003
1 test_loss=4.983763 test_acc=0.0021
step=2500 train_loss=4.483424 train_acc=0.0462 val_loss=5.025413 val_acc=0.005
5 test_loss=5.042314 test_acc=0.0023
step=3000 train_loss=4.465569 train_acc=0.0480 val_loss=5.053791 val_acc=0.002
4 test_loss=5.065077 test_acc=0.0021
step=3500 train_loss=4.462391 train_acc=0.0457 val_loss=5.055329 val_acc=0.001
6 test_loss=5.063979 test_acc=0.0017
step=4000 train_loss=4.458614 train_acc=0.0473 val_loss=5.068110 val_acc=0.000
0 test_loss=5.068289 test_acc=0.0029
step=4500 train_loss=4.438114 train_acc=0.0522 val_loss=5.080587 val_acc=0.001
6 test_loss=5.079586 test_acc=0.0018
step=5000 train_loss=4.401098 train_acc=0.0475 val_loss=4.971897 val_acc=0.003
9 test_loss=4.973165 test_acc=0.0037
step=5500 train_loss=4.277464 train_acc=0.0582 val_loss=4.721404 val_acc=0.006
3 test_loss=4.719431 test_acc=0.0082
step=6000 train_loss=3.185191 train_acc=0.1655 val_loss=3.324009 val_acc=0.093
3 test_loss=3.309755 test_acc=0.0923
step=6500 train_loss=2.963873 train_acc=0.1872 val_loss=3.104348 val_acc=0.102
7 test_loss=3.097298 test_acc=0.1058
step=7000 train_loss=2.837347 train_acc=0.2238 val_loss=2.981429 val_acc=0.117
6 test_loss=2.972851 test_acc=0.1249
step=7500 train_loss=2.801765 train_acc=0.2068 val_loss=2.926947 val_acc=0.133
2 test_loss=2.916721 test_acc=0.1311
step=8000 train_loss=2.774417 train_acc=0.2123 val_loss=2.884375 val_acc=0.152
6991 test_acc=0.1437
step=8500 train_loss=2.757061 train_acc=0.2164 val_loss=2.874609 val_acc=0.154
```

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```
4 test_loss=2.868063 test_acc=0.1468
step=9000 train_loss=2.752496 train_acc=0.2402 val_loss=2.878794 val_acc=0.157
5 test_loss=2.860074 test_acc=0.1614
step=9500 train_loss=2.744020 train_acc=0.2287 val_loss=2.857191 val_acc=0.159
1 test_loss=2.856919 test_acc=0.1458
step=10000 train_loss=2.741624 train_acc=0.2446 val_loss=2.853028 val_acc=0.16
22 test_loss=2.848761 test_acc=0.1591
step=10500 train_loss=2.738707 train_acc=0.2178 val_loss=2.850318 val_acc=0.13
79 test_loss=2.842822 test_acc=0.1356
step=11000 train_loss=2.721571 train_acc=0.2334 val_loss=2.807123 val_acc=0.14
34 test_loss=2.795562 test_acc=0.1610
step=11500 train_loss=2.708535 train_acc=0.2290 val_loss=2.767220 val_acc=0.15
28 test_loss=2.752808 test_acc=0.1668
step=12000 train_loss=2.717655 train_acc=0.1781 val_loss=2.760566 val_acc=0.15
28 test_loss=2.749789 test_acc=0.1493
step=12500 train_loss=2.697762 train_acc=0.2290 val_loss=2.734966 val_acc=0.19
20 test_loss=2.730393 test_acc=0.1912
step=13000 train_loss=2.691271 train_acc=0.2138 val_loss=2.726447 val_acc=0.17
55 test_loss=2.720224 test_acc=0.1723
step=13500 train_loss=2.689579 train_acc=0.2554 val_loss=2.727241 val_acc=0.21
63 test_loss=2.720816 test_acc=0.2208
step=14000 train_loss=2.694138 train_acc=0.2491 val_loss=2.733422 val_acc=0.20
92 test_loss=2.722416 test_acc=0.2101
step=14500 train_loss=2.682340 train_acc=0.2324 val_loss=2.715548 val_acc=0.18
57 test_loss=2.710814 test_acc=0.1899
step=15000 train_loss=2.679542 train_acc=0.2136 val_loss=2.718878 val_acc=0.18
50 test_loss=2.711825 test_acc=0.1728
step=15500 train_loss=2.680933 train_acc=0.2308 val_loss=2.715875 val_acc=0.18
10 test_loss=2.711084 test_acc=0.1907
step=16000 train_loss=2.675498 train_acc=0.2269 val_loss=2.711436 val_acc=0.19
75 test_loss=2.707911 test_acc=0.1823
step=16500 train_loss=2.678310 train_acc=0.2188 val_loss=2.723360 val_acc=0.18
65 test_loss=2.711273 test_acc=0.1762
step=17000 train_loss=2.673639 train_acc=0.2285 val_loss=2.716955 val_acc=0.16
69 test_loss=2.705913 test_acc=0.1837
step=17500 train_loss=2.669700 train_acc=0.2303 val_loss=2.711435 val_acc=0.20
22 test_loss=2.698397 test_acc=0.1984
step=18000 train_loss=2.669845 train_acc=0.2178 val_loss=2.701077 val_acc=0.18
34 test_loss=2.697455 test_acc=0.1831
step=18500 train_loss=2.669197 train_acc=0.2666 val_loss=2.705849 val_acc=0.22
34 test_loss=2.699031 test_acc=0.2230
step=19000 train_loss=2.672820 train_acc=0.2334 val_loss=2.714557 val_acc=0.19
44 test_loss=2.704682 test_acc=0.1925
step=19500 train_loss=2.673799 train_acc=0.2433 val_loss=2.711525 val_acc=0.19
98 test_loss=2.702372 test_acc=0.2110
step=20000 train_loss=2.675941 train_acc=0.2068 val_loss=2.712594 val_acc=0.17
40 test_loss=2.704226 test_acc=0.1713
step=20500 train_loss=2.666108 train_acc=0.2029 val_loss=2.707325 val_acc=0.15
13 test_loss=2.694711 test_acc=0.1661
step=21000 train_loss=2.663059 train_acc=0.2050 val_loss=2.710167 val_acc=0.16
22 test_loss=2.697365 test_acc=0.1672
step=21500 train_loss=2.671467 train_acc=0.1990 val_loss=2.710030 val_acc=0.15
52 test_loss=2.702719 test_acc=0.1589
step=22000 train_loss=2.672626 train_acc=0.2112 val_loss=2.713665 val_acc=0.17
24 test_loss=2.702158 test_acc=0.1728
step=22500 train_loss=2.663103 train_acc=0.2504 val_loss=2.706869 val_acc=0.18
73 test_loss=2.693177 test_acc=0.2091
step=23000 train_loss=2.664788 train_acc=0.2292 val_loss=2.702602 val_acc=0.18
93480 test_acc=0.1858
step=23500 train_loss=2.664125 train_acc=0.2245 val_loss=2.706337 val_acc=0.17
```

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```
95 test_loss=2.699936 test_acc=0.1861
step=24000 train_loss=2.666880 train_acc=0.1974 val_loss=2.709188 val_acc=0.16
46 test_loss=2.701922 test_acc=0.1567
step=24500 train_loss=2.661913 train_acc=0.2170 val_loss=2.695693 val_acc=0.19
04 test_loss=2.691917 test_acc=0.1848
step=25000 train_loss=2.662895 train_acc=0.2034 val_loss=2.698990 val_acc=0.17
16 test_loss=2.695406 test_acc=0.1696
step=25500 train_loss=2.664943 train_acc=0.2352 val_loss=2.705981 val_acc=0.19
36 test_loss=2.693163 test_acc=0.2023
step=26000 train_loss=2.665169 train_acc=0.2580 val_loss=2.709740 val_acc=0.21
32 test_loss=2.695460 test_acc=0.2178
step=26500 train_loss=2.665027 train_acc=0.2332 val_loss=2.703611 val_acc=0.18
26 test_loss=2.695762 test_acc=0.1930
step=27000 train_loss=2.661693 train_acc=0.2149 val_loss=2.698866 val_acc=0.18
50 test_loss=2.692390 test_acc=0.1802
step=27500 train_loss=2.662458 train_acc=0.2499 val_loss=2.695554 val_acc=0.21
16 test_loss=2.691387 test_acc=0.2098
step=28000 train_loss=2.657100 train_acc=0.2232 val_loss=2.693500 val_acc=0.18
97 test_loss=2.688688 test_acc=0.1824
step=28500 train_loss=2.665951 train_acc=0.2311 val_loss=2.704216 val_acc=0.18
10 test_loss=2.701340 test_acc=0.1860
step=29000 train_loss=2.655724 train_acc=0.2311 val_loss=2.687841 val_acc=0.19
83 test_loss=2.687735 test_acc=0.1884
step=29500 train_loss=2.663121 train_acc=0.2146 val_loss=2.701490 val_acc=0.17
32 test_loss=2.695058 test_acc=0.1729
step=30000 train_loss=2.655994 train_acc=0.2465 val_loss=2.698283 val_acc=0.21
08 test_loss=2.684870 test_acc=0.2138
step=30500 train_loss=2.660240 train_acc=0.2465 val_loss=2.689693 val_acc=0.21
00 test_loss=2.688202 test_acc=0.2089
step=31000 train_loss=2.656498 train_acc=0.2125 val_loss=2.696556 val_acc=0.16
30 test_loss=2.689218 test_acc=0.1805
step=31500 train_loss=2.659363 train_acc=0.1987 val_loss=2.703688 val_acc=0.15
83 test_loss=2.694423 test_acc=0.1647
step=32000 train_loss=2.657324 train_acc=0.2384 val_loss=2.696209 val_acc=0.21
47 test_loss=2.690024 test_acc=0.2028
step=32500 train_loss=2.660053 train_acc=0.2384 val_loss=2.701922 val_acc=0.20
22 test_loss=2.687590 test_acc=0.2063
step=33000 train_loss=2.654346 train_acc=0.2311 val_loss=2.699269 val_acc=0.16
93 test_loss=2.687052 test_acc=0.1908
step=33500 train_loss=2.651325 train_acc=0.2235 val_loss=2.691398 val_acc=0.18
34 test_loss=2.683924 test_acc=0.1801
step=34000 train_loss=2.653032 train_acc=0.2397 val_loss=2.689592 val_acc=0.21
47 test_loss=2.684226 test_acc=0.2006
step=34500 train_loss=2.650512 train_acc=0.2470 val_loss=2.694515 val_acc=0.21
00 test_loss=2.681740 test_acc=0.1982
step=35000 train_loss=2.650443 train_acc=0.2172 val_loss=2.690877 val_acc=0.18
10 test_loss=2.683682 test_acc=0.1745
step=35500 train_loss=2.647756 train_acc=0.2358 val_loss=2.684928 val_acc=0.18
81 test_loss=2.678011 test_acc=0.1939
step=36000 train_loss=2.650297 train_acc=0.2201 val_loss=2.698192 val_acc=0.17
55 test_loss=2.685913 test_acc=0.1753
step=36500 train_loss=2.651114 train_acc=0.2251 val_loss=2.687578 val_acc=0.18
34 test_loss=2.678153 test_acc=0.1900
step=37000 train_loss=2.648344 train_acc=0.2125 val_loss=2.698229 val_acc=0.16
14 test_loss=2.682470 test_acc=0.1724
step=37500 train_loss=2.642186 train_acc=0.2222 val_loss=2.683649 val_acc=0.18
97 test_loss=2.674418 test_acc=0.1805
step=38000 train_loss=2.642471 train_acc=0.2520 val_loss=2.683056 val_acc=0.19
76789 test_acc=0.2148
step=38500 train_loss=2.642524 train_acc=0.2457 val_loss=2.681228 val_acc=0.20
```

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```
85 test_loss=2.672764 test_acc=0.1998
step=39000 train_loss=2.643821 train_acc=0.2167 val_loss=2.686197 val_acc=0.18
34 test_loss=2.675081 test_acc=0.1785
step=39500 train_loss=2.642348 train_acc=0.2219 val_loss=2.684984 val_acc=0.18
73 test_loss=2.673055 test_acc=0.1882
step=40000 train_loss=2.645969 train_acc=0.2269 val_loss=2.684626 val_acc=0.19
28 test_loss=2.676752 test_acc=0.1913
step=40500 train_loss=2.637382 train_acc=0.2517 val_loss=2.678843 val_acc=0.19
12 test_loss=2.670086 test_acc=0.1987
step=41000 train_loss=2.645378 train_acc=0.2138 val_loss=2.689336 val_acc=0.17
16 test_loss=2.674056 test_acc=0.1823
step=41500 train_loss=2.639486 train_acc=0.2548 val_loss=2.681333 val_acc=0.21
08 test_loss=2.674287 test_acc=0.2074
step=42000 train_loss=2.640791 train_acc=0.2402 val_loss=2.682191 val_acc=0.19
36 test_loss=2.672300 test_acc=0.1998
step=42500 train_loss=2.634986 train_acc=0.2444 val_loss=2.679445 val_acc=0.18
65 test_loss=2.668481 test_acc=0.1968
step=43000 train_loss=2.639180 train_acc=0.2736 val_loss=2.680882 val_acc=0.22
10 test_loss=2.668731 test_acc=0.2250
step=43500 train_loss=2.638321 train_acc=0.2428 val_loss=2.675582 val_acc=0.19
67 test_loss=2.667562 test_acc=0.1994
step=44000 train_loss=2.640008 train_acc=0.2332 val_loss=2.682238 val_acc=0.19
83 test_loss=2.670836 test_acc=0.1950
step=44500 train_loss=2.640573 train_acc=0.2661 val_loss=2.682195 val_acc=0.21
32 test_loss=2.670263 test_acc=0.2161
step=45000 train_loss=2.638753 train_acc=0.2496 val_loss=2.683364 val_acc=0.19
98 test_loss=2.674481 test_acc=0.2032
step=45500 train_loss=2.636056 train_acc=0.2658 val_loss=2.675284 val_acc=0.21
47 test_loss=2.664921 test_acc=0.2195
step=46000 train_loss=2.632938 train_acc=0.2872 val_loss=2.676710 val_acc=0.23
12 test_loss=2.662154 test_acc=0.2374
step=46500 train_loss=2.634653 train_acc=0.2460 val_loss=2.679474 val_acc=0.21
16 test_loss=2.668408 test_acc=0.1956
step=47000 train_loss=2.631284 train_acc=0.2428 val_loss=2.673397 val_acc=0.20
69 test_loss=2.661289 test_acc=0.2023
step=47500 train_loss=2.633090 train_acc=0.2478 val_loss=2.674599 val_acc=0.21
08 test_loss=2.663430 test_acc=0.2078
step=48000 train_loss=2.638115 train_acc=0.2397 val_loss=2.682295 val_acc=0.20
61 test_loss=2.670763 test_acc=0.1951
step=48500 train_loss=2.631310 train_acc=0.2786 val_loss=2.667000 val_acc=0.22
73 test_loss=2.660781 test_acc=0.2276
step=49000 train_loss=2.633535 train_acc=0.2486 val_loss=2.669935 val_acc=0.20
53 test_loss=2.661848 test_acc=0.2014
step=49500 train_loss=2.636273 train_acc=0.2313 val_loss=2.678655 val_acc=0.19
91 test_loss=2.667815 test_acc=0.1894
step=50000 train_loss=2.634830 train_acc=0.2514 val_loss=2.675410 val_acc=0.20
14 test_loss=2.665842 test_acc=0.2085
step=50500 train_loss=2.630985 train_acc=0.2117 val_loss=2.673691 val_acc=0.16
69 test_loss=2.662424 test_acc=0.1733
step=51000 train_loss=2.633004 train_acc=0.2833 val_loss=2.671740 val_acc=0.24
06 test_loss=2.664684 test_acc=0.2380
step=51500 train_loss=2.629639 train_acc=0.2316 val_loss=2.664596 val_acc=0.22
10 test_loss=2.661121 test_acc=0.1913
step=52000 train_loss=2.627918 train_acc=0.2394 val_loss=2.670497 val_acc=0.17
63 test_loss=2.659206 test_acc=0.1912
step=52500 train_loss=2.628044 train_acc=0.2407 val_loss=2.667110 val_acc=0.18
81 test_loss=2.658419 test_acc=0.1937
step=53000 train_loss=2.628039 train_acc=0.2384 val_loss=2.665928 val_acc=0.20
58071 test_acc=0.1957
step=53500 train_loss=2.624786 train_acc=0.2399 val_loss=2.666713 val_acc=0.19
```

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```
59 test_loss=2.656031 test_acc=0.1929
step=54000 train_loss=2.625356 train_acc=0.2298 val_loss=2.666764 val_acc=0.17
79 test_loss=2.658104 test_acc=0.1887
step=54500 train_loss=2.623892 train_acc=0.2525 val_loss=2.663584 val_acc=0.20
69 test_loss=2.653067 test_acc=0.2062
step=55000 train_loss=2.626153 train_acc=0.2530 val_loss=2.666019 val_acc=0.20
14 test_loss=2.654907 test_acc=0.2110
step=55500 train_loss=2.622438 train_acc=0.2671 val_loss=2.652562 val_acc=0.23
51 test_loss=2.650865 test_acc=0.2201
step=56000 train_loss=2.620738 train_acc=0.2517 val_loss=2.659882 val_acc=0.19
04 test_loss=2.651143 test_acc=0.2042
step=56500 train_loss=2.620885 train_acc=0.2590 val_loss=2.662089 val_acc=0.21
47 test_loss=2.653244 test_acc=0.2136
step=57000 train_loss=2.621793 train_acc=0.2238 val_loss=2.663798 val_acc=0.17
24 test_loss=2.652328 test_acc=0.1776
step=57500 train_loss=2.620778 train_acc=0.2342 val_loss=2.659226 val_acc=0.19
44 test_loss=2.649847 test_acc=0.1952
step=58000 train_loss=2.619635 train_acc=0.2188 val_loss=2.662015 val_acc=0.16
54 test_loss=2.653333 test_acc=0.1772
step=58500 train_loss=2.619105 train_acc=0.2480 val_loss=2.657200 val_acc=0.19
98 test_loss=2.649906 test_acc=0.2071
step=59000 train_loss=2.618587 train_acc=0.2786 val_loss=2.656449 val_acc=0.21
16 test_loss=2.647828 test_acc=0.2265
step=59500 train_loss=2.615933 train_acc=0.2305 val_loss=2.658128 val_acc=0.18
65 test_loss=2.645422 test_acc=0.1867
step=60000 train_loss=2.617812 train_acc=0.2436 val_loss=2.660846 val_acc=0.19
98 test_loss=2.648831 test_acc=0.1984
step=60500 train_loss=2.614984 train_acc=0.2457 val_loss=2.656267 val_acc=0.20
22 test_loss=2.644032 test_acc=0.2029
step=61000 train_loss=2.616040 train_acc=0.2026 val_loss=2.647932 val_acc=0.16
07 test_loss=2.645381 test_acc=0.1640
step=61500 train_loss=2.616410 train_acc=0.2520 val_loss=2.655174 val_acc=0.20
14 test_loss=2.646533 test_acc=0.2062
step=62000 train_loss=2.615461 train_acc=0.2483 val_loss=2.649984 val_acc=0.20
92 test_loss=2.648470 test_acc=0.1997
step=62500 train_loss=2.615478 train_acc=0.2577 val_loss=2.648318 val_acc=0.21
63 test_loss=2.642064 test_acc=0.2124
step=63000 train_loss=2.612536 train_acc=0.2319 val_loss=2.653776 val_acc=0.17
95 test_loss=2.645432 test_acc=0.1857
step=63500 train_loss=2.612304 train_acc=0.2191 val_loss=2.654682 val_acc=0.18
10 test_loss=2.646420 test_acc=0.1747
step=64000 train_loss=2.611741 train_acc=0.2830 val_loss=2.643628 val_acc=0.23
90 test_loss=2.640103 test_acc=0.2358
step=64500 train_loss=2.610908 train_acc=0.2397 val_loss=2.650858 val_acc=0.17
48 test_loss=2.639699 test_acc=0.1947
step=65000 train_loss=2.612087 train_acc=0.2342 val_loss=2.652086 val_acc=0.20
22 test_loss=2.644810 test_acc=0.1899
step=65500 train_loss=2.611285 train_acc=0.2360 val_loss=2.654870 val_acc=0.16
77 test_loss=2.641214 test_acc=0.1891
step=66000 train_loss=2.609313 train_acc=0.2292 val_loss=2.646344 val_acc=0.18
73 test_loss=2.638448 test_acc=0.1837
step=66500 train_loss=2.610264 train_acc=0.2339 val_loss=2.654201 val_acc=0.18
03 test_loss=2.642716 test_acc=0.1909
step=67000 train_loss=2.611519 train_acc=0.2530 val_loss=2.645459 val_acc=0.21
08 test_loss=2.641184 test_acc=0.2045
step=67500 train_loss=2.609223 train_acc=0.2825 val_loss=2.648855 val_acc=0.22
73 test_loss=2.638462 test_acc=0.2286
step=68000 train_loss=2.607815 train_acc=0.2413 val_loss=2.648578 val_acc=0.18
36269 test_acc=0.1997
step=68500 train_loss=2.609175 train_acc=0.2822 val_loss=2.652226 val_acc=0.23
```

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```
20 test_loss=2.639541 test_acc=0.2349
step=69000 train_loss=2.609017 train_acc=0.2561 val_loss=2.647941 val_acc=0.22
10 test_loss=2.642751 test_acc=0.2118
step=69500 train_loss=2.607980 train_acc=0.2551 val_loss=2.642744 val_acc=0.21
71 test_loss=2.637467 test_acc=0.2034
step=70000 train_loss=2.608527 train_acc=0.2436 val_loss=2.646324 val_acc=0.19
51 test_loss=2.639470 test_acc=0.1973
step=70500 train_loss=2.609689 train_acc=0.2736 val_loss=2.646082 val_acc=0.23
04 test_loss=2.639123 test_acc=0.2289
step=71000 train_loss=2.608013 train_acc=0.2261 val_loss=2.645549 val_acc=0.18
73 test_loss=2.640412 test_acc=0.1772
step=71500 train_loss=2.607849 train_acc=0.2269 val_loss=2.648746 val_acc=0.18
97 test_loss=2.640150 test_acc=0.1775
step=72000 train_loss=2.609288 train_acc=0.2496 val_loss=2.650996 val_acc=0.20
69 test_loss=2.641138 test_acc=0.2083
step=72500 train_loss=2.610824 train_acc=0.2005 val_loss=2.654244 val_acc=0.15
67 test_loss=2.644791 test_acc=0.1558
step=73000 train_loss=2.614857 train_acc=0.2462 val_loss=2.655198 val_acc=0.22
49 test_loss=2.646681 test_acc=0.2055
step=73500 train_loss=2.612835 train_acc=0.2789 val_loss=2.652213 val_acc=0.22
96 test_loss=2.643939 test_acc=0.2268
step=74000 train_loss=2.609086 train_acc=0.2773 val_loss=2.646549 val_acc=0.22
73 test_loss=2.637256 test_acc=0.2318
step=74500 train_loss=2.611530 train_acc=0.2634 val_loss=2.653822 val_acc=0.20
53 test_loss=2.642694 test_acc=0.2149
step=75000 train_loss=2.611505 train_acc=0.2621 val_loss=2.652051 val_acc=0.21
32 test_loss=2.642689 test_acc=0.2157
step=75500 train_loss=2.609986 train_acc=0.2475 val_loss=2.650901 val_acc=0.20
06 test_loss=2.642248 test_acc=0.1999
step=76000 train_loss=2.610301 train_acc=0.2663 val_loss=2.655137 val_acc=0.20
30 test_loss=2.642825 test_acc=0.2119
step=76500 train_loss=2.610430 train_acc=0.2624 val_loss=2.650699 val_acc=0.20
85 test_loss=2.641355 test_acc=0.2131
step=77000 train_loss=2.608657 train_acc=0.2841 val_loss=2.650692 val_acc=0.22
49 test_loss=2.640885 test_acc=0.2370
step=77500 train_loss=2.609984 train_acc=0.2556 val_loss=2.651605 val_acc=0.19
91 test_loss=2.641683 test_acc=0.1999
step=78000 train_loss=2.608801 train_acc=0.2509 val_loss=2.647230 val_acc=0.20
22 test_loss=2.638735 test_acc=0.2057
step=78500 train_loss=2.609174 train_acc=0.2783 val_loss=2.651233 val_acc=0.21
08 test_loss=2.639661 test_acc=0.2248
step=79000 train_loss=2.607156 train_acc=0.2666 val_loss=2.651929 val_acc=0.19
91 test_loss=2.639116 test_acc=0.2188
step=79500 train_loss=2.607193 train_acc=0.2601 val_loss=2.646953 val_acc=0.22
18 test_loss=2.638688 test_acc=0.2050
step=80000 train_loss=2.606364 train_acc=0.2418 val_loss=2.645077 val_acc=0.18
89 test_loss=2.637641 test_acc=0.1934
step=80500 train_loss=2.605279 train_acc=0.2969 val_loss=2.644133 val_acc=0.25
16 test_loss=2.636251 test_acc=0.2460
step=81000 train_loss=2.606405 train_acc=0.2496 val_loss=2.646654 val_acc=0.21
00 test_loss=2.637942 test_acc=0.2034
step=81500 train_loss=2.605256 train_acc=0.2838 val_loss=2.645551 val_acc=0.23
12 test_loss=2.634372 test_acc=0.2288
step=82000 train_loss=2.603912 train_acc=0.2350 val_loss=2.643598 val_acc=0.18
42 test_loss=2.634134 test_acc=0.1894
step=82500 train_loss=2.605238 train_acc=0.2376 val_loss=2.641837 val_acc=0.19
83 test_loss=2.634169 test_acc=0.1951
step=83000 train_loss=2.605310 train_acc=0.2520 val_loss=2.644929 val_acc=0.19
37147 test_acc=0.2046
step=83500 train_loss=2.603529 train_acc=0.2799 val_loss=2.644826 val_acc=0.22
```

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```
88 test_loss=2.634293 test_acc=0.2325
step=84000 train_loss=2.602626 train_acc=0.2556 val_loss=2.643568 val_acc=0.21
24 test_loss=2.634362 test_acc=0.2089
step=84500 train_loss=2.601906 train_acc=0.2462 val_loss=2.644122 val_acc=0.19
75 test_loss=2.633371 test_acc=0.2032
step=85000 train_loss=2.601515 train_acc=0.2984 val_loss=2.641132 val_acc=0.24
61 test_loss=2.633051 test_acc=0.2489
step=85500 train_loss=2.601711 train_acc=0.2789 val_loss=2.640384 val_acc=0.22
88 test_loss=2.633207 test_acc=0.2294
step=86000 train_loss=2.600617 train_acc=0.2778 val_loss=2.641260 val_acc=0.21
32 test_loss=2.632816 test_acc=0.2251
step=86500 train_loss=2.599402 train_acc=0.2674 val_loss=2.640641 val_acc=0.19
98 test_loss=2.629551 test_acc=0.2149
step=87000 train_loss=2.599979 train_acc=0.2601 val_loss=2.639286 val_acc=0.20
85 test_loss=2.629584 test_acc=0.2055
step=87500 train_loss=2.598399 train_acc=0.2681 val_loss=2.639208 val_acc=0.21
24 test_loss=2.630608 test_acc=0.2135
step=88000 train_loss=2.599753 train_acc=0.2749 val_loss=2.640644 val_acc=0.21
08 test_loss=2.629796 test_acc=0.2245
step=88500 train_loss=2.599903 train_acc=0.2940 val_loss=2.636530 val_acc=0.24
61 test_loss=2.631270 test_acc=0.2365
step=89000 train_loss=2.599551 train_acc=0.2747 val_loss=2.638966 val_acc=0.20
85 test_loss=2.628844 test_acc=0.2247
step=89500 train_loss=2.598687 train_acc=0.2679 val_loss=2.638148 val_acc=0.20
77 test_loss=2.631690 test_acc=0.2168
step=90000 train_loss=2.597788 train_acc=0.2721 val_loss=2.638097 val_acc=0.21
55 test_loss=2.629758 test_acc=0.2252
step=90500 train_loss=2.597869 train_acc=0.2512 val_loss=2.635941 val_acc=0.20
45 test_loss=2.628281 test_acc=0.2053
step=91000 train_loss=2.597772 train_acc=0.2786 val_loss=2.635695 val_acc=0.24
22 test_loss=2.628739 test_acc=0.2201
step=91500 train_loss=2.597862 train_acc=0.2554 val_loss=2.638086 val_acc=0.19
91 test_loss=2.629342 test_acc=0.2025
step=92000 train_loss=2.598290 train_acc=0.2885 val_loss=2.639428 val_acc=0.23
12 test_loss=2.629075 test_acc=0.2359
step=92500 train_loss=2.597415 train_acc=0.2919 val_loss=2.637103 val_acc=0.23
43 test_loss=2.627947 test_acc=0.2359
step=93000 train_loss=2.597310 train_acc=0.2554 val_loss=2.637797 val_acc=0.19
12 test_loss=2.627090 test_acc=0.2014
step=93500 train_loss=2.596963 train_acc=0.2833 val_loss=2.636923 val_acc=0.22
10 test_loss=2.628080 test_acc=0.2312
step=94000 train_loss=2.596895 train_acc=0.2671 val_loss=2.638815 val_acc=0.20
22 test_loss=2.629289 test_acc=0.2182
step=94500 train_loss=2.596661 train_acc=0.3047 val_loss=2.637257 val_acc=0.23
75 test_loss=2.627410 test_acc=0.2452
step=95000 train_loss=2.596856 train_acc=0.2875 val_loss=2.634246 val_acc=0.22
26 test_loss=2.624935 test_acc=0.2366
step=95500 train_loss=2.595969 train_acc=0.2700 val_loss=2.631002 val_acc=0.21
94 test_loss=2.625715 test_acc=0.2171
step=96000 train_loss=2.596087 train_acc=0.2838 val_loss=2.634870 val_acc=0.22
73 test_loss=2.625297 test_acc=0.2305
step=96500 train_loss=2.595205 train_acc=0.2501 val_loss=2.633872 val_acc=0.19
98 test_loss=2.624876 test_acc=0.1965
step=97000 train_loss=2.594870 train_acc=0.2804 val_loss=2.633976 val_acc=0.22
41 test_loss=2.625386 test_acc=0.2248
step=97500 train_loss=2.595218 train_acc=0.2909 val_loss=2.634768 val_acc=0.23
75 test_loss=2.624692 test_acc=0.2383
step=98000 train_loss=2.595388 train_acc=0.2619 val_loss=2.635307 val_acc=0.20
26174 test_acc=0.2051
step=98500 train_loss=2.594862 train_acc=0.2525 val_loss=2.635958 val_acc=0.20
```

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```
22 test_loss=2.626001 test_acc=0.1955
step=99000 train_loss=2.594308 train_acc=0.2465 val_loss=2.633708 val_acc=0.20
77 test_loss=2.624675 test_acc=0.1986
step=99500 train_loss=2.594066 train_acc=0.2830 val_loss=2.635437 val_acc=0.22
02 test_loss=2.624515 test_acc=0.2333
step=100000 train_loss=2.594643 train_acc=0.2919 val_loss=2.631031 val_acc=0.2
351 test_loss=2.624440 test_acc=0.2385
best metrics: {
  "step": 95500,
  "lr": 0.0001045803511535803,
  "train_loss": 2.5959692737141102,
  "train_acc": 0.2699738903394256,
  "val_loss": 2.631002020312701,
  "val_acc": 0.219435736677116,
  "test_loss": 2.62571457735042,
  "test_acc": 0.21714733133237635,
  "elapsed_sec": 3951.7127900123596
}
checkpoint dir: /Users/shobhit/Documents/UW/CSE 493/cse493s-spring26-hw1/out/s
ubmission/part1/warmup/add_p113_l1_s42

$ /opt/homebrew/Caskroom/miniconda/base/envs/cse493/bin/python train.py --mode
modular --op + --p 113 --n_layer 2 --n_head 4 --n_embd 128 --steps 100000 --se
ed 42 --batch_size 128 --lr 0.001 --min_lr 0.0001 --lr_schedule cosine --warmu
p_steps 1000 --grad_clip 1.0 --beta1 0.9 --beta2 0.95 --weight_decay 0.01 --op
timizer adam --eval_interval 500 --log_interval 500 --out_dir '/Users/shobhit/
Documents/UW/CSE 493/cse493s-spring26-hw1/out/submission/part1/warmup/add_p113
_l2_s42' --bias --resume
number of parameters: 0.41M
[resume] model_last.pt not found in /Users/shobhit/Documents/UW/CSE 493/cse493
s-spring26-hw1/out/submission/part1/warmup/add_p113_l2_s42; starting fresh
device=mps train=3830 val=1276 test=7663 batch_size=128
step=1 train_loss=4.822444 train_acc=0.0000 val_loss=4.825297 val_acc=0.0000 t
est_loss=4.822713 test_acc=0.0000
step=500 train_loss=4.567648 train_acc=0.0507 val_loss=4.911019 val_acc=0.0047
test_loss=4.904754 test_acc=0.0039
step=1000 train_loss=4.499023 train_acc=0.0493 val_loss=4.961007 val_acc=0.005
5 test_loss=4.957599 test_acc=0.0023
step=1500 train_loss=4.582663 train_acc=0.0392 val_loss=4.908550 val_acc=0.003
1 test_loss=4.902462 test_acc=0.0038
step=2000 train_loss=4.525004 train_acc=0.0381 val_loss=4.959035 val_acc=0.003
1 test_loss=4.963707 test_acc=0.0030
step=2500 train_loss=4.499768 train_acc=0.0407 val_loss=5.000336 val_acc=0.003
9 test_loss=5.007852 test_acc=0.0034
step=3000 train_loss=4.470735 train_acc=0.0454 val_loss=5.021812 val_acc=0.004
7 test_loss=5.022482 test_acc=0.0035
step=3500 train_loss=4.453678 train_acc=0.0486 val_loss=5.021336 val_acc=0.001
6 test_loss=5.028445 test_acc=0.0034
step=4000 train_loss=3.650073 train_acc=0.0953 val_loss=3.845928 val_acc=0.036
8 test_loss=3.829029 test_acc=0.0401
step=4500 train_loss=3.010797 train_acc=0.2123 val_loss=3.182488 val_acc=0.090
1 test_loss=3.173235 test_acc=0.0992
step=5000 train_loss=2.888433 train_acc=0.2044 val_loss=3.101646 val_acc=0.091
7 test_loss=3.077904 test_acc=0.0951
step=5500 train_loss=2.843172 train_acc=0.2240 val_loss=3.048704 val_acc=0.093
3 test_loss=3.028207 test_acc=0.1070
step=6000 train_loss=2.818737 train_acc=0.2238 val_loss=2.967363 val_acc=0.120
7 test_loss=2.954593 test_acc=0.1233
step=6500 train_loss=2.788121 train_acc=0.2123 val_loss=2.879150 val_acc=0.131
7 test_loss=2.872019 test_acc=0.1468
```

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```
step=7000 train_loss=2.754834 train_acc=0.2313 val_loss=2.837752 val_acc=0.149
7 test_loss=2.827290 test_acc=0.1616
step=7500 train_loss=2.738196 train_acc=0.2178 val_loss=2.804739 val_acc=0.163
8 test_loss=2.795898 test_acc=0.1609
step=8000 train_loss=2.730843 train_acc=0.2332 val_loss=2.768891 val_acc=0.175
5 test_loss=2.760270 test_acc=0.1909
step=8500 train_loss=2.721957 train_acc=0.2300 val_loss=2.758810 val_acc=0.190
4 test_loss=2.753047 test_acc=0.1909
step=9000 train_loss=2.716458 train_acc=0.2146 val_loss=2.761148 val_acc=0.180
3 test_loss=2.750050 test_acc=0.1831
step=9500 train_loss=2.705986 train_acc=0.2399 val_loss=2.740578 val_acc=0.217
9 test_loss=2.738114 test_acc=0.1984
step=10000 train_loss=2.703051 train_acc=0.2363 val_loss=2.741678 val_acc=0.19
91 test_loss=2.735067 test_acc=0.1995
step=10500 train_loss=2.699545 train_acc=0.2319 val_loss=2.742032 val_acc=0.17
16 test_loss=2.733867 test_acc=0.1874
step=11000 train_loss=2.703735 train_acc=0.2488 val_loss=2.743160 val_acc=0.20
77 test_loss=2.731083 test_acc=0.2059
step=11500 train_loss=2.706125 train_acc=0.2128 val_loss=2.745076 val_acc=0.16
85 test_loss=2.732760 test_acc=0.1880
step=12000 train_loss=2.708908 train_acc=0.2164 val_loss=2.753022 val_acc=0.16
46 test_loss=2.741653 test_acc=0.1767
step=12500 train_loss=2.710186 train_acc=0.2034 val_loss=2.748310 val_acc=0.16
07 test_loss=2.741492 test_acc=0.1729
step=13000 train_loss=2.730074 train_acc=0.2112 val_loss=2.770224 val_acc=0.16
54 test_loss=2.762532 test_acc=0.1657
step=13500 train_loss=2.723937 train_acc=0.2219 val_loss=2.762651 val_acc=0.18
03 test_loss=2.754566 test_acc=0.1914
step=14000 train_loss=2.734764 train_acc=0.2384 val_loss=2.779288 val_acc=0.17
08 test_loss=2.764113 test_acc=0.1985
step=14500 train_loss=2.718665 train_acc=0.2010 val_loss=2.754125 val_acc=0.15
44 test_loss=2.748333 test_acc=0.1693
step=15000 train_loss=2.729106 train_acc=0.2240 val_loss=2.767186 val_acc=0.19
04 test_loss=2.761004 test_acc=0.1844
step=15500 train_loss=2.725213 train_acc=0.2593 val_loss=2.760025 val_acc=0.19
04 test_loss=2.756305 test_acc=0.2092
step=16000 train_loss=2.719964 train_acc=0.2157 val_loss=2.761479 val_acc=0.17
87 test_loss=2.756136 test_acc=0.1710
step=16500 train_loss=2.724822 train_acc=0.2110 val_loss=2.770540 val_acc=0.16
30 test_loss=2.757043 test_acc=0.1695
step=17000 train_loss=2.721132 train_acc=0.2099 val_loss=2.767775 val_acc=0.15
83 test_loss=2.755409 test_acc=0.1694
step=17500 train_loss=2.714382 train_acc=0.2136 val_loss=2.760523 val_acc=0.17
16 test_loss=2.747139 test_acc=0.1785
step=18000 train_loss=2.716616 train_acc=0.2235 val_loss=2.751917 val_acc=0.18
03 test_loss=2.746444 test_acc=0.1798
step=18500 train_loss=2.717244 train_acc=0.2441 val_loss=2.755975 val_acc=0.20
69 test_loss=2.748308 test_acc=0.2124
step=19000 train_loss=2.714280 train_acc=0.2606 val_loss=2.757355 val_acc=0.20
14 test_loss=2.746645 test_acc=0.2168
step=19500 train_loss=2.715267 train_acc=0.2303 val_loss=2.756387 val_acc=0.16
85 test_loss=2.747066 test_acc=0.1935
step=20000 train_loss=2.724073 train_acc=0.2240 val_loss=2.761712 val_acc=0.19
59 test_loss=2.753395 test_acc=0.1950
step=20500 train_loss=2.712655 train_acc=0.2078 val_loss=2.752467 val_acc=0.14
97 test_loss=2.740444 test_acc=0.1788
step=21000 train_loss=2.716588 train_acc=0.2117 val_loss=2.761590 val_acc=0.15
75 test_loss=2.750855 test_acc=0.1750
_loss=2.722811 train_acc=0.2316 val_loss=2.762143 val_acc=0.17
71 test_loss=2.756541 test_acc=0.1857
```

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```
step=22000 train_loss=2.722587 train_acc=0.2457 val_loss=2.764930 val_acc=0.19
59 test_loss=2.753009 test_acc=0.2122
step=22500 train_loss=2.717301 train_acc=0.2217 val_loss=2.763131 val_acc=0.15
60 test_loss=2.747098 test_acc=0.1837
step=23000 train_loss=2.713503 train_acc=0.2446 val_loss=2.754912 val_acc=0.19
51 test_loss=2.742486 test_acc=0.2059
step=23500 train_loss=2.716192 train_acc=0.2211 val_loss=2.758043 val_acc=0.18
03 test_loss=2.751383 test_acc=0.1828
step=24000 train_loss=2.710131 train_acc=0.2253 val_loss=2.756513 val_acc=0.16
22 test_loss=2.747123 test_acc=0.1858
step=24500 train_loss=2.705155 train_acc=0.2266 val_loss=2.741057 val_acc=0.17
48 test_loss=2.737213 test_acc=0.1852
step=25000 train_loss=2.711147 train_acc=0.2078 val_loss=2.751634 val_acc=0.18
03 test_loss=2.745688 test_acc=0.1706
step=25500 train_loss=2.709694 train_acc=0.2115 val_loss=2.751998 val_acc=0.17
55 test_loss=2.737987 test_acc=0.1853
step=26000 train_loss=2.714710 train_acc=0.2337 val_loss=2.759582 val_acc=0.19
12 test_loss=2.746459 test_acc=0.1940
step=26500 train_loss=2.710153 train_acc=0.2235 val_loss=2.747912 val_acc=0.18
26 test_loss=2.738734 test_acc=0.1942
step=27000 train_loss=2.703823 train_acc=0.2222 val_loss=2.742390 val_acc=0.17
87 test_loss=2.735805 test_acc=0.1802
step=27500 train_loss=2.706269 train_acc=0.2300 val_loss=2.740580 val_acc=0.18
34 test_loss=2.736363 test_acc=0.1896
step=28000 train_loss=2.704011 train_acc=0.2543 val_loss=2.745984 val_acc=0.19
75 test_loss=2.737641 test_acc=0.2168
step=28500 train_loss=2.701624 train_acc=0.2123 val_loss=2.742430 val_acc=0.15
91 test_loss=2.737501 test_acc=0.1755
step=29000 train_loss=2.704956 train_acc=0.2068 val_loss=2.743230 val_acc=0.17
48 test_loss=2.740108 test_acc=0.1682
step=29500 train_loss=2.704049 train_acc=0.2120 val_loss=2.744810 val_acc=0.17
79 test_loss=2.737713 test_acc=0.1787
step=30000 train_loss=2.701467 train_acc=0.2418 val_loss=2.745019 val_acc=0.18
73 test_loss=2.731506 test_acc=0.1999
step=30500 train_loss=2.700835 train_acc=0.2517 val_loss=2.733156 val_acc=0.21
39 test_loss=2.729842 test_acc=0.2128
step=31000 train_loss=2.703945 train_acc=0.2068 val_loss=2.744350 val_acc=0.15
60 test_loss=2.734726 test_acc=0.1704
step=31500 train_loss=2.710252 train_acc=0.2167 val_loss=2.755034 val_acc=0.17
08 test_loss=2.746053 test_acc=0.1743
step=32000 train_loss=2.706762 train_acc=0.2180 val_loss=2.746085 val_acc=0.17
16 test_loss=2.738106 test_acc=0.1828
step=32500 train_loss=2.700085 train_acc=0.2386 val_loss=2.745911 val_acc=0.19
98 test_loss=2.729823 test_acc=0.2066
step=33000 train_loss=2.700784 train_acc=0.2514 val_loss=2.745302 val_acc=0.20
06 test_loss=2.734872 test_acc=0.2089
step=33500 train_loss=2.700349 train_acc=0.2363 val_loss=2.743066 val_acc=0.18
73 test_loss=2.734015 test_acc=0.1940
step=34000 train_loss=2.699819 train_acc=0.2418 val_loss=2.738594 val_acc=0.18
81 test_loss=2.731459 test_acc=0.2064
step=34500 train_loss=2.698397 train_acc=0.2457 val_loss=2.745006 val_acc=0.19
83 test_loss=2.731384 test_acc=0.1999
step=35000 train_loss=2.692565 train_acc=0.2065 val_loss=2.735591 val_acc=0.15
60 test_loss=2.727907 test_acc=0.1784
step=35500 train_loss=2.696194 train_acc=0.2057 val_loss=2.735039 val_acc=0.16
38 test_loss=2.725654 test_acc=0.1656
step=36000 train_loss=2.687143 train_acc=0.2298 val_loss=2.733445 val_acc=0.18
10 test_loss=2.722142 test_acc=0.1878
step=36500 train_loss=2.691372 train_acc=0.2504 val_loss=2.730776 val_acc=0.19
67 test_loss=2.720729 test_acc=0.2118
```

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```
step=37000 train_loss=2.692032 train_acc=0.2238 val_loss=2.743624 val_acc=0.15
91 test_loss=2.725873 test_acc=0.1833
step=37500 train_loss=2.693396 train_acc=0.2658 val_loss=2.737488 val_acc=0.21
55 test_loss=2.726925 test_acc=0.2179
step=38000 train_loss=2.684904 train_acc=0.2535 val_loss=2.723755 val_acc=0.19
28 test_loss=2.718926 test_acc=0.2196
step=38500 train_loss=2.683425 train_acc=0.2572 val_loss=2.723305 val_acc=0.21
79 test_loss=2.714463 test_acc=0.2221
step=39000 train_loss=2.683195 train_acc=0.2608 val_loss=2.727119 val_acc=0.19
59 test_loss=2.715406 test_acc=0.2185
step=39500 train_loss=2.686686 train_acc=0.2373 val_loss=2.731309 val_acc=0.19
67 test_loss=2.719431 test_acc=0.2019
step=40000 train_loss=2.686445 train_acc=0.2230 val_loss=2.726295 val_acc=0.17
95 test_loss=2.719439 test_acc=0.1847
step=40500 train_loss=2.682330 train_acc=0.2582 val_loss=2.723614 val_acc=0.19
59 test_loss=2.714428 test_acc=0.2140
step=41000 train_loss=2.683000 train_acc=0.2381 val_loss=2.728243 val_acc=0.17
48 test_loss=2.711146 test_acc=0.2017
step=41500 train_loss=2.682154 train_acc=0.2316 val_loss=2.725807 val_acc=0.17
71 test_loss=2.717723 test_acc=0.1903
step=42000 train_loss=2.687157 train_acc=0.2295 val_loss=2.731021 val_acc=0.18
18 test_loss=2.720318 test_acc=0.1955
step=42500 train_loss=2.681433 train_acc=0.2413 val_loss=2.726788 val_acc=0.18
10 test_loss=2.715000 test_acc=0.2040
step=43000 train_loss=2.682354 train_acc=0.2538 val_loss=2.723573 val_acc=0.19
20 test_loss=2.713078 test_acc=0.2080
step=43500 train_loss=2.676765 train_acc=0.2115 val_loss=2.716575 val_acc=0.16
54 test_loss=2.707500 test_acc=0.1806
step=44000 train_loss=2.676548 train_acc=0.2238 val_loss=2.719913 val_acc=0.16
77 test_loss=2.708599 test_acc=0.1913
step=44500 train_loss=2.671109 train_acc=0.2334 val_loss=2.714585 val_acc=0.17
71 test_loss=2.701458 test_acc=0.1894
step=45000 train_loss=2.673493 train_acc=0.2812 val_loss=2.716986 val_acc=0.21
32 test_loss=2.708834 test_acc=0.2303
step=45500 train_loss=2.675903 train_acc=0.2313 val_loss=2.716262 val_acc=0.17
24 test_loss=2.706353 test_acc=0.1924
step=46000 train_loss=2.672588 train_acc=0.2339 val_loss=2.716292 val_acc=0.17
08 test_loss=2.702582 test_acc=0.2002
step=46500 train_loss=2.676371 train_acc=0.2486 val_loss=2.723520 val_acc=0.19
04 test_loss=2.710076 test_acc=0.2055
step=47000 train_loss=2.671483 train_acc=0.2554 val_loss=2.715127 val_acc=0.20
14 test_loss=2.702010 test_acc=0.2051
step=47500 train_loss=2.676364 train_acc=0.2415 val_loss=2.718897 val_acc=0.19
20 test_loss=2.706467 test_acc=0.1969
step=48000 train_loss=2.669752 train_acc=0.2702 val_loss=2.715918 val_acc=0.22
18 test_loss=2.703423 test_acc=0.2303
step=48500 train_loss=2.663837 train_acc=0.2546 val_loss=2.702020 val_acc=0.17
87 test_loss=2.693892 test_acc=0.2101
step=49000 train_loss=2.668200 train_acc=0.2496 val_loss=2.704410 val_acc=0.20
61 test_loss=2.696445 test_acc=0.2101
step=49500 train_loss=2.668096 train_acc=0.2376 val_loss=2.711366 val_acc=0.18
50 test_loss=2.699684 test_acc=0.1909
step=50000 train_loss=2.664490 train_acc=0.2486 val_loss=2.706982 val_acc=0.18
57 test_loss=2.696021 test_acc=0.2046
step=50500 train_loss=2.662021 train_acc=0.2363 val_loss=2.707101 val_acc=0.18
57 test_loss=2.694080 test_acc=0.1980
step=51000 train_loss=2.662664 train_acc=0.2543 val_loss=2.705124 val_acc=0.19
98 test_loss=2.696622 test_acc=0.2076
step=51500 train_loss=2.661970 train_acc=0.2345 val_loss=2.698448 val_acc=0.20
61 test_loss=2.694904 test_acc=0.1956
```

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```
step=52000 train_loss=2.660007 train_acc=0.2478 val_loss=2.703790 val_acc=0.18
18 test_loss=2.692212 test_acc=0.2050
step=52500 train_loss=2.658720 train_acc=0.2614 val_loss=2.698689 val_acc=0.21
24 test_loss=2.689624 test_acc=0.2152
step=53000 train_loss=2.658039 train_acc=0.2313 val_loss=2.700997 val_acc=0.16
85 test_loss=2.690418 test_acc=0.1944
step=53500 train_loss=2.658326 train_acc=0.2274 val_loss=2.700728 val_acc=0.16
30 test_loss=2.690416 test_acc=0.1830
step=54000 train_loss=2.659941 train_acc=0.2285 val_loss=2.698848 val_acc=0.18
57 test_loss=2.693280 test_acc=0.1890
step=54500 train_loss=2.651325 train_acc=0.2765 val_loss=2.692603 val_acc=0.20
92 test_loss=2.679514 test_acc=0.2273
step=55000 train_loss=2.652240 train_acc=0.2791 val_loss=2.692479 val_acc=0.22
73 test_loss=2.682297 test_acc=0.2367
step=55500 train_loss=2.653666 train_acc=0.2514 val_loss=2.684001 val_acc=0.18
26 test_loss=2.681777 test_acc=0.2075
step=56000 train_loss=2.653309 train_acc=0.2488 val_loss=2.698036 val_acc=0.16
30 test_loss=2.687165 test_acc=0.2062
step=56500 train_loss=2.653994 train_acc=0.2650 val_loss=2.696517 val_acc=0.19
59 test_loss=2.687812 test_acc=0.2169
step=57000 train_loss=2.653757 train_acc=0.2747 val_loss=2.697870 val_acc=0.20
38 test_loss=2.685807 test_acc=0.2327
step=57500 train_loss=2.648844 train_acc=0.2540 val_loss=2.686758 val_acc=0.20
38 test_loss=2.677671 test_acc=0.2102
step=58000 train_loss=2.649628 train_acc=0.2321 val_loss=2.693480 val_acc=0.18
34 test_loss=2.683626 test_acc=0.1871
step=58500 train_loss=2.651837 train_acc=0.2804 val_loss=2.693148 val_acc=0.20
61 test_loss=2.683424 test_acc=0.2306
step=59000 train_loss=2.646305 train_acc=0.2833 val_loss=2.684894 val_acc=0.22
02 test_loss=2.676690 test_acc=0.2333
step=59500 train_loss=2.644594 train_acc=0.2728 val_loss=2.687184 val_acc=0.20
06 test_loss=2.674697 test_acc=0.2325
step=60000 train_loss=2.644452 train_acc=0.2695 val_loss=2.688589 val_acc=0.20
06 test_loss=2.676131 test_acc=0.2225
step=60500 train_loss=2.644457 train_acc=0.2413 val_loss=2.689232 val_acc=0.18
34 test_loss=2.674694 test_acc=0.1999
step=61000 train_loss=2.641548 train_acc=0.2368 val_loss=2.675186 val_acc=0.17
87 test_loss=2.671394 test_acc=0.1892
step=61500 train_loss=2.641359 train_acc=0.2540 val_loss=2.681823 val_acc=0.20
30 test_loss=2.671611 test_acc=0.2131
step=62000 train_loss=2.641120 train_acc=0.2783 val_loss=2.676934 val_acc=0.21
71 test_loss=2.675175 test_acc=0.2298
step=62500 train_loss=2.639787 train_acc=0.3010 val_loss=2.674235 val_acc=0.24
61 test_loss=2.667454 test_acc=0.2423
step=63000 train_loss=2.642001 train_acc=0.2653 val_loss=2.685562 val_acc=0.20
69 test_loss=2.675536 test_acc=0.2207
step=63500 train_loss=2.637504 train_acc=0.2538 val_loss=2.681680 val_acc=0.19
12 test_loss=2.671735 test_acc=0.2061
step=64000 train_loss=2.635167 train_acc=0.2744 val_loss=2.669983 val_acc=0.23
59 test_loss=2.665128 test_acc=0.2348
step=64500 train_loss=2.634338 train_acc=0.2593 val_loss=2.673976 val_acc=0.19
20 test_loss=2.663522 test_acc=0.2110
step=65000 train_loss=2.636530 train_acc=0.2702 val_loss=2.678833 val_acc=0.21
08 test_loss=2.669598 test_acc=0.2239
step=65500 train_loss=2.633156 train_acc=0.2687 val_loss=2.676805 val_acc=0.18
73 test_loss=2.662930 test_acc=0.2221
step=66000 train_loss=2.632652 train_acc=0.2598 val_loss=2.670912 val_acc=0.20
92 test_loss=2.661893 test_acc=0.2121
_loss=2.635575 train_acc=0.2663 val_loss=2.680114 val_acc=0.19
75 test_loss=2.667868 test_acc=0.2225
```

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```
step=67000 train_loss=2.634700 train_acc=0.2629 val_loss=2.670989 val_acc=0.20
53 test_loss=2.665997 test_acc=0.2131
step=67500 train_loss=2.631369 train_acc=0.2822 val_loss=2.673899 val_acc=0.19
91 test_loss=2.661474 test_acc=0.2408
step=68000 train_loss=2.631618 train_acc=0.2540 val_loss=2.673251 val_acc=0.20
45 test_loss=2.661320 test_acc=0.2155
step=68500 train_loss=2.626400 train_acc=0.2611 val_loss=2.671620 val_acc=0.19
12 test_loss=2.656886 test_acc=0.2169
step=69000 train_loss=2.628937 train_acc=0.2585 val_loss=2.670236 val_acc=0.20
22 test_loss=2.663068 test_acc=0.2104
step=69500 train_loss=2.628651 train_acc=0.2935 val_loss=2.664708 val_acc=0.24
92 test_loss=2.658214 test_acc=0.2439
step=70000 train_loss=2.629786 train_acc=0.2483 val_loss=2.669325 val_acc=0.19
04 test_loss=2.660987 test_acc=0.2066
step=70500 train_loss=2.626299 train_acc=0.2930 val_loss=2.665095 val_acc=0.23
90 test_loss=2.657060 test_acc=0.2442
step=71000 train_loss=2.623456 train_acc=0.2708 val_loss=2.661968 val_acc=0.22
02 test_loss=2.655988 test_acc=0.2258
step=71500 train_loss=2.623959 train_acc=0.2862 val_loss=2.666276 val_acc=0.22
88 test_loss=2.657462 test_acc=0.2328
step=72000 train_loss=2.622713 train_acc=0.2742 val_loss=2.664550 val_acc=0.20
77 test_loss=2.654846 test_acc=0.2215
step=72500 train_loss=2.622940 train_acc=0.2731 val_loss=2.667320 val_acc=0.21
71 test_loss=2.656904 test_acc=0.2248
step=73000 train_loss=2.621093 train_acc=0.2742 val_loss=2.663871 val_acc=0.20
38 test_loss=2.654017 test_acc=0.2258
step=73500 train_loss=2.622549 train_acc=0.2554 val_loss=2.663854 val_acc=0.19
12 test_loss=2.654502 test_acc=0.2100
step=74000 train_loss=2.621537 train_acc=0.2556 val_loss=2.660623 val_acc=0.19
36 test_loss=2.650787 test_acc=0.2117
step=74500 train_loss=2.619745 train_acc=0.3065 val_loss=2.663571 val_acc=0.23
90 test_loss=2.651714 test_acc=0.2550
step=75000 train_loss=2.617640 train_acc=0.2752 val_loss=2.659436 val_acc=0.20
22 test_loss=2.649231 test_acc=0.2276
step=75500 train_loss=2.618237 train_acc=0.2742 val_loss=2.660393 val_acc=0.21
39 test_loss=2.651107 test_acc=0.2218
step=76000 train_loss=2.617688 train_acc=0.2781 val_loss=2.664326 val_acc=0.20
22 test_loss=2.651260 test_acc=0.2276
step=76500 train_loss=2.615646 train_acc=0.2843 val_loss=2.657212 val_acc=0.22
65 test_loss=2.647711 test_acc=0.2348
step=77000 train_loss=2.614999 train_acc=0.2467 val_loss=2.658690 val_acc=0.17
87 test_loss=2.648021 test_acc=0.1964
step=77500 train_loss=2.614531 train_acc=0.2893 val_loss=2.656991 val_acc=0.22
26 test_loss=2.646030 test_acc=0.2473
step=78000 train_loss=2.612583 train_acc=0.2966 val_loss=2.652236 val_acc=0.23
04 test_loss=2.643028 test_acc=0.2419
step=78500 train_loss=2.613816 train_acc=0.2796 val_loss=2.658691 val_acc=0.19
51 test_loss=2.644092 test_acc=0.2295
step=79000 train_loss=2.613520 train_acc=0.2595 val_loss=2.659800 val_acc=0.18
57 test_loss=2.646091 test_acc=0.2080
step=79500 train_loss=2.612454 train_acc=0.2773 val_loss=2.653821 val_acc=0.21
63 test_loss=2.644841 test_acc=0.2256
step=80000 train_loss=2.612156 train_acc=0.2668 val_loss=2.652685 val_acc=0.19
67 test_loss=2.643493 test_acc=0.2179
step=80500 train_loss=2.610632 train_acc=0.2940 val_loss=2.650863 val_acc=0.22
81 test_loss=2.642270 test_acc=0.2414
step=81000 train_loss=2.610503 train_acc=0.2833 val_loss=2.651877 val_acc=0.21
79 test_loss=2.642218 test_acc=0.2339
_loss=2.610086 train_acc=0.2854 val_loss=2.652330 val_acc=0.21
79 test_loss=2.639749 test_acc=0.2399
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```
step=82000 train_loss=2.610157 train_acc=0.3037 val_loss=2.651235 val_acc=0.24
29 test_loss=2.641346 test_acc=0.2533
step=82500 train_loss=2.609391 train_acc=0.2862 val_loss=2.647117 val_acc=0.22
02 test_loss=2.638967 test_acc=0.2339
step=83000 train_loss=2.609252 train_acc=0.2924 val_loss=2.650200 val_acc=0.23
82 test_loss=2.641290 test_acc=0.2402
step=83500 train_loss=2.608824 train_acc=0.2483 val_loss=2.651569 val_acc=0.17
79 test_loss=2.640119 test_acc=0.2002
step=84000 train_loss=2.607601 train_acc=0.2974 val_loss=2.650580 val_acc=0.22
73 test_loss=2.640493 test_acc=0.2439
step=84500 train_loss=2.606387 train_acc=0.2890 val_loss=2.649148 val_acc=0.21
94 test_loss=2.638125 test_acc=0.2362
step=85000 train_loss=2.606022 train_acc=0.2535 val_loss=2.647379 val_acc=0.18
42 test_loss=2.637677 test_acc=0.2042
step=85500 train_loss=2.604747 train_acc=0.2869 val_loss=2.643771 val_acc=0.22
57 test_loss=2.636544 test_acc=0.2410
step=86000 train_loss=2.605276 train_acc=0.2702 val_loss=2.647326 val_acc=0.21
08 test_loss=2.637488 test_acc=0.2200
step=86500 train_loss=2.604100 train_acc=0.2854 val_loss=2.646450 val_acc=0.21
16 test_loss=2.635035 test_acc=0.2362
step=87000 train_loss=2.604328 train_acc=0.2966 val_loss=2.645354 val_acc=0.23
90 test_loss=2.634882 test_acc=0.2443
step=87500 train_loss=2.604028 train_acc=0.2721 val_loss=2.646067 val_acc=0.20
61 test_loss=2.636958 test_acc=0.2234
step=88000 train_loss=2.605125 train_acc=0.2807 val_loss=2.647276 val_acc=0.21
39 test_loss=2.635594 test_acc=0.2306
step=88500 train_loss=2.604152 train_acc=0.3371 val_loss=2.641834 val_acc=0.27
35 test_loss=2.635816 test_acc=0.2829
step=89000 train_loss=2.603598 train_acc=0.2875 val_loss=2.644339 val_acc=0.21
71 test_loss=2.633594 test_acc=0.2416
step=89500 train_loss=2.603564 train_acc=0.2655 val_loss=2.644288 val_acc=0.19
91 test_loss=2.636589 test_acc=0.2134
step=90000 train_loss=2.602937 train_acc=0.2533 val_loss=2.644164 val_acc=0.19
04 test_loss=2.635075 test_acc=0.2040
step=90500 train_loss=2.602714 train_acc=0.2909 val_loss=2.642323 val_acc=0.22
18 test_loss=2.633878 test_acc=0.2391
step=91000 train_loss=2.602222 train_acc=0.2689 val_loss=2.642140 val_acc=0.20
69 test_loss=2.633818 test_acc=0.2203
step=91500 train_loss=2.602796 train_acc=0.2997 val_loss=2.645340 val_acc=0.24
14 test_loss=2.634338 test_acc=0.2523
step=92000 train_loss=2.602739 train_acc=0.2574 val_loss=2.645832 val_acc=0.19
91 test_loss=2.634208 test_acc=0.2074
step=92500 train_loss=2.601815 train_acc=0.3172 val_loss=2.642333 val_acc=0.24
92 test_loss=2.632669 test_acc=0.2666
step=93000 train_loss=2.601933 train_acc=0.3039 val_loss=2.643772 val_acc=0.24
22 test_loss=2.632269 test_acc=0.2571
step=93500 train_loss=2.601225 train_acc=0.2762 val_loss=2.642178 val_acc=0.22
65 test_loss=2.633045 test_acc=0.2268
step=94000 train_loss=2.601532 train_acc=0.2885 val_loss=2.643642 val_acc=0.22
26 test_loss=2.634114 test_acc=0.2369
step=94500 train_loss=2.600455 train_acc=0.2950 val_loss=2.641311 val_acc=0.22
26 test_loss=2.631672 test_acc=0.2414
step=95000 train_loss=2.601015 train_acc=0.2679 val_loss=2.640047 val_acc=0.20
61 test_loss=2.629937 test_acc=0.2175
step=95500 train_loss=2.599874 train_acc=0.2872 val_loss=2.636299 val_acc=0.22
73 test_loss=2.629678 test_acc=0.2372
step=96000 train_loss=2.600207 train_acc=0.2574 val_loss=2.639185 val_acc=0.19
36 test_loss=2.629142 test_acc=0.2101
_loss=2.599257 train_acc=0.3107 val_loss=2.639071 val_acc=0.24
22 test_loss=2.629932 test_acc=0.2567
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```
step=97000 train_loss=2.599473 train_acc=0.3144 val_loss=2.640074 val_acc=0.24
53 test_loss=2.630227 test_acc=0.2584
step=97500 train_loss=2.598986 train_acc=0.3099 val_loss=2.640303 val_acc=0.23
75 test_loss=2.629300 test_acc=0.2519
step=98000 train_loss=2.599185 train_acc=0.2932 val_loss=2.640341 val_acc=0.22
34 test_loss=2.630495 test_acc=0.2402
step=98500 train_loss=2.598508 train_acc=0.3211 val_loss=2.640838 val_acc=0.25
86 test_loss=2.630519 test_acc=0.2674
step=99000 train_loss=2.598378 train_acc=0.2825 val_loss=2.639386 val_acc=0.22
10 test_loss=2.629057 test_acc=0.2322
step=99500 train_loss=2.597917 train_acc=0.2893 val_loss=2.641021 val_acc=0.21
32 test_loss=2.628561 test_acc=0.2397
step=100000 train_loss=2.598456 train_acc=0.2898 val_loss=2.636878 val_acc=0.2
179 test_loss=2.628725 test_acc=0.2369
best metrics: {
  "step": 95500,
  "lr": 0.0001045803511535803,
  "train_loss": 2.5998740417839032,
  "train_acc": 0.28720626631853785,
  "val_loss": 2.636299366487605,
  "val_acc": 0.22727272727272727,
  "test_loss": 2.629677861567252,
  "test_acc": 0.23724389925616599,
  "elapsed_sec": 8798.434955835342
}
checkpoint dir: /Users/shobhit/Documents/UW/CSE 493/cse493s-spring26-hw1/out/s
ubmission/part1/warmup/add_p113_l2_s42

$ /opt/homebrew/Caskroom/miniconda/base/envs/cse493/bin/python train.py --mode
modular --op - --p 97 --n_layer 1 --n_head 4 --n_embd 128 --steps 100000 --see
d 42 --batch_size 128 --lr 0.001 --min_lr 0.0001 --lr_schedule cosine --warmup
_steps 1000 --grad_clip 1.0 --beta1 0.9 --beta2 0.95 --weight_decay 0.01 --opt
imizer adam --eval_interval 500 --log_interval 500 --out_dir '/Users/shobhit/D
ocuments/UW/CSE 493/cse493s-spring26-hw1/out/submission/part1/warmup/sub_p97_l
1_s42' --bias --resume
number of parameters: 0.21M
[resume] model_last.pt not found in /Users/shobhit/Documents/UW/CSE 493/cse493
s-spring26-hw1/out/submission/part1/warmup/sub_p97_l1_s42; starting fresh
device=mps train=2822 val=940 test=5647 batch_size=128
step=1 train_loss=4.686533 train_acc=0.0000 val_loss=4.686589 val_acc=0.0000 t
est_loss=4.687852 test_acc=0.0000
step=500 train_loss=4.285946 train_acc=0.0705 val_loss=4.842632 val_acc=0.0021
test_loss=4.848047 test_acc=0.0012
step=1000 train_loss=4.251603 train_acc=0.0811 val_loss=4.862812 val_acc=0.000
0 test_loss=4.875589 test_acc=0.0009
step=1500 train_loss=4.362493 train_acc=0.0471 val_loss=4.802447 val_acc=0.002
1 test_loss=4.808020 test_acc=0.0025
step=2000 train_loss=4.296999 train_acc=0.0542 val_loss=4.911804 val_acc=0.002
1 test_loss=4.914841 test_acc=0.0012
step=2500 train_loss=4.258339 train_acc=0.0599 val_loss=4.948513 val_acc=0.003
2 test_loss=4.956987 test_acc=0.0005
step=3000 train_loss=4.247303 train_acc=0.0574 val_loss=4.987138 val_acc=0.002
1 test_loss=4.990081 test_acc=0.0014
step=3500 train_loss=4.235094 train_acc=0.0581 val_loss=4.970453 val_acc=0.001
1 test_loss=4.985171 test_acc=0.0012
step=4000 train_loss=4.227363 train_acc=0.0673 val_loss=4.992991 val_acc=0.002
1 test_loss=4.990009 test_acc=0.0019
step=4500 train_loss=4.237495 train_acc=0.0648 val_loss=4.978728 val_acc=0.005
                                                  4067 test_acc=0.0030
step=5000 train_loss=4.219229 train_acc=0.0606 val_loss=5.016569 val_acc=0.002
```

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```
1 test_loss=5.018972 test_acc=0.0018
step=5500 train_loss=4.213020 train_acc=0.0659 val_loss=5.011593 val_acc=0.004
3 test_loss=5.010539 test_acc=0.0021
step=6000 train_loss=4.199878 train_acc=0.0659 val_loss=5.006874 val_acc=0.003
2 test_loss=5.005613 test_acc=0.0012
step=6500 train_loss=4.199984 train_acc=0.0666 val_loss=5.025853 val_acc=0.004
3 test_loss=5.008168 test_acc=0.0018
step=7000 train_loss=4.190961 train_acc=0.0691 val_loss=5.005318 val_acc=0.001
1 test_loss=5.002504 test_acc=0.0018
step=7500 train_loss=4.191736 train_acc=0.0687 val_loss=5.008520 val_acc=0.004
3 test_loss=4.999844 test_acc=0.0030
step=8000 train_loss=4.181098 train_acc=0.0705 val_loss=4.975169 val_acc=0.002
1 test_loss=4.989138 test_acc=0.0035
step=8500 train_loss=4.045646 train_acc=0.0858 val_loss=4.645991 val_acc=0.009
6 test_loss=4.636848 test_acc=0.0073
step=9000 train_loss=3.670777 train_acc=0.1304 val_loss=3.993578 val_acc=0.023
4 test_loss=3.989481 test_acc=0.0260
step=9500 train_loss=3.316083 train_acc=0.1513 val_loss=3.482737 val_acc=0.083
0 test_loss=3.479665 test_acc=0.0749
step=10000 train_loss=3.171404 train_acc=0.1658 val_loss=3.311205 val_acc=0.08
94 test_loss=3.309130 test_acc=0.0917
step=10500 train_loss=3.126633 train_acc=0.1527 val_loss=3.206281 val_acc=0.11
60 test_loss=3.211551 test_acc=0.1047
step=11000 train_loss=3.236333 train_acc=0.1481 val_loss=3.274568 val_acc=0.11
06 test_loss=3.283833 test_acc=0.1075
step=11500 train_loss=3.223655 train_acc=0.1375 val_loss=3.263253 val_acc=0.11
60 test_loss=3.271914 test_acc=0.0972
step=12000 train_loss=3.226084 train_acc=0.1304 val_loss=3.270266 val_acc=0.10
43 test_loss=3.278389 test_acc=0.0954
step=12500 train_loss=3.279811 train_acc=0.1077 val_loss=3.327441 val_acc=0.07
45 test_loss=3.327896 test_acc=0.0790
step=13000 train_loss=3.218481 train_acc=0.1276 val_loss=3.258063 val_acc=0.09
79 test_loss=3.265808 test_acc=0.0884
step=13500 train_loss=3.214172 train_acc=0.1460 val_loss=3.256050 val_acc=0.11
70 test_loss=3.263672 test_acc=0.1071
step=14000 train_loss=3.199042 train_acc=0.1396 val_loss=3.237095 val_acc=0.10
74 test_loss=3.248666 test_acc=0.0986
step=14500 train_loss=3.203862 train_acc=0.1442 val_loss=3.244016 val_acc=0.11
70 test_loss=3.250258 test_acc=0.1039
step=15000 train_loss=3.229890 train_acc=0.1318 val_loss=3.269213 val_acc=0.09
68 test_loss=3.273367 test_acc=0.0963
step=15500 train_loss=3.216214 train_acc=0.1502 val_loss=3.257594 val_acc=0.12
23 test_loss=3.266200 test_acc=0.1075
step=16000 train_loss=3.276144 train_acc=0.1233 val_loss=3.312320 val_acc=0.10
21 test_loss=3.321969 test_acc=0.0910
step=16500 train_loss=3.196988 train_acc=0.1570 val_loss=3.234087 val_acc=0.13
94 test_loss=3.249028 test_acc=0.1135
step=17000 train_loss=3.196409 train_acc=0.1492 val_loss=3.231502 val_acc=0.11
91 test_loss=3.244291 test_acc=0.1098
step=17500 train_loss=3.226900 train_acc=0.1339 val_loss=3.270122 val_acc=0.09
68 test_loss=3.276575 test_acc=0.0993
step=18000 train_loss=3.215822 train_acc=0.1453 val_loss=3.259603 val_acc=0.11
06 test_loss=3.266326 test_acc=0.0974
step=18500 train_loss=3.215324 train_acc=0.1364 val_loss=3.257755 val_acc=0.09
79 test_loss=3.266725 test_acc=0.1029
step=19000 train_loss=3.221603 train_acc=0.1347 val_loss=3.263692 val_acc=0.10
96 test_loss=3.270802 test_acc=0.0949
step=19500 train_loss=3.245042 train_acc=0.1502 val_loss=3.281971 val_acc=0.12
95758 test_acc=0.1047
step=20000 train_loss=3.232052 train_acc=0.1442 val_loss=3.271790 val_acc=0.12
```

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```
77 test_loss=3.283305 test_acc=0.1041
step=20500 train_loss=3.231586 train_acc=0.1162 val_loss=3.272090 val_acc=0.08
51 test_loss=3.279256 test_acc=0.0777
step=21000 train_loss=3.233680 train_acc=0.1584 val_loss=3.271976 val_acc=0.13
72 test_loss=3.281216 test_acc=0.1137
step=21500 train_loss=3.240705 train_acc=0.1407 val_loss=3.278117 val_acc=0.11
38 test_loss=3.286224 test_acc=0.1071
step=22000 train_loss=3.239454 train_acc=0.1251 val_loss=3.277300 val_acc=0.10
32 test_loss=3.291788 test_acc=0.0846
step=22500 train_loss=3.244943 train_acc=0.1517 val_loss=3.286892 val_acc=0.12
77 test_loss=3.295159 test_acc=0.1064
step=23000 train_loss=3.236494 train_acc=0.1357 val_loss=3.276629 val_acc=0.10
32 test_loss=3.284003 test_acc=0.0951
step=23500 train_loss=3.255228 train_acc=0.1517 val_loss=3.295848 val_acc=0.13
09 test_loss=3.306550 test_acc=0.1064
step=24000 train_loss=3.231909 train_acc=0.1407 val_loss=3.266701 val_acc=0.10
53 test_loss=3.277727 test_acc=0.0956
step=24500 train_loss=3.257779 train_acc=0.1304 val_loss=3.293977 val_acc=0.10
85 test_loss=3.306255 test_acc=0.0912
step=25000 train_loss=3.294673 train_acc=0.1290 val_loss=3.334910 val_acc=0.08
30 test_loss=3.345935 test_acc=0.0875
step=25500 train_loss=3.270733 train_acc=0.1421 val_loss=3.315182 val_acc=0.10
21 test_loss=3.323423 test_acc=0.1039
step=26000 train_loss=3.262298 train_acc=0.1371 val_loss=3.299431 val_acc=0.12
02 test_loss=3.310800 test_acc=0.0999
step=26500 train_loss=3.254715 train_acc=0.1297 val_loss=3.302006 val_acc=0.10
96 test_loss=3.305792 test_acc=0.0954
step=27000 train_loss=3.249473 train_acc=0.1446 val_loss=3.290321 val_acc=0.10
85 test_loss=3.299676 test_acc=0.1059
step=27500 train_loss=3.276396 train_acc=0.1354 val_loss=3.315434 val_acc=0.09
47 test_loss=3.325314 test_acc=0.0931
step=28000 train_loss=3.281464 train_acc=0.1269 val_loss=3.317153 val_acc=0.10
85 test_loss=3.331560 test_acc=0.0914
step=28500 train_loss=3.271056 train_acc=0.1410 val_loss=3.305325 val_acc=0.10
85 test_loss=3.319134 test_acc=0.0953
step=29000 train_loss=3.302071 train_acc=0.1155 val_loss=3.341193 val_acc=0.08
30 test_loss=3.350656 test_acc=0.0838
step=29500 train_loss=3.274512 train_acc=0.1471 val_loss=3.312227 val_acc=0.12
13 test_loss=3.321754 test_acc=0.1020
step=30000 train_loss=3.286392 train_acc=0.1212 val_loss=3.318827 val_acc=0.10
53 test_loss=3.332704 test_acc=0.0813
step=30500 train_loss=3.253957 train_acc=0.1499 val_loss=3.298587 val_acc=0.11
91 test_loss=3.304354 test_acc=0.1086
step=31000 train_loss=3.253539 train_acc=0.1332 val_loss=3.299183 val_acc=0.11
28 test_loss=3.305004 test_acc=0.0923
step=31500 train_loss=3.260317 train_acc=0.1350 val_loss=3.302750 val_acc=0.11
60 test_loss=3.308398 test_acc=0.0995
step=32000 train_loss=3.275918 train_acc=0.1339 val_loss=3.318730 val_acc=0.12
45 test_loss=3.327960 test_acc=0.0939
step=32500 train_loss=3.267267 train_acc=0.1371 val_loss=3.303906 val_acc=0.12
66 test_loss=3.315532 test_acc=0.1004
step=33000 train_loss=3.247476 train_acc=0.1712 val_loss=3.283014 val_acc=0.12
55 test_loss=3.293881 test_acc=0.1302
step=33500 train_loss=3.233436 train_acc=0.1417 val_loss=3.273506 val_acc=0.12
77 test_loss=3.284984 test_acc=0.1039
step=34000 train_loss=3.238098 train_acc=0.1538 val_loss=3.275456 val_acc=0.12
02 test_loss=3.286443 test_acc=0.1110
step=34500 train_loss=3.229055 train_acc=0.1648 val_loss=3.268140 val_acc=0.12
76206 test_acc=0.1176
step=35000 train_loss=3.237340 train_acc=0.1662 val_loss=3.273187 val_acc=0.13
```

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```
62 test_loss=3.286046 test_acc=0.1142
step=35500 train_loss=3.253223 train_acc=0.1439 val_loss=3.290374 val_acc=0.11
70 test_loss=3.299508 test_acc=0.1025
step=36000 train_loss=3.277209 train_acc=0.1453 val_loss=3.312955 val_acc=0.10
85 test_loss=3.326854 test_acc=0.0990
step=36500 train_loss=3.232278 train_acc=0.1474 val_loss=3.271592 val_acc=0.12
45 test_loss=3.278516 test_acc=0.1093
step=37000 train_loss=3.227226 train_acc=0.1446 val_loss=3.262225 val_acc=0.12
13 test_loss=3.273892 test_acc=0.1043
step=37500 train_loss=3.264851 train_acc=0.1499 val_loss=3.303504 val_acc=0.12
66 test_loss=3.313495 test_acc=0.1080
step=38000 train_loss=3.263471 train_acc=0.1478 val_loss=3.302822 val_acc=0.11
91 test_loss=3.312767 test_acc=0.1063
step=38500 train_loss=3.235131 train_acc=0.1368 val_loss=3.280755 val_acc=0.09
89 test_loss=3.284219 test_acc=0.0944
step=39000 train_loss=3.232013 train_acc=0.1687 val_loss=3.269649 val_acc=0.14
04 test_loss=3.283041 test_acc=0.1162
step=39500 train_loss=3.242283 train_acc=0.1446 val_loss=3.276258 val_acc=0.12
98 test_loss=3.289228 test_acc=0.1066
step=40000 train_loss=3.233532 train_acc=0.1552 val_loss=3.274733 val_acc=0.10
96 test_loss=3.282786 test_acc=0.1123
step=40500 train_loss=3.242952 train_acc=0.1226 val_loss=3.280097 val_acc=0.10
85 test_loss=3.290583 test_acc=0.0960
step=41000 train_loss=3.234102 train_acc=0.1428 val_loss=3.265578 val_acc=0.11
06 test_loss=3.278388 test_acc=0.0995
step=41500 train_loss=3.221635 train_acc=0.1350 val_loss=3.257822 val_acc=0.11
49 test_loss=3.267282 test_acc=0.1015
step=42000 train_loss=3.216543 train_acc=0.1775 val_loss=3.254817 val_acc=0.14
15 test_loss=3.266364 test_acc=0.1271
step=42500 train_loss=3.260950 train_acc=0.1272 val_loss=3.297975 val_acc=0.10
85 test_loss=3.310033 test_acc=0.0946
step=43000 train_loss=3.234140 train_acc=0.1566 val_loss=3.272059 val_acc=0.14
15 test_loss=3.282302 test_acc=0.1140
step=43500 train_loss=3.223377 train_acc=0.1531 val_loss=3.263302 val_acc=0.11
60 test_loss=3.273222 test_acc=0.1087
step=44000 train_loss=3.244457 train_acc=0.1322 val_loss=3.281743 val_acc=0.10
11 test_loss=3.290078 test_acc=0.0960
step=44500 train_loss=3.236782 train_acc=0.1449 val_loss=3.277093 val_acc=0.12
02 test_loss=3.287838 test_acc=0.1029
step=45000 train_loss=3.241234 train_acc=0.1371 val_loss=3.278339 val_acc=0.10
32 test_loss=3.294604 test_acc=0.0988
step=45500 train_loss=3.201470 train_acc=0.1665 val_loss=3.240662 val_acc=0.13
51 test_loss=3.250210 test_acc=0.1155
step=46000 train_loss=3.218247 train_acc=0.1577 val_loss=3.252227 val_acc=0.13
30 test_loss=3.267942 test_acc=0.1119
step=46500 train_loss=3.222510 train_acc=0.1439 val_loss=3.257535 val_acc=0.11
70 test_loss=3.270031 test_acc=0.1061
step=47000 train_loss=3.208780 train_acc=0.1393 val_loss=3.248382 val_acc=0.12
45 test_loss=3.258036 test_acc=0.1015
step=47500 train_loss=3.257473 train_acc=0.1212 val_loss=3.298288 val_acc=0.09
89 test_loss=3.301899 test_acc=0.0834
step=48000 train_loss=3.233958 train_acc=0.1332 val_loss=3.269960 val_acc=0.10
96 test_loss=3.279692 test_acc=0.0967
step=48500 train_loss=3.212988 train_acc=0.1375 val_loss=3.249727 val_acc=0.10
21 test_loss=3.262311 test_acc=0.1004
step=49000 train_loss=3.208826 train_acc=0.1378 val_loss=3.248001 val_acc=0.11
38 test_loss=3.256461 test_acc=0.1043
step=49500 train_loss=3.210508 train_acc=0.1534 val_loss=3.252239 val_acc=0.11
61117 test_acc=0.1059
step=50000 train_loss=3.195256 train_acc=0.1598 val_loss=3.233627 val_acc=0.14
```

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```
15 test_loss=3.243962 test_acc=0.1117
step=50500 train_loss=3.213216 train_acc=0.1478 val_loss=3.251237 val_acc=0.10
74 test_loss=3.262095 test_acc=0.1052
step=51000 train_loss=3.188899 train_acc=0.1336 val_loss=3.228499 val_acc=0.10
64 test_loss=3.237612 test_acc=0.0969
step=51500 train_loss=3.201791 train_acc=0.1545 val_loss=3.238353 val_acc=0.13
19 test_loss=3.250819 test_acc=0.1112
step=52000 train_loss=3.205014 train_acc=0.1580 val_loss=3.242098 val_acc=0.13
09 test_loss=3.252880 test_acc=0.1169
step=52500 train_loss=3.182443 train_acc=0.1736 val_loss=3.217387 val_acc=0.14
15 test_loss=3.230010 test_acc=0.1179
step=53000 train_loss=3.194416 train_acc=0.1552 val_loss=3.231203 val_acc=0.11
91 test_loss=3.244162 test_acc=0.1119
step=53500 train_loss=3.203542 train_acc=0.1393 val_loss=3.242108 val_acc=0.10
32 test_loss=3.250891 test_acc=0.1022
step=54000 train_loss=3.209497 train_acc=0.1588 val_loss=3.246798 val_acc=0.13
30 test_loss=3.260853 test_acc=0.1130
step=54500 train_loss=3.203853 train_acc=0.1623 val_loss=3.239871 val_acc=0.12
23 test_loss=3.249633 test_acc=0.1098
step=55000 train_loss=3.186873 train_acc=0.1549 val_loss=3.228849 val_acc=0.11
81 test_loss=3.236446 test_acc=0.1112
step=55500 train_loss=3.191833 train_acc=0.1605 val_loss=3.229687 val_acc=0.12
45 test_loss=3.242567 test_acc=0.1110
step=56000 train_loss=3.176938 train_acc=0.1648 val_loss=3.214519 val_acc=0.13
09 test_loss=3.226626 test_acc=0.1105
step=56500 train_loss=3.181634 train_acc=0.1538 val_loss=3.226730 val_acc=0.12
23 test_loss=3.233115 test_acc=0.1126
step=57000 train_loss=3.176412 train_acc=0.1726 val_loss=3.215811 val_acc=0.13
83 test_loss=3.227194 test_acc=0.1186
step=57500 train_loss=3.170002 train_acc=0.1747 val_loss=3.207519 val_acc=0.13
72 test_loss=3.215386 test_acc=0.1264
step=58000 train_loss=3.166701 train_acc=0.1584 val_loss=3.205708 val_acc=0.12
98 test_loss=3.218611 test_acc=0.1061
step=58500 train_loss=3.198610 train_acc=0.1673 val_loss=3.237353 val_acc=0.13
83 test_loss=3.245219 test_acc=0.1241
step=59000 train_loss=3.164719 train_acc=0.1648 val_loss=3.205951 val_acc=0.13
09 test_loss=3.214577 test_acc=0.1137
step=59500 train_loss=3.153761 train_acc=0.1800 val_loss=3.192629 val_acc=0.14
15 test_loss=3.201615 test_acc=0.1220
step=60000 train_loss=3.169741 train_acc=0.1673 val_loss=3.208799 val_acc=0.13
19 test_loss=3.217596 test_acc=0.1186
step=60500 train_loss=3.163193 train_acc=0.1605 val_loss=3.199787 val_acc=0.13
51 test_loss=3.212650 test_acc=0.1119
step=61000 train_loss=3.162806 train_acc=0.1712 val_loss=3.205169 val_acc=0.14
47 test_loss=3.213436 test_acc=0.1241
step=61500 train_loss=3.159602 train_acc=0.1623 val_loss=3.200126 val_acc=0.12
55 test_loss=3.208927 test_acc=0.1156
step=62000 train_loss=3.154423 train_acc=0.1747 val_loss=3.193849 val_acc=0.13
51 test_loss=3.201097 test_acc=0.1300
step=62500 train_loss=3.159417 train_acc=0.1577 val_loss=3.196640 val_acc=0.13
72 test_loss=3.208046 test_acc=0.1121
step=63000 train_loss=3.154230 train_acc=0.1605 val_loss=3.191894 val_acc=0.12
02 test_loss=3.203260 test_acc=0.1227
step=63500 train_loss=3.145423 train_acc=0.1729 val_loss=3.182274 val_acc=0.14
89 test_loss=3.194623 test_acc=0.1248
step=64000 train_loss=3.133252 train_acc=0.1478 val_loss=3.172945 val_acc=0.12
23 test_loss=3.182815 test_acc=0.1091
step=64500 train_loss=3.134037 train_acc=0.1729 val_loss=3.172733 val_acc=0.13
81397 test_acc=0.1220
step=65000 train_loss=3.136808 train_acc=0.1609 val_loss=3.178317 val_acc=0.12
```

Loading [MathJax]/extensions/Safe.js

```
13 test_loss=3.183716 test_acc=0.1206
step=65500 train_loss=3.129071 train_acc=0.1609 val_loss=3.169436 val_acc=0.12
66 test_loss=3.178577 test_acc=0.1167
step=66000 train_loss=3.140331 train_acc=0.1637 val_loss=3.180452 val_acc=0.12
55 test_loss=3.188759 test_acc=0.1176
step=66500 train_loss=3.121554 train_acc=0.1758 val_loss=3.160341 val_acc=0.13
83 test_loss=3.169185 test_acc=0.1222
step=67000 train_loss=3.119742 train_acc=0.1765 val_loss=3.157494 val_acc=0.14
47 test_loss=3.169733 test_acc=0.1186
step=67500 train_loss=3.124832 train_acc=0.1733 val_loss=3.164543 val_acc=0.12
98 test_loss=3.172509 test_acc=0.1238
step=68000 train_loss=3.119240 train_acc=0.1814 val_loss=3.161341 val_acc=0.14
15 test_loss=3.168864 test_acc=0.1346
step=68500 train_loss=3.129104 train_acc=0.1658 val_loss=3.168228 val_acc=0.14
04 test_loss=3.176793 test_acc=0.1162
step=69000 train_loss=3.144577 train_acc=0.1630 val_loss=3.180381 val_acc=0.14
47 test_loss=3.192035 test_acc=0.1133
step=69500 train_loss=3.122246 train_acc=0.1772 val_loss=3.158839 val_acc=0.14
15 test_loss=3.169279 test_acc=0.1185
step=70000 train_loss=3.118045 train_acc=0.1772 val_loss=3.159107 val_acc=0.13
51 test_loss=3.168433 test_acc=0.1256
step=70500 train_loss=3.118232 train_acc=0.1758 val_loss=3.160057 val_acc=0.13
62 test_loss=3.166193 test_acc=0.1236
step=71000 train_loss=3.098998 train_acc=0.1910 val_loss=3.135160 val_acc=0.15
85 test_loss=3.144564 test_acc=0.1378
step=71500 train_loss=3.106789 train_acc=0.1818 val_loss=3.145645 val_acc=0.15
11 test_loss=3.154103 test_acc=0.1307
step=72000 train_loss=3.105672 train_acc=0.1892 val_loss=3.147097 val_acc=0.15
21 test_loss=3.153380 test_acc=0.1328
step=72500 train_loss=3.100620 train_acc=0.1627 val_loss=3.143539 val_acc=0.12
98 test_loss=3.149196 test_acc=0.1167
step=73000 train_loss=3.090506 train_acc=0.1715 val_loss=3.127083 val_acc=0.13
94 test_loss=3.139105 test_acc=0.1179
step=73500 train_loss=3.090721 train_acc=0.1836 val_loss=3.129392 val_acc=0.14
57 test_loss=3.140756 test_acc=0.1284
step=74000 train_loss=3.088471 train_acc=0.1850 val_loss=3.128325 val_acc=0.14
47 test_loss=3.137306 test_acc=0.1339
step=74500 train_loss=3.087542 train_acc=0.1860 val_loss=3.126952 val_acc=0.14
89 test_loss=3.137783 test_acc=0.1305
step=75000 train_loss=3.090822 train_acc=0.1680 val_loss=3.130172 val_acc=0.13
09 test_loss=3.141374 test_acc=0.1163
step=75500 train_loss=3.088347 train_acc=0.1924 val_loss=3.127693 val_acc=0.14
89 test_loss=3.136532 test_acc=0.1330
step=76000 train_loss=3.084814 train_acc=0.1928 val_loss=3.122986 val_acc=0.15
53 test_loss=3.132603 test_acc=0.1408
step=76500 train_loss=3.083826 train_acc=0.1864 val_loss=3.121528 val_acc=0.16
06 test_loss=3.133354 test_acc=0.1310
step=77000 train_loss=3.081407 train_acc=0.1786 val_loss=3.121272 val_acc=0.14
79 test_loss=3.130594 test_acc=0.1284
step=77500 train_loss=3.077708 train_acc=0.1673 val_loss=3.114882 val_acc=0.13
51 test_loss=3.124211 test_acc=0.1186
step=78000 train_loss=3.071092 train_acc=0.1945 val_loss=3.110132 val_acc=0.15
00 test_loss=3.117647 test_acc=0.1342
step=78500 train_loss=3.066319 train_acc=0.1871 val_loss=3.103487 val_acc=0.15
32 test_loss=3.115131 test_acc=0.1349
step=79000 train_loss=3.067788 train_acc=0.1860 val_loss=3.102305 val_acc=0.15
53 test_loss=3.113474 test_acc=0.1287
step=79500 train_loss=3.064105 train_acc=0.1772 val_loss=3.102533 val_acc=0.13
13896 test_acc=0.1247
step=80000 train_loss=3.058661 train_acc=0.1782 val_loss=3.095135 val_acc=0.12
```

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```
87 test_loss=3.106485 test_acc=0.1229
step=80500 train_loss=3.070415 train_acc=0.1786 val_loss=3.110155 val_acc=0.14
36 test_loss=3.120942 test_acc=0.1229
step=81000 train_loss=3.062076 train_acc=0.1669 val_loss=3.101392 val_acc=0.12
23 test_loss=3.109219 test_acc=0.1183
step=81500 train_loss=3.057819 train_acc=0.1896 val_loss=3.098951 val_acc=0.15
11 test_loss=3.105989 test_acc=0.1406
step=82000 train_loss=3.059123 train_acc=0.1821 val_loss=3.097424 val_acc=0.15
21 test_loss=3.106743 test_acc=0.1307
step=82500 train_loss=3.060576 train_acc=0.1790 val_loss=3.102254 val_acc=0.13
19 test_loss=3.109338 test_acc=0.1312
step=83000 train_loss=3.064663 train_acc=0.1623 val_loss=3.104541 val_acc=0.12
77 test_loss=3.113333 test_acc=0.1181
step=83500 train_loss=3.058033 train_acc=0.1832 val_loss=3.097797 val_acc=0.13
94 test_loss=3.107740 test_acc=0.1240
step=84000 train_loss=3.056062 train_acc=0.1811 val_loss=3.093543 val_acc=0.13
30 test_loss=3.102256 test_acc=0.1254
step=84500 train_loss=3.042857 train_acc=0.1960 val_loss=3.082819 val_acc=0.15
11 test_loss=3.090815 test_acc=0.1387
step=85000 train_loss=3.040787 train_acc=0.1914 val_loss=3.079605 val_acc=0.14
89 test_loss=3.088418 test_acc=0.1371
step=85500 train_loss=3.036184 train_acc=0.1719 val_loss=3.076808 val_acc=0.13
30 test_loss=3.085036 test_acc=0.1181
step=86000 train_loss=3.042084 train_acc=0.1956 val_loss=3.082845 val_acc=0.15
21 test_loss=3.089592 test_acc=0.1431
step=86500 train_loss=3.044183 train_acc=0.1917 val_loss=3.083679 val_acc=0.14
36 test_loss=3.091055 test_acc=0.1339
step=87000 train_loss=3.037690 train_acc=0.1857 val_loss=3.078873 val_acc=0.14
04 test_loss=3.085699 test_acc=0.1367
step=87500 train_loss=3.036084 train_acc=0.1896 val_loss=3.075385 val_acc=0.15
00 test_loss=3.081787 test_acc=0.1387
step=88000 train_loss=3.036419 train_acc=0.1797 val_loss=3.076939 val_acc=0.14
57 test_loss=3.085072 test_acc=0.1286
step=88500 train_loss=3.029203 train_acc=0.1988 val_loss=3.069970 val_acc=0.15
64 test_loss=3.077962 test_acc=0.1415
step=89000 train_loss=3.026680 train_acc=0.1903 val_loss=3.063344 val_acc=0.15
64 test_loss=3.072635 test_acc=0.1374
step=89500 train_loss=3.021929 train_acc=0.1963 val_loss=3.059830 val_acc=0.16
70 test_loss=3.069459 test_acc=0.1449
step=90000 train_loss=3.018507 train_acc=0.1910 val_loss=3.056213 val_acc=0.15
96 test_loss=3.067097 test_acc=0.1346
step=90500 train_loss=3.014484 train_acc=0.1991 val_loss=3.052863 val_acc=0.15
74 test_loss=3.063201 test_acc=0.1418
step=91000 train_loss=3.017440 train_acc=0.1914 val_loss=3.056884 val_acc=0.15
11 test_loss=3.065526 test_acc=0.1280
step=91500 train_loss=3.018351 train_acc=0.1821 val_loss=3.056439 val_acc=0.14
79 test_loss=3.066467 test_acc=0.1277
step=92000 train_loss=3.013934 train_acc=0.1910 val_loss=3.051417 val_acc=0.15
32 test_loss=3.061558 test_acc=0.1369
step=92500 train_loss=3.017658 train_acc=0.1754 val_loss=3.058039 val_acc=0.13
72 test_loss=3.067228 test_acc=0.1190
step=93000 train_loss=3.012071 train_acc=0.2108 val_loss=3.050980 val_acc=0.17
45 test_loss=3.061462 test_acc=0.1468
step=93500 train_loss=3.013349 train_acc=0.1903 val_loss=3.052830 val_acc=0.15
21 test_loss=3.061583 test_acc=0.1349
step=94000 train_loss=3.015493 train_acc=0.1942 val_loss=3.052783 val_acc=0.14
79 test_loss=3.062913 test_acc=0.1369
step=94500 train_loss=3.006315 train_acc=0.1896 val_loss=3.046114 val_acc=0.15
53411 test_acc=0.1390
step=95000 train_loss=3.004086 train_acc=0.1846 val_loss=3.043132 val_acc=0.15
```

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```
64 test_loss=3.051086 test_acc=0.1321
step=95500 train_loss=2.998569 train_acc=0.1960 val_loss=3.036704 val_acc=0.16
28 test_loss=3.046128 test_acc=0.1404
step=96000 train_loss=2.997245 train_acc=0.1956 val_loss=3.037181 val_acc=0.14
68 test_loss=3.046659 test_acc=0.1408
step=96500 train_loss=2.998189 train_acc=0.1821 val_loss=3.037393 val_acc=0.15
53 test_loss=3.047367 test_acc=0.1342
step=97000 train_loss=2.996685 train_acc=0.2009 val_loss=3.034946 val_acc=0.16
49 test_loss=3.045098 test_acc=0.1470
step=97500 train_loss=2.996353 train_acc=0.2062 val_loss=3.033172 val_acc=0.16
06 test_loss=3.044043 test_acc=0.1410
step=98000 train_loss=2.998235 train_acc=0.2069 val_loss=3.037677 val_acc=0.16
60 test_loss=3.046018 test_acc=0.1424
step=98500 train_loss=2.998800 train_acc=0.2038 val_loss=3.038161 val_acc=0.15
64 test_loss=3.046703 test_acc=0.1440
step=99000 train_loss=3.001406 train_acc=0.2034 val_loss=3.041229 val_acc=0.17
23 test_loss=3.048837 test_acc=0.1443
step=99500 train_loss=3.002250 train_acc=0.1928 val_loss=3.041457 val_acc=0.16
06 test_loss=3.051223 test_acc=0.1418
step=100000 train_loss=2.996854 train_acc=0.1974 val_loss=3.035674 val_acc=0.1
660 test_loss=3.044565 test_acc=0.1450
best metrics: {
  "step": 97500,
  "lr": 0.00010141535082162599,
  "train_loss": 2.996353162255885,
  "train_acc": 0.20623671155209072,
  "val_loss": 3.0331718465115163,
  "val_acc": 0.16063829787234044,
  "test_loss": 3.0440432523781795,
  "test_acc": 0.14095980166460068,
  "elapsed_sec": 1970.4852628707886
}
checkpoint dir: /Users/shobhit/Documents/UW/CSE 493/cse493s-spring26-hw1/out/s
ubmission/part1/warmup/sub_p97_l1_s42

$ /opt/homebrew/Caskroom/miniconda/base/envs/cse493/bin/python train.py --mode
modular --op - --p 97 --n_layer 2 --n_head 4 --n_embd 128 --steps 100000 --see
d 42 --batch_size 128 --lr 0.001 --min_lr 0.0001 --lr_schedule cosine --warmup
_steps 1000 --grad_clip 1.0 --beta1 0.9 --beta2 0.95 --weight_decay 0.01 --opt
imizer adam --eval_interval 500 --log_interval 500 --out_dir '/Users/shobhit/D
ocuments/UW/CSE 493/cse493s-spring26-hw1/out/submission/part1/warmup/sub_p97_l
2_s42' --bias --resume
number of parameters: 0.41M
[resume] model_last.pt not found in /Users/shobhit/Documents/UW/CSE 493/cse493
s-spring26-hw1/out/submission/part1/warmup/sub_p97_l2_s42; starting fresh
device=mps train=2822 val=940 test=5647 batch_size=128
step=1 train_loss=4.687313 train_acc=0.0000 val_loss=4.684959 val_acc=0.0000 t
est_loss=4.688042 test_acc=0.0000
step=500 train_loss=4.342619 train_acc=0.0634 val_loss=4.803458 val_acc=0.0011
test_loss=4.813308 test_acc=0.0016
step=1000 train_loss=4.251218 train_acc=0.0822 val_loss=4.861333 val_acc=0.000
0 test_loss=4.878212 test_acc=0.0004
step=1500 train_loss=4.392021 train_acc=0.0514 val_loss=4.776920 val_acc=0.003
2 test_loss=4.784626 test_acc=0.0014
step=2000 train_loss=4.326712 train_acc=0.0532 val_loss=4.852279 val_acc=0.004
3 test_loss=4.877421 test_acc=0.0030
step=2500 train_loss=4.297583 train_acc=0.0539 val_loss=4.903905 val_acc=0.004
3 test_loss=4.919106 test_acc=0.0027
loss=4.280923 train_acc=0.0532 val_loss=4.942250 val_acc=0.001
1 test_loss=4.947528 test_acc=0.0021
```

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```
step=3500 train_loss=4.269431 train_acc=0.0592 val_loss=4.932942 val_acc=0.004
3 test_loss=4.946429 test_acc=0.0023
step=4000 train_loss=4.266506 train_acc=0.0524 val_loss=4.955989 val_acc=0.003
2 test_loss=4.965534 test_acc=0.0025
step=4500 train_loss=4.255906 train_acc=0.0560 val_loss=4.961520 val_acc=0.001
1 test_loss=4.974597 test_acc=0.0021
step=5000 train_loss=4.257480 train_acc=0.0571 val_loss=4.978884 val_acc=0.001
1 test_loss=4.977073 test_acc=0.0028
step=5500 train_loss=4.253541 train_acc=0.0606 val_loss=4.959382 val_acc=0.003
2 test_loss=4.952786 test_acc=0.0032
step=6000 train_loss=4.243238 train_acc=0.0539 val_loss=4.961099 val_acc=0.002
1 test_loss=4.957902 test_acc=0.0016
step=6500 train_loss=4.200474 train_acc=0.0606 val_loss=4.870351 val_acc=0.006
4 test_loss=4.846479 test_acc=0.0043
step=7000 train_loss=4.101373 train_acc=0.0680 val_loss=4.649106 val_acc=0.004
3 test_loss=4.616218 test_acc=0.0062
step=7500 train_loss=4.065409 train_acc=0.0758 val_loss=4.535031 val_acc=0.010
6 test_loss=4.519538 test_acc=0.0113
step=8000 train_loss=3.714616 train_acc=0.1137 val_loss=3.959191 val_acc=0.039
4 test_loss=3.951824 test_acc=0.0344
step=8500 train_loss=3.507028 train_acc=0.1219 val_loss=3.638165 val_acc=0.064
9 test_loss=3.621199 test_acc=0.0586
step=9000 train_loss=3.226051 train_acc=0.1276 val_loss=3.304528 val_acc=0.090
4 test_loss=3.300306 test_acc=0.0800
step=9500 train_loss=3.134294 train_acc=0.1265 val_loss=3.196599 val_acc=0.091
5 test_loss=3.189824 test_acc=0.0850
step=10000 train_loss=3.023990 train_acc=0.1609 val_loss=3.069668 val_acc=0.12
34 test_loss=3.067057 test_acc=0.1146
step=10500 train_loss=2.986901 train_acc=0.1492 val_loss=3.023825 val_acc=0.12
02 test_loss=3.026921 test_acc=0.1112
step=11000 train_loss=2.990312 train_acc=0.1577 val_loss=3.025257 val_acc=0.12
34 test_loss=3.025688 test_acc=0.1151
step=11500 train_loss=3.001606 train_acc=0.1520 val_loss=3.037337 val_acc=0.12
98 test_loss=3.039755 test_acc=0.1086
step=12000 train_loss=2.990454 train_acc=0.1634 val_loss=3.032497 val_acc=0.13
51 test_loss=3.029352 test_acc=0.1208
step=12500 train_loss=2.983861 train_acc=0.1680 val_loss=3.018983 val_acc=0.12
77 test_loss=3.017510 test_acc=0.1222
step=13000 train_loss=3.001155 train_acc=0.1439 val_loss=3.040079 val_acc=0.11
91 test_loss=3.041318 test_acc=0.0985
step=13500 train_loss=2.991650 train_acc=0.1619 val_loss=3.030793 val_acc=0.12
23 test_loss=3.032269 test_acc=0.1140
step=14000 train_loss=2.982918 train_acc=0.1736 val_loss=3.015427 val_acc=0.14
47 test_loss=3.021203 test_acc=0.1217
step=14500 train_loss=3.006935 train_acc=0.1637 val_loss=3.042061 val_acc=0.12
87 test_loss=3.041083 test_acc=0.1165
step=15000 train_loss=2.980631 train_acc=0.1811 val_loss=3.020925 val_acc=0.13
62 test_loss=3.018878 test_acc=0.1328
step=15500 train_loss=3.007699 train_acc=0.1602 val_loss=3.041484 val_acc=0.12
02 test_loss=3.042360 test_acc=0.1183
step=16000 train_loss=2.990040 train_acc=0.1690 val_loss=3.030087 val_acc=0.13
72 test_loss=3.029878 test_acc=0.1199
step=16500 train_loss=2.997878 train_acc=0.1828 val_loss=3.035096 val_acc=0.14
04 test_loss=3.035848 test_acc=0.1358
step=17000 train_loss=2.996573 train_acc=0.1584 val_loss=3.028453 val_acc=0.12
87 test_loss=3.033432 test_acc=0.1123
step=17500 train_loss=2.989254 train_acc=0.1797 val_loss=3.031650 val_acc=0.13
94 test_loss=3.027755 test_acc=0.1254
step=18000 train_loss=3.009580 train_acc=0.1432 val_loss=3.047612 val_acc=0.10
53 test_loss=3.046394 test_acc=0.0999
```

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```
step=18500 train_loss=3.021552 train_acc=0.1623 val_loss=3.053594 val_acc=0.13
09 test_loss=3.058381 test_acc=0.1133
step=19000 train_loss=3.002788 train_acc=0.1598 val_loss=3.041707 val_acc=0.12
45 test_loss=3.041367 test_acc=0.1135
step=19500 train_loss=2.998057 train_acc=0.1644 val_loss=3.032969 val_acc=0.11
70 test_loss=3.039895 test_acc=0.1162
step=20000 train_loss=3.001420 train_acc=0.1747 val_loss=3.037898 val_acc=0.12
87 test_loss=3.039225 test_acc=0.1236
step=20500 train_loss=3.024792 train_acc=0.1393 val_loss=3.062770 val_acc=0.10
32 test_loss=3.065748 test_acc=0.0981
step=21000 train_loss=3.011465 train_acc=0.1591 val_loss=3.043650 val_acc=0.14
15 test_loss=3.047466 test_acc=0.1172
step=21500 train_loss=2.992881 train_acc=0.1740 val_loss=3.029325 val_acc=0.12
55 test_loss=3.026458 test_acc=0.1302
step=22000 train_loss=3.001858 train_acc=0.1797 val_loss=3.038913 val_acc=0.15
11 test_loss=3.042588 test_acc=0.1374
step=22500 train_loss=3.004496 train_acc=0.1658 val_loss=3.044818 val_acc=0.12
45 test_loss=3.044555 test_acc=0.1160
step=23000 train_loss=3.019223 train_acc=0.1683 val_loss=3.054068 val_acc=0.13
30 test_loss=3.055602 test_acc=0.1192
step=23500 train_loss=3.018713 train_acc=0.1630 val_loss=3.055842 val_acc=0.12
77 test_loss=3.060649 test_acc=0.1178
step=24000 train_loss=3.029842 train_acc=0.1747 val_loss=3.062509 val_acc=0.14
47 test_loss=3.063655 test_acc=0.1217
step=24500 train_loss=3.015199 train_acc=0.1758 val_loss=3.053200 val_acc=0.14
26 test_loss=3.054537 test_acc=0.1287
step=25000 train_loss=3.007502 train_acc=0.1697 val_loss=3.042661 val_acc=0.13
94 test_loss=3.046613 test_acc=0.1248
step=25500 train_loss=3.038752 train_acc=0.1697 val_loss=3.080226 val_acc=0.12
55 test_loss=3.078801 test_acc=0.1149
step=26000 train_loss=3.018111 train_acc=0.1765 val_loss=3.055646 val_acc=0.12
13 test_loss=3.055566 test_acc=0.1209
step=26500 train_loss=3.014357 train_acc=0.1595 val_loss=3.052702 val_acc=0.13
09 test_loss=3.054286 test_acc=0.1048
step=27000 train_loss=3.024961 train_acc=0.1765 val_loss=3.061035 val_acc=0.13
83 test_loss=3.065476 test_acc=0.1222
step=27500 train_loss=3.014099 train_acc=0.1534 val_loss=3.052536 val_acc=0.12
23 test_loss=3.053705 test_acc=0.1098
step=28000 train_loss=3.018576 train_acc=0.1495 val_loss=3.060542 val_acc=0.11
06 test_loss=3.060197 test_acc=0.1032
step=28500 train_loss=3.024800 train_acc=0.1513 val_loss=3.060937 val_acc=0.12
02 test_loss=3.063365 test_acc=0.1064
step=29000 train_loss=3.014406 train_acc=0.1719 val_loss=3.054531 val_acc=0.13
72 test_loss=3.054696 test_acc=0.1188
step=29500 train_loss=3.037811 train_acc=0.1566 val_loss=3.074512 val_acc=0.12
55 test_loss=3.075201 test_acc=0.1137
step=30000 train_loss=3.024851 train_acc=0.1605 val_loss=3.061959 val_acc=0.13
30 test_loss=3.062151 test_acc=0.1234
step=30500 train_loss=3.027991 train_acc=0.1775 val_loss=3.063845 val_acc=0.15
21 test_loss=3.065021 test_acc=0.1355
step=31000 train_loss=3.026800 train_acc=0.1623 val_loss=3.067203 val_acc=0.12
34 test_loss=3.067603 test_acc=0.1202
step=31500 train_loss=3.023925 train_acc=0.1751 val_loss=3.061681 val_acc=0.13
62 test_loss=3.062211 test_acc=0.1261
step=32000 train_loss=3.041352 train_acc=0.1591 val_loss=3.078487 val_acc=0.13
19 test_loss=3.080193 test_acc=0.1181
step=32500 train_loss=3.038961 train_acc=0.1797 val_loss=3.072989 val_acc=0.13
30 test_loss=3.077219 test_acc=0.1279
step=33000 train_loss=3.028769 train_acc=0.1811 val_loss=3.063593 val_acc=0.14
89 test_loss=3.065187 test_acc=0.1344
```

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```
step=33500 train_loss=3.044949 train_acc=0.1552 val_loss=3.081813 val_acc=0.12
13 test_loss=3.085541 test_acc=0.1116
step=34000 train_loss=3.040696 train_acc=0.1818 val_loss=3.077899 val_acc=0.14
04 test_loss=3.079344 test_acc=0.1312
step=34500 train_loss=3.022007 train_acc=0.1839 val_loss=3.062723 val_acc=0.13
83 test_loss=3.061835 test_acc=0.1275
step=35000 train_loss=3.041236 train_acc=0.1697 val_loss=3.076645 val_acc=0.11
60 test_loss=3.080097 test_acc=0.1261
step=35500 train_loss=3.044453 train_acc=0.1782 val_loss=3.081154 val_acc=0.14
26 test_loss=3.081402 test_acc=0.1284
step=36000 train_loss=3.038877 train_acc=0.1715 val_loss=3.071863 val_acc=0.12
98 test_loss=3.074610 test_acc=0.1215
step=36500 train_loss=3.035664 train_acc=0.1637 val_loss=3.068660 val_acc=0.12
34 test_loss=3.066867 test_acc=0.1190
step=37000 train_loss=3.053598 train_acc=0.1598 val_loss=3.090055 val_acc=0.12
23 test_loss=3.090673 test_acc=0.1133
step=37500 train_loss=3.060731 train_acc=0.1793 val_loss=3.096769 val_acc=0.13
09 test_loss=3.099236 test_acc=0.1252
step=38000 train_loss=3.024067 train_acc=0.1669 val_loss=3.064941 val_acc=0.12
45 test_loss=3.063066 test_acc=0.1130
step=38500 train_loss=3.021066 train_acc=0.1765 val_loss=3.063376 val_acc=0.13
62 test_loss=3.059469 test_acc=0.1346
step=39000 train_loss=3.029878 train_acc=0.1772 val_loss=3.065537 val_acc=0.13
72 test_loss=3.069871 test_acc=0.1222
step=39500 train_loss=3.029251 train_acc=0.1797 val_loss=3.063564 val_acc=0.14
15 test_loss=3.068993 test_acc=0.1227
step=40000 train_loss=3.043877 train_acc=0.1793 val_loss=3.088436 val_acc=0.13
19 test_loss=3.081739 test_acc=0.1310
step=40500 train_loss=3.033034 train_acc=0.1712 val_loss=3.069236 val_acc=0.12
45 test_loss=3.073400 test_acc=0.1195
step=41000 train_loss=3.048084 train_acc=0.1492 val_loss=3.082367 val_acc=0.11
60 test_loss=3.084930 test_acc=0.1038
step=41500 train_loss=3.037567 train_acc=0.1712 val_loss=3.070343 val_acc=0.12
66 test_loss=3.073593 test_acc=0.1220
step=42000 train_loss=3.029299 train_acc=0.1839 val_loss=3.067718 val_acc=0.14
57 test_loss=3.069570 test_acc=0.1335
step=42500 train_loss=3.015187 train_acc=0.1743 val_loss=3.048489 val_acc=0.14
57 test_loss=3.054403 test_acc=0.1211
step=43000 train_loss=3.058188 train_acc=0.1960 val_loss=3.092267 val_acc=0.13
83 test_loss=3.094043 test_acc=0.1411
step=43500 train_loss=3.049172 train_acc=0.1860 val_loss=3.086317 val_acc=0.13
40 test_loss=3.087068 test_acc=0.1378
step=44000 train_loss=3.039928 train_acc=0.1768 val_loss=3.075283 val_acc=0.14
15 test_loss=3.076365 test_acc=0.1227
step=44500 train_loss=3.035161 train_acc=0.1807 val_loss=3.072251 val_acc=0.14
89 test_loss=3.075131 test_acc=0.1266
step=45000 train_loss=3.042803 train_acc=0.1779 val_loss=3.080836 val_acc=0.14
57 test_loss=3.082945 test_acc=0.1254
step=45500 train_loss=3.056550 train_acc=0.1814 val_loss=3.093944 val_acc=0.12
77 test_loss=3.094621 test_acc=0.1241
step=46000 train_loss=3.050275 train_acc=0.1616 val_loss=3.084202 val_acc=0.12
77 test_loss=3.090259 test_acc=0.1126
step=46500 train_loss=3.049713 train_acc=0.1843 val_loss=3.083523 val_acc=0.15
74 test_loss=3.086944 test_acc=0.1294
step=47000 train_loss=3.034286 train_acc=0.1779 val_loss=3.072627 val_acc=0.14
57 test_loss=3.073942 test_acc=0.1241
step=47500 train_loss=3.045914 train_acc=0.1850 val_loss=3.080972 val_acc=0.15
21 test_loss=3.084618 test_acc=0.1302
step=48000 train_loss=3.049934 train_acc=0.1747 val_loss=3.083396 val_acc=0.14
26 test_loss=3.086225 test_acc=0.1155
```

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```
step=48500 train_loss=3.032231 train_acc=0.1797 val_loss=3.067426 val_acc=0.14
47 test_loss=3.069764 test_acc=0.1222
step=49000 train_loss=3.050121 train_acc=0.1828 val_loss=3.086158 val_acc=0.14
57 test_loss=3.089105 test_acc=0.1294
step=49500 train_loss=3.049530 train_acc=0.1729 val_loss=3.085095 val_acc=0.14
36 test_loss=3.089011 test_acc=0.1209
step=50000 train_loss=3.036499 train_acc=0.1765 val_loss=3.075806 val_acc=0.13
83 test_loss=3.076875 test_acc=0.1259
step=50500 train_loss=3.042742 train_acc=0.1797 val_loss=3.079972 val_acc=0.15
43 test_loss=3.081166 test_acc=0.1263
step=51000 train_loss=3.049817 train_acc=0.1775 val_loss=3.089911 val_acc=0.13
51 test_loss=3.088859 test_acc=0.1256
step=51500 train_loss=3.056792 train_acc=0.1722 val_loss=3.088729 val_acc=0.13
51 test_loss=3.093277 test_acc=0.1167
step=52000 train_loss=3.061137 train_acc=0.1970 val_loss=3.094841 val_acc=0.14
68 test_loss=3.098525 test_acc=0.1431
step=52500 train_loss=3.048552 train_acc=0.1619 val_loss=3.084564 val_acc=0.13
51 test_loss=3.087147 test_acc=0.1126
step=53000 train_loss=3.049088 train_acc=0.1704 val_loss=3.082703 val_acc=0.13
09 test_loss=3.087237 test_acc=0.1201
step=53500 train_loss=3.042271 train_acc=0.1729 val_loss=3.079803 val_acc=0.14
15 test_loss=3.083261 test_acc=0.1213
step=54000 train_loss=3.022842 train_acc=0.1786 val_loss=3.060401 val_acc=0.13
94 test_loss=3.066281 test_acc=0.1245
step=54500 train_loss=3.027660 train_acc=0.1761 val_loss=3.062406 val_acc=0.13
19 test_loss=3.063791 test_acc=0.1238
step=55000 train_loss=3.026442 train_acc=0.1921 val_loss=3.066613 val_acc=0.14
47 test_loss=3.065924 test_acc=0.1378
step=55500 train_loss=3.028714 train_acc=0.1917 val_loss=3.063240 val_acc=0.14
89 test_loss=3.067201 test_acc=0.1314
step=56000 train_loss=3.024691 train_acc=0.1619 val_loss=3.061155 val_acc=0.12
45 test_loss=3.065312 test_acc=0.1093
step=56500 train_loss=3.039521 train_acc=0.1836 val_loss=3.077512 val_acc=0.15
11 test_loss=3.078916 test_acc=0.1259
step=57000 train_loss=3.038277 train_acc=0.1726 val_loss=3.073872 val_acc=0.14
36 test_loss=3.077577 test_acc=0.1201
step=57500 train_loss=3.035878 train_acc=0.1860 val_loss=3.071569 val_acc=0.13
83 test_loss=3.074342 test_acc=0.1321
step=58000 train_loss=3.031307 train_acc=0.1736 val_loss=3.067300 val_acc=0.14
57 test_loss=3.072269 test_acc=0.1176
step=58500 train_loss=3.025263 train_acc=0.1871 val_loss=3.060880 val_acc=0.13
72 test_loss=3.063574 test_acc=0.1298
step=59000 train_loss=3.030451 train_acc=0.1903 val_loss=3.065106 val_acc=0.15
43 test_loss=3.067170 test_acc=0.1395
step=59500 train_loss=3.022160 train_acc=0.1867 val_loss=3.059276 val_acc=0.14
04 test_loss=3.060332 test_acc=0.1250
step=60000 train_loss=3.022653 train_acc=0.1914 val_loss=3.057309 val_acc=0.15
00 test_loss=3.059038 test_acc=0.1378
step=60500 train_loss=3.018258 train_acc=0.1790 val_loss=3.053217 val_acc=0.13
83 test_loss=3.058710 test_acc=0.1263
step=61000 train_loss=3.020151 train_acc=0.2013 val_loss=3.057051 val_acc=0.15
32 test_loss=3.057593 test_acc=0.1429
step=61500 train_loss=3.025183 train_acc=0.1790 val_loss=3.060250 val_acc=0.13
83 test_loss=3.064434 test_acc=0.1252
step=62000 train_loss=3.025478 train_acc=0.1903 val_loss=3.060606 val_acc=0.15
11 test_loss=3.063492 test_acc=0.1342
step=62500 train_loss=3.017992 train_acc=0.1694 val_loss=3.052768 val_acc=0.14
15 test_loss=3.057529 test_acc=0.1156
_loss=3.023788 train_acc=0.1758 val_loss=3.057784 val_acc=0.13
40 test_loss=3.063405 test_acc=0.1263
```

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```
step=63500 train_loss=3.014983 train_acc=0.1818 val_loss=3.052681 val_acc=0.14
04 test_loss=3.053872 test_acc=0.1298
step=64000 train_loss=3.022733 train_acc=0.1857 val_loss=3.060530 val_acc=0.14
89 test_loss=3.062425 test_acc=0.1302
step=64500 train_loss=2.996143 train_acc=0.1953 val_loss=3.034173 val_acc=0.15
00 test_loss=3.036071 test_acc=0.1364
step=65000 train_loss=3.015418 train_acc=0.1860 val_loss=3.050784 val_acc=0.13
94 test_loss=3.052571 test_acc=0.1339
step=65500 train_loss=3.017616 train_acc=0.1726 val_loss=3.052353 val_acc=0.13
62 test_loss=3.055574 test_acc=0.1181
step=66000 train_loss=3.015880 train_acc=0.2062 val_loss=3.053345 val_acc=0.15
74 test_loss=3.055739 test_acc=0.1456
step=66500 train_loss=2.988239 train_acc=0.1999 val_loss=3.025860 val_acc=0.15
85 test_loss=3.028647 test_acc=0.1436
step=67000 train_loss=3.003115 train_acc=0.1786 val_loss=3.037808 val_acc=0.13
83 test_loss=3.041643 test_acc=0.1213
step=67500 train_loss=3.003183 train_acc=0.1878 val_loss=3.041106 val_acc=0.16
17 test_loss=3.040338 test_acc=0.1241
step=68000 train_loss=3.003547 train_acc=0.1775 val_loss=3.040605 val_acc=0.14
36 test_loss=3.040899 test_acc=0.1204
step=68500 train_loss=3.007661 train_acc=0.1910 val_loss=3.043142 val_acc=0.14
47 test_loss=3.045951 test_acc=0.1355
step=69000 train_loss=3.005148 train_acc=0.1821 val_loss=3.041045 val_acc=0.15
32 test_loss=3.043887 test_acc=0.1271
step=69500 train_loss=2.994311 train_acc=0.1821 val_loss=3.028935 val_acc=0.15
32 test_loss=3.032211 test_acc=0.1291
step=70000 train_loss=2.988924 train_acc=0.1779 val_loss=3.025098 val_acc=0.14
57 test_loss=3.028486 test_acc=0.1248
step=70500 train_loss=2.991093 train_acc=0.1942 val_loss=3.028239 val_acc=0.15
43 test_loss=3.029168 test_acc=0.1360
step=71000 train_loss=2.986552 train_acc=0.1878 val_loss=3.021988 val_acc=0.15
21 test_loss=3.022836 test_acc=0.1364
step=71500 train_loss=2.995766 train_acc=0.1938 val_loss=3.030278 val_acc=0.15
43 test_loss=3.032660 test_acc=0.1348
step=72000 train_loss=2.989034 train_acc=0.1885 val_loss=3.025311 val_acc=0.15
43 test_loss=3.026378 test_acc=0.1312
step=72500 train_loss=2.988104 train_acc=0.1935 val_loss=3.025244 val_acc=0.15
21 test_loss=3.027943 test_acc=0.1353
step=73000 train_loss=2.983712 train_acc=0.1889 val_loss=3.017110 val_acc=0.15
53 test_loss=3.022748 test_acc=0.1307
step=73500 train_loss=2.979950 train_acc=0.2130 val_loss=3.013616 val_acc=0.18
09 test_loss=3.019060 test_acc=0.1486
step=74000 train_loss=2.985316 train_acc=0.1935 val_loss=3.020009 val_acc=0.14
89 test_loss=3.023502 test_acc=0.1348
step=74500 train_loss=2.977040 train_acc=0.2080 val_loss=3.012725 val_acc=0.16
81 test_loss=3.017333 test_acc=0.1468
step=75000 train_loss=2.961473 train_acc=0.1832 val_loss=2.999298 val_acc=0.13
19 test_loss=3.001881 test_acc=0.1280
step=75500 train_loss=2.966293 train_acc=0.2310 val_loss=3.003575 val_acc=0.18
30 test_loss=3.004395 test_acc=0.1588
step=76000 train_loss=2.967488 train_acc=0.2151 val_loss=3.001560 val_acc=0.16
38 test_loss=3.005503 test_acc=0.1488
step=76500 train_loss=2.960631 train_acc=0.1917 val_loss=2.995373 val_acc=0.15
74 test_loss=3.000824 test_acc=0.1339
step=77000 train_loss=2.956633 train_acc=0.1984 val_loss=2.992092 val_acc=0.16
28 test_loss=2.996201 test_acc=0.1406
step=77500 train_loss=2.954013 train_acc=0.2009 val_loss=2.989196 val_acc=0.15
74 test_loss=2.991560 test_acc=0.1415
step=78000 train_loss=2.955398 train_acc=0.2069 val_loss=2.991845 val_acc=0.16
60 test_loss=2.993377 test_acc=0.1500
```

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```
step=78500 train_loss=2.956955 train_acc=0.2055 val_loss=2.991965 val_acc=0.16
28 test_loss=2.994543 test_acc=0.1426
step=79000 train_loss=2.953716 train_acc=0.2052 val_loss=2.986572 val_acc=0.15
74 test_loss=2.990804 test_acc=0.1472
step=79500 train_loss=2.946295 train_acc=0.2013 val_loss=2.980914 val_acc=0.15
53 test_loss=2.986789 test_acc=0.1381
step=80000 train_loss=2.944843 train_acc=0.2020 val_loss=2.979102 val_acc=0.15
21 test_loss=2.983341 test_acc=0.1470
step=80500 train_loss=2.940816 train_acc=0.2062 val_loss=2.976966 val_acc=0.16
81 test_loss=2.980260 test_acc=0.1440
step=81000 train_loss=2.939085 train_acc=0.2055 val_loss=2.976646 val_acc=0.15
53 test_loss=2.977589 test_acc=0.1503
step=81500 train_loss=2.937722 train_acc=0.2356 val_loss=2.975365 val_acc=0.17
77 test_loss=2.975763 test_acc=0.1716
step=82000 train_loss=2.945418 train_acc=0.1999 val_loss=2.981398 val_acc=0.16
49 test_loss=2.984347 test_acc=0.1450
step=82500 train_loss=2.938183 train_acc=0.2247 val_loss=2.974675 val_acc=0.17
77 test_loss=2.977105 test_acc=0.1617
step=83000 train_loss=2.930485 train_acc=0.1953 val_loss=2.966760 val_acc=0.15
00 test_loss=2.970851 test_acc=0.1392
step=83500 train_loss=2.929351 train_acc=0.1963 val_loss=2.965105 val_acc=0.15
21 test_loss=2.970248 test_acc=0.1328
step=84000 train_loss=2.930179 train_acc=0.1991 val_loss=2.965505 val_acc=0.16
17 test_loss=2.968091 test_acc=0.1404
step=84500 train_loss=2.926110 train_acc=0.2041 val_loss=2.962270 val_acc=0.15
74 test_loss=2.964818 test_acc=0.1422
step=85000 train_loss=2.920172 train_acc=0.2006 val_loss=2.956638 val_acc=0.14
79 test_loss=2.959796 test_acc=0.1403
step=85500 train_loss=2.928875 train_acc=0.2009 val_loss=2.964837 val_acc=0.14
79 test_loss=2.968912 test_acc=0.1401
step=86000 train_loss=2.924607 train_acc=0.2094 val_loss=2.962170 val_acc=0.16
49 test_loss=2.963059 test_acc=0.1516
step=86500 train_loss=2.920105 train_acc=0.2155 val_loss=2.956623 val_acc=0.16
91 test_loss=2.958325 test_acc=0.1541
step=87000 train_loss=2.914916 train_acc=0.2002 val_loss=2.953022 val_acc=0.16
17 test_loss=2.953105 test_acc=0.1404
step=87500 train_loss=2.913712 train_acc=0.2059 val_loss=2.950314 val_acc=0.16
49 test_loss=2.951159 test_acc=0.1480
step=88000 train_loss=2.904843 train_acc=0.2066 val_loss=2.941894 val_acc=0.17
02 test_loss=2.943917 test_acc=0.1537
step=88500 train_loss=2.904903 train_acc=0.1984 val_loss=2.942386 val_acc=0.15
96 test_loss=2.944333 test_acc=0.1397
step=89000 train_loss=2.905278 train_acc=0.1995 val_loss=2.940198 val_acc=0.16
17 test_loss=2.943927 test_acc=0.1394
step=89500 train_loss=2.904432 train_acc=0.2098 val_loss=2.939213 val_acc=0.15
74 test_loss=2.943258 test_acc=0.1472
step=90000 train_loss=2.902675 train_acc=0.2123 val_loss=2.937809 val_acc=0.17
45 test_loss=2.942146 test_acc=0.1477
step=90500 train_loss=2.902858 train_acc=0.1928 val_loss=2.939213 val_acc=0.16
06 test_loss=2.943023 test_acc=0.1356
step=91000 train_loss=2.899182 train_acc=0.2162 val_loss=2.934872 val_acc=0.16
17 test_loss=2.938545 test_acc=0.1564
step=91500 train_loss=2.903134 train_acc=0.2080 val_loss=2.936422 val_acc=0.16
28 test_loss=2.941122 test_acc=0.1484
step=92000 train_loss=2.901711 train_acc=0.2030 val_loss=2.935488 val_acc=0.16
28 test_loss=2.940036 test_acc=0.1450
step=92500 train_loss=2.896087 train_acc=0.1999 val_loss=2.932520 val_acc=0.16
17 test_loss=2.936265 test_acc=0.1406
_loss=2.895631 train_acc=0.1942 val_loss=2.931678 val_acc=0.15
32 test_loss=2.934980 test_acc=0.1374
```

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```
step=93500 train_loss=2.893938 train_acc=0.2186 val_loss=2.930316 val_acc=0.18
40 test_loss=2.933499 test_acc=0.1617
step=94000 train_loss=2.892070 train_acc=0.2232 val_loss=2.928052 val_acc=0.17
77 test_loss=2.931270 test_acc=0.1574
step=94500 train_loss=2.886902 train_acc=0.2240 val_loss=2.923212 val_acc=0.17
77 test_loss=2.925838 test_acc=0.1571
step=95000 train_loss=2.885348 train_acc=0.2105 val_loss=2.921329 val_acc=0.16
38 test_loss=2.924224 test_acc=0.1479
step=95500 train_loss=2.881246 train_acc=0.2179 val_loss=2.916527 val_acc=0.16
60 test_loss=2.920241 test_acc=0.1507
step=96000 train_loss=2.885870 train_acc=0.2140 val_loss=2.922886 val_acc=0.16
60 test_loss=2.926183 test_acc=0.1562
step=96500 train_loss=2.887364 train_acc=0.1984 val_loss=2.923342 val_acc=0.15
74 test_loss=2.926674 test_acc=0.1404
step=97000 train_loss=2.887834 train_acc=0.1981 val_loss=2.923433 val_acc=0.15
32 test_loss=2.926000 test_acc=0.1369
step=97500 train_loss=2.890280 train_acc=0.2030 val_loss=2.924613 val_acc=0.15
64 test_loss=2.928653 test_acc=0.1408
step=98000 train_loss=2.886614 train_acc=0.2130 val_loss=2.923099 val_acc=0.16
70 test_loss=2.924967 test_acc=0.1539
step=98500 train_loss=2.884095 train_acc=0.2062 val_loss=2.920275 val_acc=0.16
06 test_loss=2.922866 test_acc=0.1447
step=99000 train_loss=2.881734 train_acc=0.2126 val_loss=2.916993 val_acc=0.17
87 test_loss=2.920052 test_acc=0.1528
step=99500 train_loss=2.883871 train_acc=0.2133 val_loss=2.918899 val_acc=0.17
34 test_loss=2.923285 test_acc=0.1526
step=100000 train_loss=2.881334 train_acc=0.2094 val_loss=2.917386 val_acc=0.1
713 test_loss=2.920144 test_acc=0.1521
best metrics: {
  "step": 95500,
  "lr": 0.0001045803511535803,
  "train_loss": 2.881245730026321,
  "train_acc": 0.2179305457122608,
  "val_loss": 2.9165273656236366,
  "val_acc": 0.16595744680851063,
  "test_loss": 2.920240738507138,
  "test_acc": 0.1506994864529839,
  "elapsed_sec": 2722.7166368961334
}
checkpoint dir: /Users/shobhit/Documents/UW/CSE 493/cse493s-spring26-hw1/out/s
ubmission/part1/warmup/sub_p97_l2_s42

$ /opt/homebrew/Caskroom/miniconda/base/envs/cse493/bin/python train.py --mode
modular --op - --p 113 --n_layer 1 --n_head 4 --n_embd 128 --steps 100000 --se
ed 42 --batch_size 128 --lr 0.001 --min_lr 0.0001 --lr_schedule cosine --warmu
p_steps 1000 --grad_clip 1.0 --beta1 0.9 --beta2 0.95 --weight_decay 0.01 --op
timizer adam --eval_interval 500 --log_interval 500 --out_dir '/Users/shobhit/
Documents/UW/CSE 493/cse493s-spring26-hw1/out/submission/part1/warmup/sub_p113
_l1_s42' --bias --resume
number of parameters: 0.21M
[resume] model_last.pt not found in /Users/shobhit/Documents/UW/CSE 493/cse493
s-spring26-hw1/out/submission/part1/warmup/sub_p113_l1_s42; starting fresh
device=mps train=3830 val=1276 test=7663 batch_size=128
step=1 train_loss=4.811426 train_acc=0.0010 val_loss=4.813471 val_acc=0.0016 t
est_loss=4.812796 test_acc=0.0022
step=500 train_loss=4.517925 train_acc=0.0603 val_loss=4.939927 val_acc=0.0008
test_loss=4.927378 test_acc=0.0014
step=1000 train_loss=4.482533 train_acc=0.0551 val_loss=4.966699 val_acc=0.000
3529 test_acc=0.0018
step=1500 train_loss=4.576023 train_acc=0.0428 val_loss=4.909896 val_acc=0.002
```

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```
4 test_loss=4.894646 test_acc=0.0020
step=2000 train_loss=4.510382 train_acc=0.0366 val_loss=5.002697 val_acc=0.001
6 test_loss=4.978921 test_acc=0.0029
step=2500 train_loss=4.474453 train_acc=0.0499 val_loss=5.046264 val_acc=0.001
6 test_loss=5.021083 test_acc=0.0020
step=3000 train_loss=4.469542 train_acc=0.0467 val_loss=5.038883 val_acc=0.002
4 test_loss=5.025213 test_acc=0.0016
step=3500 train_loss=4.455566 train_acc=0.0467 val_loss=5.058893 val_acc=0.003
9 test_loss=5.043916 test_acc=0.0012
step=4000 train_loss=4.453749 train_acc=0.0551 val_loss=5.064499 val_acc=0.003
1 test_loss=5.037697 test_acc=0.0012
step=4500 train_loss=4.443668 train_acc=0.0530 val_loss=5.075622 val_acc=0.000
8 test_loss=5.049559 test_acc=0.0020
step=5000 train_loss=4.442656 train_acc=0.0548 val_loss=5.077247 val_acc=0.001
6 test_loss=5.053110 test_acc=0.0026
step=5500 train_loss=4.433983 train_acc=0.0527 val_loss=5.072636 val_acc=0.001
6 test_loss=5.059478 test_acc=0.0013
step=6000 train_loss=4.445576 train_acc=0.0509 val_loss=5.062601 val_acc=0.003
1 test_loss=5.057435 test_acc=0.0025
step=6500 train_loss=4.425714 train_acc=0.0548 val_loss=5.066946 val_acc=0.003
9 test_loss=5.054012 test_acc=0.0021
step=7000 train_loss=4.343296 train_acc=0.0486 val_loss=4.861259 val_acc=0.007
1 test_loss=4.832373 test_acc=0.0059
step=7500 train_loss=4.304689 train_acc=0.0499 val_loss=4.733187 val_acc=0.004
7 test_loss=4.701689 test_acc=0.0086
step=8000 train_loss=4.067657 train_acc=0.0726 val_loss=4.318091 val_acc=0.020
4 test_loss=4.296697 test_acc=0.0224
step=8500 train_loss=3.864664 train_acc=0.0705 val_loss=4.027538 val_acc=0.040
8 test_loss=3.991565 test_acc=0.0414
step=9000 train_loss=3.431412 train_acc=0.1157 val_loss=3.531838 val_acc=0.079
9 test_loss=3.502142 test_acc=0.0758
step=9500 train_loss=3.123803 train_acc=0.1569 val_loss=3.169662 val_acc=0.123
8 test_loss=3.158684 test_acc=0.1236
step=10000 train_loss=3.020684 train_acc=0.1718 val_loss=3.063040 val_acc=0.12
70 test_loss=3.051617 test_acc=0.1488
step=10500 train_loss=2.994458 train_acc=0.1809 val_loss=3.029230 val_acc=0.16
54 test_loss=3.025846 test_acc=0.1528
step=11000 train_loss=2.970840 train_acc=0.1726 val_loss=3.011401 val_acc=0.14
26 test_loss=3.002215 test_acc=0.1451
step=11500 train_loss=2.978133 train_acc=0.1760 val_loss=3.019097 val_acc=0.14
89 test_loss=3.011477 test_acc=0.1415
step=12000 train_loss=2.978800 train_acc=0.1877 val_loss=3.021902 val_acc=0.15
99 test_loss=3.011533 test_acc=0.1570
step=12500 train_loss=2.970051 train_acc=0.1640 val_loss=3.016098 val_acc=0.12
93 test_loss=3.002083 test_acc=0.1411
step=13000 train_loss=2.952052 train_acc=0.1862 val_loss=2.992333 val_acc=0.15
36 test_loss=2.981550 test_acc=0.1542
step=13500 train_loss=2.962309 train_acc=0.1783 val_loss=3.004149 val_acc=0.15
36 test_loss=2.993255 test_acc=0.1536
step=14000 train_loss=2.929470 train_acc=0.1603 val_loss=2.969672 val_acc=0.12
77 test_loss=2.957801 test_acc=0.1385
step=14500 train_loss=2.925828 train_acc=0.1851 val_loss=2.962475 val_acc=0.14
81 test_loss=2.955394 test_acc=0.1567
step=15000 train_loss=2.932098 train_acc=0.1982 val_loss=2.967249 val_acc=0.16
61 test_loss=2.963341 test_acc=0.1681
step=15500 train_loss=2.937739 train_acc=0.1726 val_loss=2.980151 val_acc=0.15
67 test_loss=2.968614 test_acc=0.1446
step=16000 train_loss=2.932940 train_acc=0.1708 val_loss=2.965112 val_acc=0.14
61678 test_acc=0.1460
step=16500 train_loss=2.920356 train_acc=0.1742 val_loss=2.957714 val_acc=0.13
```

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```
71 test_loss=2.951371 test_acc=0.1501
step=17000 train_loss=2.908415 train_acc=0.1736 val_loss=2.948630 val_acc=0.14
89 test_loss=2.938381 test_acc=0.1503
step=17500 train_loss=2.913716 train_acc=0.1872 val_loss=2.963541 val_acc=0.15
05 test_loss=2.948117 test_acc=0.1526
step=18000 train_loss=2.906973 train_acc=0.1992 val_loss=2.948233 val_acc=0.15
99 test_loss=2.934166 test_acc=0.1732
step=18500 train_loss=2.903845 train_acc=0.1914 val_loss=2.948649 val_acc=0.18
65 test_loss=2.937076 test_acc=0.1621
step=19000 train_loss=2.895746 train_acc=0.1621 val_loss=2.933548 val_acc=0.12
15 test_loss=2.927023 test_acc=0.1325
step=19500 train_loss=2.901919 train_acc=0.1742 val_loss=2.942210 val_acc=0.13
95 test_loss=2.934413 test_acc=0.1411
step=20000 train_loss=2.898695 train_acc=0.2222 val_loss=2.935098 val_acc=0.19
67 test_loss=2.930141 test_acc=0.1948
step=20500 train_loss=2.898095 train_acc=0.1940 val_loss=2.932234 val_acc=0.17
63 test_loss=2.923796 test_acc=0.1747
step=21000 train_loss=2.901405 train_acc=0.1969 val_loss=2.941575 val_acc=0.16
38 test_loss=2.930719 test_acc=0.1715
step=21500 train_loss=2.893160 train_acc=0.1969 val_loss=2.928753 val_acc=0.17
16 test_loss=2.922027 test_acc=0.1706
step=22000 train_loss=2.888026 train_acc=0.2183 val_loss=2.935070 val_acc=0.18
03 test_loss=2.918253 test_acc=0.1895
step=22500 train_loss=2.887496 train_acc=0.1817 val_loss=2.926093 val_acc=0.14
42 test_loss=2.919884 test_acc=0.1563
step=23000 train_loss=2.888974 train_acc=0.2021 val_loss=2.929727 val_acc=0.16
85 test_loss=2.918000 test_acc=0.1715
step=23500 train_loss=2.883441 train_acc=0.1859 val_loss=2.919941 val_acc=0.15
83 test_loss=2.916100 test_acc=0.1549
step=24000 train_loss=2.884923 train_acc=0.1898 val_loss=2.919284 val_acc=0.17
63 test_loss=2.914113 test_acc=0.1528
step=24500 train_loss=2.885532 train_acc=0.2021 val_loss=2.928748 val_acc=0.17
08 test_loss=2.915125 test_acc=0.1763
step=25000 train_loss=2.885225 train_acc=0.1789 val_loss=2.924512 val_acc=0.15
99 test_loss=2.917116 test_acc=0.1502
step=25500 train_loss=2.882247 train_acc=0.1982 val_loss=2.912951 val_acc=0.16
93 test_loss=2.909829 test_acc=0.1730
step=26000 train_loss=2.877212 train_acc=0.1747 val_loss=2.912886 val_acc=0.16
77 test_loss=2.906714 test_acc=0.1519
step=26500 train_loss=2.883301 train_acc=0.1634 val_loss=2.912623 val_acc=0.14
97 test_loss=2.912535 test_acc=0.1409
step=27000 train_loss=2.869517 train_acc=0.1953 val_loss=2.907275 val_acc=0.18
34 test_loss=2.899925 test_acc=0.1720
step=27500 train_loss=2.876655 train_acc=0.1755 val_loss=2.913162 val_acc=0.14
73 test_loss=2.904246 test_acc=0.1507
step=28000 train_loss=2.881273 train_acc=0.1982 val_loss=2.920968 val_acc=0.16
93 test_loss=2.908424 test_acc=0.1729
step=28500 train_loss=2.864893 train_acc=0.2104 val_loss=2.900774 val_acc=0.18
42 test_loss=2.894009 test_acc=0.1753
step=29000 train_loss=2.876192 train_acc=0.1864 val_loss=2.912138 val_acc=0.16
22 test_loss=2.906286 test_acc=0.1576
step=29500 train_loss=2.869480 train_acc=0.1995 val_loss=2.906025 val_acc=0.15
91 test_loss=2.899451 test_acc=0.1760
step=30000 train_loss=2.866150 train_acc=0.1791 val_loss=2.910247 val_acc=0.16
38 test_loss=2.896509 test_acc=0.1553
step=30500 train_loss=2.868618 train_acc=0.2117 val_loss=2.906309 val_acc=0.17
95 test_loss=2.898121 test_acc=0.1844
step=31000 train_loss=2.866270 train_acc=0.1935 val_loss=2.903913 val_acc=0.16
96536 test_acc=0.1639
step=31500 train_loss=2.863550 train_acc=0.1794 val_loss=2.907083 val_acc=0.15
```

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```
75 test_loss=2.895047 test_acc=0.1597
step=32000 train_loss=2.866666 train_acc=0.2084 val_loss=2.904857 val_acc=0.15
83 test_loss=2.893856 test_acc=0.1785
step=32500 train_loss=2.861151 train_acc=0.1909 val_loss=2.902133 val_acc=0.16
07 test_loss=2.893410 test_acc=0.1622
step=33000 train_loss=2.855875 train_acc=0.2235 val_loss=2.899729 val_acc=0.19
44 test_loss=2.888823 test_acc=0.1922
step=33500 train_loss=2.862376 train_acc=0.1789 val_loss=2.893476 val_acc=0.16
14 test_loss=2.891454 test_acc=0.1550
step=34000 train_loss=2.855116 train_acc=0.1807 val_loss=2.896841 val_acc=0.15
20 test_loss=2.886416 test_acc=0.1583
step=34500 train_loss=2.853231 train_acc=0.1927 val_loss=2.900613 val_acc=0.15
91 test_loss=2.885182 test_acc=0.1616
step=35000 train_loss=2.851141 train_acc=0.2107 val_loss=2.892199 val_acc=0.17
79 test_loss=2.880200 test_acc=0.1770
step=35500 train_loss=2.858334 train_acc=0.1796 val_loss=2.894110 val_acc=0.16
54 test_loss=2.885556 test_acc=0.1533
step=36000 train_loss=2.852663 train_acc=0.2076 val_loss=2.891951 val_acc=0.15
91 test_loss=2.882546 test_acc=0.1745
step=36500 train_loss=2.848582 train_acc=0.2104 val_loss=2.886359 val_acc=0.18
73 test_loss=2.881562 test_acc=0.1815
step=37000 train_loss=2.843216 train_acc=0.2055 val_loss=2.881404 val_acc=0.17
55 test_loss=2.870567 test_acc=0.1807
step=37500 train_loss=2.847582 train_acc=0.2102 val_loss=2.892711 val_acc=0.17
32 test_loss=2.877113 test_acc=0.1790
step=38000 train_loss=2.845829 train_acc=0.1963 val_loss=2.884245 val_acc=0.18
34 test_loss=2.876191 test_acc=0.1666
step=38500 train_loss=2.841302 train_acc=0.2144 val_loss=2.885102 val_acc=0.17
16 test_loss=2.870221 test_acc=0.1890
step=39000 train_loss=2.838386 train_acc=0.2225 val_loss=2.877698 val_acc=0.20
38 test_loss=2.867363 test_acc=0.1926
step=39500 train_loss=2.843282 train_acc=0.1956 val_loss=2.887661 val_acc=0.15
36 test_loss=2.873401 test_acc=0.1704
step=40000 train_loss=2.837803 train_acc=0.1830 val_loss=2.882525 val_acc=0.14
34 test_loss=2.867449 test_acc=0.1601
step=40500 train_loss=2.838115 train_acc=0.1890 val_loss=2.877485 val_acc=0.15
05 test_loss=2.868762 test_acc=0.1649
step=41000 train_loss=2.837868 train_acc=0.2193 val_loss=2.881317 val_acc=0.18
73 test_loss=2.866949 test_acc=0.1847
step=41500 train_loss=2.837014 train_acc=0.1836 val_loss=2.878889 val_acc=0.15
99 test_loss=2.867786 test_acc=0.1561
step=42000 train_loss=2.832589 train_acc=0.2123 val_loss=2.870882 val_acc=0.17
24 test_loss=2.860669 test_acc=0.1861
step=42500 train_loss=2.837398 train_acc=0.2008 val_loss=2.870569 val_acc=0.16
77 test_loss=2.866767 test_acc=0.1733
step=43000 train_loss=2.831061 train_acc=0.2107 val_loss=2.871901 val_acc=0.16
14 test_loss=2.862209 test_acc=0.1794
step=43500 train_loss=2.832396 train_acc=0.1880 val_loss=2.873577 val_acc=0.14
89 test_loss=2.863873 test_acc=0.1563
step=44000 train_loss=2.828877 train_acc=0.2413 val_loss=2.864904 val_acc=0.21
55 test_loss=2.857055 test_acc=0.2126
step=44500 train_loss=2.832562 train_acc=0.2162 val_loss=2.875186 val_acc=0.18
57 test_loss=2.864425 test_acc=0.1820
step=45000 train_loss=2.830634 train_acc=0.2144 val_loss=2.869892 val_acc=0.16
77 test_loss=2.859547 test_acc=0.1903
step=45500 train_loss=2.819429 train_acc=0.2097 val_loss=2.860118 val_acc=0.18
18 test_loss=2.850774 test_acc=0.1899
step=46000 train_loss=2.820857 train_acc=0.2222 val_loss=2.863721 val_acc=0.17
50532 test_acc=0.1944
step=46500 train_loss=2.812890 train_acc=0.2191 val_loss=2.849092 val_acc=0.19
```

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```
91 test_loss=2.843556 test_acc=0.1858
step=47000 train_loss=2.817103 train_acc=0.2269 val_loss=2.859412 val_acc=0.19
83 test_loss=2.848161 test_acc=0.1982
step=47500 train_loss=2.818897 train_acc=0.2180 val_loss=2.861411 val_acc=0.19
67 test_loss=2.851054 test_acc=0.1895
step=48000 train_loss=2.815503 train_acc=0.2010 val_loss=2.861685 val_acc=0.17
16 test_loss=2.846587 test_acc=0.1755
step=48500 train_loss=2.813143 train_acc=0.2178 val_loss=2.861184 val_acc=0.17
79 test_loss=2.843182 test_acc=0.1907
step=49000 train_loss=2.817139 train_acc=0.2044 val_loss=2.854964 val_acc=0.17
71 test_loss=2.848024 test_acc=0.1733
step=49500 train_loss=2.809083 train_acc=0.2018 val_loss=2.846033 val_acc=0.17
40 test_loss=2.840589 test_acc=0.1720
step=50000 train_loss=2.805005 train_acc=0.1953 val_loss=2.850087 val_acc=0.16
30 test_loss=2.836376 test_acc=0.1690
step=50500 train_loss=2.808838 train_acc=0.2188 val_loss=2.848559 val_acc=0.19
51 test_loss=2.838600 test_acc=0.1886
step=51000 train_loss=2.810442 train_acc=0.2193 val_loss=2.849339 val_acc=0.18
65 test_loss=2.837673 test_acc=0.1870
step=51500 train_loss=2.803191 train_acc=0.2131 val_loss=2.840962 val_acc=0.19
12 test_loss=2.832413 test_acc=0.1883
step=52000 train_loss=2.805617 train_acc=0.2076 val_loss=2.843231 val_acc=0.17
95 test_loss=2.835728 test_acc=0.1732
step=52500 train_loss=2.797862 train_acc=0.2316 val_loss=2.836458 val_acc=0.21
00 test_loss=2.826075 test_acc=0.1964
step=53000 train_loss=2.802189 train_acc=0.2188 val_loss=2.843997 val_acc=0.17
55 test_loss=2.829844 test_acc=0.1891
step=53500 train_loss=2.800368 train_acc=0.2144 val_loss=2.839569 val_acc=0.18
34 test_loss=2.831187 test_acc=0.1820
step=54000 train_loss=2.791622 train_acc=0.1990 val_loss=2.834158 val_acc=0.16
22 test_loss=2.822350 test_acc=0.1690
step=54500 train_loss=2.788769 train_acc=0.2052 val_loss=2.828290 val_acc=0.17
95 test_loss=2.818684 test_acc=0.1776
step=55000 train_loss=2.795070 train_acc=0.2334 val_loss=2.827302 val_acc=0.20
69 test_loss=2.823723 test_acc=0.2007
step=55500 train_loss=2.790908 train_acc=0.2402 val_loss=2.834100 val_acc=0.18
73 test_loss=2.820118 test_acc=0.2051
step=56000 train_loss=2.787582 train_acc=0.2329 val_loss=2.824953 val_acc=0.19
12 test_loss=2.817960 test_acc=0.2002
step=56500 train_loss=2.783369 train_acc=0.2298 val_loss=2.824329 val_acc=0.18
10 test_loss=2.812966 test_acc=0.2031
step=57000 train_loss=2.785425 train_acc=0.2444 val_loss=2.828290 val_acc=0.20
38 test_loss=2.817417 test_acc=0.2068
step=57500 train_loss=2.788673 train_acc=0.2258 val_loss=2.822774 val_acc=0.20
30 test_loss=2.817647 test_acc=0.1927
step=58000 train_loss=2.784713 train_acc=0.2642 val_loss=2.826253 val_acc=0.21
71 test_loss=2.815120 test_acc=0.2322
step=58500 train_loss=2.778269 train_acc=0.2381 val_loss=2.813054 val_acc=0.20
22 test_loss=2.807300 test_acc=0.2040
step=59000 train_loss=2.778230 train_acc=0.2397 val_loss=2.821330 val_acc=0.20
38 test_loss=2.811419 test_acc=0.2093
step=59500 train_loss=2.777146 train_acc=0.2196 val_loss=2.815641 val_acc=0.16
93 test_loss=2.805588 test_acc=0.1901
step=60000 train_loss=2.771991 train_acc=0.2514 val_loss=2.813365 val_acc=0.19
59 test_loss=2.801496 test_acc=0.2205
step=60500 train_loss=2.766754 train_acc=0.2428 val_loss=2.804688 val_acc=0.19
44 test_loss=2.796151 test_acc=0.2055
step=61000 train_loss=2.772300 train_acc=0.2433 val_loss=2.810358 val_acc=0.19
00316 test_acc=0.2126
step=61500 train_loss=2.769223 train_acc=0.2258 val_loss=2.808484 val_acc=0.19
```

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```
12 test_loss=2.798546 test_acc=0.1969
step=62000 train_loss=2.769071 train_acc=0.2196 val_loss=2.805639 val_acc=0.19
36 test_loss=2.797766 test_acc=0.1870
step=62500 train_loss=2.764496 train_acc=0.2504 val_loss=2.805456 val_acc=0.21
63 test_loss=2.794876 test_acc=0.2139
step=63000 train_loss=2.763139 train_acc=0.2339 val_loss=2.806472 val_acc=0.20
06 test_loss=2.790900 test_acc=0.2088
step=63500 train_loss=2.759717 train_acc=0.2621 val_loss=2.798526 val_acc=0.21
47 test_loss=2.786328 test_acc=0.2308
step=64000 train_loss=2.760022 train_acc=0.2274 val_loss=2.794777 val_acc=0.17
48 test_loss=2.786345 test_acc=0.1977
step=64500 train_loss=2.763181 train_acc=0.2188 val_loss=2.802057 val_acc=0.16
22 test_loss=2.791144 test_acc=0.1847
step=65000 train_loss=2.759713 train_acc=0.2543 val_loss=2.800237 val_acc=0.20
38 test_loss=2.787523 test_acc=0.2139
step=65500 train_loss=2.758086 train_acc=0.2457 val_loss=2.798539 val_acc=0.19
44 test_loss=2.786415 test_acc=0.2098
step=66000 train_loss=2.752767 train_acc=0.2903 val_loss=2.790578 val_acc=0.23
43 test_loss=2.779878 test_acc=0.2479
step=66500 train_loss=2.752280 train_acc=0.2110 val_loss=2.792620 val_acc=0.16
93 test_loss=2.782024 test_acc=0.1771
step=67000 train_loss=2.752691 train_acc=0.2164 val_loss=2.791647 val_acc=0.19
44 test_loss=2.781571 test_acc=0.1897
step=67500 train_loss=2.749562 train_acc=0.2663 val_loss=2.787515 val_acc=0.23
51 test_loss=2.778456 test_acc=0.2273
step=68000 train_loss=2.750048 train_acc=0.2457 val_loss=2.790567 val_acc=0.20
30 test_loss=2.777877 test_acc=0.2102
step=68500 train_loss=2.750819 train_acc=0.2815 val_loss=2.793573 val_acc=0.24
37 test_loss=2.780498 test_acc=0.2443
step=69000 train_loss=2.744631 train_acc=0.2708 val_loss=2.785779 val_acc=0.22
26 test_loss=2.772950 test_acc=0.2376
step=69500 train_loss=2.747926 train_acc=0.2493 val_loss=2.785818 val_acc=0.21
16 test_loss=2.777024 test_acc=0.2148
step=70000 train_loss=2.746217 train_acc=0.2379 val_loss=2.785864 val_acc=0.20
22 test_loss=2.775493 test_acc=0.2070
step=70500 train_loss=2.744486 train_acc=0.2420 val_loss=2.785053 val_acc=0.18
81 test_loss=2.775275 test_acc=0.2087
step=71000 train_loss=2.742578 train_acc=0.2373 val_loss=2.781142 val_acc=0.19
28 test_loss=2.772447 test_acc=0.2031
step=71500 train_loss=2.740488 train_acc=0.2632 val_loss=2.779672 val_acc=0.21
39 test_loss=2.769775 test_acc=0.2258
step=72000 train_loss=2.737042 train_acc=0.2352 val_loss=2.780241 val_acc=0.18
34 test_loss=2.767714 test_acc=0.2002
step=72500 train_loss=2.736473 train_acc=0.2334 val_loss=2.778576 val_acc=0.18
18 test_loss=2.766210 test_acc=0.2001
step=73000 train_loss=2.736040 train_acc=0.2757 val_loss=2.775837 val_acc=0.23
43 test_loss=2.765402 test_acc=0.2340
step=73500 train_loss=2.733850 train_acc=0.2384 val_loss=2.774859 val_acc=0.20
69 test_loss=2.763849 test_acc=0.2017
step=74000 train_loss=2.732018 train_acc=0.2580 val_loss=2.771979 val_acc=0.21
47 test_loss=2.759531 test_acc=0.2199
step=74500 train_loss=2.730471 train_acc=0.2334 val_loss=2.772918 val_acc=0.18
18 test_loss=2.760602 test_acc=0.1985
step=75000 train_loss=2.731794 train_acc=0.2253 val_loss=2.768802 val_acc=0.19
12 test_loss=2.762380 test_acc=0.1875
step=75500 train_loss=2.729790 train_acc=0.2514 val_loss=2.770976 val_acc=0.19
51 test_loss=2.759539 test_acc=0.2136
step=76000 train_loss=2.728448 train_acc=0.2460 val_loss=2.768604 val_acc=0.19
58315 test_acc=0.2092
step=76500 train_loss=2.727107 train_acc=0.2324 val_loss=2.769257 val_acc=0.18
```

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```
50 test_loss=2.759249 test_acc=0.1887
step=77000 train_loss=2.723977 train_acc=0.2598 val_loss=2.764319 val_acc=0.21
94 test_loss=2.752694 test_acc=0.2250
step=77500 train_loss=2.724216 train_acc=0.2115 val_loss=2.762674 val_acc=0.16
93 test_loss=2.753935 test_acc=0.1803
step=78000 train_loss=2.721765 train_acc=0.2726 val_loss=2.763611 val_acc=0.22
73 test_loss=2.750784 test_acc=0.2346
step=78500 train_loss=2.719826 train_acc=0.2433 val_loss=2.759312 val_acc=0.18
18 test_loss=2.747920 test_acc=0.2104
step=79000 train_loss=2.718125 train_acc=0.2433 val_loss=2.759348 val_acc=0.20
38 test_loss=2.748720 test_acc=0.2085
step=79500 train_loss=2.717531 train_acc=0.2744 val_loss=2.751426 val_acc=0.22
26 test_loss=2.744432 test_acc=0.2318
step=80000 train_loss=2.719761 train_acc=0.2684 val_loss=2.760393 val_acc=0.21
79 test_loss=2.750044 test_acc=0.2324
step=80500 train_loss=2.718319 train_acc=0.2728 val_loss=2.758444 val_acc=0.22
73 test_loss=2.748129 test_acc=0.2363
step=81000 train_loss=2.713423 train_acc=0.2295 val_loss=2.748945 val_acc=0.19
51 test_loss=2.742621 test_acc=0.1882
step=81500 train_loss=2.711218 train_acc=0.2906 val_loss=2.750278 val_acc=0.22
81 test_loss=2.739598 test_acc=0.2432
step=82000 train_loss=2.711388 train_acc=0.2804 val_loss=2.747011 val_acc=0.23
35 test_loss=2.739365 test_acc=0.2443
step=82500 train_loss=2.712126 train_acc=0.2956 val_loss=2.752374 val_acc=0.23
35 test_loss=2.741516 test_acc=0.2486
step=83000 train_loss=2.711681 train_acc=0.2786 val_loss=2.751362 val_acc=0.24
29 test_loss=2.742046 test_acc=0.2352
step=83500 train_loss=2.710038 train_acc=0.2789 val_loss=2.750844 val_acc=0.23
04 test_loss=2.739660 test_acc=0.2422
step=84000 train_loss=2.709141 train_acc=0.2520 val_loss=2.750290 val_acc=0.19
28 test_loss=2.738053 test_acc=0.2147
step=84500 train_loss=2.706124 train_acc=0.2486 val_loss=2.747229 val_acc=0.19
83 test_loss=2.735487 test_acc=0.2100
step=85000 train_loss=2.706079 train_acc=0.2642 val_loss=2.744371 val_acc=0.21
16 test_loss=2.734821 test_acc=0.2251
step=85500 train_loss=2.705663 train_acc=0.2836 val_loss=2.744681 val_acc=0.22
26 test_loss=2.732962 test_acc=0.2451
step=86000 train_loss=2.703026 train_acc=0.2420 val_loss=2.744311 val_acc=0.19
51 test_loss=2.733642 test_acc=0.2066
step=86500 train_loss=2.702882 train_acc=0.2747 val_loss=2.741236 val_acc=0.22
57 test_loss=2.730440 test_acc=0.2299
step=87000 train_loss=2.701224 train_acc=0.2561 val_loss=2.744833 val_acc=0.20
61 test_loss=2.730764 test_acc=0.2195
step=87500 train_loss=2.700279 train_acc=0.2877 val_loss=2.738766 val_acc=0.24
84 test_loss=2.729516 test_acc=0.2483
step=88000 train_loss=2.700382 train_acc=0.2791 val_loss=2.739142 val_acc=0.23
90 test_loss=2.730087 test_acc=0.2402
step=88500 train_loss=2.700657 train_acc=0.2702 val_loss=2.737963 val_acc=0.21
55 test_loss=2.728179 test_acc=0.2299
step=89000 train_loss=2.700170 train_acc=0.2846 val_loss=2.740053 val_acc=0.23
82 test_loss=2.729774 test_acc=0.2435
step=89500 train_loss=2.700456 train_acc=0.2775 val_loss=2.741319 val_acc=0.22
10 test_loss=2.730238 test_acc=0.2345
step=90000 train_loss=2.699621 train_acc=0.2446 val_loss=2.740743 val_acc=0.19
91 test_loss=2.730678 test_acc=0.2101
step=90500 train_loss=2.698339 train_acc=0.2640 val_loss=2.735170 val_acc=0.22
34 test_loss=2.727309 test_acc=0.2239
step=91000 train_loss=2.696613 train_acc=0.2802 val_loss=2.733725 val_acc=0.23
25937 test_acc=0.2409
step=91500 train_loss=2.695707 train_acc=0.2634 val_loss=2.735053 val_acc=0.21
```

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```
08 test_loss=2.724769 test_acc=0.2235
step=92000 train_loss=2.695208 train_acc=0.2888 val_loss=2.736746 val_acc=0.22
41 test_loss=2.724984 test_acc=0.2490
step=92500 train_loss=2.694685 train_acc=0.2781 val_loss=2.733478 val_acc=0.23
75 test_loss=2.724506 test_acc=0.2389
step=93000 train_loss=2.693297 train_acc=0.2877 val_loss=2.733155 val_acc=0.23
59 test_loss=2.721616 test_acc=0.2481
step=93500 train_loss=2.692557 train_acc=0.2473 val_loss=2.732184 val_acc=0.20
45 test_loss=2.722218 test_acc=0.2106
step=94000 train_loss=2.691985 train_acc=0.3003 val_loss=2.729911 val_acc=0.23
51 test_loss=2.720566 test_acc=0.2538
step=94500 train_loss=2.690810 train_acc=0.3003 val_loss=2.729696 val_acc=0.24
45 test_loss=2.719609 test_acc=0.2576
step=95000 train_loss=2.690134 train_acc=0.2666 val_loss=2.730854 val_acc=0.21
16 test_loss=2.719785 test_acc=0.2277
step=95500 train_loss=2.689674 train_acc=0.2950 val_loss=2.728665 val_acc=0.24
61 test_loss=2.718514 test_acc=0.2627
step=96000 train_loss=2.688370 train_acc=0.2590 val_loss=2.728621 val_acc=0.19
98 test_loss=2.718496 test_acc=0.2222
step=96500 train_loss=2.687902 train_acc=0.2828 val_loss=2.727542 val_acc=0.22
73 test_loss=2.717810 test_acc=0.2417
step=97000 train_loss=2.688309 train_acc=0.2624 val_loss=2.728329 val_acc=0.21
79 test_loss=2.717864 test_acc=0.2247
step=97500 train_loss=2.688002 train_acc=0.2984 val_loss=2.727227 val_acc=0.24
45 test_loss=2.717089 test_acc=0.2558
step=98000 train_loss=2.688122 train_acc=0.2867 val_loss=2.727887 val_acc=0.22
57 test_loss=2.717121 test_acc=0.2461
step=98500 train_loss=2.688499 train_acc=0.2697 val_loss=2.731817 val_acc=0.21
55 test_loss=2.718600 test_acc=0.2350
step=99000 train_loss=2.687543 train_acc=0.2744 val_loss=2.728212 val_acc=0.23
67 test_loss=2.718253 test_acc=0.2327
step=99500 train_loss=2.688607 train_acc=0.2689 val_loss=2.728905 val_acc=0.23
35 test_loss=2.718025 test_acc=0.2312
step=100000 train_loss=2.687208 train_acc=0.2940 val_loss=2.727973 val_acc=0.2
422 test_loss=2.715703 test_acc=0.2556
best metrics: {
  "step": 97500,
  "lr": 0.00010141535082162599,
  "train_loss": 2.6880017932961566,
  "train_acc": 0.2984334203655353,
  "val_loss": 2.7272274008365263,
  "val_acc": 0.2445141065830721,
  "test_loss": 2.717088606748664,
  "test_acc": 0.2557745008482318,
  "elapsed_sec": 5108.33687877655
}
checkpoint dir: /Users/shobhit/Documents/UW/CSE 493/cse493s-spring26-hw1/out/s
ubmission/part1/warmup/sub_p113_l1_s42

$ /opt/homebrew/Caskroom/miniconda/base/envs/cse493/bin/python train.py --mode
modular --op - --p 113 --n_layer 2 --n_head 4 --n_embd 128 --steps 100000 --se
ed 42 --batch_size 128 --lr 0.001 --min_lr 0.0001 --lr_schedule cosine --warmu
p_steps 1000 --grad_clip 1.0 --beta1 0.9 --beta2 0.95 --weight_decay 0.01 --op
timizer adam --eval_interval 500 --log_interval 500 --out_dir '/Users/shobhit/
Documents/UW/CSE 493/cse493s-spring26-hw1/out/submission/part1/warmup/sub_p113
_l2_s42' --bias --resume
number of parameters: 0.41M
[resume] model_last.pt not found in /Users/shobhit/Documents/UW/CSE 493/cse493
ut/submission/part1/warmup/sub_p113_l2_s42; starting fresh
device=mps train=3830 val=1276 test=7663 batch_size=128
```

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```
step=1 train_loss=4.823739 train_acc=0.0000 val_loss=4.819319 val_acc=0.0000 t
est_loss=4.821909 test_acc=0.0000
step=500 train_loss=4.565738 train_acc=0.0457 val_loss=4.904173 val_acc=0.0024
test_loss=4.899907 test_acc=0.0020
step=1000 train_loss=4.506654 train_acc=0.0420 val_loss=4.953138 val_acc=0.002
4 test_loss=4.949226 test_acc=0.0010
step=1500 train_loss=4.603322 train_acc=0.0405 val_loss=4.877910 val_acc=0.002
4 test_loss=4.865556 test_acc=0.0013
step=2000 train_loss=4.547039 train_acc=0.0407 val_loss=4.951850 val_acc=0.004
7 test_loss=4.949429 test_acc=0.0027
step=2500 train_loss=4.505602 train_acc=0.0397 val_loss=5.013038 val_acc=0.001
6 test_loss=5.005966 test_acc=0.0022
step=3000 train_loss=4.494287 train_acc=0.0478 val_loss=5.020369 val_acc=0.000
8 test_loss=5.009915 test_acc=0.0025
step=3500 train_loss=4.484295 train_acc=0.0423 val_loss=5.043341 val_acc=0.003
9 test_loss=5.025386 test_acc=0.0025
step=4000 train_loss=4.484891 train_acc=0.0470 val_loss=5.025831 val_acc=0.003
9 test_loss=5.006254 test_acc=0.0030
step=4500 train_loss=4.362829 train_acc=0.0478 val_loss=4.806924 val_acc=0.003
9 test_loss=4.800757 test_acc=0.0072
step=5000 train_loss=3.753655 train_acc=0.1123 val_loss=3.905387 val_acc=0.053
3 test_loss=3.895352 test_acc=0.0506
step=5500 train_loss=3.434297 train_acc=0.1191 val_loss=3.538398 val_acc=0.070
5 test_loss=3.523420 test_acc=0.0706
step=6000 train_loss=3.367085 train_acc=0.1154 val_loss=3.451694 val_acc=0.098
7 test_loss=3.437892 test_acc=0.0774
step=6500 train_loss=3.286746 train_acc=0.1266 val_loss=3.360567 val_acc=0.094
8 test_loss=3.344983 test_acc=0.0906
step=7000 train_loss=3.134467 train_acc=0.1574 val_loss=3.173531 val_acc=0.147
3 test_loss=3.166703 test_acc=0.1249
step=7500 train_loss=3.100914 train_acc=0.1540 val_loss=3.137560 val_acc=0.140
3 test_loss=3.129303 test_acc=0.1314
step=8000 train_loss=3.094097 train_acc=0.1580 val_loss=3.124360 val_acc=0.164
6 test_loss=3.120677 test_acc=0.1344
step=8500 train_loss=3.089000 train_acc=0.1585 val_loss=3.118533 val_acc=0.148
9 test_loss=3.111466 test_acc=0.1300
step=9000 train_loss=3.100698 train_acc=0.1577 val_loss=3.135053 val_acc=0.146
6 test_loss=3.128483 test_acc=0.1289
step=9500 train_loss=3.094897 train_acc=0.1642 val_loss=3.126875 val_acc=0.142
6 test_loss=3.119713 test_acc=0.1421
step=10000 train_loss=3.098921 train_acc=0.1454 val_loss=3.135278 val_acc=0.11
83 test_loss=3.125122 test_acc=0.1240
step=10500 train_loss=3.093174 train_acc=0.1603 val_loss=3.121313 val_acc=0.14
89 test_loss=3.118000 test_acc=0.1403
step=11000 train_loss=3.168396 train_acc=0.1595 val_loss=3.201621 val_acc=0.13
48 test_loss=3.192618 test_acc=0.1412
step=11500 train_loss=3.290767 train_acc=0.1389 val_loss=3.325045 val_acc=0.12
23 test_loss=3.317921 test_acc=0.1130
step=12000 train_loss=3.349405 train_acc=0.1261 val_loss=3.380191 val_acc=0.11
13 test_loss=3.373725 test_acc=0.1043
step=12500 train_loss=3.408047 train_acc=0.1308 val_loss=3.438546 val_acc=0.10
74 test_loss=3.430203 test_acc=0.1086
step=13000 train_loss=3.393623 train_acc=0.1295 val_loss=3.423691 val_acc=0.10
66 test_loss=3.415426 test_acc=0.1049
step=13500 train_loss=3.435784 train_acc=0.0956 val_loss=3.464991 val_acc=0.07
68 test_loss=3.454849 test_acc=0.0771
step=14000 train_loss=3.403498 train_acc=0.1329 val_loss=3.431757 val_acc=0.11
52 test_loss=3.424494 test_acc=0.1152
_loss=3.421517 train_acc=0.1316 val_loss=3.449625 val_acc=0.11
29 test_loss=3.441902 test_acc=0.1148
```

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```
step=15000 train_loss=3.482929 train_acc=0.1131 val_loss=3.506702 val_acc=0.09
80 test_loss=3.504166 test_acc=0.0916
step=15500 train_loss=3.451758 train_acc=0.1159 val_loss=3.484095 val_acc=0.09
80 test_loss=3.471895 test_acc=0.0955
step=16000 train_loss=3.418437 train_acc=0.1366 val_loss=3.443732 val_acc=0.11
83 test_loss=3.437287 test_acc=0.1194
step=16500 train_loss=3.414481 train_acc=0.1240 val_loss=3.443206 val_acc=0.10
58 test_loss=3.439588 test_acc=0.1018
step=17000 train_loss=3.451996 train_acc=0.1334 val_loss=3.477582 val_acc=0.11
44 test_loss=3.470291 test_acc=0.1159
step=17500 train_loss=3.395355 train_acc=0.1480 val_loss=3.430022 val_acc=0.12
54 test_loss=3.418859 test_acc=0.1293
step=18000 train_loss=3.446444 train_acc=0.1277 val_loss=3.476460 val_acc=0.10
58 test_loss=3.465916 test_acc=0.1131
step=18500 train_loss=3.443980 train_acc=0.1102 val_loss=3.473727 val_acc=0.10
66 test_loss=3.468318 test_acc=0.0927
step=19000 train_loss=3.466547 train_acc=0.1240 val_loss=3.492084 val_acc=0.10
82 test_loss=3.487203 test_acc=0.1014
step=19500 train_loss=3.430573 train_acc=0.1178 val_loss=3.458942 val_acc=0.09
72 test_loss=3.454033 test_acc=0.0994
step=20000 train_loss=3.457718 train_acc=0.1191 val_loss=3.478146 val_acc=0.09
87 test_loss=3.479116 test_acc=0.1037
step=20500 train_loss=3.430364 train_acc=0.1277 val_loss=3.453659 val_acc=0.12
15 test_loss=3.448933 test_acc=0.1077
step=21000 train_loss=3.450738 train_acc=0.1384 val_loss=3.478084 val_acc=0.13
32 test_loss=3.470897 test_acc=0.1181
step=21500 train_loss=3.483160 train_acc=0.1217 val_loss=3.508288 val_acc=0.10
82 test_loss=3.502086 test_acc=0.0967
step=22000 train_loss=3.493160 train_acc=0.1366 val_loss=3.526418 val_acc=0.10
66 test_loss=3.514484 test_acc=0.1133
step=22500 train_loss=3.473951 train_acc=0.1360 val_loss=3.502975 val_acc=0.11
68 test_loss=3.496705 test_acc=0.1133
step=23000 train_loss=3.468547 train_acc=0.1347 val_loss=3.492704 val_acc=0.12
15 test_loss=3.488114 test_acc=0.1109
step=23500 train_loss=3.465996 train_acc=0.1167 val_loss=3.492234 val_acc=0.11
52 test_loss=3.486826 test_acc=0.0993
step=24000 train_loss=3.448771 train_acc=0.1243 val_loss=3.472722 val_acc=0.09
56 test_loss=3.469073 test_acc=0.0993
step=24500 train_loss=3.492392 train_acc=0.1444 val_loss=3.518166 val_acc=0.12
38 test_loss=3.511488 test_acc=0.1229
step=25000 train_loss=3.474225 train_acc=0.1352 val_loss=3.500632 val_acc=0.11
05 test_loss=3.495890 test_acc=0.1116
step=25500 train_loss=3.451562 train_acc=0.1303 val_loss=3.477543 val_acc=0.10
34 test_loss=3.472246 test_acc=0.1135
step=26000 train_loss=3.470429 train_acc=0.1136 val_loss=3.496194 val_acc=0.10
03 test_loss=3.491459 test_acc=0.0953
step=26500 train_loss=3.476532 train_acc=0.1206 val_loss=3.498903 val_acc=0.10
97 test_loss=3.497024 test_acc=0.1064
step=27000 train_loss=3.476633 train_acc=0.1235 val_loss=3.500633 val_acc=0.11
68 test_loss=3.498503 test_acc=0.1015
step=27500 train_loss=3.470425 train_acc=0.1457 val_loss=3.494634 val_acc=0.11
91 test_loss=3.487116 test_acc=0.1155
step=28000 train_loss=3.579583 train_acc=0.0822 val_loss=3.600992 val_acc=0.07
92 test_loss=3.595590 test_acc=0.0689
step=28500 train_loss=3.461342 train_acc=0.1151 val_loss=3.483537 val_acc=0.11
60 test_loss=3.480855 test_acc=0.0987
step=29000 train_loss=3.483512 train_acc=0.1345 val_loss=3.509279 val_acc=0.12
62 test_loss=3.504906 test_acc=0.1139
_loss=3.454574 train_acc=0.1232 val_loss=3.477549 val_acc=0.10
66 test_loss=3.475350 test_acc=0.1024
```

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```
step=30000 train_loss=3.475934 train_acc=0.1358 val_loss=3.502960 val_acc=0.12
62 test_loss=3.495501 test_acc=0.1130
step=30500 train_loss=3.471940 train_acc=0.1269 val_loss=3.502497 val_acc=0.11
36 test_loss=3.495500 test_acc=0.1118
step=31000 train_loss=3.485479 train_acc=0.1104 val_loss=3.510852 val_acc=0.10
11 test_loss=3.505873 test_acc=0.0940
step=31500 train_loss=3.473733 train_acc=0.1005 val_loss=3.501453 val_acc=0.09
64 test_loss=3.494424 test_acc=0.0962
step=32000 train_loss=3.458322 train_acc=0.1303 val_loss=3.485941 val_acc=0.10
74 test_loss=3.478034 test_acc=0.1134
step=32500 train_loss=3.468766 train_acc=0.1287 val_loss=3.499080 val_acc=0.10
74 test_loss=3.492418 test_acc=0.1079
step=33000 train_loss=3.492147 train_acc=0.1167 val_loss=3.520798 val_acc=0.10
50 test_loss=3.514756 test_acc=0.0963
step=33500 train_loss=3.463777 train_acc=0.1431 val_loss=3.485496 val_acc=0.12
93 test_loss=3.484824 test_acc=0.1208
step=34000 train_loss=3.445712 train_acc=0.1243 val_loss=3.473818 val_acc=0.10
89 test_loss=3.469102 test_acc=0.1028
step=34500 train_loss=3.461418 train_acc=0.1261 val_loss=3.492144 val_acc=0.11
36 test_loss=3.482734 test_acc=0.1075
step=35000 train_loss=3.455767 train_acc=0.1405 val_loss=3.481962 val_acc=0.12
70 test_loss=3.475108 test_acc=0.1191
step=35500 train_loss=3.463292 train_acc=0.1460 val_loss=3.486648 val_acc=0.13
87 test_loss=3.483085 test_acc=0.1240
step=36000 train_loss=3.472630 train_acc=0.1214 val_loss=3.500555 val_acc=0.10
03 test_loss=3.494295 test_acc=0.1057
step=36500 train_loss=3.449396 train_acc=0.1376 val_loss=3.475956 val_acc=0.12
30 test_loss=3.471618 test_acc=0.1174
step=37000 train_loss=3.430465 train_acc=0.1358 val_loss=3.458131 val_acc=0.13
01 test_loss=3.452007 test_acc=0.1152
step=37500 train_loss=3.520354 train_acc=0.1298 val_loss=3.545659 val_acc=0.11
05 test_loss=3.538026 test_acc=0.1138
step=38000 train_loss=3.481480 train_acc=0.1415 val_loss=3.506629 val_acc=0.12
30 test_loss=3.500616 test_acc=0.1081
step=38500 train_loss=3.468394 train_acc=0.1329 val_loss=3.494710 val_acc=0.10
34 test_loss=3.488222 test_acc=0.1052
step=39000 train_loss=3.462866 train_acc=0.1342 val_loss=3.489266 val_acc=0.12
46 test_loss=3.481126 test_acc=0.1118
step=39500 train_loss=3.457538 train_acc=0.1245 val_loss=3.488681 val_acc=0.09
01 test_loss=3.478022 test_acc=0.1024
step=40000 train_loss=3.499637 train_acc=0.1175 val_loss=3.529074 val_acc=0.09
25 test_loss=3.520047 test_acc=0.1007
step=40500 train_loss=3.492472 train_acc=0.1222 val_loss=3.517209 val_acc=0.10
19 test_loss=3.512962 test_acc=0.1044
step=41000 train_loss=3.484057 train_acc=0.1316 val_loss=3.512013 val_acc=0.10
42 test_loss=3.505831 test_acc=0.1111
step=41500 train_loss=3.463434 train_acc=0.1326 val_loss=3.491682 val_acc=0.11
52 test_loss=3.487596 test_acc=0.1064
step=42000 train_loss=3.469626 train_acc=0.1368 val_loss=3.495153 val_acc=0.13
32 test_loss=3.492808 test_acc=0.1174
step=42500 train_loss=3.478300 train_acc=0.1206 val_loss=3.505797 val_acc=0.10
74 test_loss=3.502350 test_acc=0.0985
step=43000 train_loss=3.419141 train_acc=0.1282 val_loss=3.449095 val_acc=0.11
83 test_loss=3.445346 test_acc=0.1053
step=43500 train_loss=3.434221 train_acc=0.1133 val_loss=3.463682 val_acc=0.09
48 test_loss=3.462566 test_acc=0.0921
step=44000 train_loss=3.456033 train_acc=0.1444 val_loss=3.482115 val_acc=0.12
62 test_loss=3.481054 test_acc=0.1180
_loss=3.447560 train_acc=0.1003 val_loss=3.479700 val_acc=0.08
86 test_loss=3.474999 test_acc=0.0852
```

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```
step=45000 train_loss=3.417716 train_acc=0.1460 val_loss=3.445144 val_acc=0.12
70 test_loss=3.441349 test_acc=0.1219
step=45500 train_loss=3.387053 train_acc=0.1473 val_loss=3.421168 val_acc=0.13
01 test_loss=3.415080 test_acc=0.1248
step=46000 train_loss=3.447203 train_acc=0.1418 val_loss=3.479418 val_acc=0.13
56 test_loss=3.472894 test_acc=0.1141
step=46500 train_loss=3.409701 train_acc=0.1493 val_loss=3.441096 val_acc=0.14
18 test_loss=3.435946 test_acc=0.1248
step=47000 train_loss=3.435907 train_acc=0.1478 val_loss=3.467323 val_acc=0.13
48 test_loss=3.463720 test_acc=0.1176
step=47500 train_loss=3.396684 train_acc=0.1446 val_loss=3.432837 val_acc=0.13
56 test_loss=3.425705 test_acc=0.1180
step=48000 train_loss=3.409064 train_acc=0.1136 val_loss=3.444366 val_acc=0.08
93 test_loss=3.436661 test_acc=0.0899
step=48500 train_loss=3.436158 train_acc=0.1180 val_loss=3.471818 val_acc=0.09
33 test_loss=3.460854 test_acc=0.1009
step=49000 train_loss=3.416448 train_acc=0.1272 val_loss=3.447594 val_acc=0.11
21 test_loss=3.443804 test_acc=0.1018
step=49500 train_loss=3.398517 train_acc=0.1347 val_loss=3.427900 val_acc=0.11
13 test_loss=3.427443 test_acc=0.1074
step=50000 train_loss=3.406103 train_acc=0.1261 val_loss=3.442380 val_acc=0.10
11 test_loss=3.435299 test_acc=0.1013
step=50500 train_loss=3.413061 train_acc=0.1159 val_loss=3.445750 val_acc=0.09
95 test_loss=3.439984 test_acc=0.0980
step=51000 train_loss=3.403621 train_acc=0.1454 val_loss=3.433913 val_acc=0.12
07 test_loss=3.429408 test_acc=0.1203
step=51500 train_loss=3.381522 train_acc=0.1491 val_loss=3.408318 val_acc=0.13
32 test_loss=3.404345 test_acc=0.1267
step=52000 train_loss=3.396167 train_acc=0.1251 val_loss=3.425990 val_acc=0.12
38 test_loss=3.421832 test_acc=0.1023
step=52500 train_loss=3.403511 train_acc=0.1530 val_loss=3.434959 val_acc=0.13
87 test_loss=3.429018 test_acc=0.1218
step=53000 train_loss=3.390802 train_acc=0.1360 val_loss=3.421985 val_acc=0.12
54 test_loss=3.415594 test_acc=0.1103
step=53500 train_loss=3.385232 train_acc=0.1415 val_loss=3.414821 val_acc=0.12
46 test_loss=3.410822 test_acc=0.1181
step=54000 train_loss=3.386221 train_acc=0.1251 val_loss=3.417682 val_acc=0.12
38 test_loss=3.412068 test_acc=0.1028
step=54500 train_loss=3.368634 train_acc=0.1253 val_loss=3.401124 val_acc=0.10
34 test_loss=3.395764 test_acc=0.0974
step=55000 train_loss=3.375289 train_acc=0.1415 val_loss=3.399854 val_acc=0.12
15 test_loss=3.400001 test_acc=0.1156
step=55500 train_loss=3.379305 train_acc=0.1507 val_loss=3.413185 val_acc=0.13
40 test_loss=3.406157 test_acc=0.1245
step=56000 train_loss=3.392782 train_acc=0.1337 val_loss=3.420339 val_acc=0.11
68 test_loss=3.419443 test_acc=0.1074
step=56500 train_loss=3.380859 train_acc=0.1616 val_loss=3.410240 val_acc=0.15
60 test_loss=3.406922 test_acc=0.1272
step=57000 train_loss=3.384928 train_acc=0.1217 val_loss=3.418202 val_acc=0.10
82 test_loss=3.412960 test_acc=0.0997
step=57500 train_loss=3.374151 train_acc=0.1488 val_loss=3.400085 val_acc=0.14
81 test_loss=3.399960 test_acc=0.1270
step=58000 train_loss=3.362544 train_acc=0.1355 val_loss=3.394265 val_acc=0.13
40 test_loss=3.390447 test_acc=0.1103
step=58500 train_loss=3.378283 train_acc=0.1316 val_loss=3.406431 val_acc=0.12
07 test_loss=3.403728 test_acc=0.1095
step=59000 train_loss=3.378182 train_acc=0.1164 val_loss=3.409038 val_acc=0.11
05 test_loss=3.405606 test_acc=0.0983
step=59500 train_loss=3.358223 train_acc=0.1405 val_loss=3.387639 val_acc=0.13
24 test_loss=3.385170 test_acc=0.1120
```

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```
step=60000 train_loss=3.354915 train_acc=0.1397 val_loss=3.386286 val_acc=0.11
29 test_loss=3.380586 test_acc=0.1065
step=60500 train_loss=3.355967 train_acc=0.1465 val_loss=3.384215 val_acc=0.12
30 test_loss=3.381880 test_acc=0.1168
step=61000 train_loss=3.360010 train_acc=0.1350 val_loss=3.390540 val_acc=0.12
15 test_loss=3.385368 test_acc=0.1108
step=61500 train_loss=3.354715 train_acc=0.1650 val_loss=3.386034 val_acc=0.13
09 test_loss=3.380127 test_acc=0.1301
step=62000 train_loss=3.335070 train_acc=0.1407 val_loss=3.363762 val_acc=0.12
85 test_loss=3.361179 test_acc=0.1168
step=62500 train_loss=3.354361 train_acc=0.1371 val_loss=3.383134 val_acc=0.11
29 test_loss=3.379927 test_acc=0.1121
step=63000 train_loss=3.343044 train_acc=0.1444 val_loss=3.373231 val_acc=0.13
17 test_loss=3.366242 test_acc=0.1225
step=63500 train_loss=3.336962 train_acc=0.1577 val_loss=3.366317 val_acc=0.13
87 test_loss=3.361351 test_acc=0.1296
step=64000 train_loss=3.353655 train_acc=0.1418 val_loss=3.383869 val_acc=0.12
46 test_loss=3.378087 test_acc=0.1216
step=64500 train_loss=3.326073 train_acc=0.1561 val_loss=3.356447 val_acc=0.11
99 test_loss=3.351431 test_acc=0.1283
step=65000 train_loss=3.329940 train_acc=0.1512 val_loss=3.358193 val_acc=0.13
48 test_loss=3.355370 test_acc=0.1278
step=65500 train_loss=3.323720 train_acc=0.1376 val_loss=3.356539 val_acc=0.11
68 test_loss=3.349615 test_acc=0.1167
step=66000 train_loss=3.330977 train_acc=0.1621 val_loss=3.361541 val_acc=0.14
66 test_loss=3.356147 test_acc=0.1331
step=66500 train_loss=3.320330 train_acc=0.1480 val_loss=3.351721 val_acc=0.13
64 test_loss=3.345829 test_acc=0.1275
step=67000 train_loss=3.328744 train_acc=0.1598 val_loss=3.361216 val_acc=0.13
40 test_loss=3.355512 test_acc=0.1275
step=67500 train_loss=3.316636 train_acc=0.1522 val_loss=3.346464 val_acc=0.13
64 test_loss=3.341405 test_acc=0.1245
step=68000 train_loss=3.333732 train_acc=0.1540 val_loss=3.362747 val_acc=0.12
23 test_loss=3.357765 test_acc=0.1265
step=68500 train_loss=3.334802 train_acc=0.1540 val_loss=3.367678 val_acc=0.13
09 test_loss=3.361426 test_acc=0.1278
step=69000 train_loss=3.332001 train_acc=0.1533 val_loss=3.364203 val_acc=0.14
03 test_loss=3.357639 test_acc=0.1220
step=69500 train_loss=3.312144 train_acc=0.1559 val_loss=3.342038 val_acc=0.12
93 test_loss=3.338652 test_acc=0.1219
step=70000 train_loss=3.314003 train_acc=0.1619 val_loss=3.345850 val_acc=0.14
73 test_loss=3.340612 test_acc=0.1305
step=70500 train_loss=3.302526 train_acc=0.1520 val_loss=3.333806 val_acc=0.12
38 test_loss=3.329700 test_acc=0.1193
step=71000 train_loss=3.318957 train_acc=0.1493 val_loss=3.349444 val_acc=0.13
71 test_loss=3.345931 test_acc=0.1215
step=71500 train_loss=3.301583 train_acc=0.1684 val_loss=3.332412 val_acc=0.13
09 test_loss=3.327874 test_acc=0.1317
step=72000 train_loss=3.303588 train_acc=0.1533 val_loss=3.338436 val_acc=0.12
70 test_loss=3.332010 test_acc=0.1241
step=72500 train_loss=3.306383 train_acc=0.1598 val_loss=3.338207 val_acc=0.14
66 test_loss=3.332902 test_acc=0.1331
step=73000 train_loss=3.304351 train_acc=0.1713 val_loss=3.335067 val_acc=0.13
56 test_loss=3.331738 test_acc=0.1347
step=73500 train_loss=3.302207 train_acc=0.1535 val_loss=3.333815 val_acc=0.13
01 test_loss=3.328968 test_acc=0.1224
step=74000 train_loss=3.300346 train_acc=0.1530 val_loss=3.331714 val_acc=0.12
77 test_loss=3.326037 test_acc=0.1246
_loss=3.297741 train_acc=0.1757 val_loss=3.330090 val_acc=0.15
05 test_loss=3.324442 test_acc=0.1432
```

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```
step=75000 train_loss=3.294204 train_acc=0.1611 val_loss=3.325043 val_acc=0.14
58 test_loss=3.321747 test_acc=0.1342
step=75500 train_loss=3.287183 train_acc=0.1786 val_loss=3.321112 val_acc=0.14
81 test_loss=3.313718 test_acc=0.1544
step=76000 train_loss=3.290790 train_acc=0.1632 val_loss=3.324944 val_acc=0.14
66 test_loss=3.318077 test_acc=0.1381
step=76500 train_loss=3.277993 train_acc=0.1849 val_loss=3.311449 val_acc=0.16
54 test_loss=3.306270 test_acc=0.1446
step=77000 train_loss=3.287616 train_acc=0.1627 val_loss=3.319022 val_acc=0.14
50 test_loss=3.314353 test_acc=0.1319
step=77500 train_loss=3.285186 train_acc=0.1180 val_loss=3.316240 val_acc=0.10
19 test_loss=3.311932 test_acc=0.0970
step=78000 train_loss=3.286104 train_acc=0.1564 val_loss=3.318905 val_acc=0.13
40 test_loss=3.312678 test_acc=0.1285
step=78500 train_loss=3.285132 train_acc=0.1514 val_loss=3.316208 val_acc=0.12
07 test_loss=3.310325 test_acc=0.1223
step=79000 train_loss=3.274791 train_acc=0.1611 val_loss=3.305340 val_acc=0.14
18 test_loss=3.301780 test_acc=0.1352
step=79500 train_loss=3.262670 train_acc=0.1935 val_loss=3.290949 val_acc=0.17
01 test_loss=3.287792 test_acc=0.1593
step=80000 train_loss=3.263265 train_acc=0.1739 val_loss=3.294995 val_acc=0.15
52 test_loss=3.290130 test_acc=0.1416
step=80500 train_loss=3.258638 train_acc=0.1648 val_loss=3.290803 val_acc=0.14
97 test_loss=3.285055 test_acc=0.1369
step=81000 train_loss=3.254215 train_acc=0.1572 val_loss=3.283900 val_acc=0.14
11 test_loss=3.280791 test_acc=0.1319
step=81500 train_loss=3.251787 train_acc=0.1705 val_loss=3.282305 val_acc=0.15
36 test_loss=3.277783 test_acc=0.1385
step=82000 train_loss=3.245191 train_acc=0.1768 val_loss=3.275243 val_acc=0.15
91 test_loss=3.271027 test_acc=0.1400
step=82500 train_loss=3.239465 train_acc=0.1796 val_loss=3.273349 val_acc=0.16
69 test_loss=3.266055 test_acc=0.1419
step=83000 train_loss=3.229475 train_acc=0.1564 val_loss=3.261595 val_acc=0.14
11 test_loss=3.256963 test_acc=0.1270
step=83500 train_loss=3.224225 train_acc=0.1927 val_loss=3.257431 val_acc=0.17
79 test_loss=3.251179 test_acc=0.1532
step=84000 train_loss=3.219977 train_acc=0.1854 val_loss=3.252274 val_acc=0.16
69 test_loss=3.246257 test_acc=0.1526
step=84500 train_loss=3.221844 train_acc=0.1527 val_loss=3.255400 val_acc=0.11
99 test_loss=3.248894 test_acc=0.1253
step=85000 train_loss=3.215010 train_acc=0.1676 val_loss=3.248658 val_acc=0.13
64 test_loss=3.242187 test_acc=0.1334
step=85500 train_loss=3.214206 train_acc=0.1689 val_loss=3.248436 val_acc=0.13
56 test_loss=3.240960 test_acc=0.1411
step=86000 train_loss=3.209366 train_acc=0.1903 val_loss=3.242843 val_acc=0.17
63 test_loss=3.236401 test_acc=0.1570
step=86500 train_loss=3.207802 train_acc=0.1969 val_loss=3.239931 val_acc=0.17
01 test_loss=3.233461 test_acc=0.1535
step=87000 train_loss=3.206663 train_acc=0.1650 val_loss=3.241085 val_acc=0.14
66 test_loss=3.233854 test_acc=0.1328
step=87500 train_loss=3.205559 train_acc=0.1728 val_loss=3.237415 val_acc=0.15
44 test_loss=3.232277 test_acc=0.1407
step=88000 train_loss=3.207719 train_acc=0.1909 val_loss=3.238740 val_acc=0.16
61 test_loss=3.234766 test_acc=0.1550
step=88500 train_loss=3.206489 train_acc=0.1901 val_loss=3.236457 val_acc=0.16
77 test_loss=3.231854 test_acc=0.1563
step=89000 train_loss=3.197967 train_acc=0.1950 val_loss=3.231024 val_acc=0.17
95 test_loss=3.225242 test_acc=0.1636
_loss=3.195755 train_acc=0.1940 val_loss=3.229988 val_acc=0.16
46 test_loss=3.222990 test_acc=0.1586
```

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```
step=90000 train_loss=3.189603 train_acc=0.1796 val_loss=3.224551 val_acc=0.16
46 test_loss=3.218990 test_acc=0.1452
step=90500 train_loss=3.192996 train_acc=0.1807 val_loss=3.226225 val_acc=0.15
75 test_loss=3.221153 test_acc=0.1451
step=91000 train_loss=3.193021 train_acc=0.1995 val_loss=3.226003 val_acc=0.17
71 test_loss=3.220871 test_acc=0.1597
step=91500 train_loss=3.189097 train_acc=0.1997 val_loss=3.222322 val_acc=0.16
14 test_loss=3.216185 test_acc=0.1665
step=92000 train_loss=3.184803 train_acc=0.2010 val_loss=3.219869 val_acc=0.16
54 test_loss=3.213012 test_acc=0.1613
step=92500 train_loss=3.175669 train_acc=0.2167 val_loss=3.209886 val_acc=0.19
59 test_loss=3.204020 test_acc=0.1726
step=93000 train_loss=3.173324 train_acc=0.1802 val_loss=3.208549 val_acc=0.15
91 test_loss=3.201522 test_acc=0.1446
step=93500 train_loss=3.173537 train_acc=0.1854 val_loss=3.208639 val_acc=0.16
77 test_loss=3.201513 test_acc=0.1494
step=94000 train_loss=3.171834 train_acc=0.2031 val_loss=3.207591 val_acc=0.18
65 test_loss=3.199910 test_acc=0.1673
step=94500 train_loss=3.176231 train_acc=0.1930 val_loss=3.212118 val_acc=0.18
03 test_loss=3.204277 test_acc=0.1592
step=95000 train_loss=3.169217 train_acc=0.1958 val_loss=3.206239 val_acc=0.17
16 test_loss=3.198528 test_acc=0.1574
step=95500 train_loss=3.166844 train_acc=0.1969 val_loss=3.203439 val_acc=0.16
69 test_loss=3.196141 test_acc=0.1596
step=96000 train_loss=3.169175 train_acc=0.1966 val_loss=3.206070 val_acc=0.17
87 test_loss=3.199116 test_acc=0.1567
step=96500 train_loss=3.164378 train_acc=0.2050 val_loss=3.201889 val_acc=0.17
87 test_loss=3.194816 test_acc=0.1677
step=97000 train_loss=3.163169 train_acc=0.1833 val_loss=3.203357 val_acc=0.16
07 test_loss=3.194468 test_acc=0.1458
step=97500 train_loss=3.162400 train_acc=0.1833 val_loss=3.200722 val_acc=0.16
30 test_loss=3.192431 test_acc=0.1486
step=98000 train_loss=3.163780 train_acc=0.1987 val_loss=3.200997 val_acc=0.18
57 test_loss=3.193514 test_acc=0.1623
step=98500 train_loss=3.163573 train_acc=0.2133 val_loss=3.202839 val_acc=0.18
73 test_loss=3.193655 test_acc=0.1819
step=99000 train_loss=3.158572 train_acc=0.1830 val_loss=3.197823 val_acc=0.15
91 test_loss=3.189630 test_acc=0.1429
step=99500 train_loss=3.158044 train_acc=0.2076 val_loss=3.196467 val_acc=0.18
18 test_loss=3.188179 test_acc=0.1676
step=100000 train_loss=3.159801 train_acc=0.2131 val_loss=3.199273 val_acc=0.1
904 test_loss=3.189511 test_acc=0.1766
best metrics: {
  "step": 8500,
  "lr": 0.0009873152057455939,
  "train_loss": 3.088999981917538,
  "train_acc": 0.15848563968668408,
  "val_loss": 3.118533242085137,
  "val_acc": 0.14890282131661442,
  "test_loss": 3.111466132052866,
  "test_acc": 0.12997520553308103,
  "elapsed_sec": 98.5770058631897
}
checkpoint dir: /Users/shobhit/Documents/UW/CSE 493/cse493s-spring26-hw1/out/s
ubmission/part1/warmup/sub_p113_l2_s42

$ /opt/homebrew/Caskroom/miniconda/base/envs/cse493/bin/python train.py --mode
modular --op + --p 97 --n_layer 1 --n_head 4 --n_embd 128 --steps 100000 --see
d 42 --batch_size 128 --lr 0.001 --min_lr 0.0001 --lr_schedule cosine --warmu
p_steps 1000 --grad_clip 1.0 --beta1 0.9 --beta2 0.95 --weight_decay 0.01 --op
```

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```
timizer adam --eval_interval 500 --log_interval 500 --out_dir '/Users/shobhit/
Documents/UW/CSE 493/cse493s-spring26-hw1/out/submission/part1/warmup_restart
s/add_p97_l1_s101' --bias --resume
number of parameters: 0.21M
[resume] model_last.pt not found in /Users/shobhit/Documents/UW/CSE 493/cse493
s-spring26-hw1/out/submission/part1/warmup_restarts/add_p97_l1_s101; starting
fresh
device=mps train=2822 val=940 test=5647 batch_size=128
step=1 train_loss=4.685918 train_acc=0.0000 val_loss=4.680871 val_acc=0.0000 t
est_loss=4.684292 test_acc=0.0000
step=500 train_loss=4.304925 train_acc=0.0776 val_loss=4.829797 val_acc=0.0021
test_loss=4.824545 test_acc=0.0021
step=1000 train_loss=4.255857 train_acc=0.0840 val_loss=4.855812 val_acc=0.002
1 test_loss=4.874784 test_acc=0.0004
step=1500 train_loss=4.365523 train_acc=0.0556 val_loss=4.793590 val_acc=0.001
1 test_loss=4.798390 test_acc=0.0014
step=2000 train_loss=4.289088 train_acc=0.0553 val_loss=4.901622 val_acc=0.003
2 test_loss=4.908147 test_acc=0.0025
step=2500 train_loss=4.247770 train_acc=0.0638 val_loss=4.954677 val_acc=0.001
1 test_loss=4.958348 test_acc=0.0018
step=3000 train_loss=4.241819 train_acc=0.0638 val_loss=5.000682 val_acc=0.001
1 test_loss=4.995160 test_acc=0.0018
step=3500 train_loss=4.229653 train_acc=0.0609 val_loss=5.006864 val_acc=0.000
0 test_loss=4.993348 test_acc=0.0025
step=4000 train_loss=4.217166 train_acc=0.0709 val_loss=5.011121 val_acc=0.000
0 test_loss=4.999111 test_acc=0.0025
step=4500 train_loss=4.089119 train_acc=0.0773 val_loss=4.848764 val_acc=0.002
1 test_loss=4.807810 test_acc=0.0053
step=5000 train_loss=3.042309 train_acc=0.2101 val_loss=3.248956 val_acc=0.089
4 test_loss=3.254582 test_acc=0.0813
step=5500 train_loss=2.812880 train_acc=0.2413 val_loss=3.033160 val_acc=0.108
5 test_loss=3.042822 test_acc=0.0990
step=6000 train_loss=2.716115 train_acc=0.2417 val_loss=2.879741 val_acc=0.128
7 test_loss=2.875566 test_acc=0.1185
step=6500 train_loss=2.597559 train_acc=0.2459 val_loss=2.701144 val_acc=0.131
9 test_loss=2.709579 test_acc=0.1427
step=7000 train_loss=2.539266 train_acc=0.2349 val_loss=2.644736 val_acc=0.136
2 test_loss=2.647193 test_acc=0.1401
step=7500 train_loss=2.521253 train_acc=0.2682 val_loss=2.597237 val_acc=0.190
4 test_loss=2.611823 test_acc=0.1806
step=8000 train_loss=2.501766 train_acc=0.2736 val_loss=2.583419 val_acc=0.162
8 test_loss=2.592048 test_acc=0.1673
step=8500 train_loss=2.498994 train_acc=0.2707 val_loss=2.582920 val_acc=0.186
2 test_loss=2.588042 test_acc=0.1755
step=9000 train_loss=2.492183 train_acc=0.2923 val_loss=2.580399 val_acc=0.204
3 test_loss=2.582265 test_acc=0.1852
step=9500 train_loss=2.496783 train_acc=0.2566 val_loss=2.578567 val_acc=0.163
8 test_loss=2.577466 test_acc=0.1663
step=10000 train_loss=2.492543 train_acc=0.2931 val_loss=2.572361 val_acc=0.18
51 test_loss=2.565019 test_acc=0.2005
step=10500 train_loss=2.497367 train_acc=0.2729 val_loss=2.541149 val_acc=0.22
23 test_loss=2.550046 test_acc=0.2088
step=11000 train_loss=2.489072 train_acc=0.2658 val_loss=2.533090 val_acc=0.21
81 test_loss=2.537364 test_acc=0.2054
step=11500 train_loss=2.493873 train_acc=0.2509 val_loss=2.534596 val_acc=0.20
32 test_loss=2.541557 test_acc=0.1974
step=12000 train_loss=2.485896 train_acc=0.2902 val_loss=2.529994 val_acc=0.23
30 test_loss=2.536181 test_acc=0.2114
step=12500 train_loss=2.475524 train_acc=0.2551 val_loss=2.512483 val_acc=0.21
49 test_loss=2.524360 test_acc=0.1951
```

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```
step=13000 train_loss=2.485605 train_acc=0.2566 val_loss=2.523110 val_acc=0.22
34 test_loss=2.529334 test_acc=0.2091
step=13500 train_loss=2.480966 train_acc=0.3065 val_loss=2.524045 val_acc=0.25
11 test_loss=2.527069 test_acc=0.2493
step=14000 train_loss=2.482483 train_acc=0.2576 val_loss=2.524896 val_acc=0.22
13 test_loss=2.531976 test_acc=0.2063
step=14500 train_loss=2.479561 train_acc=0.2902 val_loss=2.532007 val_acc=0.22
13 test_loss=2.528446 test_acc=0.2304
step=15000 train_loss=2.477673 train_acc=0.2789 val_loss=2.529378 val_acc=0.22
13 test_loss=2.530947 test_acc=0.2152
step=15500 train_loss=2.478677 train_acc=0.2697 val_loss=2.515988 val_acc=0.22
34 test_loss=2.522750 test_acc=0.2074
step=16000 train_loss=2.474827 train_acc=0.2856 val_loss=2.514186 val_acc=0.23
19 test_loss=2.520261 test_acc=0.2198
step=16500 train_loss=2.472741 train_acc=0.2796 val_loss=2.510528 val_acc=0.23
51 test_loss=2.515983 test_acc=0.2253
step=17000 train_loss=2.482107 train_acc=0.2753 val_loss=2.521671 val_acc=0.22
87 test_loss=2.528233 test_acc=0.2228
step=17500 train_loss=2.472949 train_acc=0.3016 val_loss=2.511794 val_acc=0.23
83 test_loss=2.516599 test_acc=0.2424
step=18000 train_loss=2.468213 train_acc=0.2658 val_loss=2.505008 val_acc=0.21
17 test_loss=2.518121 test_acc=0.1936
step=18500 train_loss=2.468086 train_acc=0.2612 val_loss=2.512112 val_acc=0.20
96 test_loss=2.517643 test_acc=0.2035
step=19000 train_loss=2.479390 train_acc=0.2668 val_loss=2.529217 val_acc=0.20
85 test_loss=2.528204 test_acc=0.2102
step=19500 train_loss=2.466529 train_acc=0.2721 val_loss=2.514120 val_acc=0.19
57 test_loss=2.513204 test_acc=0.2120
step=20000 train_loss=2.464127 train_acc=0.2686 val_loss=2.506260 val_acc=0.22
66 test_loss=2.509654 test_acc=0.2060
step=20500 train_loss=2.470357 train_acc=0.2693 val_loss=2.517167 val_acc=0.22
45 test_loss=2.520628 test_acc=0.2097
step=21000 train_loss=2.469575 train_acc=0.2356 val_loss=2.512584 val_acc=0.17
23 test_loss=2.518095 test_acc=0.1881
step=21500 train_loss=2.469239 train_acc=0.2636 val_loss=2.519677 val_acc=0.20
21 test_loss=2.515431 test_acc=0.2074
step=22000 train_loss=2.468257 train_acc=0.3044 val_loss=2.516699 val_acc=0.24
47 test_loss=2.521227 test_acc=0.2405
step=22500 train_loss=2.469946 train_acc=0.2445 val_loss=2.521278 val_acc=0.17
98 test_loss=2.523821 test_acc=0.1845
step=23000 train_loss=2.462767 train_acc=0.2569 val_loss=2.510271 val_acc=0.20
53 test_loss=2.515551 test_acc=0.1920
step=23500 train_loss=2.456867 train_acc=0.2452 val_loss=2.505174 val_acc=0.19
26 test_loss=2.509417 test_acc=0.1875
step=24000 train_loss=2.467484 train_acc=0.2686 val_loss=2.512928 val_acc=0.21
91 test_loss=2.509444 test_acc=0.2061
step=24500 train_loss=2.461815 train_acc=0.3083 val_loss=2.510777 val_acc=0.23
94 test_loss=2.511902 test_acc=0.2408
step=25000 train_loss=2.470456 train_acc=0.2902 val_loss=2.512027 val_acc=0.22
13 test_loss=2.514889 test_acc=0.2272
step=25500 train_loss=2.466516 train_acc=0.2693 val_loss=2.506614 val_acc=0.22
87 test_loss=2.508673 test_acc=0.2194
step=26000 train_loss=2.461323 train_acc=0.2831 val_loss=2.505899 val_acc=0.22
13 test_loss=2.506476 test_acc=0.2261
step=26500 train_loss=2.461590 train_acc=0.2952 val_loss=2.506445 val_acc=0.24
26 test_loss=2.511723 test_acc=0.2315
step=27000 train_loss=2.466496 train_acc=0.2488 val_loss=2.499637 val_acc=0.21
28 test_loss=2.515053 test_acc=0.1817
                _loss=2.460045 train_acc=0.2541 val_loss=2.498888 val_acc=0.21
28 test_loss=2.505150 test_acc=0.1980
```

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```
step=28000 train_loss=2.460701 train_acc=0.2594 val_loss=2.505985 val_acc=0.20
00 test_loss=2.511584 test_acc=0.2001
step=28500 train_loss=2.460474 train_acc=0.3129 val_loss=2.504038 val_acc=0.26
06 test_loss=2.504982 test_acc=0.2463
step=29000 train_loss=2.454850 train_acc=0.2714 val_loss=2.502231 val_acc=0.21
70 test_loss=2.500386 test_acc=0.2081
step=29500 train_loss=2.461139 train_acc=0.2782 val_loss=2.505016 val_acc=0.20
00 test_loss=2.507247 test_acc=0.2148
step=30000 train_loss=2.460288 train_acc=0.2665 val_loss=2.499672 val_acc=0.20
11 test_loss=2.498891 test_acc=0.2091
step=30500 train_loss=2.457736 train_acc=0.2622 val_loss=2.505523 val_acc=0.18
62 test_loss=2.504374 test_acc=0.2100
step=31000 train_loss=2.461341 train_acc=0.2955 val_loss=2.512846 val_acc=0.23
40 test_loss=2.508359 test_acc=0.2277
step=31500 train_loss=2.459538 train_acc=0.2838 val_loss=2.499213 val_acc=0.24
04 test_loss=2.497963 test_acc=0.2291
step=32000 train_loss=2.460123 train_acc=0.2636 val_loss=2.507614 val_acc=0.19
89 test_loss=2.509649 test_acc=0.2026
step=32500 train_loss=2.462603 train_acc=0.3136 val_loss=2.504311 val_acc=0.24
36 test_loss=2.507550 test_acc=0.2403
step=33000 train_loss=2.464455 train_acc=0.2899 val_loss=2.510243 val_acc=0.22
66 test_loss=2.508687 test_acc=0.2263
step=33500 train_loss=2.468777 train_acc=0.3101 val_loss=2.505192 val_acc=0.25
11 test_loss=2.513200 test_acc=0.2414
step=34000 train_loss=2.480854 train_acc=0.2388 val_loss=2.518967 val_acc=0.20
53 test_loss=2.527340 test_acc=0.1866
step=34500 train_loss=2.482078 train_acc=0.2810 val_loss=2.518845 val_acc=0.23
30 test_loss=2.523654 test_acc=0.2231
step=35000 train_loss=2.483084 train_acc=0.2796 val_loss=2.531026 val_acc=0.21
49 test_loss=2.532131 test_acc=0.2191
step=35500 train_loss=2.479294 train_acc=0.2665 val_loss=2.522579 val_acc=0.22
02 test_loss=2.530638 test_acc=0.2036
step=36000 train_loss=2.484274 train_acc=0.2931 val_loss=2.530477 val_acc=0.25
11 test_loss=2.528816 test_acc=0.2410
step=36500 train_loss=2.481671 train_acc=0.2566 val_loss=2.523032 val_acc=0.20
53 test_loss=2.529350 test_acc=0.1962
step=37000 train_loss=2.480524 train_acc=0.2721 val_loss=2.519787 val_acc=0.21
70 test_loss=2.529238 test_acc=0.2040
step=37500 train_loss=2.483078 train_acc=0.3249 val_loss=2.530228 val_acc=0.25
64 test_loss=2.529905 test_acc=0.2575
step=38000 train_loss=2.485315 train_acc=0.2853 val_loss=2.525388 val_acc=0.22
02 test_loss=2.528377 test_acc=0.2316
step=38500 train_loss=2.476916 train_acc=0.2445 val_loss=2.517432 val_acc=0.20
00 test_loss=2.520864 test_acc=0.1877
step=39000 train_loss=2.480889 train_acc=0.2658 val_loss=2.508093 val_acc=0.21
70 test_loss=2.526253 test_acc=0.2036
step=39500 train_loss=2.477246 train_acc=0.2690 val_loss=2.517897 val_acc=0.22
13 test_loss=2.521344 test_acc=0.2058
step=40000 train_loss=2.483160 train_acc=0.2651 val_loss=2.526734 val_acc=0.20
85 test_loss=2.529227 test_acc=0.2077
step=40500 train_loss=2.481229 train_acc=0.2700 val_loss=2.521644 val_acc=0.21
91 test_loss=2.530004 test_acc=0.2065
step=41000 train_loss=2.478171 train_acc=0.2534 val_loss=2.523026 val_acc=0.19
26 test_loss=2.527799 test_acc=0.1944
step=41500 train_loss=2.475114 train_acc=0.2835 val_loss=2.509624 val_acc=0.21
81 test_loss=2.520764 test_acc=0.2311
step=42000 train_loss=2.475403 train_acc=0.2558 val_loss=2.521436 val_acc=0.20
32 test_loss=2.528612 test_acc=0.1900
step=42500 train_loss=2.476075 train_acc=0.3147 val_loss=2.518314 val_acc=0.23
51 test_loss=2.520513 test_acc=0.2485
```

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```
step=43000 train_loss=2.471863 train_acc=0.2768 val_loss=2.509983 val_acc=0.21
70 test_loss=2.515436 test_acc=0.2145
step=43500 train_loss=2.473939 train_acc=0.2721 val_loss=2.517927 val_acc=0.20
64 test_loss=2.515434 test_acc=0.2198
step=44000 train_loss=2.477234 train_acc=0.2484 val_loss=2.518293 val_acc=0.20
00 test_loss=2.523109 test_acc=0.1930
step=44500 train_loss=2.473100 train_acc=0.2945 val_loss=2.522959 val_acc=0.23
19 test_loss=2.522977 test_acc=0.2318
step=45000 train_loss=2.474241 train_acc=0.2636 val_loss=2.518939 val_acc=0.18
94 test_loss=2.519889 test_acc=0.2153
step=45500 train_loss=2.472563 train_acc=0.2675 val_loss=2.517692 val_acc=0.20
85 test_loss=2.520039 test_acc=0.2079
step=46000 train_loss=2.470826 train_acc=0.3079 val_loss=2.512721 val_acc=0.24
89 test_loss=2.517193 test_acc=0.2444
step=46500 train_loss=2.469424 train_acc=0.3225 val_loss=2.510979 val_acc=0.26
38 test_loss=2.515129 test_acc=0.2580
step=47000 train_loss=2.470606 train_acc=0.2817 val_loss=2.509814 val_acc=0.22
23 test_loss=2.519204 test_acc=0.2160
step=47500 train_loss=2.468262 train_acc=0.2753 val_loss=2.512111 val_acc=0.20
21 test_loss=2.517113 test_acc=0.2054
step=48000 train_loss=2.463892 train_acc=0.3016 val_loss=2.498310 val_acc=0.25
11 test_loss=2.509473 test_acc=0.2423
step=48500 train_loss=2.463392 train_acc=0.2863 val_loss=2.514224 val_acc=0.20
64 test_loss=2.511294 test_acc=0.2208
step=49000 train_loss=2.460647 train_acc=0.3005 val_loss=2.503751 val_acc=0.25
53 test_loss=2.505344 test_acc=0.2430
step=49500 train_loss=2.464817 train_acc=0.2679 val_loss=2.507552 val_acc=0.20
74 test_loss=2.513746 test_acc=0.2072
step=50000 train_loss=2.473001 train_acc=0.2807 val_loss=2.518350 val_acc=0.22
34 test_loss=2.520314 test_acc=0.2194
step=50500 train_loss=2.465309 train_acc=0.2750 val_loss=2.504445 val_acc=0.21
81 test_loss=2.507544 test_acc=0.2130
step=51000 train_loss=2.455537 train_acc=0.2527 val_loss=2.493242 val_acc=0.20
11 test_loss=2.503114 test_acc=0.1936
step=51500 train_loss=2.458919 train_acc=0.2658 val_loss=2.504972 val_acc=0.19
26 test_loss=2.509254 test_acc=0.2118
step=52000 train_loss=2.464134 train_acc=0.3072 val_loss=2.501890 val_acc=0.26
81 test_loss=2.508705 test_acc=0.2458
step=52500 train_loss=2.459579 train_acc=0.2856 val_loss=2.506750 val_acc=0.20
96 test_loss=2.506818 test_acc=0.2230
step=53000 train_loss=2.455912 train_acc=0.3094 val_loss=2.494414 val_acc=0.24
36 test_loss=2.498084 test_acc=0.2385
step=53500 train_loss=2.458545 train_acc=0.3012 val_loss=2.496484 val_acc=0.24
15 test_loss=2.505134 test_acc=0.2375
step=54000 train_loss=2.455045 train_acc=0.2778 val_loss=2.502667 val_acc=0.21
60 test_loss=2.507779 test_acc=0.2072
step=54500 train_loss=2.457455 train_acc=0.2473 val_loss=2.496428 val_acc=0.21
70 test_loss=2.505950 test_acc=0.1824
step=55000 train_loss=2.453395 train_acc=0.3104 val_loss=2.496465 val_acc=0.24
36 test_loss=2.498363 test_acc=0.2407
step=55500 train_loss=2.449153 train_acc=0.2775 val_loss=2.493717 val_acc=0.21
60 test_loss=2.495167 test_acc=0.2162
step=56000 train_loss=2.450581 train_acc=0.2449 val_loss=2.495115 val_acc=0.19
26 test_loss=2.497756 test_acc=0.1863
step=56500 train_loss=2.453510 train_acc=0.3175 val_loss=2.501852 val_acc=0.26
06 test_loss=2.502618 test_acc=0.2513
step=57000 train_loss=2.451417 train_acc=0.3037 val_loss=2.492870 val_acc=0.22
77 test_loss=2.498456 test_acc=0.2286
_loss=2.445745 train_acc=0.2994 val_loss=2.484386 val_acc=0.22
13 test_loss=2.491158 test_acc=0.2254
```

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```
step=58000 train_loss=2.446054 train_acc=0.2807 val_loss=2.484867 val_acc=0.20
74 test_loss=2.493137 test_acc=0.2152
step=58500 train_loss=2.443776 train_acc=0.2906 val_loss=2.484497 val_acc=0.23
62 test_loss=2.493543 test_acc=0.2198
step=59000 train_loss=2.446766 train_acc=0.3083 val_loss=2.485122 val_acc=0.24
89 test_loss=2.490786 test_acc=0.2384
step=59500 train_loss=2.445406 train_acc=0.2867 val_loss=2.498227 val_acc=0.21
28 test_loss=2.493170 test_acc=0.2251
step=60000 train_loss=2.443330 train_acc=0.3154 val_loss=2.486326 val_acc=0.24
26 test_loss=2.489139 test_acc=0.2449
step=60500 train_loss=2.441477 train_acc=0.3097 val_loss=2.479427 val_acc=0.23
19 test_loss=2.487317 test_acc=0.2414
step=61000 train_loss=2.440331 train_acc=0.3172 val_loss=2.478572 val_acc=0.27
23 test_loss=2.485225 test_acc=0.2449
step=61500 train_loss=2.436496 train_acc=0.3122 val_loss=2.477524 val_acc=0.25
11 test_loss=2.483628 test_acc=0.2394
step=62000 train_loss=2.435023 train_acc=0.3345 val_loss=2.477866 val_acc=0.25
85 test_loss=2.484415 test_acc=0.2555
step=62500 train_loss=2.435748 train_acc=0.2750 val_loss=2.480087 val_acc=0.20
53 test_loss=2.482701 test_acc=0.2090
step=63000 train_loss=2.433910 train_acc=0.3062 val_loss=2.472794 val_acc=0.25
53 test_loss=2.479952 test_acc=0.2346
step=63500 train_loss=2.434014 train_acc=0.3097 val_loss=2.481964 val_acc=0.23
94 test_loss=2.481106 test_acc=0.2380
step=64000 train_loss=2.434153 train_acc=0.2757 val_loss=2.476631 val_acc=0.20
32 test_loss=2.479133 test_acc=0.2093
step=64500 train_loss=2.431901 train_acc=0.3037 val_loss=2.472600 val_acc=0.25
96 test_loss=2.479536 test_acc=0.2355
step=65000 train_loss=2.428833 train_acc=0.2959 val_loss=2.473845 val_acc=0.23
30 test_loss=2.479119 test_acc=0.2189
step=65500 train_loss=2.429541 train_acc=0.3377 val_loss=2.473257 val_acc=0.27
55 test_loss=2.477541 test_acc=0.2596
step=66000 train_loss=2.431709 train_acc=0.2892 val_loss=2.470819 val_acc=0.23
40 test_loss=2.478746 test_acc=0.2253
step=66500 train_loss=2.429152 train_acc=0.2828 val_loss=2.467229 val_acc=0.23
51 test_loss=2.474330 test_acc=0.2129
step=67000 train_loss=2.428960 train_acc=0.3104 val_loss=2.470603 val_acc=0.24
89 test_loss=2.477454 test_acc=0.2400
step=67500 train_loss=2.427734 train_acc=0.2842 val_loss=2.472603 val_acc=0.22
55 test_loss=2.474341 test_acc=0.2222
step=68000 train_loss=2.425183 train_acc=0.3047 val_loss=2.466952 val_acc=0.25
21 test_loss=2.472542 test_acc=0.2309
step=68500 train_loss=2.427176 train_acc=0.3108 val_loss=2.468795 val_acc=0.24
57 test_loss=2.472019 test_acc=0.2396
step=69000 train_loss=2.426707 train_acc=0.2945 val_loss=2.466840 val_acc=0.23
51 test_loss=2.474857 test_acc=0.2231
step=69500 train_loss=2.423338 train_acc=0.3281 val_loss=2.464217 val_acc=0.25
43 test_loss=2.470183 test_acc=0.2564
step=70000 train_loss=2.420600 train_acc=0.2845 val_loss=2.459619 val_acc=0.23
19 test_loss=2.463473 test_acc=0.2251
step=70500 train_loss=2.422332 train_acc=0.3193 val_loss=2.465806 val_acc=0.24
15 test_loss=2.467593 test_acc=0.2504
step=71000 train_loss=2.417941 train_acc=0.2860 val_loss=2.457916 val_acc=0.23
19 test_loss=2.464619 test_acc=0.2168
step=71500 train_loss=2.419914 train_acc=0.3072 val_loss=2.457085 val_acc=0.25
00 test_loss=2.465378 test_acc=0.2430
step=72000 train_loss=2.417935 train_acc=0.2941 val_loss=2.456353 val_acc=0.23
72 test_loss=2.464331 test_acc=0.2260
step=72500 train_loss=2.417339 train_acc=0.3296 val_loss=2.456621 val_acc=0.26
38 test_loss=2.464065 test_acc=0.2538
```

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```
step=73000 train_loss=2.416452 train_acc=0.3242 val_loss=2.454009 val_acc=0.25
32 test_loss=2.463953 test_acc=0.2561
step=73500 train_loss=2.416225 train_acc=0.2962 val_loss=2.456065 val_acc=0.23
62 test_loss=2.465508 test_acc=0.2254
step=74000 train_loss=2.415515 train_acc=0.3352 val_loss=2.455064 val_acc=0.27
66 test_loss=2.462648 test_acc=0.2577
step=74500 train_loss=2.414202 train_acc=0.3111 val_loss=2.459453 val_acc=0.24
47 test_loss=2.462684 test_acc=0.2369
step=75000 train_loss=2.413800 train_acc=0.3175 val_loss=2.455232 val_acc=0.25
96 test_loss=2.457726 test_acc=0.2470
step=75500 train_loss=2.411463 train_acc=0.3274 val_loss=2.451737 val_acc=0.27
02 test_loss=2.459398 test_acc=0.2520
step=76000 train_loss=2.410234 train_acc=0.3324 val_loss=2.452175 val_acc=0.27
34 test_loss=2.456500 test_acc=0.2584
step=76500 train_loss=2.409541 train_acc=0.2838 val_loss=2.448387 val_acc=0.22
66 test_loss=2.453209 test_acc=0.2196
step=77000 train_loss=2.409211 train_acc=0.3327 val_loss=2.454856 val_acc=0.25
53 test_loss=2.460005 test_acc=0.2566
step=77500 train_loss=2.409056 train_acc=0.3009 val_loss=2.452193 val_acc=0.23
09 test_loss=2.456707 test_acc=0.2336
step=78000 train_loss=2.408612 train_acc=0.3175 val_loss=2.449801 val_acc=0.25
11 test_loss=2.452951 test_acc=0.2405
step=78500 train_loss=2.409798 train_acc=0.3055 val_loss=2.452823 val_acc=0.26
06 test_loss=2.456665 test_acc=0.2364
step=79000 train_loss=2.409448 train_acc=0.3515 val_loss=2.451593 val_acc=0.28
83 test_loss=2.459190 test_acc=0.2718
step=79500 train_loss=2.407060 train_acc=0.3228 val_loss=2.449627 val_acc=0.26
81 test_loss=2.451858 test_acc=0.2453
step=80000 train_loss=2.407723 train_acc=0.2991 val_loss=2.453330 val_acc=0.23
62 test_loss=2.454419 test_acc=0.2295
step=80500 train_loss=2.406246 train_acc=0.3125 val_loss=2.448588 val_acc=0.25
11 test_loss=2.451971 test_acc=0.2371
step=81000 train_loss=2.405742 train_acc=0.3189 val_loss=2.445631 val_acc=0.24
68 test_loss=2.453017 test_acc=0.2428
step=81500 train_loss=2.405351 train_acc=0.3249 val_loss=2.447305 val_acc=0.25
43 test_loss=2.449595 test_acc=0.2506
step=82000 train_loss=2.404510 train_acc=0.3218 val_loss=2.444999 val_acc=0.26
38 test_loss=2.452741 test_acc=0.2465
step=82500 train_loss=2.403482 train_acc=0.3338 val_loss=2.445074 val_acc=0.27
45 test_loss=2.449388 test_acc=0.2593
step=83000 train_loss=2.402363 train_acc=0.3434 val_loss=2.443544 val_acc=0.25
96 test_loss=2.448559 test_acc=0.2676
step=83500 train_loss=2.401881 train_acc=0.2913 val_loss=2.445094 val_acc=0.22
45 test_loss=2.449289 test_acc=0.2221
step=84000 train_loss=2.400199 train_acc=0.3356 val_loss=2.440718 val_acc=0.26
17 test_loss=2.445162 test_acc=0.2630
step=84500 train_loss=2.400193 train_acc=0.3150 val_loss=2.443048 val_acc=0.24
57 test_loss=2.447157 test_acc=0.2415
step=85000 train_loss=2.400612 train_acc=0.3210 val_loss=2.441257 val_acc=0.25
53 test_loss=2.446927 test_acc=0.2467
step=85500 train_loss=2.400602 train_acc=0.3012 val_loss=2.441860 val_acc=0.24
47 test_loss=2.445642 test_acc=0.2281
step=86000 train_loss=2.400188 train_acc=0.3540 val_loss=2.442474 val_acc=0.27
87 test_loss=2.448269 test_acc=0.2747
step=86500 train_loss=2.399191 train_acc=0.3451 val_loss=2.442017 val_acc=0.27
13 test_loss=2.447406 test_acc=0.2646
step=87000 train_loss=2.399457 train_acc=0.3111 val_loss=2.443202 val_acc=0.25
21 test_loss=2.448838 test_acc=0.2355
step=87500 train_loss=2.399468 train_acc=0.3402 val_loss=2.443526 val_acc=0.26
17 test_loss=2.447331 test_acc=0.2635
```

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```
step=88000 train_loss=2.398836 train_acc=0.3522 val_loss=2.441094 val_acc=0.28
30 test_loss=2.444615 test_acc=0.2681
step=88500 train_loss=2.398236 train_acc=0.3724 val_loss=2.437321 val_acc=0.29
68 test_loss=2.443170 test_acc=0.2904
step=89000 train_loss=2.397348 train_acc=0.3193 val_loss=2.439779 val_acc=0.25
74 test_loss=2.442702 test_acc=0.2460
step=89500 train_loss=2.397570 train_acc=0.3370 val_loss=2.438286 val_acc=0.26
17 test_loss=2.443113 test_acc=0.2557
step=90000 train_loss=2.398016 train_acc=0.3611 val_loss=2.440004 val_acc=0.29
04 test_loss=2.445009 test_acc=0.2752
step=90500 train_loss=2.395923 train_acc=0.3437 val_loss=2.434398 val_acc=0.28
30 test_loss=2.440503 test_acc=0.2679
step=91000 train_loss=2.395121 train_acc=0.3111 val_loss=2.437251 val_acc=0.24
68 test_loss=2.441097 test_acc=0.2353
step=91500 train_loss=2.395638 train_acc=0.3210 val_loss=2.436763 val_acc=0.25
96 test_loss=2.441416 test_acc=0.2426
step=92000 train_loss=2.395699 train_acc=0.3434 val_loss=2.436240 val_acc=0.26
70 test_loss=2.440068 test_acc=0.2665
step=92500 train_loss=2.395284 train_acc=0.3565 val_loss=2.435786 val_acc=0.29
15 test_loss=2.440388 test_acc=0.2803
step=93000 train_loss=2.394573 train_acc=0.3359 val_loss=2.437330 val_acc=0.25
85 test_loss=2.440685 test_acc=0.2555
step=93500 train_loss=2.394299 train_acc=0.3505 val_loss=2.438309 val_acc=0.26
17 test_loss=2.441324 test_acc=0.2686
step=94000 train_loss=2.394747 train_acc=0.3508 val_loss=2.433994 val_acc=0.27
45 test_loss=2.440186 test_acc=0.2617
step=94500 train_loss=2.394474 train_acc=0.3462 val_loss=2.434569 val_acc=0.27
55 test_loss=2.441113 test_acc=0.2681
step=95000 train_loss=2.394178 train_acc=0.3009 val_loss=2.434219 val_acc=0.23
83 test_loss=2.439136 test_acc=0.2258
step=95500 train_loss=2.393783 train_acc=0.3373 val_loss=2.436092 val_acc=0.26
17 test_loss=2.440795 test_acc=0.2593
step=96000 train_loss=2.393148 train_acc=0.3894 val_loss=2.434953 val_acc=0.30
96 test_loss=2.440134 test_acc=0.2973
step=96500 train_loss=2.392751 train_acc=0.3274 val_loss=2.435579 val_acc=0.26
38 test_loss=2.440361 test_acc=0.2527
step=97000 train_loss=2.392533 train_acc=0.3342 val_loss=2.435727 val_acc=0.27
13 test_loss=2.439514 test_acc=0.2504
step=97500 train_loss=2.392734 train_acc=0.3575 val_loss=2.438255 val_acc=0.26
60 test_loss=2.438943 test_acc=0.2747
step=98000 train_loss=2.392996 train_acc=0.3310 val_loss=2.436890 val_acc=0.27
02 test_loss=2.440307 test_acc=0.2628
step=98500 train_loss=2.392647 train_acc=0.3009 val_loss=2.435899 val_acc=0.22
45 test_loss=2.440201 test_acc=0.2283
step=99000 train_loss=2.391924 train_acc=0.3182 val_loss=2.433253 val_acc=0.25
96 test_loss=2.439249 test_acc=0.2428
step=99500 train_loss=2.391789 train_acc=0.3274 val_loss=2.436510 val_acc=0.26
60 test_loss=2.440534 test_acc=0.2492
step=100000 train_loss=2.392066 train_acc=0.3228 val_loss=2.434216 val_acc=0.2
436 test_loss=2.437761 test_acc=0.2461
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  "lr": 0.00010022655592756672,
  "train_loss": 2.391924000901588,
  "train_acc": 0.3182140326009922,
  "val_loss": 2.4332525588096456,
  "val_acc": 0.25957446808510637,
  "test_loss": 2.4392487487266057,
  "test_acc": 0.2427837789976979,
  "elapsed_sec": 3182.0560672283173
```

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```
}
checkpoint dir: /Users/shobhit/Documents/UW/CSE 493/cse493s-spring26-hw1/out/s
ubmission/part1/warmup_restarts/add_p97_l1_s101

$ /opt/homebrew/Caskroom/miniconda/base/envs/cse493/bin/python train.py --mode
modular --op + --p 97 --n_layer 1 --n_head 4 --n_embd 128 --steps 100000 --see
d 202 --batch_size 128 --lr 0.001 --min_lr 0.0001 --lr_schedule cosine --warmu
p_steps 1000 --grad_clip 1.0 --beta1 0.9 --beta2 0.95 --weight_decay 0.01 --op
timizer adam --eval_interval 500 --log_interval 500 --out_dir '/Users/shobhit/
Documents/UW/CSE 493/cse493s-spring26-hw1/out/submission/part1/warmup_restart
s/add_p97_l1_s202' --bias --resume
number of parameters: 0.21M
[resume] model_last.pt not found in /Users/shobhit/Documents/UW/CSE 493/cse493
s-spring26-hw1/out/submission/part1/warmup_restarts/add_p97_l1_s202; starting
fresh
device=mps train=2822 val=940 test=5647 batch_size=128
step=1 train_loss=4.683427 train_acc=0.0000 val_loss=4.671990 val_acc=0.0000 t
est_loss=4.687884 test_acc=0.0000
step=500 train_loss=4.296852 train_acc=0.0819 val_loss=4.841604 val_acc=0.0021
test_loss=4.842768 test_acc=0.0025
step=1000 train_loss=4.251988 train_acc=0.0875 val_loss=4.860036 val_acc=0.001
1 test_loss=4.870018 test_acc=0.0005
step=1500 train_loss=4.354639 train_acc=0.0617 val_loss=4.798033 val_acc=0.003
2 test_loss=4.812067 test_acc=0.0011
step=2000 train_loss=4.276234 train_acc=0.0613 val_loss=4.912685 val_acc=0.003
2 test_loss=4.923159 test_acc=0.0023
step=2500 train_loss=4.230754 train_acc=0.0684 val_loss=4.966713 val_acc=0.004
3 test_loss=4.976593 test_acc=0.0019
step=3000 train_loss=4.211825 train_acc=0.0719 val_loss=4.976516 val_acc=0.002
1 test_loss=4.987891 test_acc=0.0012
step=3500 train_loss=4.202381 train_acc=0.0666 val_loss=4.963167 val_acc=0.005
3 test_loss=4.988796 test_acc=0.0023
step=4000 train_loss=4.176246 train_acc=0.0790 val_loss=4.975243 val_acc=0.003
2 test_loss=4.982263 test_acc=0.0027
step=4500 train_loss=4.105239 train_acc=0.0797 val_loss=4.814343 val_acc=0.006
4 test_loss=4.820667 test_acc=0.0051
step=5000 train_loss=2.974661 train_acc=0.2215 val_loss=3.160200 val_acc=0.103
2 test_loss=3.180597 test_acc=0.0859
step=5500 train_loss=2.764239 train_acc=0.2541 val_loss=2.985439 val_acc=0.120
2 test_loss=2.994016 test_acc=0.1139
step=6000 train_loss=2.687011 train_acc=0.2491 val_loss=2.930803 val_acc=0.126
6 test_loss=2.939645 test_acc=0.1155
step=6500 train_loss=2.661383 train_acc=0.2566 val_loss=2.933136 val_acc=0.112
8 test_loss=2.939599 test_acc=0.1140
step=7000 train_loss=2.649234 train_acc=0.2700 val_loss=2.919513 val_acc=0.126
6 test_loss=2.938102 test_acc=0.1078
step=7500 train_loss=2.611207 train_acc=0.2775 val_loss=2.878187 val_acc=0.129
8 test_loss=2.906408 test_acc=0.1153
step=8000 train_loss=2.597105 train_acc=0.2686 val_loss=2.816343 val_acc=0.137
2 test_loss=2.843894 test_acc=0.1233
step=8500 train_loss=2.571837 train_acc=0.2835 val_loss=2.776581 val_acc=0.145
7 test_loss=2.789815 test_acc=0.1390
step=9000 train_loss=2.534943 train_acc=0.2916 val_loss=2.723740 val_acc=0.153
2 test_loss=2.739005 test_acc=0.1443
step=9500 train_loss=2.518225 train_acc=0.2931 val_loss=2.695446 val_acc=0.146
8 test_loss=2.703282 test_acc=0.1542
step=10000 train_loss=2.512879 train_acc=0.2909 val_loss=2.690265 val_acc=0.16
06 test_loss=2.688290 test_acc=0.1498
_loss=2.501901 train_acc=0.2994 val_loss=2.664035 val_acc=0.18
09 test_loss=2.679962 test_acc=0.1468
```

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```
step=11000 train_loss=2.494789 train_acc=0.3047 val_loss=2.656854 val_acc=0.16
49 test_loss=2.660808 test_acc=0.1727
step=11500 train_loss=2.500916 train_acc=0.2821 val_loss=2.657099 val_acc=0.17
45 test_loss=2.679675 test_acc=0.1486
step=12000 train_loss=2.490786 train_acc=0.2962 val_loss=2.663157 val_acc=0.18
19 test_loss=2.678079 test_acc=0.1564
step=12500 train_loss=2.483815 train_acc=0.3094 val_loss=2.649465 val_acc=0.17
77 test_loss=2.656042 test_acc=0.1585
step=13000 train_loss=2.484288 train_acc=0.3076 val_loss=2.650659 val_acc=0.17
34 test_loss=2.660914 test_acc=0.1675
step=13500 train_loss=2.488839 train_acc=0.3072 val_loss=2.641652 val_acc=0.18
62 test_loss=2.667025 test_acc=0.1592
step=14000 train_loss=2.493838 train_acc=0.2838 val_loss=2.630412 val_acc=0.17
66 test_loss=2.642869 test_acc=0.1762
step=14500 train_loss=2.486142 train_acc=0.3072 val_loss=2.610700 val_acc=0.20
96 test_loss=2.627516 test_acc=0.1808
step=15000 train_loss=2.487713 train_acc=0.2760 val_loss=2.627692 val_acc=0.18
51 test_loss=2.641059 test_acc=0.1689
step=15500 train_loss=2.480587 train_acc=0.2743 val_loss=2.642421 val_acc=0.18
09 test_loss=2.650581 test_acc=0.1592
step=16000 train_loss=2.486322 train_acc=0.2714 val_loss=2.620124 val_acc=0.20
96 test_loss=2.645094 test_acc=0.1472
step=16500 train_loss=2.474686 train_acc=0.2771 val_loss=2.630053 val_acc=0.17
34 test_loss=2.641666 test_acc=0.1569
step=17000 train_loss=2.478879 train_acc=0.2923 val_loss=2.616796 val_acc=0.19
68 test_loss=2.637977 test_acc=0.1606
step=17500 train_loss=2.481087 train_acc=0.2874 val_loss=2.631688 val_acc=0.18
09 test_loss=2.637766 test_acc=0.1654
step=18000 train_loss=2.471042 train_acc=0.3129 val_loss=2.627194 val_acc=0.21
17 test_loss=2.633814 test_acc=0.1785
step=18500 train_loss=2.475556 train_acc=0.2814 val_loss=2.629981 val_acc=0.17
77 test_loss=2.644221 test_acc=0.1723
step=19000 train_loss=2.475293 train_acc=0.2955 val_loss=2.621362 val_acc=0.20
43 test_loss=2.638214 test_acc=0.1780
step=19500 train_loss=2.468928 train_acc=0.2923 val_loss=2.620318 val_acc=0.17
77 test_loss=2.637858 test_acc=0.1723
step=20000 train_loss=2.473852 train_acc=0.2934 val_loss=2.627986 val_acc=0.20
00 test_loss=2.637879 test_acc=0.1719
step=20500 train_loss=2.468425 train_acc=0.2828 val_loss=2.621778 val_acc=0.17
13 test_loss=2.631927 test_acc=0.1711
step=21000 train_loss=2.465621 train_acc=0.3086 val_loss=2.618389 val_acc=0.20
43 test_loss=2.632211 test_acc=0.1842
step=21500 train_loss=2.472730 train_acc=0.2945 val_loss=2.628928 val_acc=0.19
57 test_loss=2.645433 test_acc=0.1679
step=22000 train_loss=2.470866 train_acc=0.3030 val_loss=2.636224 val_acc=0.19
68 test_loss=2.638751 test_acc=0.1728
step=22500 train_loss=2.464380 train_acc=0.3030 val_loss=2.624110 val_acc=0.18
62 test_loss=2.638585 test_acc=0.1748
step=23000 train_loss=2.466180 train_acc=0.3062 val_loss=2.636068 val_acc=0.19
04 test_loss=2.646375 test_acc=0.1764
step=23500 train_loss=2.468243 train_acc=0.2906 val_loss=2.634446 val_acc=0.18
30 test_loss=2.645604 test_acc=0.1705
step=24000 train_loss=2.469013 train_acc=0.2838 val_loss=2.625510 val_acc=0.19
15 test_loss=2.641972 test_acc=0.1620
step=24500 train_loss=2.465779 train_acc=0.2980 val_loss=2.612604 val_acc=0.18
09 test_loss=2.629385 test_acc=0.1718
step=25000 train_loss=2.463666 train_acc=0.3033 val_loss=2.632510 val_acc=0.20
85 test_loss=2.641329 test_acc=0.1748
_loss=2.467315 train_acc=0.2789 val_loss=2.636889 val_acc=0.18
19 test_loss=2.641525 test_acc=0.1613
```

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```
step=26000 train_loss=2.467398 train_acc=0.2860 val_loss=2.625257 val_acc=0.17
77 test_loss=2.642700 test_acc=0.1732
step=26500 train_loss=2.462556 train_acc=0.3094 val_loss=2.635861 val_acc=0.19
57 test_loss=2.638118 test_acc=0.1778
step=27000 train_loss=2.466740 train_acc=0.2916 val_loss=2.623746 val_acc=0.17
23 test_loss=2.634021 test_acc=0.1693
step=27500 train_loss=2.468393 train_acc=0.3069 val_loss=2.638178 val_acc=0.19
04 test_loss=2.648291 test_acc=0.1836
step=28000 train_loss=2.460464 train_acc=0.2941 val_loss=2.618334 val_acc=0.18
72 test_loss=2.631113 test_acc=0.1643
step=28500 train_loss=2.466100 train_acc=0.3009 val_loss=2.620497 val_acc=0.19
26 test_loss=2.633410 test_acc=0.1734
step=29000 train_loss=2.464960 train_acc=0.3065 val_loss=2.635528 val_acc=0.19
89 test_loss=2.638565 test_acc=0.1900
step=29500 train_loss=2.462280 train_acc=0.2867 val_loss=2.631681 val_acc=0.17
55 test_loss=2.637844 test_acc=0.1661
step=30000 train_loss=2.463997 train_acc=0.2874 val_loss=2.642753 val_acc=0.17
66 test_loss=2.639080 test_acc=0.1771
step=30500 train_loss=2.459558 train_acc=0.2892 val_loss=2.619787 val_acc=0.19
79 test_loss=2.636937 test_acc=0.1719
step=31000 train_loss=2.460338 train_acc=0.2895 val_loss=2.638349 val_acc=0.18
72 test_loss=2.641727 test_acc=0.1645
step=31500 train_loss=2.462008 train_acc=0.3125 val_loss=2.626229 val_acc=0.22
02 test_loss=2.636904 test_acc=0.1812
step=32000 train_loss=2.461361 train_acc=0.2945 val_loss=2.634665 val_acc=0.19
57 test_loss=2.642146 test_acc=0.1714
step=32500 train_loss=2.472843 train_acc=0.2966 val_loss=2.642333 val_acc=0.19
79 test_loss=2.647981 test_acc=0.1764
step=33000 train_loss=2.454757 train_acc=0.2952 val_loss=2.623631 val_acc=0.20
43 test_loss=2.640747 test_acc=0.1675
step=33500 train_loss=2.458203 train_acc=0.2877 val_loss=2.627148 val_acc=0.19
15 test_loss=2.643363 test_acc=0.1688
step=34000 train_loss=2.457791 train_acc=0.2955 val_loss=2.620389 val_acc=0.20
96 test_loss=2.639601 test_acc=0.1666
step=34500 train_loss=2.458704 train_acc=0.2952 val_loss=2.616716 val_acc=0.18
51 test_loss=2.628226 test_acc=0.1719
step=35000 train_loss=2.457104 train_acc=0.3076 val_loss=2.629410 val_acc=0.19
26 test_loss=2.631331 test_acc=0.1817
step=35500 train_loss=2.455010 train_acc=0.2948 val_loss=2.634713 val_acc=0.17
45 test_loss=2.636135 test_acc=0.1700
step=36000 train_loss=2.459208 train_acc=0.3161 val_loss=2.624071 val_acc=0.21
49 test_loss=2.635414 test_acc=0.1866
step=36500 train_loss=2.457115 train_acc=0.3069 val_loss=2.623890 val_acc=0.19
04 test_loss=2.633410 test_acc=0.1831
step=37000 train_loss=2.459334 train_acc=0.3076 val_loss=2.637449 val_acc=0.18
62 test_loss=2.647853 test_acc=0.1746
step=37500 train_loss=2.461194 train_acc=0.2934 val_loss=2.642663 val_acc=0.17
77 test_loss=2.650994 test_acc=0.1666
step=38000 train_loss=2.454597 train_acc=0.3115 val_loss=2.637262 val_acc=0.20
64 test_loss=2.646409 test_acc=0.1822
step=38500 train_loss=2.467192 train_acc=0.3026 val_loss=2.630216 val_acc=0.20
21 test_loss=2.642023 test_acc=0.1739
step=39000 train_loss=2.460478 train_acc=0.2984 val_loss=2.625673 val_acc=0.20
96 test_loss=2.641259 test_acc=0.1652
step=39500 train_loss=2.458374 train_acc=0.3249 val_loss=2.633088 val_acc=0.19
15 test_loss=2.636295 test_acc=0.1884
step=40000 train_loss=2.452430 train_acc=0.3104 val_loss=2.620661 val_acc=0.18
62 test_loss=2.630012 test_acc=0.1813
_loss=2.459145 train_acc=0.2998 val_loss=2.627427 val_acc=0.18
72 test_loss=2.637674 test_acc=0.1776
```

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```
step=41000 train_loss=2.455577 train_acc=0.3104 val_loss=2.627369 val_acc=0.18
40 test_loss=2.630642 test_acc=0.1851
step=41500 train_loss=2.450973 train_acc=0.3104 val_loss=2.616523 val_acc=0.20
74 test_loss=2.631941 test_acc=0.1753
step=42000 train_loss=2.453531 train_acc=0.3065 val_loss=2.630081 val_acc=0.18
19 test_loss=2.635078 test_acc=0.1718
step=42500 train_loss=2.451137 train_acc=0.3221 val_loss=2.629392 val_acc=0.20
74 test_loss=2.643851 test_acc=0.1760
step=43000 train_loss=2.454339 train_acc=0.3154 val_loss=2.631710 val_acc=0.20
74 test_loss=2.639258 test_acc=0.1836
step=43500 train_loss=2.454836 train_acc=0.3111 val_loss=2.621863 val_acc=0.20
43 test_loss=2.637140 test_acc=0.1840
step=44000 train_loss=2.452535 train_acc=0.3133 val_loss=2.632044 val_acc=0.19
36 test_loss=2.643003 test_acc=0.1789
step=44500 train_loss=2.453778 train_acc=0.2931 val_loss=2.620505 val_acc=0.19
04 test_loss=2.632140 test_acc=0.1723
step=45000 train_loss=2.452633 train_acc=0.3076 val_loss=2.632609 val_acc=0.19
36 test_loss=2.639978 test_acc=0.1787
step=45500 train_loss=2.455835 train_acc=0.3083 val_loss=2.631027 val_acc=0.18
72 test_loss=2.637857 test_acc=0.1822
step=46000 train_loss=2.451201 train_acc=0.3161 val_loss=2.625405 val_acc=0.18
83 test_loss=2.630176 test_acc=0.1796
step=46500 train_loss=2.446665 train_acc=0.2931 val_loss=2.624158 val_acc=0.20
11 test_loss=2.635648 test_acc=0.1696
step=47000 train_loss=2.450834 train_acc=0.3324 val_loss=2.626996 val_acc=0.20
64 test_loss=2.632771 test_acc=0.1941
step=47500 train_loss=2.455443 train_acc=0.3147 val_loss=2.627165 val_acc=0.19
04 test_loss=2.636173 test_acc=0.1794
step=48000 train_loss=2.451162 train_acc=0.2984 val_loss=2.623418 val_acc=0.19
36 test_loss=2.637047 test_acc=0.1682
step=48500 train_loss=2.449540 train_acc=0.3009 val_loss=2.630283 val_acc=0.17
87 test_loss=2.637148 test_acc=0.1780
step=49000 train_loss=2.451191 train_acc=0.3143 val_loss=2.630101 val_acc=0.19
57 test_loss=2.634163 test_acc=0.1875
step=49500 train_loss=2.448828 train_acc=0.3239 val_loss=2.635389 val_acc=0.20
53 test_loss=2.637671 test_acc=0.1913
step=50000 train_loss=2.450868 train_acc=0.3019 val_loss=2.624927 val_acc=0.17
87 test_loss=2.630534 test_acc=0.1744
step=50500 train_loss=2.451677 train_acc=0.3257 val_loss=2.643435 val_acc=0.18
19 test_loss=2.644587 test_acc=0.1854
step=51000 train_loss=2.452621 train_acc=0.3168 val_loss=2.632412 val_acc=0.18
83 test_loss=2.636596 test_acc=0.1810
step=51500 train_loss=2.445702 train_acc=0.3090 val_loss=2.625549 val_acc=0.21
60 test_loss=2.640640 test_acc=0.1804
step=52000 train_loss=2.451717 train_acc=0.3210 val_loss=2.620934 val_acc=0.20
96 test_loss=2.632412 test_acc=0.1845
step=52500 train_loss=2.451707 train_acc=0.3218 val_loss=2.611303 val_acc=0.20
32 test_loss=2.627940 test_acc=0.1870
step=53000 train_loss=2.454072 train_acc=0.3076 val_loss=2.632777 val_acc=0.19
04 test_loss=2.640057 test_acc=0.1792
step=53500 train_loss=2.450808 train_acc=0.3072 val_loss=2.625510 val_acc=0.20
43 test_loss=2.639481 test_acc=0.1771
step=54000 train_loss=2.449793 train_acc=0.3090 val_loss=2.644852 val_acc=0.18
51 test_loss=2.645485 test_acc=0.1810
step=54500 train_loss=2.450086 train_acc=0.3196 val_loss=2.640246 val_acc=0.18
72 test_loss=2.643710 test_acc=0.1813
step=55000 train_loss=2.446245 train_acc=0.3133 val_loss=2.631269 val_acc=0.18
94 test_loss=2.643433 test_acc=0.1773
step=55500 train_loss=2.445821 train_acc=0.3062 val_loss=2.631702 val_acc=0.20
64 test_loss=2.639042 test_acc=0.1824
```

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```
step=56000 train_loss=2.450698 train_acc=0.3026 val_loss=2.629550 val_acc=0.18
51 test_loss=2.637540 test_acc=0.1746
step=56500 train_loss=2.447002 train_acc=0.3193 val_loss=2.629695 val_acc=0.20
53 test_loss=2.640398 test_acc=0.1845
step=57000 train_loss=2.442901 train_acc=0.3189 val_loss=2.631461 val_acc=0.20
11 test_loss=2.639988 test_acc=0.1854
step=57500 train_loss=2.447582 train_acc=0.3136 val_loss=2.629174 val_acc=0.19
04 test_loss=2.639767 test_acc=0.1732
step=58000 train_loss=2.442682 train_acc=0.3200 val_loss=2.634993 val_acc=0.20
43 test_loss=2.637083 test_acc=0.1882
step=58500 train_loss=2.444235 train_acc=0.3271 val_loss=2.634338 val_acc=0.21
91 test_loss=2.642235 test_acc=0.1868
step=59000 train_loss=2.445143 train_acc=0.3356 val_loss=2.637908 val_acc=0.22
13 test_loss=2.645505 test_acc=0.1893
step=59500 train_loss=2.444988 train_acc=0.3299 val_loss=2.639631 val_acc=0.20
96 test_loss=2.642013 test_acc=0.1831
step=60000 train_loss=2.445186 train_acc=0.3168 val_loss=2.639845 val_acc=0.19
68 test_loss=2.641710 test_acc=0.1902
step=60500 train_loss=2.440429 train_acc=0.3225 val_loss=2.633200 val_acc=0.21
28 test_loss=2.642135 test_acc=0.1893
step=61000 train_loss=2.445614 train_acc=0.3030 val_loss=2.646365 val_acc=0.18
51 test_loss=2.647616 test_acc=0.1806
step=61500 train_loss=2.443987 train_acc=0.3235 val_loss=2.639047 val_acc=0.21
17 test_loss=2.642827 test_acc=0.1806
step=62000 train_loss=2.447941 train_acc=0.3140 val_loss=2.643871 val_acc=0.21
49 test_loss=2.653599 test_acc=0.1925
step=62500 train_loss=2.446082 train_acc=0.3079 val_loss=2.640865 val_acc=0.19
26 test_loss=2.647169 test_acc=0.1776
step=63000 train_loss=2.445610 train_acc=0.3232 val_loss=2.630304 val_acc=0.19
15 test_loss=2.642123 test_acc=0.1828
step=63500 train_loss=2.444546 train_acc=0.3108 val_loss=2.634134 val_acc=0.19
57 test_loss=2.642268 test_acc=0.1826
step=64000 train_loss=2.446627 train_acc=0.3044 val_loss=2.641376 val_acc=0.20
32 test_loss=2.651022 test_acc=0.1787
step=64500 train_loss=2.444862 train_acc=0.3069 val_loss=2.645483 val_acc=0.18
19 test_loss=2.650800 test_acc=0.1633
step=65000 train_loss=2.445847 train_acc=0.3108 val_loss=2.642748 val_acc=0.19
04 test_loss=2.642658 test_acc=0.1787
step=65500 train_loss=2.442726 train_acc=0.3228 val_loss=2.638287 val_acc=0.20
96 test_loss=2.644207 test_acc=0.1813
step=66000 train_loss=2.444062 train_acc=0.3182 val_loss=2.649870 val_acc=0.19
36 test_loss=2.645844 test_acc=0.1889
step=66500 train_loss=2.443176 train_acc=0.3434 val_loss=2.638757 val_acc=0.21
60 test_loss=2.650330 test_acc=0.1936
step=67000 train_loss=2.442709 train_acc=0.3505 val_loss=2.633098 val_acc=0.21
91 test_loss=2.642100 test_acc=0.2008
step=67500 train_loss=2.441854 train_acc=0.3285 val_loss=2.639417 val_acc=0.20
64 test_loss=2.643720 test_acc=0.1937
step=68000 train_loss=2.443237 train_acc=0.3214 val_loss=2.641328 val_acc=0.19
15 test_loss=2.642090 test_acc=0.1840
step=68500 train_loss=2.443453 train_acc=0.3193 val_loss=2.651280 val_acc=0.19
36 test_loss=2.651815 test_acc=0.1804
step=69000 train_loss=2.440768 train_acc=0.3221 val_loss=2.636079 val_acc=0.19
68 test_loss=2.643783 test_acc=0.1836
step=69500 train_loss=2.445281 train_acc=0.3253 val_loss=2.644857 val_acc=0.20
11 test_loss=2.652190 test_acc=0.1787
step=70000 train_loss=2.441217 train_acc=0.3246 val_loss=2.641895 val_acc=0.20
21 test_loss=2.648868 test_acc=0.1913
_loss=2.440429 train_acc=0.3274 val_loss=2.642186 val_acc=0.19
89 test_loss=2.647242 test_acc=0.1861
```

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```
step=71000 train_loss=2.440765 train_acc=0.3228 val_loss=2.639171 val_acc=0.18
83 test_loss=2.644162 test_acc=0.1820
step=71500 train_loss=2.444563 train_acc=0.3264 val_loss=2.652331 val_acc=0.20
43 test_loss=2.650547 test_acc=0.1927
step=72000 train_loss=2.441779 train_acc=0.3285 val_loss=2.637936 val_acc=0.20
53 test_loss=2.643142 test_acc=0.2006
step=72500 train_loss=2.440662 train_acc=0.3281 val_loss=2.648624 val_acc=0.19
47 test_loss=2.645344 test_acc=0.1916
step=73000 train_loss=2.439772 train_acc=0.3303 val_loss=2.649636 val_acc=0.19
15 test_loss=2.647168 test_acc=0.1951
step=73500 train_loss=2.439497 train_acc=0.3182 val_loss=2.648629 val_acc=0.18
94 test_loss=2.654543 test_acc=0.1796
step=74000 train_loss=2.440445 train_acc=0.3246 val_loss=2.640723 val_acc=0.20
32 test_loss=2.647587 test_acc=0.1845
step=74500 train_loss=2.439017 train_acc=0.3242 val_loss=2.651964 val_acc=0.19
68 test_loss=2.649369 test_acc=0.1920
step=75000 train_loss=2.438942 train_acc=0.3310 val_loss=2.641571 val_acc=0.19
26 test_loss=2.642737 test_acc=0.1902
step=75500 train_loss=2.439983 train_acc=0.3338 val_loss=2.649675 val_acc=0.21
28 test_loss=2.652115 test_acc=0.1872
step=76000 train_loss=2.437937 train_acc=0.3363 val_loss=2.640504 val_acc=0.21
70 test_loss=2.648725 test_acc=0.1937
step=76500 train_loss=2.437831 train_acc=0.3317 val_loss=2.653330 val_acc=0.18
94 test_loss=2.652090 test_acc=0.1865
step=77000 train_loss=2.437863 train_acc=0.3342 val_loss=2.655687 val_acc=0.19
57 test_loss=2.653207 test_acc=0.1944
step=77500 train_loss=2.437152 train_acc=0.3274 val_loss=2.645867 val_acc=0.20
74 test_loss=2.648321 test_acc=0.1879
step=78000 train_loss=2.438060 train_acc=0.3391 val_loss=2.645270 val_acc=0.20
43 test_loss=2.647235 test_acc=0.1944
step=78500 train_loss=2.437721 train_acc=0.3246 val_loss=2.647823 val_acc=0.20
32 test_loss=2.646155 test_acc=0.1928
step=79000 train_loss=2.438668 train_acc=0.3398 val_loss=2.645851 val_acc=0.21
38 test_loss=2.646481 test_acc=0.1957
step=79500 train_loss=2.440332 train_acc=0.3249 val_loss=2.651186 val_acc=0.21
28 test_loss=2.646514 test_acc=0.1920
step=80000 train_loss=2.440011 train_acc=0.3133 val_loss=2.648657 val_acc=0.19
57 test_loss=2.654233 test_acc=0.1843
step=80500 train_loss=2.439327 train_acc=0.3257 val_loss=2.646046 val_acc=0.20
11 test_loss=2.651151 test_acc=0.1884
step=81000 train_loss=2.437453 train_acc=0.3292 val_loss=2.649847 val_acc=0.20
43 test_loss=2.653077 test_acc=0.1861
step=81500 train_loss=2.440214 train_acc=0.3292 val_loss=2.654611 val_acc=0.21
17 test_loss=2.658533 test_acc=0.1974
step=82000 train_loss=2.439096 train_acc=0.3363 val_loss=2.644736 val_acc=0.20
85 test_loss=2.652006 test_acc=0.1874
step=82500 train_loss=2.437830 train_acc=0.3235 val_loss=2.655032 val_acc=0.19
36 test_loss=2.655802 test_acc=0.1865
step=83000 train_loss=2.438245 train_acc=0.3388 val_loss=2.648709 val_acc=0.21
49 test_loss=2.654906 test_acc=0.1879
step=83500 train_loss=2.437385 train_acc=0.3246 val_loss=2.656024 val_acc=0.20
11 test_loss=2.654992 test_acc=0.1879
step=84000 train_loss=2.434904 train_acc=0.3260 val_loss=2.652284 val_acc=0.21
17 test_loss=2.652087 test_acc=0.1866
step=84500 train_loss=2.435362 train_acc=0.3473 val_loss=2.647064 val_acc=0.22
13 test_loss=2.653201 test_acc=0.1998
step=85000 train_loss=2.436484 train_acc=0.3430 val_loss=2.651383 val_acc=0.20
21 test_loss=2.652427 test_acc=0.1898
_loss=2.436621 train_acc=0.3331 val_loss=2.654722 val_acc=0.19
57 test_loss=2.655987 test_acc=0.1835
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```
step=86000 train_loss=2.438738 train_acc=0.3377 val_loss=2.650473 val_acc=0.19
79 test_loss=2.652576 test_acc=0.1897
step=86500 train_loss=2.436932 train_acc=0.3310 val_loss=2.652849 val_acc=0.20
64 test_loss=2.656695 test_acc=0.1847
step=87000 train_loss=2.437811 train_acc=0.3281 val_loss=2.654507 val_acc=0.20
11 test_loss=2.656281 test_acc=0.1881
step=87500 train_loss=2.436725 train_acc=0.3522 val_loss=2.654994 val_acc=0.21
70 test_loss=2.653819 test_acc=0.1937
step=88000 train_loss=2.436341 train_acc=0.3402 val_loss=2.660786 val_acc=0.20
43 test_loss=2.660901 test_acc=0.1909
step=88500 train_loss=2.438162 train_acc=0.3476 val_loss=2.654763 val_acc=0.21
91 test_loss=2.658809 test_acc=0.1936
step=89000 train_loss=2.437653 train_acc=0.3412 val_loss=2.657463 val_acc=0.21
06 test_loss=2.660256 test_acc=0.1898
step=89500 train_loss=2.437665 train_acc=0.3310 val_loss=2.656793 val_acc=0.20
11 test_loss=2.658731 test_acc=0.1904
step=90000 train_loss=2.437989 train_acc=0.3281 val_loss=2.656361 val_acc=0.19
89 test_loss=2.655928 test_acc=0.1858
step=90500 train_loss=2.438987 train_acc=0.3327 val_loss=2.654558 val_acc=0.20
11 test_loss=2.657730 test_acc=0.1874
step=91000 train_loss=2.436972 train_acc=0.3349 val_loss=2.658222 val_acc=0.20
00 test_loss=2.654786 test_acc=0.1916
step=91500 train_loss=2.436815 train_acc=0.3373 val_loss=2.656215 val_acc=0.21
91 test_loss=2.660592 test_acc=0.1902
step=92000 train_loss=2.437445 train_acc=0.3430 val_loss=2.654412 val_acc=0.22
23 test_loss=2.658326 test_acc=0.1907
step=92500 train_loss=2.435550 train_acc=0.3572 val_loss=2.652199 val_acc=0.22
77 test_loss=2.654188 test_acc=0.2006
step=93000 train_loss=2.436099 train_acc=0.3476 val_loss=2.655461 val_acc=0.22
13 test_loss=2.657140 test_acc=0.1921
step=93500 train_loss=2.436831 train_acc=0.3487 val_loss=2.655945 val_acc=0.20
96 test_loss=2.659151 test_acc=0.1960
step=94000 train_loss=2.435247 train_acc=0.3420 val_loss=2.658725 val_acc=0.21
60 test_loss=2.658107 test_acc=0.1960
step=94500 train_loss=2.434842 train_acc=0.3434 val_loss=2.651301 val_acc=0.20
32 test_loss=2.651590 test_acc=0.1967
step=95000 train_loss=2.435214 train_acc=0.3370 val_loss=2.659818 val_acc=0.20
32 test_loss=2.657711 test_acc=0.1913
step=95500 train_loss=2.434704 train_acc=0.3278 val_loss=2.659977 val_acc=0.20
43 test_loss=2.659651 test_acc=0.1856
step=96000 train_loss=2.434816 train_acc=0.3359 val_loss=2.657640 val_acc=0.20
96 test_loss=2.662551 test_acc=0.1927
step=96500 train_loss=2.435428 train_acc=0.3320 val_loss=2.659569 val_acc=0.20
43 test_loss=2.661561 test_acc=0.1934
step=97000 train_loss=2.436282 train_acc=0.3370 val_loss=2.655956 val_acc=0.21
06 test_loss=2.660465 test_acc=0.1974
step=97500 train_loss=2.436058 train_acc=0.3487 val_loss=2.656306 val_acc=0.20
11 test_loss=2.658247 test_acc=0.1932
step=98000 train_loss=2.435820 train_acc=0.3320 val_loss=2.657573 val_acc=0.21
70 test_loss=2.662671 test_acc=0.1819
step=98500 train_loss=2.436506 train_acc=0.3366 val_loss=2.662472 val_acc=0.19
57 test_loss=2.664275 test_acc=0.1905
step=99000 train_loss=2.436334 train_acc=0.3303 val_loss=2.662896 val_acc=0.19
04 test_loss=2.663771 test_acc=0.1813
step=99500 train_loss=2.436445 train_acc=0.3377 val_loss=2.661094 val_acc=0.21
06 test_loss=2.662849 test_acc=0.1902
step=100000 train_loss=2.435805 train_acc=0.3466 val_loss=2.657598 val_acc=0.2
138 test_loss=2.659163 test_acc=0.1957
```

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```
    "step": 14500,
```

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  "lr": 0.0009593343979095333,
  "train_loss": 2.4861421808125193,
  "train_acc": 0.3072289156626506,
  "val_loss": 2.61069966377096,
  "val_acc": 0.20957446808510638,
  "test_loss": 2.627515656897003,
  "test_acc": 0.18080396670798654,
  "elapsed_sec": 115.69451189041138
}
checkpoint dir: /Users/shobhit/Documents/UW/CSE 493/cse493s-spring26-hw1/out/s
ubmission/part1/warmup_restarts/add_p97_l1_s202

$ /opt/homebrew/Caskroom/miniconda/base/envs/cse493/bin/python train.py --mode
modular --op + --p 97 --n_layer 1 --n_head 4 --n_embd 128 --steps 100000 --see
d 303 --batch_size 128 --lr 0.001 --min_lr 0.0001 --lr_schedule cosine --warmu
p_steps 1000 --grad_clip 1.0 --beta1 0.9 --beta2 0.95 --weight_decay 0.01 --op
timizer adam --eval_interval 500 --log_interval 500 --out_dir '/Users/shobhit/
Documents/UW/CSE 493/cse493s-spring26-hw1/out/submission/part1/warmup_restart
s/add_p97_l1_s303' --bias --resume
number of parameters: 0.21M
[resume] model_last.pt not found in /Users/shobhit/Documents/UW/CSE 493/cse493
s-spring26-hw1/out/submission/part1/warmup_restarts/add_p97_l1_s303; starting
fresh
device=mps train=2822 val=940 test=5647 batch_size=128
step=1 train_loss=4.684333 train_acc=0.0000 val_loss=4.696957 val_acc=0.0000 t
est_loss=4.682514 test_acc=0.0000
step=500 train_loss=4.299879 train_acc=0.0819 val_loss=4.852598 val_acc=0.0043
test_loss=4.834099 test_acc=0.0023
step=1000 train_loss=4.242141 train_acc=0.0808 val_loss=4.893535 val_acc=0.000
0 test_loss=4.881743 test_acc=0.0005
step=1500 train_loss=4.380462 train_acc=0.0595 val_loss=4.822823 val_acc=0.002
1 test_loss=4.799471 test_acc=0.0025
step=2000 train_loss=4.298407 train_acc=0.0599 val_loss=4.937778 val_acc=0.004
3 test_loss=4.932034 test_acc=0.0028
step=2500 train_loss=4.275171 train_acc=0.0553 val_loss=4.986227 val_acc=0.001
1 test_loss=4.963571 test_acc=0.0027
step=3000 train_loss=4.254736 train_acc=0.0606 val_loss=5.003011 val_acc=0.001
1 test_loss=4.983030 test_acc=0.0030
step=3500 train_loss=4.236790 train_acc=0.0563 val_loss=5.006038 val_acc=0.001
1 test_loss=4.991944 test_acc=0.0028
step=4000 train_loss=4.226550 train_acc=0.0648 val_loss=5.041981 val_acc=0.000
0 test_loss=5.016415 test_acc=0.0037
step=4500 train_loss=4.212915 train_acc=0.0666 val_loss=5.046050 val_acc=0.004
3 test_loss=5.008354 test_acc=0.0027
step=5000 train_loss=4.197561 train_acc=0.0695 val_loss=5.018239 val_acc=0.005
3 test_loss=5.004246 test_acc=0.0037
step=5500 train_loss=4.170520 train_acc=0.0709 val_loss=5.009009 val_acc=0.006
4 test_loss=4.989244 test_acc=0.0030
step=6000 train_loss=4.126690 train_acc=0.0758 val_loss=4.918030 val_acc=0.004
3 test_loss=4.889693 test_acc=0.0048
step=6500 train_loss=3.990821 train_acc=0.0918 val_loss=4.598209 val_acc=0.008
5 test_loss=4.569633 test_acc=0.0124
step=7000 train_loss=3.924522 train_acc=0.0822 val_loss=4.378582 val_acc=0.019
1 test_loss=4.351162 test_acc=0.0200
step=7500 train_loss=3.708772 train_acc=0.1006 val_loss=4.011522 val_acc=0.035
1 test_loss=4.002261 test_acc=0.0384
step=8000 train_loss=2.796214 train_acc=0.1889 val_loss=2.992939 val_acc=0.106
4 test_loss=2.965782 test_acc=0.1158
step=8500 train_loss=2.238331 train_acc=0.3933 val_loss=2.325480 val_acc=0.308
5 test_loss=2.332610 test_acc=0.2980
```

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```
step=9000 train_loss=1.776350 train_acc=0.5762 val_loss=1.847196 val_acc=0.505
3 test_loss=1.850990 test_acc=0.4879
step=9500 train_loss=1.698653 train_acc=0.6056 val_loss=1.773928 val_acc=0.506
4 test_loss=1.768862 test_acc=0.5196
step=10000 train_loss=1.714460 train_acc=0.6343 val_loss=1.799021 val_acc=0.55
32 test_loss=1.790425 test_acc=0.5428
step=10500 train_loss=1.688356 train_acc=0.6141 val_loss=1.768487 val_acc=0.50
85 test_loss=1.761344 test_acc=0.5235
step=11000 train_loss=1.668236 train_acc=0.6435 val_loss=1.739081 val_acc=0.53
83 test_loss=1.733332 test_acc=0.5630
step=11500 train_loss=1.698805 train_acc=0.6262 val_loss=1.771446 val_acc=0.52
87 test_loss=1.763411 test_acc=0.5344
step=12000 train_loss=1.683066 train_acc=0.6378 val_loss=1.766771 val_acc=0.53
94 test_loss=1.751796 test_acc=0.5686
step=12500 train_loss=1.679775 train_acc=0.6134 val_loss=1.747044 val_acc=0.51
60 test_loss=1.747240 test_acc=0.5300
step=13000 train_loss=1.696404 train_acc=0.6361 val_loss=1.775640 val_acc=0.54
79 test_loss=1.766745 test_acc=0.5534
step=13500 train_loss=1.661931 train_acc=0.6449 val_loss=1.726636 val_acc=0.55
53 test_loss=1.731341 test_acc=0.5603
step=14000 train_loss=1.667164 train_acc=0.6386 val_loss=1.747025 val_acc=0.53
72 test_loss=1.734987 test_acc=0.5537
step=14500 train_loss=1.658238 train_acc=0.6166 val_loss=1.740381 val_acc=0.49
89 test_loss=1.722835 test_acc=0.5371
step=15000 train_loss=1.667321 train_acc=0.6577 val_loss=1.737944 val_acc=0.55
21 test_loss=1.728536 test_acc=0.5780
step=15500 train_loss=1.682820 train_acc=0.6368 val_loss=1.753779 val_acc=0.55
11 test_loss=1.743215 test_acc=0.5619
step=16000 train_loss=1.695128 train_acc=0.6357 val_loss=1.770219 val_acc=0.55
00 test_loss=1.757067 test_acc=0.5601
step=16500 train_loss=1.673923 train_acc=0.6152 val_loss=1.758636 val_acc=0.50
53 test_loss=1.741090 test_acc=0.5270
step=17000 train_loss=1.661951 train_acc=0.6368 val_loss=1.756808 val_acc=0.54
04 test_loss=1.734009 test_acc=0.5569
step=17500 train_loss=1.620833 train_acc=0.6460 val_loss=1.698723 val_acc=0.54
47 test_loss=1.692653 test_acc=0.5543
step=18000 train_loss=1.642431 train_acc=0.6368 val_loss=1.717741 val_acc=0.54
36 test_loss=1.708928 test_acc=0.5651
step=18500 train_loss=1.650140 train_acc=0.6627 val_loss=1.727703 val_acc=0.55
53 test_loss=1.716351 test_acc=0.5690
step=19000 train_loss=1.675930 train_acc=0.6311 val_loss=1.771434 val_acc=0.51
49 test_loss=1.747018 test_acc=0.5396
step=19500 train_loss=1.644862 train_acc=0.6279 val_loss=1.725946 val_acc=0.52
45 test_loss=1.717496 test_acc=0.5390
step=20000 train_loss=1.635483 train_acc=0.6471 val_loss=1.712597 val_acc=0.54
79 test_loss=1.703436 test_acc=0.5722
step=20500 train_loss=1.638406 train_acc=0.6478 val_loss=1.709873 val_acc=0.54
79 test_loss=1.708166 test_acc=0.5514
step=21000 train_loss=1.618152 train_acc=0.6655 val_loss=1.690298 val_acc=0.57
34 test_loss=1.685705 test_acc=0.5727
step=21500 train_loss=1.611121 train_acc=0.6545 val_loss=1.694213 val_acc=0.53
51 test_loss=1.674660 test_acc=0.5690
step=22000 train_loss=1.639237 train_acc=0.6488 val_loss=1.715848 val_acc=0.53
09 test_loss=1.705953 test_acc=0.5555
step=22500 train_loss=1.633162 train_acc=0.6765 val_loss=1.724149 val_acc=0.54
79 test_loss=1.702033 test_acc=0.5692
step=23000 train_loss=1.626782 train_acc=0.6318 val_loss=1.696291 val_acc=0.54
57 test_loss=1.694491 test_acc=0.5454
step=23500 train_loss=1.629364 train_acc=0.6687 val_loss=1.705850 val_acc=0.56
17 test_loss=1.696820 test_acc=0.5899
```

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```
step=24000 train_loss=1.619836 train_acc=0.6329 val_loss=1.688397 val_acc=0.57
23 test_loss=1.687759 test_acc=0.5576
step=24500 train_loss=1.619186 train_acc=0.6439 val_loss=1.695742 val_acc=0.54
15 test_loss=1.687420 test_acc=0.5578
step=25000 train_loss=1.631044 train_acc=0.6265 val_loss=1.715470 val_acc=0.51
91 test_loss=1.702881 test_acc=0.5444
step=25500 train_loss=1.628383 train_acc=0.6683 val_loss=1.717739 val_acc=0.59
26 test_loss=1.698269 test_acc=0.5908
step=26000 train_loss=1.624266 train_acc=0.6605 val_loss=1.696293 val_acc=0.57
55 test_loss=1.688579 test_acc=0.5803
step=26500 train_loss=1.635388 train_acc=0.6527 val_loss=1.712443 val_acc=0.54
47 test_loss=1.703452 test_acc=0.5672
step=27000 train_loss=1.621556 train_acc=0.6254 val_loss=1.706415 val_acc=0.52
98 test_loss=1.695743 test_acc=0.5360
step=27500 train_loss=1.597879 train_acc=0.6743 val_loss=1.678493 val_acc=0.58
72 test_loss=1.668222 test_acc=0.5853
step=28000 train_loss=1.606932 train_acc=0.6683 val_loss=1.682421 val_acc=0.57
23 test_loss=1.675439 test_acc=0.5814
step=28500 train_loss=1.596307 train_acc=0.6704 val_loss=1.673231 val_acc=0.57
45 test_loss=1.665290 test_acc=0.5787
step=29000 train_loss=1.625363 train_acc=0.6425 val_loss=1.699413 val_acc=0.55
43 test_loss=1.693863 test_acc=0.5582
step=29500 train_loss=1.597725 train_acc=0.6782 val_loss=1.673610 val_acc=0.57
23 test_loss=1.670505 test_acc=0.5893
step=30000 train_loss=1.604084 train_acc=0.6616 val_loss=1.680822 val_acc=0.53
72 test_loss=1.678294 test_acc=0.5637
step=30500 train_loss=1.600271 train_acc=0.6772 val_loss=1.666568 val_acc=0.58
72 test_loss=1.665005 test_acc=0.5961
step=31000 train_loss=1.575247 train_acc=0.7062 val_loss=1.643266 val_acc=0.60
85 test_loss=1.643610 test_acc=0.6191
step=31500 train_loss=1.600552 train_acc=0.6896 val_loss=1.666399 val_acc=0.61
06 test_loss=1.668539 test_acc=0.6021
step=32000 train_loss=1.590715 train_acc=0.6449 val_loss=1.663788 val_acc=0.56
81 test_loss=1.657712 test_acc=0.5651
step=32500 train_loss=1.587579 train_acc=0.6637 val_loss=1.656580 val_acc=0.59
15 test_loss=1.650735 test_acc=0.5842
step=33000 train_loss=1.585562 train_acc=0.6513 val_loss=1.666376 val_acc=0.55
96 test_loss=1.653351 test_acc=0.5730
step=33500 train_loss=1.567911 train_acc=0.6719 val_loss=1.647292 val_acc=0.53
51 test_loss=1.640815 test_acc=0.5777
step=34000 train_loss=1.571329 train_acc=0.6772 val_loss=1.640692 val_acc=0.56
91 test_loss=1.638604 test_acc=0.5964
step=34500 train_loss=1.571252 train_acc=0.6474 val_loss=1.651638 val_acc=0.55
43 test_loss=1.642908 test_acc=0.5546
step=35000 train_loss=1.578471 train_acc=0.6435 val_loss=1.661158 val_acc=0.55
21 test_loss=1.650178 test_acc=0.5605
step=35500 train_loss=1.569441 train_acc=0.6754 val_loss=1.645044 val_acc=0.56
70 test_loss=1.639922 test_acc=0.5858
step=36000 train_loss=1.565205 train_acc=0.6864 val_loss=1.638976 val_acc=0.60
96 test_loss=1.633999 test_acc=0.5993
step=36500 train_loss=1.587064 train_acc=0.6761 val_loss=1.660185 val_acc=0.59
36 test_loss=1.656183 test_acc=0.5950
step=37000 train_loss=1.568808 train_acc=0.6853 val_loss=1.633937 val_acc=0.57
87 test_loss=1.633736 test_acc=0.5902
step=37500 train_loss=1.584875 train_acc=0.6935 val_loss=1.655734 val_acc=0.59
89 test_loss=1.652635 test_acc=0.5932
step=38000 train_loss=1.554575 train_acc=0.6821 val_loss=1.624874 val_acc=0.60
11 test_loss=1.621651 test_acc=0.5959
step=38500 train_loss=1.558116 train_acc=0.6765 val_loss=1.624957 val_acc=0.58
51 test_loss=1.624714 test_acc=0.5881
```

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```
step=39000 train_loss=1.569825 train_acc=0.6747 val_loss=1.636286 val_acc=0.55
64 test_loss=1.643384 test_acc=0.5771
step=39500 train_loss=1.562153 train_acc=0.6754 val_loss=1.627271 val_acc=0.57
34 test_loss=1.629672 test_acc=0.5819
step=40000 train_loss=1.562407 train_acc=0.6832 val_loss=1.632146 val_acc=0.59
68 test_loss=1.632126 test_acc=0.5964
step=40500 train_loss=1.562750 train_acc=0.6942 val_loss=1.637499 val_acc=0.60
11 test_loss=1.632713 test_acc=0.5968
step=41000 train_loss=1.564765 train_acc=0.6818 val_loss=1.634984 val_acc=0.58
83 test_loss=1.635141 test_acc=0.5913
step=41500 train_loss=1.552479 train_acc=0.6999 val_loss=1.622396 val_acc=0.60
11 test_loss=1.618156 test_acc=0.6148
step=42000 train_loss=1.577421 train_acc=0.6407 val_loss=1.643643 val_acc=0.54
89 test_loss=1.643357 test_acc=0.5534
step=42500 train_loss=1.562170 train_acc=0.7045 val_loss=1.626364 val_acc=0.60
00 test_loss=1.627448 test_acc=0.6145
step=43000 train_loss=1.557854 train_acc=0.6637 val_loss=1.628277 val_acc=0.56
70 test_loss=1.625706 test_acc=0.5743
step=43500 train_loss=1.559397 train_acc=0.6921 val_loss=1.629895 val_acc=0.57
45 test_loss=1.629471 test_acc=0.5945
step=44000 train_loss=1.551673 train_acc=0.6942 val_loss=1.621643 val_acc=0.59
68 test_loss=1.621539 test_acc=0.6003
step=44500 train_loss=1.539690 train_acc=0.6899 val_loss=1.615252 val_acc=0.59
04 test_loss=1.614516 test_acc=0.5879
step=45000 train_loss=1.542019 train_acc=0.6967 val_loss=1.606057 val_acc=0.62
34 test_loss=1.608036 test_acc=0.6134
step=45500 train_loss=1.544213 train_acc=0.7105 val_loss=1.614703 val_acc=0.61
70 test_loss=1.611905 test_acc=0.6226
step=46000 train_loss=1.539893 train_acc=0.6850 val_loss=1.611638 val_acc=0.59
36 test_loss=1.608618 test_acc=0.6023
step=46500 train_loss=1.539444 train_acc=0.7232 val_loss=1.611889 val_acc=0.60
43 test_loss=1.609449 test_acc=0.6258
step=47000 train_loss=1.546839 train_acc=0.7009 val_loss=1.627482 val_acc=0.59
47 test_loss=1.618039 test_acc=0.5996
step=47500 train_loss=1.547768 train_acc=0.6669 val_loss=1.623428 val_acc=0.55
64 test_loss=1.616428 test_acc=0.5736
step=48000 train_loss=1.532739 train_acc=0.7066 val_loss=1.610940 val_acc=0.62
87 test_loss=1.603312 test_acc=0.6117
step=48500 train_loss=1.533698 train_acc=0.7190 val_loss=1.594799 val_acc=0.61
06 test_loss=1.600957 test_acc=0.6262
step=49000 train_loss=1.525460 train_acc=0.7162 val_loss=1.605664 val_acc=0.61
28 test_loss=1.596790 test_acc=0.6210
step=49500 train_loss=1.540346 train_acc=0.6928 val_loss=1.627719 val_acc=0.59
26 test_loss=1.611240 test_acc=0.6016
step=50000 train_loss=1.530096 train_acc=0.7186 val_loss=1.597651 val_acc=0.61
49 test_loss=1.593163 test_acc=0.6386
step=50500 train_loss=1.544030 train_acc=0.7009 val_loss=1.626314 val_acc=0.61
17 test_loss=1.615371 test_acc=0.6086
step=51000 train_loss=1.526462 train_acc=0.7077 val_loss=1.602175 val_acc=0.60
53 test_loss=1.595166 test_acc=0.6177
step=51500 train_loss=1.536408 train_acc=0.7023 val_loss=1.605437 val_acc=0.59
89 test_loss=1.609328 test_acc=0.6136
step=52000 train_loss=1.530366 train_acc=0.7147 val_loss=1.598841 val_acc=0.63
19 test_loss=1.597724 test_acc=0.6239
step=52500 train_loss=1.526840 train_acc=0.7349 val_loss=1.593779 val_acc=0.62
45 test_loss=1.591184 test_acc=0.6396
step=53000 train_loss=1.516384 train_acc=0.7066 val_loss=1.591600 val_acc=0.61
17 test_loss=1.584037 test_acc=0.6117
_loss=1.513885 train_acc=0.6963 val_loss=1.592806 val_acc=0.58
30 test_loss=1.586779 test_acc=0.5966
```

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```
step=54000 train_loss=1.524793 train_acc=0.7055 val_loss=1.589146 val_acc=0.59
89 test_loss=1.590334 test_acc=0.6150
step=54500 train_loss=1.511706 train_acc=0.7077 val_loss=1.588168 val_acc=0.61
60 test_loss=1.581776 test_acc=0.6113
step=55000 train_loss=1.494252 train_acc=0.7101 val_loss=1.566613 val_acc=0.59
26 test_loss=1.564960 test_acc=0.6189
step=55500 train_loss=1.519034 train_acc=0.6977 val_loss=1.588620 val_acc=0.59
26 test_loss=1.589043 test_acc=0.6046
step=56000 train_loss=1.503982 train_acc=0.7367 val_loss=1.569598 val_acc=0.61
91 test_loss=1.564769 test_acc=0.6472
step=56500 train_loss=1.517268 train_acc=0.7240 val_loss=1.588245 val_acc=0.63
19 test_loss=1.589144 test_acc=0.6361
step=57000 train_loss=1.500201 train_acc=0.7016 val_loss=1.562268 val_acc=0.62
77 test_loss=1.569572 test_acc=0.6194
step=57500 train_loss=1.512173 train_acc=0.7229 val_loss=1.583597 val_acc=0.61
91 test_loss=1.578863 test_acc=0.6285
step=58000 train_loss=1.502311 train_acc=0.7257 val_loss=1.580647 val_acc=0.63
09 test_loss=1.570901 test_acc=0.6414
step=58500 train_loss=1.508129 train_acc=0.7247 val_loss=1.578397 val_acc=0.62
45 test_loss=1.574785 test_acc=0.6347
step=59000 train_loss=1.505924 train_acc=0.7275 val_loss=1.577736 val_acc=0.64
36 test_loss=1.571565 test_acc=0.6419
step=59500 train_loss=1.501893 train_acc=0.7232 val_loss=1.572490 val_acc=0.64
04 test_loss=1.567127 test_acc=0.6310
step=60000 train_loss=1.502984 train_acc=0.7307 val_loss=1.584024 val_acc=0.61
28 test_loss=1.570590 test_acc=0.6347
step=60500 train_loss=1.508788 train_acc=0.7023 val_loss=1.585023 val_acc=0.60
21 test_loss=1.577973 test_acc=0.6109
step=61000 train_loss=1.509653 train_acc=0.7420 val_loss=1.577553 val_acc=0.61
49 test_loss=1.574156 test_acc=0.6434
step=61500 train_loss=1.500841 train_acc=0.7208 val_loss=1.584221 val_acc=0.61
28 test_loss=1.574893 test_acc=0.6173
step=62000 train_loss=1.500939 train_acc=0.7325 val_loss=1.569989 val_acc=0.64
57 test_loss=1.564827 test_acc=0.6534
step=62500 train_loss=1.515115 train_acc=0.6864 val_loss=1.593958 val_acc=0.59
79 test_loss=1.587407 test_acc=0.5975
step=63000 train_loss=1.505349 train_acc=0.6761 val_loss=1.572092 val_acc=0.60
96 test_loss=1.574378 test_acc=0.6039
step=63500 train_loss=1.501843 train_acc=0.7293 val_loss=1.575741 val_acc=0.63
40 test_loss=1.570045 test_acc=0.6389
step=64000 train_loss=1.504502 train_acc=0.7279 val_loss=1.583630 val_acc=0.63
30 test_loss=1.573133 test_acc=0.6380
step=64500 train_loss=1.493878 train_acc=0.7654 val_loss=1.560557 val_acc=0.66
81 test_loss=1.559494 test_acc=0.6761
step=65000 train_loss=1.499916 train_acc=0.7360 val_loss=1.567615 val_acc=0.63
72 test_loss=1.565395 test_acc=0.6517
step=65500 train_loss=1.498272 train_acc=0.7041 val_loss=1.586181 val_acc=0.60
85 test_loss=1.573258 test_acc=0.6170
step=66000 train_loss=1.488280 train_acc=0.7385 val_loss=1.557288 val_acc=0.63
62 test_loss=1.555878 test_acc=0.6340
step=66500 train_loss=1.483419 train_acc=0.7342 val_loss=1.550262 val_acc=0.63
19 test_loss=1.552131 test_acc=0.6419
step=67000 train_loss=1.479874 train_acc=0.7601 val_loss=1.556417 val_acc=0.66
60 test_loss=1.548431 test_acc=0.6754
step=67500 train_loss=1.478426 train_acc=0.7495 val_loss=1.546864 val_acc=0.63
30 test_loss=1.546108 test_acc=0.6595
step=68000 train_loss=1.485129 train_acc=0.7470 val_loss=1.561362 val_acc=0.63
09 test_loss=1.553770 test_acc=0.6476
_loss=1.490089 train_acc=0.7470 val_loss=1.555570 val_acc=0.61
70 test_loss=1.552690 test_acc=0.6538
```

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```
step=69000 train_loss=1.470383 train_acc=0.7473 val_loss=1.536936 val_acc=0.63
62 test_loss=1.538478 test_acc=0.6556
step=69500 train_loss=1.474776 train_acc=0.7449 val_loss=1.545880 val_acc=0.66
49 test_loss=1.543258 test_acc=0.6586
step=70000 train_loss=1.477859 train_acc=0.7378 val_loss=1.550507 val_acc=0.64
57 test_loss=1.550171 test_acc=0.6556
step=70500 train_loss=1.479640 train_acc=0.7410 val_loss=1.548966 val_acc=0.62
45 test_loss=1.545316 test_acc=0.6561
step=71000 train_loss=1.469166 train_acc=0.7307 val_loss=1.540783 val_acc=0.62
98 test_loss=1.538258 test_acc=0.6451
step=71500 train_loss=1.474374 train_acc=0.7289 val_loss=1.546342 val_acc=0.63
72 test_loss=1.544439 test_acc=0.6428
step=72000 train_loss=1.461395 train_acc=0.7505 val_loss=1.533497 val_acc=0.64
15 test_loss=1.529662 test_acc=0.6545
step=72500 train_loss=1.451405 train_acc=0.7775 val_loss=1.515408 val_acc=0.68
62 test_loss=1.519835 test_acc=0.6878
step=73000 train_loss=1.461585 train_acc=0.7636 val_loss=1.536176 val_acc=0.64
57 test_loss=1.529763 test_acc=0.6708
step=73500 train_loss=1.461423 train_acc=0.7541 val_loss=1.532768 val_acc=0.63
83 test_loss=1.529049 test_acc=0.6646
step=74000 train_loss=1.459297 train_acc=0.7725 val_loss=1.531120 val_acc=0.66
17 test_loss=1.529330 test_acc=0.6802
step=74500 train_loss=1.453224 train_acc=0.7580 val_loss=1.518730 val_acc=0.65
74 test_loss=1.519530 test_acc=0.6658
step=75000 train_loss=1.464105 train_acc=0.7356 val_loss=1.536631 val_acc=0.63
51 test_loss=1.531309 test_acc=0.6407
step=75500 train_loss=1.462991 train_acc=0.7619 val_loss=1.538226 val_acc=0.63
40 test_loss=1.531649 test_acc=0.6674
step=76000 train_loss=1.457651 train_acc=0.7835 val_loss=1.525837 val_acc=0.68
62 test_loss=1.523908 test_acc=0.6864
step=76500 train_loss=1.457565 train_acc=0.7473 val_loss=1.529328 val_acc=0.63
09 test_loss=1.525183 test_acc=0.6623
step=77000 train_loss=1.453060 train_acc=0.7707 val_loss=1.530272 val_acc=0.65
85 test_loss=1.523400 test_acc=0.6648
step=77500 train_loss=1.451332 train_acc=0.7704 val_loss=1.517702 val_acc=0.66
28 test_loss=1.519178 test_acc=0.6775
step=78000 train_loss=1.453764 train_acc=0.7406 val_loss=1.524994 val_acc=0.65
11 test_loss=1.521272 test_acc=0.6628
step=78500 train_loss=1.446202 train_acc=0.7594 val_loss=1.516476 val_acc=0.66
81 test_loss=1.514949 test_acc=0.6627
step=79000 train_loss=1.445898 train_acc=0.7562 val_loss=1.519025 val_acc=0.65
53 test_loss=1.514023 test_acc=0.6641
step=79500 train_loss=1.439633 train_acc=0.7679 val_loss=1.513111 val_acc=0.67
13 test_loss=1.510205 test_acc=0.6827
step=80000 train_loss=1.440733 train_acc=0.7821 val_loss=1.508955 val_acc=0.68
30 test_loss=1.508649 test_acc=0.6892
step=80500 train_loss=1.442506 train_acc=0.7704 val_loss=1.516259 val_acc=0.65
32 test_loss=1.510424 test_acc=0.6735
step=81000 train_loss=1.439565 train_acc=0.7746 val_loss=1.511139 val_acc=0.66
49 test_loss=1.509445 test_acc=0.6690
step=81500 train_loss=1.435981 train_acc=0.7782 val_loss=1.504983 val_acc=0.66
38 test_loss=1.500625 test_acc=0.6871
step=82000 train_loss=1.438087 train_acc=0.7576 val_loss=1.518514 val_acc=0.66
06 test_loss=1.511317 test_acc=0.6761
step=82500 train_loss=1.435010 train_acc=0.7672 val_loss=1.501565 val_acc=0.67
87 test_loss=1.501993 test_acc=0.6802
step=83000 train_loss=1.432866 train_acc=0.7622 val_loss=1.505197 val_acc=0.65
32 test_loss=1.502701 test_acc=0.6568
step=83500 train_loss=1.434015 train_acc=0.7721 val_loss=1.500646 val_acc=0.66
91 test_loss=1.501099 test_acc=0.6789
```

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```
step=84000 train_loss=1.429951 train_acc=0.7668 val_loss=1.499673 val_acc=0.67
55 test_loss=1.498116 test_acc=0.6752
step=84500 train_loss=1.429916 train_acc=0.7952 val_loss=1.496366 val_acc=0.68
83 test_loss=1.493644 test_acc=0.6968
step=85000 train_loss=1.428146 train_acc=0.7682 val_loss=1.505700 val_acc=0.66
49 test_loss=1.501141 test_acc=0.6674
step=85500 train_loss=1.422785 train_acc=0.7746 val_loss=1.493502 val_acc=0.66
81 test_loss=1.492432 test_acc=0.6789
step=86000 train_loss=1.426005 train_acc=0.7835 val_loss=1.491373 val_acc=0.68
83 test_loss=1.491002 test_acc=0.6986
step=86500 train_loss=1.425780 train_acc=0.7884 val_loss=1.490427 val_acc=0.69
04 test_loss=1.492620 test_acc=0.6919
step=87000 train_loss=1.421513 train_acc=0.7920 val_loss=1.483487 val_acc=0.71
91 test_loss=1.487480 test_acc=0.7064
step=87500 train_loss=1.418743 train_acc=0.7824 val_loss=1.487724 val_acc=0.68
30 test_loss=1.485996 test_acc=0.6977
step=88000 train_loss=1.428567 train_acc=0.7768 val_loss=1.501456 val_acc=0.67
34 test_loss=1.499348 test_acc=0.6850
step=88500 train_loss=1.418795 train_acc=0.8001 val_loss=1.486599 val_acc=0.71
38 test_loss=1.486198 test_acc=0.7078
step=89000 train_loss=1.420133 train_acc=0.7721 val_loss=1.488058 val_acc=0.68
72 test_loss=1.486566 test_acc=0.6924
step=89500 train_loss=1.419011 train_acc=0.7736 val_loss=1.489480 val_acc=0.68
09 test_loss=1.486913 test_acc=0.6832
step=90000 train_loss=1.417564 train_acc=0.7845 val_loss=1.486226 val_acc=0.67
34 test_loss=1.485097 test_acc=0.6910
step=90500 train_loss=1.418163 train_acc=0.7874 val_loss=1.483093 val_acc=0.68
40 test_loss=1.483858 test_acc=0.6915
step=91000 train_loss=1.414436 train_acc=0.7803 val_loss=1.480062 val_acc=0.68
51 test_loss=1.478615 test_acc=0.6889
step=91500 train_loss=1.411612 train_acc=0.7845 val_loss=1.482010 val_acc=0.67
87 test_loss=1.477730 test_acc=0.6945
step=92000 train_loss=1.411184 train_acc=0.7959 val_loss=1.483037 val_acc=0.68
72 test_loss=1.479191 test_acc=0.6958
step=92500 train_loss=1.410930 train_acc=0.7807 val_loss=1.482711 val_acc=0.67
98 test_loss=1.480751 test_acc=0.6860
step=93000 train_loss=1.406161 train_acc=0.7704 val_loss=1.478019 val_acc=0.66
91 test_loss=1.474841 test_acc=0.6825
step=93500 train_loss=1.408438 train_acc=0.7693 val_loss=1.483396 val_acc=0.67
34 test_loss=1.479790 test_acc=0.6782
step=94000 train_loss=1.407195 train_acc=0.7821 val_loss=1.478378 val_acc=0.68
19 test_loss=1.475616 test_acc=0.6967
step=94500 train_loss=1.407206 train_acc=0.7775 val_loss=1.475840 val_acc=0.68
09 test_loss=1.473820 test_acc=0.6869
step=95000 train_loss=1.405253 train_acc=0.7909 val_loss=1.476693 val_acc=0.69
36 test_loss=1.475090 test_acc=0.7016
step=95500 train_loss=1.406043 train_acc=0.7966 val_loss=1.475083 val_acc=0.68
83 test_loss=1.472748 test_acc=0.7069
step=96000 train_loss=1.406308 train_acc=0.7984 val_loss=1.473206 val_acc=0.68
83 test_loss=1.473218 test_acc=0.7087
step=96500 train_loss=1.403428 train_acc=0.7888 val_loss=1.474675 val_acc=0.68
51 test_loss=1.471737 test_acc=0.6897
step=97000 train_loss=1.404167 train_acc=0.7913 val_loss=1.472714 val_acc=0.68
51 test_loss=1.472858 test_acc=0.6958
step=97500 train_loss=1.401501 train_acc=0.7902 val_loss=1.466020 val_acc=0.68
62 test_loss=1.466610 test_acc=0.7057
step=98000 train_loss=1.404171 train_acc=0.7877 val_loss=1.476384 val_acc=0.69
36 test_loss=1.472037 test_acc=0.6968
step=98500 train_loss=1.405418 train_acc=0.7895 val_loss=1.473810 val_acc=0.70
32 test_loss=1.473196 test_acc=0.7052
```

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```
step=99000 train_loss=1.406189 train_acc=0.7884 val_loss=1.473666 val_acc=0.68
19 test_loss=1.472868 test_acc=0.6913
step=99500 train_loss=1.402620 train_acc=0.7810 val_loss=1.476213 val_acc=0.68
72 test_loss=1.471858 test_acc=0.6866
step=100000 train_loss=1.404571 train_acc=0.7920 val_loss=1.471884 val_acc=0.7
000 test_loss=1.471230 test_acc=0.7053
best metrics: {
  "step": 97500,
  "lr": 0.00010141535082162599,
  "train_loss": 1.4015012105080695,
  "train_acc": 0.7902197023387668,
  "val_loss": 1.466020172707578,
  "val_acc": 0.6861702127659575,
  "test_loss": 1.4666095178951009,
  "test_acc": 0.7056844342128564,
  "elapsed_sec": 849.6231439113617
}
checkpoint dir: /Users/shobhit/Documents/UW/CSE 493/cse493s-spring26-hw1/out/s
ubmission/part1/warmup_restarts/add_p97_l1_s303
warmup runs configured: 8 executed now: 8
restart runs configured: 3 executed now: 3
```

In [8]:

```python
summary_rows = []
for r in warmup_runs + restart_runs:
    m = load_metrics(r["run_dir"])
    summary_rows.append({
        "name": r["name"],
        "op": r["op"],
        "p": r["p"],
        "n_layer": r["n_layer"],
        "seed": r["seed"],
        "best_step": m.get("step"),
        "train_loss": m.get("train_loss"),
        "train_acc": m.get("train_acc"),
        "val_loss": m.get("val_loss"),
        "val_acc": m.get("val_acc"),
        "test_loss": m.get("test_loss"),
        "test_acc": m.get("test_acc"),
    })

warmup_summary_df = pd.DataFrame(summary_rows)
warmup_summary_df.to_csv(TABLES_DIR / "1_2_warmup_summary.csv", index=False)
warmup_summary_df
```

Loading [MathJax]/extensions/Safe.js

Out[8]:

	name	op	p	n_layer	seed	best_step	train_loss	train_acc	val_loss	val_a
0	add_p97_l1_s42	+	97	1	42	99000	2.399661	0.323529	2.439905	0.2521
1	add_p97_l2_s42	+	97	2	42	99000	2.398784	0.339476	2.439876	0.2606
2	add_p113_l1_s42	+	113	1	42	95500	2.595969	0.269974	2.631002	0.2194
3	add_p113_l2_s42	+	113	2	42	95500	2.599874	0.287206	2.636299	0.2272
4	sub_p97_l1_s42	-	97	1	42	97500	2.996353	0.206237	3.033172	0.1606
5	sub_p97_l2_s42	-	97	2	42	95500	2.881246	0.217931	2.916527	0.1659
6	sub_p113_l1_s42	-	113	1	42	97500	2.688002	0.298433	2.727227	0.2445
7	sub_p113_l2_s42	-	113	2	42	8500	3.089000	0.158486	3.118533	0.1489
8	add_p97_l1_s101	+	97	1	101	99000	2.391924	0.318214	2.433253	0.2595
9	add_p97_l1_s202	+	97	1	202	14500	2.486142	0.307229	2.610700	0.2095
10	add_p97_l1_s303	+	97	1	303	97500	1.401501	0.790220	1.466020	0.6861

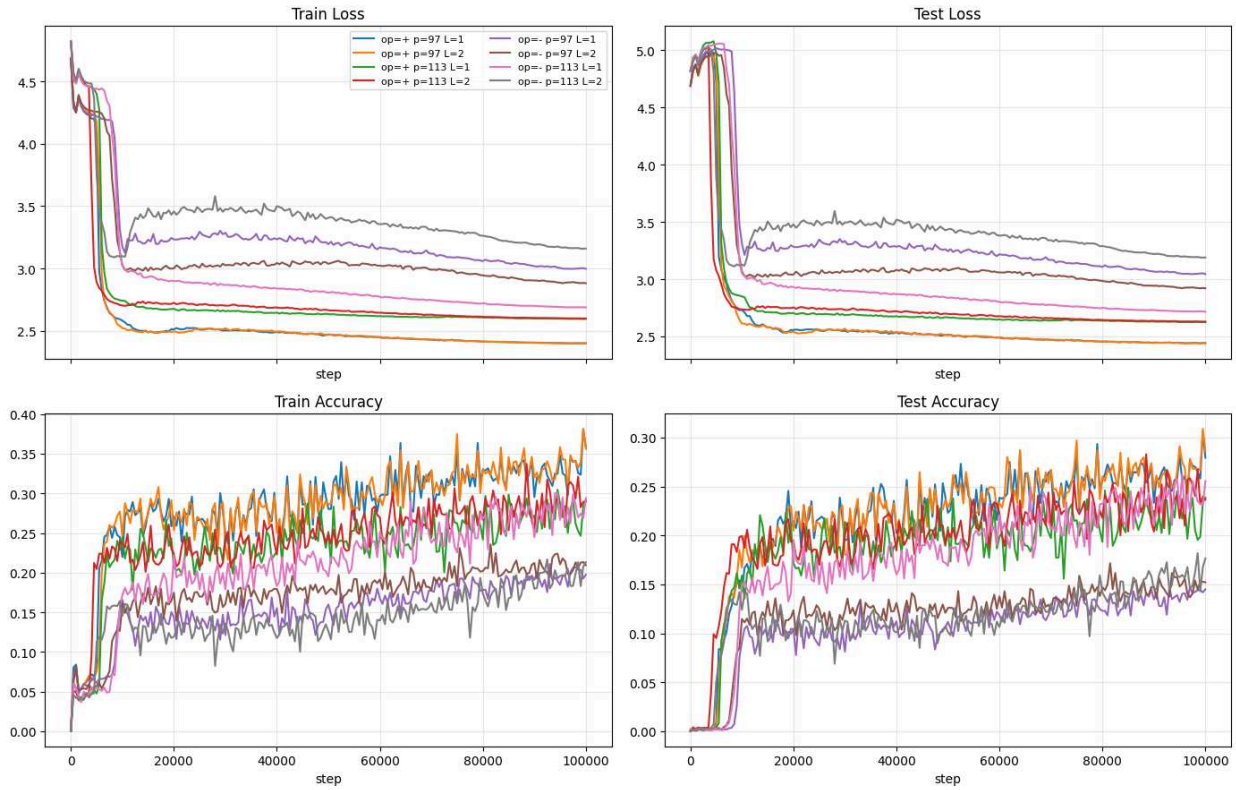
In [ ]:

```
fig, axes = plt.subplots(2, 2, figsize=(14, 9), sharex=True)
for r in warmup_runs:
    h = load_history(r["run_dir"])
    label = f"op={r['op']} p={r['p']} L={r['n_layer']}"
    axes[0, 0].plot(h["step"], h["train_loss"], label=label)
    axes[0, 1].plot(h["step"], h["test_loss"], label=label)
    axes[1, 0].plot(h["step"], h["train_acc"], label=label)
    axes[1, 1].plot(h["step"], h["test_acc"], label=label)

axes[0, 0].set_title("Train Loss")
axes[0, 1].set_title("Test Loss")
axes[1, 0].set_title("Train Accuracy")
axes[1, 1].set_title("Test Accuracy")
for ax in axes.ravel():
    ax.set_xlabel("step")
    ax.grid(alpha=0.3)

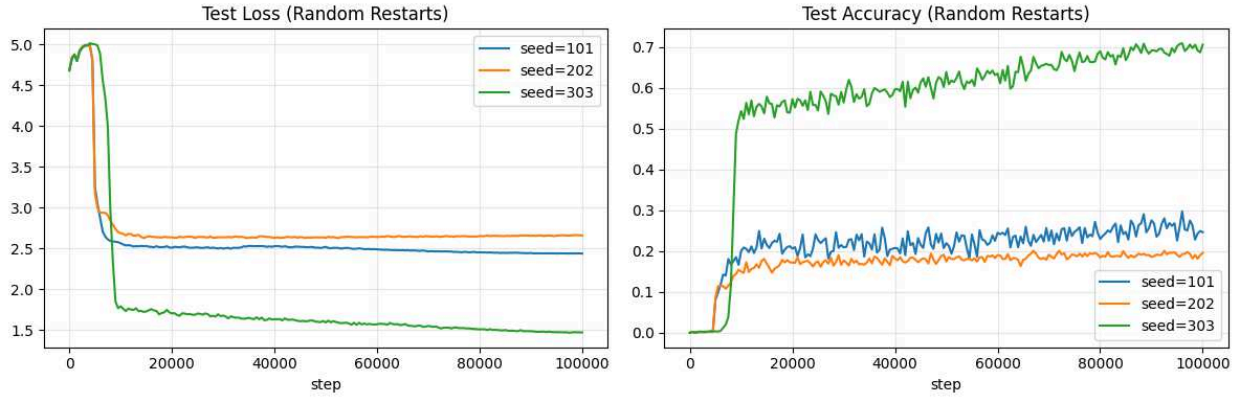
axes[0, 0].legend(fontsize=8, ncol=2)
fig.tight_layout()
fig.savefig(PLOTS_DIR / "1_2_warmup_curves.png", dpi=180)
plt.show()
```

Loading [MathJax]/extensions/Safe.js



In [ ]:
```
fig, axes = plt.subplots(1, 2, figsize=(12, 4), sharex=True)
for r in restart_runs:
    h = load_history(r["run_dir"])
    label = f"seed={r['seed']}"
    axes[0].plot(h["step"], h["test_loss"], label=label)
    axes[1].plot(h["step"], h["test_acc"], label=label)

axes[0].set_title("Test Loss (Random Restarts)")
axes[1].set_title("Test Accuracy (Random Restarts)")
for ax in axes:
    ax.set_xlabel("step")
    ax.grid(alpha=0.3)
    ax.legend()
fig.tight_layout()
fig.savefig(PLOTS_DIR / "1_2_random_restarts.png", dpi=180)
plt.show()
```



1.2 Write-Up

Loading [MathJax]/extensions/Safe.js

Setup used for warmup experiments:

- Model configs: 1-layer and 2-layer transformers, `n_embd=128`, `n_head=4`, FFN dimension `4*128=512`
- Tasks: modular addition and subtraction
- Moduli: `p=97` and `p=113`
- Optimizer: Adam, up to `100,000` steps with answer-token-only loss masking

Main observations:

- Addition generally outperformed subtraction for `p=97`:
  - `add_p97_l2_s42` test accuracy `0.2667` compared to `sub_p97_l2_s42` `0.1507`
- Increasing depth from 1 to 2 layers improved addition in both moduli:
  - `p=97`: `0.2580 -> 0.2667`
  - `p=113`: `0.2171 -> 0.2372`
- Subtraction was less stable across different settings:
  - `p=97`: `sub_p97_l1_s42=0.1410`, `sub_p97_l2_s42=0.1507`
  - `p=113`: `sub_p113_l1_s42=0.2558`, `sub_p113_l2_s42=0.1300`

Random restart `(op=+, p=97, 1-layer)` showed high variance:

- Seed 101: test accuracy `0.2428`
- Seed 202: test accuracy `0.1808`
- Seed 303: test accuracy `0.7057`

The seed choice can substantially affect generalization so in this case the variance is important to understand and get the same model architecture for training.

1.3 Grokking

In [16]:

```
import json
import shlex
import sys

SKIP_EXISTING_GROKKING_RUNS = True

def _latest_logged_step(run_dir: Path):
    history_path = run_dir / "history.json"
    if not history_path.exists():
        return None
    try:
        hist = json.loads(history_path.read_text())
    except Exception:
        return None

    if not hist:
        return None

    return hist[-1].get("step")

def _is_completed_run(run_dir: Path, target_steps: int):
```

Loading [MathJax]/extensions/Safe.js

```
    latest = _latest_logged_step(run_dir)
    ckpt_exists = (run_dir / "model_last.pt").exists() or (run_dir / "model.pt"

    if isinstance(latest, int) and latest >= target_steps and ckpt_exists:
        return True

    return False

baseline_div_dir = PART1_DIR / "grokking" / "baseline_div_p97"
if SKIP_EXISTING_GROKKING_RUNS and _is_completed_run(baseline_div_dir, GROKKIN
    print(f"[skip] baseline already complete: {baseline_div_dir}")
else:
    cmd = (
        f"{shlex.quote(sys.executable)} train.py "
        "--mode modular --op / --p 97 "
        "--n_layer 2 --n_head 4 --n_embd 128 "
        f"--steps {GROKKING_STEPS} --seed 42 "
        f"--batch_size 128 --lr {GROKKING_LR} --min_lr {GROKKING_MIN_LR} "
        f"--lr_schedule {GROKKING_LR_SCHEDULE} --warmup_steps {GROKKING_WARMUP_
        "--beta1 0.9 --beta2 0.95 "
        f"--weight_decay {GROKKING_WEIGHT_DECAY} --optimizer {BASE_OPTIMIZER} '
        f"--eval_interval {GROKKING_EVAL_INTERVAL} --log_interval {GROKKING_LO
        f"--out_dir {shlex.quote(str(baseline_div_dir))} --bias --resume"
    )
    run(cmd)
```

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```
$ /opt/homebrew/Caskroom/miniconda/base/envs/cse493/bin/python train.py --mode
modular --op / --p 97 --n_layer 2 --n_head 4 --n_embd 128 --steps 500000 --see
d 42 --batch_size 128 --lr 0.003 --min_lr 0.0001 --lr_schedule cosine --warmup
_steps 1000 --grad_clip 1.0 --beta1 0.9 --beta2 0.95 --weight_decay 0.01 --opt
imizer adam --eval_interval 2000 --log_interval 2000 --out_dir '/Users/shobhi
t/Documents/UW/CSE 493/cse493s-spring26-hw1/out/submission/part1/grokking/base
line_div_p97' --bias --resume
number of parameters: 0.41M
[resume] loaded model from /Users/shobhit/Documents/UW/CSE 493/cse493s-spring2
6-hw1/out/submission/part1/grokking/baseline_div_p97/model_last.pt
[resume] loaded optimizer from /Users/shobhit/Documents/UW/CSE 493/cse493s-spr
ing26-hw1/out/submission/part1/grokking/baseline_div_p97/optimizer_last.pt
[resume] history found with last step=500
device=mps train=2793 val=931 test=5588 batch_size=128
step=2000 train_loss=2.783151 train_acc=0.1722 val_loss=2.855428 val_acc=0.122
4 test_loss=2.862373 test_acc=0.1192
step=4000 train_loss=2.731882 train_acc=0.1554 val_loss=2.766952 val_acc=0.117
1 test_loss=2.776234 test_acc=0.1251
step=6000 train_loss=2.729654 train_acc=0.1729 val_loss=2.780178 val_acc=0.134
3 test_loss=2.787556 test_acc=0.1235
step=8000 train_loss=2.725424 train_acc=0.1783 val_loss=2.770299 val_acc=0.140
7 test_loss=2.780966 test_acc=0.1417
step=10000 train_loss=2.740086 train_acc=0.1679 val_loss=2.786099 val_acc=0.11
39 test_loss=2.789514 test_acc=0.1283
step=12000 train_loss=2.720596 train_acc=0.1701 val_loss=2.749016 val_acc=0.14
50 test_loss=2.759340 test_acc=0.1439
step=14000 train_loss=2.736606 train_acc=0.1690 val_loss=2.777647 val_acc=0.13
64 test_loss=2.783446 test_acc=0.1346
step=16000 train_loss=2.714332 train_acc=0.1758 val_loss=2.755982 val_acc=0.14
82 test_loss=2.761321 test_acc=0.1308
step=18000 train_loss=2.721647 train_acc=0.1633 val_loss=2.757821 val_acc=0.14
18 test_loss=2.768817 test_acc=0.1335
step=20000 train_loss=2.732059 train_acc=0.1636 val_loss=2.772660 val_acc=0.11
82 test_loss=2.780000 test_acc=0.1340
step=22000 train_loss=2.728304 train_acc=0.1611 val_loss=2.769687 val_acc=0.13
43 test_loss=2.776571 test_acc=0.1335
step=24000 train_loss=2.738392 train_acc=0.1676 val_loss=2.768558 val_acc=0.13
21 test_loss=2.787713 test_acc=0.1303
step=26000 train_loss=2.723892 train_acc=0.1740 val_loss=2.762372 val_acc=0.13
86 test_loss=2.774527 test_acc=0.1296
step=28000 train_loss=2.719937 train_acc=0.1779 val_loss=2.750111 val_acc=0.15
25 test_loss=2.766990 test_acc=0.1303
step=30000 train_loss=2.733674 train_acc=0.1701 val_loss=2.762856 val_acc=0.12
67 test_loss=2.791114 test_acc=0.1215
step=32000 train_loss=2.724825 train_acc=0.1658 val_loss=2.771734 val_acc=0.14
18 test_loss=2.778922 test_acc=0.1326
step=34000 train_loss=2.725858 train_acc=0.1744 val_loss=2.771424 val_acc=0.12
57 test_loss=2.778058 test_acc=0.1280
step=36000 train_loss=2.752070 train_acc=0.1640 val_loss=2.787783 val_acc=0.12
89 test_loss=2.801822 test_acc=0.1321
step=38000 train_loss=2.714923 train_acc=0.1651 val_loss=2.756161 val_acc=0.13
21 test_loss=2.758618 test_acc=0.1460
step=40000 train_loss=2.826290 train_acc=0.1429 val_loss=2.870867 val_acc=0.11
92 test_loss=2.872954 test_acc=0.1151
step=42000 train_loss=2.854529 train_acc=0.1396 val_loss=2.910906 val_acc=0.10
42 test_loss=2.908852 test_acc=0.1161
step=44000 train_loss=2.851142 train_acc=0.1461 val_loss=2.882385 val_acc=0.12
57 test_loss=2.899680 test_acc=0.1126
_loss=2.835328 train_acc=0.1375 val_loss=2.875668 val_acc=0.10
42 test_loss=2.885811 test_acc=0.1170
```

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```
step=48000 train_loss=2.829046 train_acc=0.1518 val_loss=2.883647 val_acc=0.11
17 test_loss=2.879931 test_acc=0.1133
step=50000 train_loss=2.787319 train_acc=0.1368 val_loss=2.845367 val_acc=0.11
60 test_loss=2.834500 test_acc=0.1095
step=52000 train_loss=2.770460 train_acc=0.1497 val_loss=2.824694 val_acc=0.11
17 test_loss=2.817853 test_acc=0.1208
step=54000 train_loss=2.766334 train_acc=0.1550 val_loss=2.820507 val_acc=0.11
28 test_loss=2.818240 test_acc=0.1297
step=56000 train_loss=2.751170 train_acc=0.1475 val_loss=2.799965 val_acc=0.10
85 test_loss=2.796355 test_acc=0.1188
step=58000 train_loss=2.771231 train_acc=0.1421 val_loss=2.829825 val_acc=0.12
57 test_loss=2.819908 test_acc=0.1124
step=60000 train_loss=2.752636 train_acc=0.1651 val_loss=2.808254 val_acc=0.12
24 test_loss=2.803128 test_acc=0.1249
step=62000 train_loss=2.776454 train_acc=0.1532 val_loss=2.832206 val_acc=0.13
32 test_loss=2.824539 test_acc=0.1110
step=64000 train_loss=2.770729 train_acc=0.1507 val_loss=2.827951 val_acc=0.12
24 test_loss=2.826607 test_acc=0.1127
step=66000 train_loss=2.756270 train_acc=0.1658 val_loss=2.806020 val_acc=0.12
57 test_loss=2.800811 test_acc=0.1308
step=68000 train_loss=2.738600 train_acc=0.1529 val_loss=2.797948 val_acc=0.10
10 test_loss=2.791516 test_acc=0.1254
step=70000 train_loss=2.767022 train_acc=0.1421 val_loss=2.820046 val_acc=0.11
60 test_loss=2.815930 test_acc=0.1253
step=72000 train_loss=2.723409 train_acc=0.1625 val_loss=2.773101 val_acc=0.11
82 test_loss=2.766849 test_acc=0.1358
step=74000 train_loss=2.736088 train_acc=0.1629 val_loss=2.785520 val_acc=0.13
64 test_loss=2.780378 test_acc=0.1285
step=76000 train_loss=2.765351 train_acc=0.1694 val_loss=2.826408 val_acc=0.12
03 test_loss=2.811871 test_acc=0.1303
step=78000 train_loss=2.761282 train_acc=0.1579 val_loss=2.812083 val_acc=0.11
82 test_loss=2.816529 test_acc=0.1262
step=80000 train_loss=2.741833 train_acc=0.1579 val_loss=2.792333 val_acc=0.13
00 test_loss=2.790360 test_acc=0.1231
step=82000 train_loss=2.743060 train_acc=0.1500 val_loss=2.782006 val_acc=0.13
96 test_loss=2.789790 test_acc=0.1190
step=84000 train_loss=2.767677 train_acc=0.1511 val_loss=2.806045 val_acc=0.14
29 test_loss=2.812363 test_acc=0.1170
step=86000 train_loss=2.794901 train_acc=0.1443 val_loss=2.834341 val_acc=0.10
74 test_loss=2.839618 test_acc=0.1206
step=88000 train_loss=2.748370 train_acc=0.1518 val_loss=2.797334 val_acc=0.12
03 test_loss=2.796459 test_acc=0.1210
step=90000 train_loss=2.816516 train_acc=0.1511 val_loss=2.867533 val_acc=0.10
85 test_loss=2.863063 test_acc=0.1088
step=92000 train_loss=2.771350 train_acc=0.1486 val_loss=2.821384 val_acc=0.11
28 test_loss=2.818245 test_acc=0.1186
step=94000 train_loss=2.765143 train_acc=0.1640 val_loss=2.802755 val_acc=0.14
07 test_loss=2.808677 test_acc=0.1278
step=96000 train_loss=2.854490 train_acc=0.1532 val_loss=2.902464 val_acc=0.10
96 test_loss=2.901549 test_acc=0.1140
step=98000 train_loss=2.780124 train_acc=0.1493 val_loss=2.817155 val_acc=0.12
67 test_loss=2.829775 test_acc=0.1220
step=100000 train_loss=2.800012 train_acc=0.1497 val_loss=2.847302 val_acc=0.1
203 test_loss=2.844809 test_acc=0.1281
step=102000 train_loss=2.778935 train_acc=0.1511 val_loss=2.830552 val_acc=0.1
407 test_loss=2.831464 test_acc=0.1181
step=104000 train_loss=2.778785 train_acc=0.1647 val_loss=2.830998 val_acc=0.1
364 test_loss=2.829495 test_acc=0.1272
n_loss=2.788726 train_acc=0.1611 val_loss=2.831497 val_acc=0.1
128 test_loss=2.834792 test_acc=0.1233
```

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```
step=108000 train_loss=2.765580 train_acc=0.1489 val_loss=2.824122 val_acc=0.1
310 test_loss=2.817613 test_acc=0.1235
step=110000 train_loss=2.788899 train_acc=0.1622 val_loss=2.838564 val_acc=0.1
117 test_loss=2.842418 test_acc=0.1280
step=112000 train_loss=2.740292 train_acc=0.1661 val_loss=2.785451 val_acc=0.1
300 test_loss=2.794148 test_acc=0.1229
step=114000 train_loss=2.784830 train_acc=0.1565 val_loss=2.852245 val_acc=0.1
171 test_loss=2.840151 test_acc=0.1281
step=116000 train_loss=2.749659 train_acc=0.1565 val_loss=2.807559 val_acc=0.1
289 test_loss=2.801006 test_acc=0.1199
step=118000 train_loss=2.778817 train_acc=0.1425 val_loss=2.831609 val_acc=0.1
149 test_loss=2.824134 test_acc=0.1213
step=120000 train_loss=2.764658 train_acc=0.1547 val_loss=2.814352 val_acc=0.1
246 test_loss=2.812698 test_acc=0.1222
step=122000 train_loss=2.746641 train_acc=0.1507 val_loss=2.797877 val_acc=0.1
289 test_loss=2.797859 test_acc=0.1247
step=124000 train_loss=2.760344 train_acc=0.1500 val_loss=2.803630 val_acc=0.1
106 test_loss=2.807638 test_acc=0.1233
step=126000 train_loss=2.815847 train_acc=0.1554 val_loss=2.861373 val_acc=0.1
235 test_loss=2.864910 test_acc=0.1176
step=128000 train_loss=2.773960 train_acc=0.1593 val_loss=2.824054 val_acc=0.1
203 test_loss=2.816845 test_acc=0.1256
step=130000 train_loss=2.758622 train_acc=0.1651 val_loss=2.802395 val_acc=0.1
353 test_loss=2.805893 test_acc=0.1240
step=132000 train_loss=2.780468 train_acc=0.1500 val_loss=2.831228 val_acc=0.1
321 test_loss=2.831926 test_acc=0.1154
step=134000 train_loss=2.753175 train_acc=0.1744 val_loss=2.799658 val_acc=0.1
515 test_loss=2.809974 test_acc=0.1331
step=136000 train_loss=2.733432 train_acc=0.1504 val_loss=2.788256 val_acc=0.1
235 test_loss=2.781494 test_acc=0.1145
step=138000 train_loss=2.742897 train_acc=0.1550 val_loss=2.796820 val_acc=0.1
203 test_loss=2.792306 test_acc=0.1305
step=140000 train_loss=2.751234 train_acc=0.1518 val_loss=2.796785 val_acc=0.1
235 test_loss=2.799155 test_acc=0.1246
step=142000 train_loss=2.749408 train_acc=0.1600 val_loss=2.801463 val_acc=0.1
267 test_loss=2.798233 test_acc=0.1224
step=144000 train_loss=2.754491 train_acc=0.1579 val_loss=2.805707 val_acc=0.1
128 test_loss=2.797146 test_acc=0.1262
step=146000 train_loss=2.763729 train_acc=0.1472 val_loss=2.824591 val_acc=0.1
139 test_loss=2.814118 test_acc=0.1181
step=148000 train_loss=2.738720 train_acc=0.1604 val_loss=2.784484 val_acc=0.1
214 test_loss=2.789658 test_acc=0.1194
step=150000 train_loss=2.735737 train_acc=0.1590 val_loss=2.799090 val_acc=0.1
289 test_loss=2.788090 test_acc=0.1263
step=152000 train_loss=2.743923 train_acc=0.1479 val_loss=2.794303 val_acc=0.1
257 test_loss=2.791746 test_acc=0.1197
step=154000 train_loss=2.769629 train_acc=0.1472 val_loss=2.816356 val_acc=0.1
171 test_loss=2.825231 test_acc=0.1145
step=156000 train_loss=2.766883 train_acc=0.1686 val_loss=2.819991 val_acc=0.1
353 test_loss=2.818886 test_acc=0.1303
step=158000 train_loss=2.745313 train_acc=0.1450 val_loss=2.786008 val_acc=0.1
224 test_loss=2.792698 test_acc=0.1151
step=160000 train_loss=2.751434 train_acc=0.1529 val_loss=2.797718 val_acc=0.1
246 test_loss=2.796073 test_acc=0.1233
step=162000 train_loss=2.770863 train_acc=0.1550 val_loss=2.816376 val_acc=0.1
160 test_loss=2.824437 test_acc=0.1163
step=164000 train_loss=2.779582 train_acc=0.1500 val_loss=2.829804 val_acc=0.1
139 test_loss=2.837300 test_acc=0.1163
n_loss=2.770920 train_acc=0.1593 val_loss=2.833348 val_acc=0.1
203 test_loss=2.820545 test_acc=0.1276
```

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```
step=168000 train_loss=2.758834 train_acc=0.1554 val_loss=2.809720 val_acc=0.1
235 test_loss=2.809267 test_acc=0.1201
step=170000 train_loss=2.761346 train_acc=0.1586 val_loss=2.811143 val_acc=0.1
439 test_loss=2.799346 test_acc=0.1312
step=172000 train_loss=2.724052 train_acc=0.1672 val_loss=2.769953 val_acc=0.1
343 test_loss=2.773534 test_acc=0.1355
step=174000 train_loss=2.742816 train_acc=0.1629 val_loss=2.803534 val_acc=0.1
160 test_loss=2.799290 test_acc=0.1260
step=176000 train_loss=2.739789 train_acc=0.1762 val_loss=2.783295 val_acc=0.1
429 test_loss=2.782578 test_acc=0.1419
step=178000 train_loss=2.738063 train_acc=0.1672 val_loss=2.790382 val_acc=0.1
364 test_loss=2.787420 test_acc=0.1351
step=180000 train_loss=2.732465 train_acc=0.1529 val_loss=2.794919 val_acc=0.1
074 test_loss=2.779244 test_acc=0.1229
step=182000 train_loss=2.752495 train_acc=0.1608 val_loss=2.799929 val_acc=0.1
332 test_loss=2.795222 test_acc=0.1310
step=184000 train_loss=2.730013 train_acc=0.1493 val_loss=2.780442 val_acc=0.1
031 test_loss=2.775375 test_acc=0.1217
step=186000 train_loss=2.797218 train_acc=0.1468 val_loss=2.850020 val_acc=0.1
160 test_loss=2.843949 test_acc=0.1160
step=188000 train_loss=2.725130 train_acc=0.1668 val_loss=2.775984 val_acc=0.1
386 test_loss=2.774419 test_acc=0.1283
step=190000 train_loss=2.729098 train_acc=0.1690 val_loss=2.775825 val_acc=0.1
332 test_loss=2.779337 test_acc=0.1317
step=192000 train_loss=2.763758 train_acc=0.1711 val_loss=2.813422 val_acc=0.1
289 test_loss=2.814406 test_acc=0.1355
step=194000 train_loss=2.768723 train_acc=0.1608 val_loss=2.825041 val_acc=0.1
224 test_loss=2.815376 test_acc=0.1260
step=196000 train_loss=2.740894 train_acc=0.1554 val_loss=2.796336 val_acc=0.1
257 test_loss=2.785105 test_acc=0.1322
step=198000 train_loss=2.728341 train_acc=0.1636 val_loss=2.775331 val_acc=0.1
353 test_loss=2.774566 test_acc=0.1274
step=200000 train_loss=2.734716 train_acc=0.1658 val_loss=2.773420 val_acc=0.1
439 test_loss=2.780881 test_acc=0.1331
step=202000 train_loss=2.773775 train_acc=0.1683 val_loss=2.820895 val_acc=0.1
300 test_loss=2.819878 test_acc=0.1305
step=204000 train_loss=2.752761 train_acc=0.1611 val_loss=2.800679 val_acc=0.1
343 test_loss=2.796571 test_acc=0.1305
step=206000 train_loss=2.749314 train_acc=0.1636 val_loss=2.801113 val_acc=0.1
257 test_loss=2.801870 test_acc=0.1260
step=208000 train_loss=2.735696 train_acc=0.1726 val_loss=2.804484 val_acc=0.1
160 test_loss=2.787422 test_acc=0.1340
step=210000 train_loss=2.733987 train_acc=0.1522 val_loss=2.784953 val_acc=0.1
214 test_loss=2.783083 test_acc=0.1165
step=212000 train_loss=2.770800 train_acc=0.1532 val_loss=2.813696 val_acc=0.1
472 test_loss=2.817989 test_acc=0.1265
step=214000 train_loss=2.765354 train_acc=0.1633 val_loss=2.812410 val_acc=0.1
235 test_loss=2.812287 test_acc=0.1256
step=216000 train_loss=2.731807 train_acc=0.1651 val_loss=2.780619 val_acc=0.1
289 test_loss=2.779841 test_acc=0.1314
step=218000 train_loss=2.766277 train_acc=0.1572 val_loss=2.808399 val_acc=0.1
343 test_loss=2.814234 test_acc=0.1242
step=220000 train_loss=2.747209 train_acc=0.1586 val_loss=2.804845 val_acc=0.1
246 test_loss=2.800598 test_acc=0.1208
step=222000 train_loss=2.731361 train_acc=0.1586 val_loss=2.778793 val_acc=0.1
332 test_loss=2.777460 test_acc=0.1287
step=224000 train_loss=2.706209 train_acc=0.1683 val_loss=2.753768 val_acc=0.1
375 test_loss=2.755023 test_acc=0.1380
n_loss=2.740104 train_acc=0.1497 val_loss=2.792444 val_acc=0.1
149 test_loss=2.783333 test_acc=0.1172
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```
step=228000 train_loss=2.727331 train_acc=0.1708 val_loss=2.778801 val_acc=0.1
364 test_loss=2.780036 test_acc=0.1314
step=230000 train_loss=2.720895 train_acc=0.1636 val_loss=2.774144 val_acc=0.1
224 test_loss=2.767470 test_acc=0.1288
step=232000 train_loss=2.771741 train_acc=0.1611 val_loss=2.826170 val_acc=0.1
182 test_loss=2.813399 test_acc=0.1333
step=234000 train_loss=2.713121 train_acc=0.1726 val_loss=2.761246 val_acc=0.1
310 test_loss=2.763517 test_acc=0.1303
step=236000 train_loss=2.721783 train_acc=0.1597 val_loss=2.774446 val_acc=0.1
203 test_loss=2.778138 test_acc=0.1229
step=238000 train_loss=2.729417 train_acc=0.1790 val_loss=2.776665 val_acc=0.1
407 test_loss=2.776162 test_acc=0.1348
step=240000 train_loss=2.719995 train_acc=0.1704 val_loss=2.770930 val_acc=0.1
396 test_loss=2.768215 test_acc=0.1288
step=242000 train_loss=2.733751 train_acc=0.1661 val_loss=2.783534 val_acc=0.1
300 test_loss=2.782538 test_acc=0.1283
step=244000 train_loss=2.719503 train_acc=0.1618 val_loss=2.765786 val_acc=0.1
214 test_loss=2.767073 test_acc=0.1296
step=246000 train_loss=2.698386 train_acc=0.1719 val_loss=2.749893 val_acc=0.1
300 test_loss=2.750036 test_acc=0.1340
step=248000 train_loss=2.707158 train_acc=0.1747 val_loss=2.750481 val_acc=0.1
429 test_loss=2.753213 test_acc=0.1387
step=250000 train_loss=2.738788 train_acc=0.1615 val_loss=2.794585 val_acc=0.1
278 test_loss=2.793697 test_acc=0.1233
step=252000 train_loss=2.712561 train_acc=0.1729 val_loss=2.769359 val_acc=0.1
375 test_loss=2.767659 test_acc=0.1321
step=254000 train_loss=2.715670 train_acc=0.1590 val_loss=2.770791 val_acc=0.1
246 test_loss=2.765741 test_acc=0.1315
step=256000 train_loss=2.719901 train_acc=0.1747 val_loss=2.773332 val_acc=0.1
429 test_loss=2.775698 test_acc=0.1322
step=258000 train_loss=2.727269 train_acc=0.1661 val_loss=2.770649 val_acc=0.1
310 test_loss=2.774903 test_acc=0.1278
step=260000 train_loss=2.707493 train_acc=0.1812 val_loss=2.755877 val_acc=0.1
536 test_loss=2.753891 test_acc=0.1390
step=262000 train_loss=2.704310 train_acc=0.1704 val_loss=2.758960 val_acc=0.1
192 test_loss=2.749553 test_acc=0.1324
step=264000 train_loss=2.706997 train_acc=0.1615 val_loss=2.753257 val_acc=0.1
300 test_loss=2.752897 test_acc=0.1237
step=266000 train_loss=2.718491 train_acc=0.1701 val_loss=2.764851 val_acc=0.1
257 test_loss=2.768447 test_acc=0.1301
step=268000 train_loss=2.698844 train_acc=0.1633 val_loss=2.746871 val_acc=0.1
396 test_loss=2.750618 test_acc=0.1283
step=270000 train_loss=2.706079 train_acc=0.1679 val_loss=2.752386 val_acc=0.1
289 test_loss=2.754636 test_acc=0.1299
step=272000 train_loss=2.700949 train_acc=0.1672 val_loss=2.754547 val_acc=0.1
321 test_loss=2.756325 test_acc=0.1333
step=274000 train_loss=2.696371 train_acc=0.1758 val_loss=2.749393 val_acc=0.1
439 test_loss=2.744396 test_acc=0.1333
step=276000 train_loss=2.698544 train_acc=0.1740 val_loss=2.746145 val_acc=0.1
429 test_loss=2.747500 test_acc=0.1331
step=278000 train_loss=2.701777 train_acc=0.1719 val_loss=2.766824 val_acc=0.1
310 test_loss=2.760613 test_acc=0.1276
step=280000 train_loss=2.688060 train_acc=0.1819 val_loss=2.745129 val_acc=0.1
407 test_loss=2.744337 test_acc=0.1428
step=282000 train_loss=2.684582 train_acc=0.1783 val_loss=2.729130 val_acc=0.1
450 test_loss=2.731111 test_acc=0.1315
step=284000 train_loss=2.730603 train_acc=0.1668 val_loss=2.779338 val_acc=0.1
353 test_loss=2.779482 test_acc=0.1224
n_loss=2.682266 train_acc=0.1765 val_loss=2.726562 val_acc=0.1
321 test_loss=2.727487 test_acc=0.1349
```

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```
step=288000 train_loss=2.687754 train_acc=0.1633 val_loss=2.735815 val_acc=0.1
300 test_loss=2.734987 test_acc=0.1292
step=290000 train_loss=2.688495 train_acc=0.1751 val_loss=2.743223 val_acc=0.1
343 test_loss=2.737414 test_acc=0.1410
step=292000 train_loss=2.693403 train_acc=0.1736 val_loss=2.730812 val_acc=0.1
407 test_loss=2.736410 test_acc=0.1339
step=294000 train_loss=2.672361 train_acc=0.1740 val_loss=2.720452 val_acc=0.1
375 test_loss=2.722070 test_acc=0.1324
step=296000 train_loss=2.674225 train_acc=0.1837 val_loss=2.724989 val_acc=0.1
547 test_loss=2.725575 test_acc=0.1476
step=298000 train_loss=2.672757 train_acc=0.1729 val_loss=2.725737 val_acc=0.1
310 test_loss=2.726276 test_acc=0.1283
step=300000 train_loss=2.667359 train_acc=0.1808 val_loss=2.713209 val_acc=0.1
461 test_loss=2.716743 test_acc=0.1337
step=302000 train_loss=2.668137 train_acc=0.1794 val_loss=2.721179 val_acc=0.1
235 test_loss=2.719016 test_acc=0.1303
step=304000 train_loss=2.682301 train_acc=0.1711 val_loss=2.735134 val_acc=0.1
439 test_loss=2.732720 test_acc=0.1394
step=306000 train_loss=2.688617 train_acc=0.1790 val_loss=2.736040 val_acc=0.1
300 test_loss=2.734270 test_acc=0.1365
step=308000 train_loss=2.687493 train_acc=0.1729 val_loss=2.737717 val_acc=0.1
332 test_loss=2.736076 test_acc=0.1389
step=310000 train_loss=2.681664 train_acc=0.1779 val_loss=2.737039 val_acc=0.1
439 test_loss=2.729725 test_acc=0.1337
step=312000 train_loss=2.666451 train_acc=0.1683 val_loss=2.714371 val_acc=0.1
310 test_loss=2.715496 test_acc=0.1303
step=314000 train_loss=2.669130 train_acc=0.1769 val_loss=2.727310 val_acc=0.1
364 test_loss=2.717606 test_acc=0.1337
step=316000 train_loss=2.683221 train_acc=0.1815 val_loss=2.729462 val_acc=0.1
590 test_loss=2.729881 test_acc=0.1385
step=318000 train_loss=2.673425 train_acc=0.1679 val_loss=2.723856 val_acc=0.1
429 test_loss=2.721952 test_acc=0.1333
step=320000 train_loss=2.664141 train_acc=0.1744 val_loss=2.714430 val_acc=0.1
386 test_loss=2.714588 test_acc=0.1390
step=322000 train_loss=2.674708 train_acc=0.1794 val_loss=2.726012 val_acc=0.1
289 test_loss=2.726148 test_acc=0.1421
step=324000 train_loss=2.666637 train_acc=0.1744 val_loss=2.718274 val_acc=0.1
375 test_loss=2.716431 test_acc=0.1373
step=326000 train_loss=2.679329 train_acc=0.1815 val_loss=2.724503 val_acc=0.1
504 test_loss=2.728569 test_acc=0.1364
step=328000 train_loss=2.672947 train_acc=0.1758 val_loss=2.719540 val_acc=0.1
364 test_loss=2.722914 test_acc=0.1389
step=330000 train_loss=2.668107 train_acc=0.1787 val_loss=2.708963 val_acc=0.1
364 test_loss=2.716354 test_acc=0.1412
step=332000 train_loss=2.670513 train_acc=0.1765 val_loss=2.722101 val_acc=0.1
429 test_loss=2.718984 test_acc=0.1383
step=334000 train_loss=2.693276 train_acc=0.1840 val_loss=2.747742 val_acc=0.1
439 test_loss=2.744260 test_acc=0.1423
step=336000 train_loss=2.697599 train_acc=0.1801 val_loss=2.750704 val_acc=0.1
343 test_loss=2.747650 test_acc=0.1410
step=338000 train_loss=2.683875 train_acc=0.1762 val_loss=2.731309 val_acc=0.1
343 test_loss=2.732599 test_acc=0.1399
step=340000 train_loss=2.684328 train_acc=0.1858 val_loss=2.731561 val_acc=0.1
461 test_loss=2.732910 test_acc=0.1405
step=342000 train_loss=2.690238 train_acc=0.1744 val_loss=2.738488 val_acc=0.1
257 test_loss=2.743381 test_acc=0.1331
step=344000 train_loss=2.692183 train_acc=0.1808 val_loss=2.738876 val_acc=0.1
353 test_loss=2.735528 test_acc=0.1455
n_loss=2.731158 train_acc=0.1758 val_loss=2.774158 val_acc=0.1
568 test_loss=2.776234 test_acc=0.1374
```

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```
step=348000 train_loss=2.725084 train_acc=0.1779 val_loss=2.770700 val_acc=0.1
461 test_loss=2.768614 test_acc=0.1349
step=350000 train_loss=2.723834 train_acc=0.1697 val_loss=2.778365 val_acc=0.1
332 test_loss=2.773703 test_acc=0.1339
step=352000 train_loss=2.709536 train_acc=0.1686 val_loss=2.766132 val_acc=0.1
407 test_loss=2.760637 test_acc=0.1348
step=354000 train_loss=2.726663 train_acc=0.1608 val_loss=2.778497 val_acc=0.1
310 test_loss=2.773362 test_acc=0.1319
step=356000 train_loss=2.739894 train_acc=0.1686 val_loss=2.792151 val_acc=0.1
375 test_loss=2.786696 test_acc=0.1315
step=358000 train_loss=2.754995 train_acc=0.1651 val_loss=2.803528 val_acc=0.1
267 test_loss=2.799858 test_acc=0.1321
step=360000 train_loss=2.747933 train_acc=0.1654 val_loss=2.798397 val_acc=0.1
257 test_loss=2.795277 test_acc=0.1308
step=362000 train_loss=2.759559 train_acc=0.1747 val_loss=2.810524 val_acc=0.1
396 test_loss=2.809089 test_acc=0.1339
step=364000 train_loss=2.774455 train_acc=0.1665 val_loss=2.827443 val_acc=0.1
139 test_loss=2.821537 test_acc=0.1339
step=366000 train_loss=2.757117 train_acc=0.1751 val_loss=2.809014 val_acc=0.1
364 test_loss=2.805996 test_acc=0.1371
step=368000 train_loss=2.749200 train_acc=0.1711 val_loss=2.804308 val_acc=0.1
289 test_loss=2.798505 test_acc=0.1310
step=370000 train_loss=2.761893 train_acc=0.1572 val_loss=2.813440 val_acc=0.1
106 test_loss=2.807845 test_acc=0.1237
step=372000 train_loss=2.760130 train_acc=0.1665 val_loss=2.808129 val_acc=0.1
267 test_loss=2.807913 test_acc=0.1265
step=374000 train_loss=2.758025 train_acc=0.1554 val_loss=2.803939 val_acc=0.1
321 test_loss=2.801593 test_acc=0.1265
step=376000 train_loss=2.745882 train_acc=0.1751 val_loss=2.795684 val_acc=0.1
364 test_loss=2.794170 test_acc=0.1342
step=378000 train_loss=2.774013 train_acc=0.1633 val_loss=2.820833 val_acc=0.1
278 test_loss=2.823169 test_acc=0.1219
step=380000 train_loss=2.754896 train_acc=0.1751 val_loss=2.801238 val_acc=0.1
310 test_loss=2.803348 test_acc=0.1356
step=382000 train_loss=2.759785 train_acc=0.1568 val_loss=2.811832 val_acc=0.1
214 test_loss=2.808750 test_acc=0.1240
step=384000 train_loss=2.751176 train_acc=0.1629 val_loss=2.798041 val_acc=0.1
493 test_loss=2.799628 test_acc=0.1288
step=386000 train_loss=2.759838 train_acc=0.1654 val_loss=2.807806 val_acc=0.1
235 test_loss=2.805269 test_acc=0.1342
step=388000 train_loss=2.751032 train_acc=0.1651 val_loss=2.796124 val_acc=0.1
160 test_loss=2.799175 test_acc=0.1301
step=390000 train_loss=2.739420 train_acc=0.1611 val_loss=2.787159 val_acc=0.1
278 test_loss=2.785308 test_acc=0.1276
step=392000 train_loss=2.729958 train_acc=0.1651 val_loss=2.784572 val_acc=0.1
257 test_loss=2.780103 test_acc=0.1319
step=394000 train_loss=2.731924 train_acc=0.1654 val_loss=2.779482 val_acc=0.1
364 test_loss=2.778240 test_acc=0.1310
step=396000 train_loss=2.745330 train_acc=0.1704 val_loss=2.798196 val_acc=0.1
224 test_loss=2.794882 test_acc=0.1288
step=398000 train_loss=2.739731 train_acc=0.1654 val_loss=2.787779 val_acc=0.1
106 test_loss=2.789560 test_acc=0.1292
step=400000 train_loss=2.755338 train_acc=0.1672 val_loss=2.803635 val_acc=0.1
257 test_loss=2.803969 test_acc=0.1212
step=402000 train_loss=2.740537 train_acc=0.1611 val_loss=2.784420 val_acc=0.1
235 test_loss=2.787362 test_acc=0.1203
step=404000 train_loss=2.727055 train_acc=0.1697 val_loss=2.774053 val_acc=0.1
310 test_loss=2.774915 test_acc=0.1276
n_loss=2.726005 train_acc=0.1633 val_loss=2.779762 val_acc=0.1
278 test_loss=2.773269 test_acc=0.1296
```

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```
step=408000 train_loss=2.722091 train_acc=0.1769 val_loss=2.774058 val_acc=0.1
235 test_loss=2.772956 test_acc=0.1322
step=410000 train_loss=2.741256 train_acc=0.1697 val_loss=2.787656 val_acc=0.1
418 test_loss=2.790918 test_acc=0.1288
step=412000 train_loss=2.699432 train_acc=0.1711 val_loss=2.748336 val_acc=0.1
321 test_loss=2.748673 test_acc=0.1342
step=414000 train_loss=2.713076 train_acc=0.1779 val_loss=2.767391 val_acc=0.1
235 test_loss=2.763157 test_acc=0.1348
step=416000 train_loss=2.708043 train_acc=0.1686 val_loss=2.757835 val_acc=0.1
289 test_loss=2.758116 test_acc=0.1315
step=418000 train_loss=2.697503 train_acc=0.1708 val_loss=2.746871 val_acc=0.1
214 test_loss=2.747226 test_acc=0.1310
step=420000 train_loss=2.705036 train_acc=0.1701 val_loss=2.752959 val_acc=0.1
332 test_loss=2.754080 test_acc=0.1335
step=422000 train_loss=2.718026 train_acc=0.1701 val_loss=2.772156 val_acc=0.1
429 test_loss=2.767791 test_acc=0.1335
step=424000 train_loss=2.698721 train_acc=0.1697 val_loss=2.747722 val_acc=0.1
257 test_loss=2.746195 test_acc=0.1321
step=426000 train_loss=2.702193 train_acc=0.1690 val_loss=2.750799 val_acc=0.1
267 test_loss=2.748320 test_acc=0.1305
step=428000 train_loss=2.679715 train_acc=0.1679 val_loss=2.727605 val_acc=0.1
246 test_loss=2.731132 test_acc=0.1321
step=430000 train_loss=2.692523 train_acc=0.1822 val_loss=2.743450 val_acc=0.1
364 test_loss=2.741405 test_acc=0.1389
step=432000 train_loss=2.689398 train_acc=0.1733 val_loss=2.739758 val_acc=0.1
310 test_loss=2.736550 test_acc=0.1353
step=434000 train_loss=2.677714 train_acc=0.1729 val_loss=2.728900 val_acc=0.1
439 test_loss=2.726917 test_acc=0.1365
step=436000 train_loss=2.674392 train_acc=0.1672 val_loss=2.724354 val_acc=0.1
278 test_loss=2.720690 test_acc=0.1322
step=438000 train_loss=2.688696 train_acc=0.1701 val_loss=2.734438 val_acc=0.1
300 test_loss=2.737709 test_acc=0.1296
step=440000 train_loss=2.675412 train_acc=0.1686 val_loss=2.722206 val_acc=0.1
321 test_loss=2.722943 test_acc=0.1382
step=442000 train_loss=2.665679 train_acc=0.1729 val_loss=2.712644 val_acc=0.1
332 test_loss=2.715226 test_acc=0.1324
step=444000 train_loss=2.669758 train_acc=0.1719 val_loss=2.723435 val_acc=0.1
267 test_loss=2.717068 test_acc=0.1339
step=446000 train_loss=2.660156 train_acc=0.1754 val_loss=2.711825 val_acc=0.1
246 test_loss=2.709569 test_acc=0.1362
step=448000 train_loss=2.668262 train_acc=0.1758 val_loss=2.716130 val_acc=0.1
429 test_loss=2.717271 test_acc=0.1344
step=450000 train_loss=2.651270 train_acc=0.1715 val_loss=2.703465 val_acc=0.1
364 test_loss=2.699407 test_acc=0.1360
step=452000 train_loss=2.659591 train_acc=0.1697 val_loss=2.708984 val_acc=0.1
289 test_loss=2.709162 test_acc=0.1319
step=454000 train_loss=2.652009 train_acc=0.1719 val_loss=2.702416 val_acc=0.1
461 test_loss=2.699929 test_acc=0.1360
step=456000 train_loss=2.653848 train_acc=0.1801 val_loss=2.699906 val_acc=0.1
300 test_loss=2.701355 test_acc=0.1365
step=458000 train_loss=2.650687 train_acc=0.1701 val_loss=2.700174 val_acc=0.1
214 test_loss=2.698898 test_acc=0.1340
step=460000 train_loss=2.645743 train_acc=0.1722 val_loss=2.694797 val_acc=0.1
396 test_loss=2.694195 test_acc=0.1322
step=462000 train_loss=2.640873 train_acc=0.1808 val_loss=2.690638 val_acc=0.1
375 test_loss=2.688495 test_acc=0.1362
step=464000 train_loss=2.642821 train_acc=0.1758 val_loss=2.694882 val_acc=0.1
375 test_loss=2.691936 test_acc=0.1401
n_loss=2.647655 train_acc=0.1790 val_loss=2.697386 val_acc=0.1
310 test_loss=2.695031 test_acc=0.1394
```

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```
step=468000 train_loss=2.634187 train_acc=0.1790 val_loss=2.684235 val_acc=0.1
343 test_loss=2.681395 test_acc=0.1369
step=470000 train_loss=2.642124 train_acc=0.1751 val_loss=2.691380 val_acc=0.1
332 test_loss=2.689145 test_acc=0.1326
step=472000 train_loss=2.628636 train_acc=0.1783 val_loss=2.677835 val_acc=0.1
429 test_loss=2.675680 test_acc=0.1360
step=474000 train_loss=2.627754 train_acc=0.1733 val_loss=2.677889 val_acc=0.1
396 test_loss=2.674691 test_acc=0.1331
step=476000 train_loss=2.626592 train_acc=0.1794 val_loss=2.676725 val_acc=0.1
343 test_loss=2.674636 test_acc=0.1390
step=478000 train_loss=2.616607 train_acc=0.1787 val_loss=2.666414 val_acc=0.1
386 test_loss=2.663293 test_acc=0.1401
step=480000 train_loss=2.613615 train_acc=0.1815 val_loss=2.662484 val_acc=0.1
332 test_loss=2.661407 test_acc=0.1355
step=482000 train_loss=2.617546 train_acc=0.1783 val_loss=2.666152 val_acc=0.1
332 test_loss=2.664762 test_acc=0.1362
step=484000 train_loss=2.613771 train_acc=0.1779 val_loss=2.660816 val_acc=0.1
343 test_loss=2.661111 test_acc=0.1365
step=486000 train_loss=2.618254 train_acc=0.1815 val_loss=2.668170 val_acc=0.1
375 test_loss=2.666720 test_acc=0.1365
step=488000 train_loss=2.615988 train_acc=0.1801 val_loss=2.666983 val_acc=0.1
407 test_loss=2.663926 test_acc=0.1401
step=490000 train_loss=2.611371 train_acc=0.1790 val_loss=2.658642 val_acc=0.1
203 test_loss=2.658437 test_acc=0.1339
step=492000 train_loss=2.610251 train_acc=0.1765 val_loss=2.658296 val_acc=0.1
267 test_loss=2.656401 test_acc=0.1376
step=494000 train_loss=2.609975 train_acc=0.1762 val_loss=2.659153 val_acc=0.1
267 test_loss=2.657351 test_acc=0.1382
step=496000 train_loss=2.608365 train_acc=0.1805 val_loss=2.656339 val_acc=0.1
364 test_loss=2.656784 test_acc=0.1337
step=498000 train_loss=2.610251 train_acc=0.1754 val_loss=2.658742 val_acc=0.1
364 test_loss=2.658618 test_acc=0.1356
step=500000 train_loss=2.613492 train_acc=0.1762 val_loss=2.660872 val_acc=0.1
364 test_loss=2.661496 test_acc=0.1365
best metrics: {
  "step": 496000,
  "lr": 0.00010045976265407269,
  "train_loss": 2.608364959109696,
  "train_acc": 0.18045112781954886,
  "val_loss": 2.6563387618797294,
  "val_acc": 0.1364124597207304,
  "test_loss": 2.656784374882856,
  "test_acc": 0.13367931281317108,
  "elapsed_sec": 12798.242565870285
}
checkpoint dir: /Users/shobhit/Documents/UW/CSE 493/cse493s-spring26-hw1/out/s
ubmission/part1/grokking/baseline_div_p97
```

In [17]:

```
h = load_history(baseline_div_dir)

def first_step_at_or_above(df: pd.DataFrame, col: str, threshold: float):
    s = df.loc[df[col] >= threshold, "step"]
    return int(s.iloc[0]) if not s.empty else None

train_99_step = first_step_at_or_above(h, "train_acc", 0.99)
val_99_step = first_step_at_or_above(h, "val_acc", 0.99)
test_99_step = first_step_at_or_above(h, "test_acc", 0.99)

[illegible] is None or val_99_step is None:
    val_delay_after_train_99 = None
```

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```
else:
    val_delay_after_train_99 = val_99_step - train_99_step

if train_99_step is None or test_99_step is None:
    test_delay_after_train_99 = None
else:
    test_delay_after_train_99 = test_99_step - train_99_step

grokking_summary = pd.DataFrame([
    {
        "run": "baseline_div_p97",
        "train_acc_99_step": train_99_step,
        "val_acc_99_step": val_99_step,
        "test_acc_99_step": test_99_step,
        "val_delay_after_train_99": val_delay_after_train_99,
        "test_delay_after_train_99": test_delay_after_train_99,
    }
])
grokking_summary.to_csv(TABLES_DIR / "1_3_grokking_summary.csv", index=False)
display(grokking_summary)

fig, ax = plt.subplots(figsize=(10, 4.8))
ax.plot(h["step"], h["train_acc"], label="train_acc", linewidth=2.0)
ax.plot(h["step"], h["val_acc"], label="val_acc", linewidth=2.0)
ax.plot(h["step"], h["test_acc"], label="test_acc", linewidth=1.8, alpha=0.9)

if train_99_step is not None:
    ax.axvline(train_99_step, linestyle="--", alpha=0.7, label="train acc>=99%"
if val_99_step is not None:
    ax.axvline(val_99_step, linestyle=":", alpha=0.7, label="val acc>=99%")
if test_99_step is not None:
    ax.axvline(test_99_step, linestyle="-.", alpha=0.7, label="test acc>=99%")

ax.set_xscale("log")
ax.set_xlabel("step (log scale)")
ax.set_ylabel("accuracy")
ax.set_title("Division p=97: Grokking Curves (Train/Val/Test Accuracy)")
ax.grid(alpha=0.3)
ax.legend(fontsize=8)

fig.tight_layout()
fig.savefig(PLOTS_DIR / "1_3_grokking_baseline.png", dpi=180)
plt.show()
```

	run	train_acc_99_step	val_acc_99_step	test_acc_99_step	val_delay_after_trai
0	baseline_div_p97	None	None	None	

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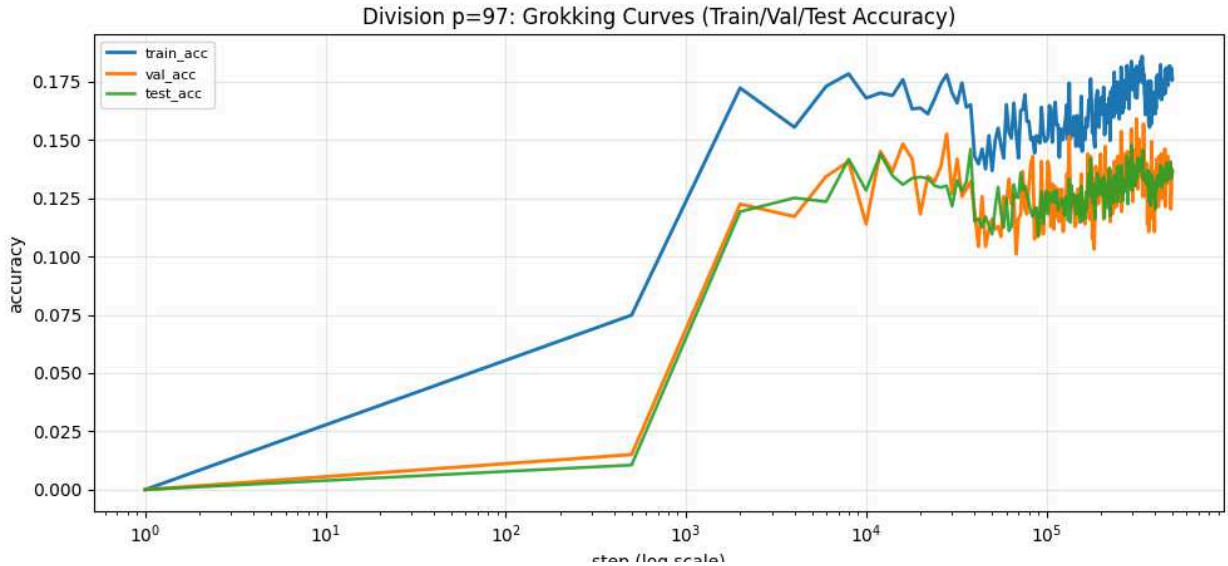

1.3 Write-Up

Results for (Adam, `weight_decay=0.01`, `train_frac=0.3`, `500k` steps):

- No grokking transition was observed within `500,000` steps. Didn't do `1,000,000` steps due to time and compute restrictions.
- `train_acc` never reached `1.0` so train->test grokking delay is undefined.
- Best test accuracy was `0.1476` at step `296,000`
- Best validation accuracy was `0.1590` at step `316,000`
- Final metrics at step `500,000`: `train_acc=0.1762`, `val_acc=0.1364`, `test_acc=0.1365`

These results show us that under this exact baseline optimzer/regularization setting, the model remains in a low accuracy regime and doesn't show any delayed generalization behavior in this budget.

1.4 Ablations Analysis

Ablation A (paper-style optimizer):

- switch to `AdamW`
- set `weight_decay=1.0`
- keep train split at baseline (`train_frac=0.3`, `val_frac=0.1`)

Ablation B (data-size with paper optimizer):

- keep `AdamW` + `weight_decay=1.0`
- increase training fraction (`train_frac=0.5`, `val_frac=0.1`)

In [18]:
```
import shlex
```

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```
ABLATION_OPTIMIZER = "adamw"
ABLATION_WEIGHT_DECAY = 1.0

ablation_runs = [
    {
        "name": "abl_adamw_wd1_trainfrac0p3",
        "weight_decay": ABLATION_WEIGHT_DECAY,
        "optimizer": ABLATION_OPTIMIZER,
        "train_frac": 0.3,
        "val_frac": 0.1,
        "lr": GROKKING_LR,
        "min_lr": GROKKING_MIN_LR,
        "lr_schedule": GROKKING_LR_SCHEDULE,
        "warmup_steps": GROKKING_WARMUP_STEPS,
        "grad_clip": GROKKING_GRAD_CLIP,
    },
    {
        "name": "abl_adamw_wd1_trainfrac0p5",
        "weight_decay": ABLATION_WEIGHT_DECAY,
        "optimizer": ABLATION_OPTIMIZER,
        "train_frac": 0.5,
        "val_frac": 0.1,
        "lr": GROKKING_LR,
        "min_lr": GROKKING_MIN_LR,
        "lr_schedule": GROKKING_LR_SCHEDULE,
        "warmup_steps": GROKKING_WARMUP_STEPS,
        "grad_clip": GROKKING_GRAD_CLIP,
    },
]

executed_ablations = 0
for ab in ablation_runs:
    run_dir = PART1_DIR / "ablations" / ab["name"]
    if SKIP_EXISTING_GROKKING_RUNS and _is_completed_run(run_dir, ABLATION_STE
        print(f"[skip] ablation already complete: {run_dir}")
        continue

    cmd = (
        f"{shlex.quote(sys.executable)} train.py "
        "--mode modular --op / --p 97 "
        "--n_layer 2 --n_head 4 --n_embd 128 "
        f"--steps {ABLATION_STEPS} --seed 42 "
        "--batch_size 128 "
        f"--lr {ab['lr']} --min_lr {ab['min_lr']} "
        f"--lr_schedule {ab['lr_schedule']} --warmup_steps {ab['warmup_steps']}
        "--beta1 0.9 --beta2 0.95 "
        f"--weight_decay {ab['weight_decay']} --optimizer {ab['optimizer']} "
        f"--train_frac {ab['train_frac']} --val_frac {ab['val_frac']} "
        f"--eval_interval {GROKKING_EVAL_INTERVAL} --log_interval {GROKKING_LO
        f"--out_dir {shlex.quote(str(run_dir))} --bias --resume"
    )
    run(cmd)
    executed_ablations += 1

print("ablation runs configured:", len(ablation_runs), "executed now:", execut
```

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```
$ /opt/homebrew/Caskroom/miniconda/base/envs/cse493/bin/python train.py --mode
modular --op / --p 97 --n_layer 2 --n_head 4 --n_embd 128 --steps 500000 --see
d 42 --batch_size 128 --lr 0.003 --min_lr 0.0001 --lr_schedule cosine --warmup
_steps 1000 --grad_clip 1.0 --beta1 0.9 --beta2 0.95 --weight_decay 1.0 --opti
mizer adamw --train_frac 0.3 --val_frac 0.1 --eval_interval 2000 --log_interva
l 2000 --out_dir '/Users/shobhit/Documents/UW/CSE 493/cse493s-spring26-hw1/ou
t/submission/part1/ablations/abl_adamw_wd1_trainfrac0p3' --bias --resume
number of parameters: 0.41M
[resume] model_last.pt not found in /Users/shobhit/Documents/UW/CSE 493/cse493
s-spring26-hw1/out/submission/part1/ablations/abl_adamw_wd1_trainfrac0p3; star
ting fresh
device=mps train=2793 val=931 test=5588 batch_size=128
step=1 train_loss=4.694038 train_acc=0.0000 val_loss=4.679460 val_acc=0.0000 t
est_loss=4.683803 test_acc=0.0000
step=2000 train_loss=0.444923 train_acc=0.8797 val_loss=0.486024 val_acc=0.861
4 test_loss=0.481083 test_acc=0.8618
step=4000 train_loss=0.286278 train_acc=0.9363 val_loss=0.300884 val_acc=0.928
0 test_loss=0.297183 test_acc=0.9304
step=6000 train_loss=0.161376 train_acc=0.9760 val_loss=0.177490 val_acc=0.967
8 test_loss=0.169166 test_acc=0.9723
step=8000 train_loss=0.158058 train_acc=0.9832 val_loss=0.167827 val_acc=0.980
7 test_loss=0.166340 test_acc=0.9810
step=10000 train_loss=0.182248 train_acc=0.9710 val_loss=0.181633 val_acc=0.97
21 test_loss=0.184189 test_acc=0.9724
step=12000 train_loss=0.224083 train_acc=0.9549 val_loss=0.233558 val_acc=0.95
70 test_loss=0.237449 test_acc=0.9511
step=14000 train_loss=0.192764 train_acc=0.9671 val_loss=0.208968 val_acc=0.96
56 test_loss=0.200043 test_acc=0.9633
step=16000 train_loss=0.152921 train_acc=0.9875 val_loss=0.165606 val_acc=0.98
82 test_loss=0.159586 test_acc=0.9851
step=18000 train_loss=0.187830 train_acc=0.9753 val_loss=0.191786 val_acc=0.97
64 test_loss=0.197132 test_acc=0.9712
step=20000 train_loss=0.219427 train_acc=0.9606 val_loss=0.237046 val_acc=0.95
60 test_loss=0.234968 test_acc=0.9553
step=22000 train_loss=0.146935 train_acc=0.9850 val_loss=0.149386 val_acc=0.98
82 test_loss=0.147796 test_acc=0.9875
step=24000 train_loss=0.199305 train_acc=0.9681 val_loss=0.213767 val_acc=0.95
27 test_loss=0.208895 test_acc=0.9617
step=26000 train_loss=0.193055 train_acc=0.9789 val_loss=0.220270 val_acc=0.96
24 test_loss=0.203508 test_acc=0.9721
step=28000 train_loss=0.221466 train_acc=0.9599 val_loss=0.222920 val_acc=0.95
81 test_loss=0.244956 test_acc=0.9490
step=30000 train_loss=0.172285 train_acc=0.9778 val_loss=0.182545 val_acc=0.97
42 test_loss=0.181039 test_acc=0.9762
step=32000 train_loss=0.239946 train_acc=0.9492 val_loss=0.261929 val_acc=0.93
66 test_loss=0.252832 test_acc=0.9470
step=34000 train_loss=0.182814 train_acc=0.9717 val_loss=0.199860 val_acc=0.96
03 test_loss=0.198938 test_acc=0.9662
step=36000 train_loss=0.159621 train_acc=0.9803 val_loss=0.170238 val_acc=0.97
21 test_loss=0.162719 test_acc=0.9787
step=38000 train_loss=0.166365 train_acc=0.9789 val_loss=0.166203 val_acc=0.98
28 test_loss=0.165870 test_acc=0.9851
step=40000 train_loss=0.179307 train_acc=0.9764 val_loss=0.187106 val_acc=0.97
85 test_loss=0.187710 test_acc=0.9744
step=42000 train_loss=0.212011 train_acc=0.9563 val_loss=0.207536 val_acc=0.96
35 test_loss=0.213247 test_acc=0.9646
step=44000 train_loss=0.226872 train_acc=0.9474 val_loss=0.230380 val_acc=0.94
41 test_loss=0.240771 test_acc=0.9463
                _loss=0.202017 train_acc=0.9703 val_loss=0.228441 val_acc=0.95
70 test_loss=0.213193 test_acc=0.9603
```

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```
step=48000 train_loss=0.190657 train_acc=0.9674 val_loss=0.206531 val_acc=0.96
35 test_loss=0.202029 test_acc=0.9692
step=50000 train_loss=0.153558 train_acc=0.9864 val_loss=0.154494 val_acc=0.98
39 test_loss=0.165508 test_acc=0.9837
step=52000 train_loss=0.158701 train_acc=0.9810 val_loss=0.172460 val_acc=0.97
21 test_loss=0.179710 test_acc=0.9708
step=54000 train_loss=0.203617 train_acc=0.9663 val_loss=0.198904 val_acc=0.96
89 test_loss=0.208263 test_acc=0.9628
step=56000 train_loss=0.132012 train_acc=0.9889 val_loss=0.136918 val_acc=0.99
25 test_loss=0.137843 test_acc=0.9889
step=58000 train_loss=0.123394 train_acc=0.9907 val_loss=0.124106 val_acc=0.98
93 test_loss=0.131570 test_acc=0.9871
step=60000 train_loss=0.158095 train_acc=0.9871 val_loss=0.162094 val_acc=0.98
60 test_loss=0.166991 test_acc=0.9841
step=62000 train_loss=0.215394 train_acc=0.9524 val_loss=0.219573 val_acc=0.95
17 test_loss=0.235906 test_acc=0.9442
step=64000 train_loss=0.159561 train_acc=0.9853 val_loss=0.171515 val_acc=0.98
07 test_loss=0.167999 test_acc=0.9850
step=66000 train_loss=0.178117 train_acc=0.9714 val_loss=0.184431 val_acc=0.97
31 test_loss=0.188070 test_acc=0.9696
step=68000 train_loss=0.159270 train_acc=0.9774 val_loss=0.167323 val_acc=0.97
42 test_loss=0.163634 test_acc=0.9773
step=70000 train_loss=0.138056 train_acc=0.9903 val_loss=0.143014 val_acc=0.98
71 test_loss=0.146761 test_acc=0.9889
step=72000 train_loss=0.151432 train_acc=0.9835 val_loss=0.154783 val_acc=0.97
96 test_loss=0.156829 test_acc=0.9814
step=74000 train_loss=0.172095 train_acc=0.9710 val_loss=0.184311 val_acc=0.97
10 test_loss=0.170461 test_acc=0.9744
step=76000 train_loss=0.167648 train_acc=0.9757 val_loss=0.187485 val_acc=0.96
89 test_loss=0.178474 test_acc=0.9744
step=78000 train_loss=0.159548 train_acc=0.9817 val_loss=0.169858 val_acc=0.97
74 test_loss=0.162749 test_acc=0.9805
step=80000 train_loss=0.188856 train_acc=0.9674 val_loss=0.196423 val_acc=0.96
03 test_loss=0.198901 test_acc=0.9631
step=82000 train_loss=0.191556 train_acc=0.9681 val_loss=0.199476 val_acc=0.96
24 test_loss=0.195750 test_acc=0.9671
step=84000 train_loss=0.158173 train_acc=0.9782 val_loss=0.169583 val_acc=0.97
64 test_loss=0.162795 test_acc=0.9785
step=86000 train_loss=0.190050 train_acc=0.9689 val_loss=0.205410 val_acc=0.96
89 test_loss=0.199153 test_acc=0.9615
step=88000 train_loss=0.168866 train_acc=0.9774 val_loss=0.168643 val_acc=0.97
42 test_loss=0.179331 test_acc=0.9701
step=90000 train_loss=0.135393 train_acc=0.9846 val_loss=0.145997 val_acc=0.98
28 test_loss=0.142009 test_acc=0.9860
step=92000 train_loss=0.194143 train_acc=0.9531 val_loss=0.190995 val_acc=0.96
56 test_loss=0.200122 test_acc=0.9563
step=94000 train_loss=0.139996 train_acc=0.9896 val_loss=0.153269 val_acc=0.98
82 test_loss=0.146054 test_acc=0.9866
step=96000 train_loss=0.183055 train_acc=0.9699 val_loss=0.187767 val_acc=0.96
99 test_loss=0.197345 test_acc=0.9667
step=98000 train_loss=0.152453 train_acc=0.9832 val_loss=0.153310 val_acc=0.98
39 test_loss=0.150403 test_acc=0.9848
step=100000 train_loss=0.193353 train_acc=0.9660 val_loss=0.203753 val_acc=0.9
646 test_loss=0.201610 test_acc=0.9667
step=102000 train_loss=0.135262 train_acc=0.9918 val_loss=0.140154 val_acc=0.9
957 test_loss=0.138889 test_acc=0.9909
step=104000 train_loss=0.138839 train_acc=0.9842 val_loss=0.142139 val_acc=0.9
828 test_loss=0.138189 test_acc=0.9844
n_loss=0.191368 train_acc=0.9595 val_loss=0.189618 val_acc=0.9
678 test_loss=0.190604 test_acc=0.9626
```

Loading [MathJax]/extensions/Safe.js

```
step=108000 train_loss=0.133679 train_acc=0.9842 val_loss=0.132800 val_acc=0.9
903 test_loss=0.135201 test_acc=0.9887
step=110000 train_loss=0.145744 train_acc=0.9821 val_loss=0.151856 val_acc=0.9
871 test_loss=0.148579 test_acc=0.9851
step=112000 train_loss=0.182153 train_acc=0.9774 val_loss=0.185332 val_acc=0.9
753 test_loss=0.186158 test_acc=0.9769
step=114000 train_loss=0.145618 train_acc=0.9825 val_loss=0.146565 val_acc=0.9
882 test_loss=0.155502 test_acc=0.9830
step=116000 train_loss=0.183499 train_acc=0.9710 val_loss=0.185780 val_acc=0.9
689 test_loss=0.195335 test_acc=0.9673
step=118000 train_loss=0.142555 train_acc=0.9846 val_loss=0.152685 val_acc=0.9
785 test_loss=0.150337 test_acc=0.9805
step=120000 train_loss=0.130328 train_acc=0.9885 val_loss=0.143916 val_acc=0.9
839 test_loss=0.142956 test_acc=0.9800
step=122000 train_loss=0.148142 train_acc=0.9864 val_loss=0.150807 val_acc=0.9
850 test_loss=0.153889 test_acc=0.9851
step=124000 train_loss=0.166559 train_acc=0.9764 val_loss=0.166900 val_acc=0.9
774 test_loss=0.179347 test_acc=0.9719
step=126000 train_loss=0.131406 train_acc=0.9842 val_loss=0.139453 val_acc=0.9
774 test_loss=0.137911 test_acc=0.9832
step=128000 train_loss=0.141688 train_acc=0.9799 val_loss=0.152093 val_acc=0.9
710 test_loss=0.146267 test_acc=0.9785
step=130000 train_loss=0.114312 train_acc=0.9928 val_loss=0.126034 val_acc=0.9
871 test_loss=0.124281 test_acc=0.9918
step=132000 train_loss=0.157799 train_acc=0.9760 val_loss=0.159283 val_acc=0.9
785 test_loss=0.161056 test_acc=0.9809
step=134000 train_loss=0.163015 train_acc=0.9846 val_loss=0.171968 val_acc=0.9
828 test_loss=0.165794 test_acc=0.9817
step=136000 train_loss=0.133038 train_acc=0.9918 val_loss=0.132057 val_acc=0.9
957 test_loss=0.138494 test_acc=0.9911
step=138000 train_loss=0.141073 train_acc=0.9835 val_loss=0.142911 val_acc=0.9
817 test_loss=0.148851 test_acc=0.9828
step=140000 train_loss=0.116251 train_acc=0.9946 val_loss=0.119232 val_acc=0.9
925 test_loss=0.125070 test_acc=0.9914
step=142000 train_loss=0.141665 train_acc=0.9799 val_loss=0.145890 val_acc=0.9
774 test_loss=0.150086 test_acc=0.9769
step=144000 train_loss=0.103283 train_acc=0.9975 val_loss=0.107363 val_acc=0.9
957 test_loss=0.105733 test_acc=0.9966
step=146000 train_loss=0.205997 train_acc=0.9620 val_loss=0.217111 val_acc=0.9
581 test_loss=0.201862 test_acc=0.9631
step=148000 train_loss=0.123033 train_acc=0.9907 val_loss=0.124919 val_acc=0.9
871 test_loss=0.128453 test_acc=0.9900
step=150000 train_loss=0.097329 train_acc=0.9975 val_loss=0.102120 val_acc=0.9
979 test_loss=0.103931 test_acc=0.9980
step=152000 train_loss=0.131808 train_acc=0.9853 val_loss=0.138486 val_acc=0.9
807 test_loss=0.132583 test_acc=0.9860
step=154000 train_loss=0.107875 train_acc=0.9946 val_loss=0.116196 val_acc=0.9
925 test_loss=0.115356 test_acc=0.9930
step=156000 train_loss=0.159310 train_acc=0.9728 val_loss=0.164558 val_acc=0.9
721 test_loss=0.164341 test_acc=0.9742
step=158000 train_loss=0.115151 train_acc=0.9939 val_loss=0.116618 val_acc=0.9
925 test_loss=0.122529 test_acc=0.9930
step=160000 train_loss=0.123234 train_acc=0.9950 val_loss=0.130986 val_acc=0.9
925 test_loss=0.129941 test_acc=0.9909
step=162000 train_loss=0.136061 train_acc=0.9871 val_loss=0.149692 val_acc=0.9
882 test_loss=0.140808 test_acc=0.9862
step=164000 train_loss=0.122860 train_acc=0.9928 val_loss=0.122705 val_acc=0.9
968 test_loss=0.124894 test_acc=0.9941
n_loss=0.098104 train_acc=0.9943 val_loss=0.099020 val_acc=0.9
979 test_loss=0.100172 test_acc=0.9955
```

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```
step=168000 train_loss=0.137023 train_acc=0.9864 val_loss=0.138111 val_acc=0.9
871 test_loss=0.144238 test_acc=0.9805
step=170000 train_loss=0.101648 train_acc=0.9928 val_loss=0.101202 val_acc=0.9
957 test_loss=0.105881 test_acc=0.9921
step=172000 train_loss=0.100238 train_acc=0.9964 val_loss=0.108079 val_acc=0.9
957 test_loss=0.109780 test_acc=0.9955
step=174000 train_loss=0.113224 train_acc=0.9950 val_loss=0.113826 val_acc=0.9
936 test_loss=0.119056 test_acc=0.9934
step=176000 train_loss=0.101919 train_acc=0.9946 val_loss=0.104875 val_acc=0.9
946 test_loss=0.108838 test_acc=0.9921
step=178000 train_loss=0.121379 train_acc=0.9882 val_loss=0.129599 val_acc=0.9
871 test_loss=0.129352 test_acc=0.9857
step=180000 train_loss=0.133091 train_acc=0.9857 val_loss=0.132352 val_acc=0.9
860 test_loss=0.134059 test_acc=0.9864
step=182000 train_loss=0.121319 train_acc=0.9900 val_loss=0.119473 val_acc=0.9
893 test_loss=0.127454 test_acc=0.9894
step=184000 train_loss=0.169202 train_acc=0.9778 val_loss=0.181759 val_acc=0.9
731 test_loss=0.183884 test_acc=0.9726
step=186000 train_loss=0.121337 train_acc=0.9882 val_loss=0.119261 val_acc=0.9
903 test_loss=0.125234 test_acc=0.9855
step=188000 train_loss=0.086741 train_acc=0.9968 val_loss=0.091881 val_acc=0.9
946 test_loss=0.092135 test_acc=0.9962
step=190000 train_loss=0.120638 train_acc=0.9968 val_loss=0.123678 val_acc=0.9
957 test_loss=0.123995 test_acc=0.9957
step=192000 train_loss=0.090445 train_acc=0.9961 val_loss=0.096017 val_acc=0.9
979 test_loss=0.092568 test_acc=0.9945
step=194000 train_loss=0.100268 train_acc=0.9928 val_loss=0.104357 val_acc=0.9
914 test_loss=0.101304 test_acc=0.9934
step=196000 train_loss=0.125205 train_acc=0.9828 val_loss=0.137547 val_acc=0.9
785 test_loss=0.127509 test_acc=0.9848
step=198000 train_loss=0.117351 train_acc=0.9853 val_loss=0.133027 val_acc=0.9
785 test_loss=0.127509 test_acc=0.9791
step=200000 train_loss=0.084758 train_acc=0.9971 val_loss=0.082206 val_acc=0.9
968 test_loss=0.082820 test_acc=0.9977
step=202000 train_loss=0.122987 train_acc=0.9835 val_loss=0.118409 val_acc=0.9
839 test_loss=0.121880 test_acc=0.9902
step=204000 train_loss=0.103623 train_acc=0.9950 val_loss=0.114726 val_acc=0.9
882 test_loss=0.111268 test_acc=0.9919
step=206000 train_loss=0.100929 train_acc=0.9957 val_loss=0.103341 val_acc=0.9
968 test_loss=0.106008 test_acc=0.9932
step=208000 train_loss=0.069802 train_acc=0.9993 val_loss=0.069729 val_acc=1.0
000 test_loss=0.072723 test_acc=0.9989
step=210000 train_loss=0.087135 train_acc=0.9982 val_loss=0.085468 val_acc=0.9
989 test_loss=0.090566 test_acc=0.9966
step=212000 train_loss=0.088916 train_acc=0.9953 val_loss=0.090969 val_acc=0.9
989 test_loss=0.091502 test_acc=0.9952
step=214000 train_loss=0.082024 train_acc=0.9975 val_loss=0.087110 val_acc=0.9
989 test_loss=0.084808 test_acc=0.9987
step=216000 train_loss=0.070366 train_acc=0.9982 val_loss=0.069639 val_acc=1.0
000 test_loss=0.072783 test_acc=0.9971
step=218000 train_loss=0.065868 train_acc=0.9993 val_loss=0.072951 val_acc=0.9
968 test_loss=0.068664 test_acc=0.9995
step=220000 train_loss=0.080054 train_acc=0.9953 val_loss=0.079268 val_acc=0.9
979 test_loss=0.080607 test_acc=0.9962
step=222000 train_loss=0.058049 train_acc=1.0000 val_loss=0.063537 val_acc=0.9
979 test_loss=0.062791 test_acc=0.9995
step=224000 train_loss=0.084312 train_acc=0.9975 val_loss=0.085934 val_acc=0.9
989 test_loss=0.089345 test_acc=0.9962
n_loss=0.064307 train_acc=0.9986 val_loss=0.065907 val_acc=0.9
989 test_loss=0.065683 test_acc=0.9993
```

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```
step=228000 train_loss=0.068464 train_acc=0.9993 val_loss=0.070654 val_acc=0.9
968 test_loss=0.069552 test_acc=0.9989
step=230000 train_loss=0.096796 train_acc=0.9903 val_loss=0.093237 val_acc=0.9
946 test_loss=0.101035 test_acc=0.9912
step=232000 train_loss=0.082420 train_acc=0.9993 val_loss=0.084281 val_acc=1.0
000 test_loss=0.086795 test_acc=0.9982
step=234000 train_loss=0.059098 train_acc=0.9993 val_loss=0.064572 val_acc=0.9
979 test_loss=0.059611 test_acc=0.9995
step=236000 train_loss=0.065558 train_acc=0.9986 val_loss=0.064574 val_acc=0.9
989 test_loss=0.064041 test_acc=0.9993
step=238000 train_loss=0.063943 train_acc=0.9989 val_loss=0.072314 val_acc=0.9
968 test_loss=0.066295 test_acc=0.9986
step=240000 train_loss=0.092258 train_acc=0.9953 val_loss=0.101881 val_acc=0.9
903 test_loss=0.097543 test_acc=0.9923
step=242000 train_loss=0.094148 train_acc=0.9928 val_loss=0.095291 val_acc=0.9
957 test_loss=0.092267 test_acc=0.9975
step=244000 train_loss=0.071218 train_acc=0.9971 val_loss=0.070822 val_acc=0.9
979 test_loss=0.069298 test_acc=0.9984
step=246000 train_loss=0.074688 train_acc=0.9979 val_loss=0.077559 val_acc=1.0
000 test_loss=0.075406 test_acc=0.9989
step=248000 train_loss=0.069845 train_acc=0.9979 val_loss=0.077119 val_acc=0.9
979 test_loss=0.074795 test_acc=0.9975
step=250000 train_loss=0.066982 train_acc=0.9982 val_loss=0.073688 val_acc=0.9
979 test_loss=0.070455 test_acc=0.9982
step=252000 train_loss=0.084404 train_acc=0.9961 val_loss=0.092825 val_acc=0.9
893 test_loss=0.088458 test_acc=0.9953
step=254000 train_loss=0.060681 train_acc=0.9993 val_loss=0.063275 val_acc=0.9
989 test_loss=0.064838 test_acc=0.9989
step=256000 train_loss=0.058974 train_acc=0.9996 val_loss=0.060877 val_acc=1.0
000 test_loss=0.062202 test_acc=0.9991
step=258000 train_loss=0.059429 train_acc=1.0000 val_loss=0.059607 val_acc=0.9
989 test_loss=0.061565 test_acc=0.9987
step=260000 train_loss=0.069502 train_acc=0.9979 val_loss=0.074370 val_acc=0.9
979 test_loss=0.069070 test_acc=0.9984
step=262000 train_loss=0.058041 train_acc=1.0000 val_loss=0.059611 val_acc=0.9
989 test_loss=0.060400 test_acc=0.9989
step=264000 train_loss=0.081976 train_acc=0.9921 val_loss=0.090606 val_acc=0.9
882 test_loss=0.092124 test_acc=0.9889
step=266000 train_loss=0.104772 train_acc=0.9910 val_loss=0.104193 val_acc=0.9
936 test_loss=0.107131 test_acc=0.9902
step=268000 train_loss=0.070978 train_acc=0.9979 val_loss=0.073398 val_acc=0.9
968 test_loss=0.073882 test_acc=0.9977
step=270000 train_loss=0.050112 train_acc=1.0000 val_loss=0.050582 val_acc=0.9
989 test_loss=0.049197 test_acc=1.0000
step=272000 train_loss=0.045635 train_acc=0.9996 val_loss=0.047541 val_acc=0.9
989 test_loss=0.047336 test_acc=0.9993
step=274000 train_loss=0.061306 train_acc=0.9968 val_loss=0.064415 val_acc=0.9
946 test_loss=0.066060 test_acc=0.9946
step=276000 train_loss=0.055244 train_acc=1.0000 val_loss=0.060268 val_acc=1.0
000 test_loss=0.056304 test_acc=1.0000
step=278000 train_loss=0.049590 train_acc=0.9996 val_loss=0.049912 val_acc=1.0
000 test_loss=0.051378 test_acc=0.9996
step=280000 train_loss=0.067510 train_acc=0.9982 val_loss=0.067688 val_acc=0.9
989 test_loss=0.066588 test_acc=0.9986
step=282000 train_loss=0.071895 train_acc=0.9943 val_loss=0.071505 val_acc=0.9
979 test_loss=0.074691 test_acc=0.9950
step=284000 train_loss=0.057058 train_acc=0.9996 val_loss=0.059306 val_acc=1.0
000 test_loss=0.060308 test_acc=0.9991
n_loss=0.056128 train_acc=0.9993 val_loss=0.056983 val_acc=0.9
989 test_loss=0.057646 test_acc=0.9987
```

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```
step=288000 train_loss=0.041897 train_acc=1.0000 val_loss=0.043558 val_acc=0.9
989 test_loss=0.042447 test_acc=1.0000
step=290000 train_loss=0.040603 train_acc=1.0000 val_loss=0.042236 val_acc=1.0
000 test_loss=0.043770 test_acc=0.9998
step=292000 train_loss=0.040874 train_acc=1.0000 val_loss=0.044709 val_acc=1.0
000 test_loss=0.041828 test_acc=1.0000
step=294000 train_loss=0.047598 train_acc=1.0000 val_loss=0.047521 val_acc=1.0
000 test_loss=0.050889 test_acc=0.9998
step=296000 train_loss=0.048664 train_acc=0.9993 val_loss=0.052008 val_acc=1.0
000 test_loss=0.051337 test_acc=0.9995
step=298000 train_loss=0.067953 train_acc=0.9932 val_loss=0.069537 val_acc=0.9
936 test_loss=0.066994 test_acc=0.9945
step=300000 train_loss=0.028169 train_acc=1.0000 val_loss=0.029511 val_acc=1.0
000 test_loss=0.029299 test_acc=1.0000
step=302000 train_loss=0.035065 train_acc=1.0000 val_loss=0.036259 val_acc=1.0
000 test_loss=0.036884 test_acc=1.0000
step=304000 train_loss=0.043241 train_acc=1.0000 val_loss=0.044107 val_acc=1.0
000 test_loss=0.045601 test_acc=1.0000
step=306000 train_loss=0.032670 train_acc=1.0000 val_loss=0.034872 val_acc=1.0
000 test_loss=0.034910 test_acc=1.0000
step=308000 train_loss=0.038212 train_acc=1.0000 val_loss=0.038838 val_acc=1.0
000 test_loss=0.038788 test_acc=1.0000
step=310000 train_loss=0.029312 train_acc=1.0000 val_loss=0.029633 val_acc=1.0
000 test_loss=0.030227 test_acc=1.0000
step=312000 train_loss=0.034741 train_acc=1.0000 val_loss=0.037806 val_acc=1.0
000 test_loss=0.037471 test_acc=1.0000
step=314000 train_loss=0.036940 train_acc=0.9996 val_loss=0.038792 val_acc=1.0
000 test_loss=0.038455 test_acc=1.0000
step=316000 train_loss=0.069542 train_acc=0.9921 val_loss=0.067018 val_acc=0.9
968 test_loss=0.073757 test_acc=0.9934
step=318000 train_loss=0.044697 train_acc=0.9996 val_loss=0.044845 val_acc=1.0
000 test_loss=0.047491 test_acc=0.9993
step=320000 train_loss=0.033962 train_acc=0.9996 val_loss=0.035596 val_acc=1.0
000 test_loss=0.035210 test_acc=0.9998
step=322000 train_loss=0.033124 train_acc=1.0000 val_loss=0.035511 val_acc=1.0
000 test_loss=0.034634 test_acc=1.0000
step=324000 train_loss=0.035724 train_acc=0.9996 val_loss=0.036971 val_acc=0.9
989 test_loss=0.036936 test_acc=0.9996
step=326000 train_loss=0.026456 train_acc=1.0000 val_loss=0.027363 val_acc=1.0
000 test_loss=0.027534 test_acc=1.0000
step=328000 train_loss=0.037574 train_acc=0.9996 val_loss=0.037126 val_acc=1.0
000 test_loss=0.038903 test_acc=0.9996
step=330000 train_loss=0.019646 train_acc=1.0000 val_loss=0.020866 val_acc=1.0
000 test_loss=0.020714 test_acc=1.0000
step=332000 train_loss=0.022104 train_acc=1.0000 val_loss=0.023158 val_acc=1.0
000 test_loss=0.023579 test_acc=1.0000
step=334000 train_loss=0.015432 train_acc=1.0000 val_loss=0.015947 val_acc=1.0
000 test_loss=0.016169 test_acc=1.0000
step=336000 train_loss=0.029672 train_acc=1.0000 val_loss=0.032416 val_acc=0.9
989 test_loss=0.033690 test_acc=0.9995
step=338000 train_loss=0.040394 train_acc=0.9964 val_loss=0.045311 val_acc=0.9
957 test_loss=0.042983 test_acc=0.9961
step=340000 train_loss=0.027153 train_acc=0.9993 val_loss=0.031108 val_acc=1.0
000 test_loss=0.028775 test_acc=0.9995
step=342000 train_loss=0.019759 train_acc=1.0000 val_loss=0.021086 val_acc=0.9
989 test_loss=0.022129 test_acc=0.9995
step=344000 train_loss=0.018917 train_acc=0.9996 val_loss=0.018082 val_acc=1.0
000 test_loss=0.019946 test_acc=0.9996
n_loss=0.021006 train_acc=0.9986 val_loss=0.022551 val_acc=0.9
979 test_loss=0.020564 test_acc=0.9993
```

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```
step=348000 train_loss=0.028000 train_acc=0.9996 val_loss=0.030898 val_acc=0.9
979 test_loss=0.030750 test_acc=0.9982
step=350000 train_loss=0.049013 train_acc=0.9889 val_loss=0.055042 val_acc=0.9
871 test_loss=0.053657 test_acc=0.9862
step=352000 train_loss=0.011760 train_acc=0.9996 val_loss=0.010915 val_acc=1.0
000 test_loss=0.011714 test_acc=0.9998
step=354000 train_loss=0.009500 train_acc=1.0000 val_loss=0.009621 val_acc=1.0
000 test_loss=0.009765 test_acc=1.0000
step=356000 train_loss=0.018352 train_acc=0.9993 val_loss=0.023476 val_acc=0.9
989 test_loss=0.020516 test_acc=0.9986
step=358000 train_loss=0.008589 train_acc=1.0000 val_loss=0.008934 val_acc=1.0
000 test_loss=0.009114 test_acc=1.0000
step=360000 train_loss=0.003472 train_acc=1.0000 val_loss=0.003609 val_acc=1.0
000 test_loss=0.003628 test_acc=1.0000
step=362000 train_loss=0.012447 train_acc=1.0000 val_loss=0.013394 val_acc=1.0
000 test_loss=0.013143 test_acc=1.0000
step=364000 train_loss=0.004929 train_acc=1.0000 val_loss=0.005125 val_acc=1.0
000 test_loss=0.005176 test_acc=1.0000
step=366000 train_loss=0.007115 train_acc=1.0000 val_loss=0.007267 val_acc=1.0
000 test_loss=0.007552 test_acc=1.0000
step=368000 train_loss=0.019988 train_acc=0.9975 val_loss=0.025569 val_acc=0.9
946 test_loss=0.021888 test_acc=0.9968
step=370000 train_loss=0.026650 train_acc=0.9946 val_loss=0.029280 val_acc=0.9
946 test_loss=0.030042 test_acc=0.9941
step=372000 train_loss=0.007364 train_acc=1.0000 val_loss=0.008455 val_acc=1.0
000 test_loss=0.008122 test_acc=1.0000
step=374000 train_loss=0.003378 train_acc=1.0000 val_loss=0.003518 val_acc=1.0
000 test_loss=0.003520 test_acc=1.0000
step=376000 train_loss=0.003820 train_acc=1.0000 val_loss=0.003969 val_acc=1.0
000 test_loss=0.003980 test_acc=1.0000
step=378000 train_loss=0.003045 train_acc=1.0000 val_loss=0.003153 val_acc=1.0
000 test_loss=0.003167 test_acc=1.0000
step=380000 train_loss=0.008748 train_acc=1.0000 val_loss=0.009589 val_acc=1.0
000 test_loss=0.009038 test_acc=1.0000
step=382000 train_loss=0.014993 train_acc=1.0000 val_loss=0.017882 val_acc=1.0
000 test_loss=0.016303 test_acc=0.9996
step=384000 train_loss=0.015968 train_acc=0.9957 val_loss=0.014870 val_acc=0.9
979 test_loss=0.014799 test_acc=0.9971
step=386000 train_loss=0.002033 train_acc=1.0000 val_loss=0.002116 val_acc=1.0
000 test_loss=0.002121 test_acc=1.0000
step=388000 train_loss=0.049503 train_acc=0.9889 val_loss=0.064936 val_acc=0.9
828 test_loss=0.064370 test_acc=0.9841
step=390000 train_loss=0.004179 train_acc=1.0000 val_loss=0.004619 val_acc=1.0
000 test_loss=0.004778 test_acc=1.0000
step=392000 train_loss=0.001667 train_acc=1.0000 val_loss=0.001734 val_acc=1.0
000 test_loss=0.001733 test_acc=1.0000
step=394000 train_loss=0.001850 train_acc=1.0000 val_loss=0.001933 val_acc=1.0
000 test_loss=0.001938 test_acc=1.0000
step=396000 train_loss=0.004205 train_acc=0.9996 val_loss=0.004900 val_acc=1.0
000 test_loss=0.004441 test_acc=0.9998
step=398000 train_loss=0.001497 train_acc=1.0000 val_loss=0.001557 val_acc=1.0
000 test_loss=0.001566 test_acc=1.0000
step=400000 train_loss=0.015805 train_acc=0.9993 val_loss=0.018721 val_acc=0.9
979 test_loss=0.018008 test_acc=0.9986
step=402000 train_loss=0.001429 train_acc=1.0000 val_loss=0.001497 val_acc=1.0
000 test_loss=0.001491 test_acc=1.0000
step=404000 train_loss=0.206484 train_acc=0.9356 val_loss=0.224877 val_acc=0.9
313 test_loss=0.190737 test_acc=0.9350
n_loss=0.001070 train_acc=1.0000 val_loss=0.001118 val_acc=1.0
000 test_loss=0.001120 test_acc=1.0000
```

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```
step=408000 train_loss=0.004957 train_acc=0.9996 val_loss=0.005710 val_acc=1.0
000 test_loss=0.005220 test_acc=1.0000
step=410000 train_loss=0.001028 train_acc=1.0000 val_loss=0.001068 val_acc=1.0
000 test_loss=0.001075 test_acc=1.0000
step=412000 train_loss=0.002058 train_acc=1.0000 val_loss=0.002163 val_acc=1.0
000 test_loss=0.002278 test_acc=1.0000
step=414000 train_loss=0.000829 train_acc=1.0000 val_loss=0.000869 val_acc=1.0
000 test_loss=0.000868 test_acc=1.0000
step=416000 train_loss=0.000632 train_acc=1.0000 val_loss=0.000661 val_acc=1.0
000 test_loss=0.000663 test_acc=1.0000
step=418000 train_loss=0.000447 train_acc=1.0000 val_loss=0.000457 val_acc=1.0
000 test_loss=0.000466 test_acc=1.0000
step=420000 train_loss=0.001913 train_acc=0.9996 val_loss=0.004289 val_acc=0.9
979 test_loss=0.002732 test_acc=0.9993
step=422000 train_loss=0.000434 train_acc=1.0000 val_loss=0.000451 val_acc=1.0
000 test_loss=0.000455 test_acc=1.0000
step=424000 train_loss=0.000311 train_acc=1.0000 val_loss=0.000324 val_acc=1.0
000 test_loss=0.000324 test_acc=1.0000
step=426000 train_loss=0.000773 train_acc=1.0000 val_loss=0.000674 val_acc=1.0
000 test_loss=0.000706 test_acc=1.0000
step=428000 train_loss=0.000311 train_acc=1.0000 val_loss=0.000330 val_acc=1.0
000 test_loss=0.000329 test_acc=1.0000
step=430000 train_loss=0.000661 train_acc=1.0000 val_loss=0.000628 val_acc=1.0
000 test_loss=0.000973 test_acc=0.9998
step=432000 train_loss=0.008431 train_acc=0.9982 val_loss=0.012448 val_acc=0.9
946 test_loss=0.008886 test_acc=0.9968
step=434000 train_loss=0.000213 train_acc=1.0000 val_loss=0.000235 val_acc=1.0
000 test_loss=0.000222 test_acc=1.0000
step=436000 train_loss=0.002148 train_acc=0.9993 val_loss=0.001419 val_acc=1.0
000 test_loss=0.001797 test_acc=1.0000
step=438000 train_loss=0.000162 train_acc=1.0000 val_loss=0.000170 val_acc=1.0
000 test_loss=0.000171 test_acc=1.0000
step=440000 train_loss=0.001126 train_acc=1.0000 val_loss=0.001060 val_acc=1.0
000 test_loss=0.001150 test_acc=1.0000
step=442000 train_loss=0.000824 train_acc=1.0000 val_loss=0.000911 val_acc=1.0
000 test_loss=0.001002 test_acc=1.0000
step=444000 train_loss=0.000054 train_acc=1.0000 val_loss=0.000057 val_acc=1.0
000 test_loss=0.000057 test_acc=1.0000
step=446000 train_loss=0.000107 train_acc=1.0000 val_loss=0.000114 val_acc=1.0
000 test_loss=0.000114 test_acc=1.0000
step=448000 train_loss=0.000118 train_acc=1.0000 val_loss=0.000122 val_acc=1.0
000 test_loss=0.000124 test_acc=1.0000
step=450000 train_loss=0.001161 train_acc=1.0000 val_loss=0.001062 val_acc=1.0
000 test_loss=0.001219 test_acc=1.0000
step=452000 train_loss=0.000058 train_acc=1.0000 val_loss=0.000061 val_acc=1.0
000 test_loss=0.000061 test_acc=1.0000
step=454000 train_loss=0.000065 train_acc=1.0000 val_loss=0.000070 val_acc=1.0
000 test_loss=0.000069 test_acc=1.0000
step=456000 train_loss=0.000011 train_acc=1.0000 val_loss=0.000011 val_acc=1.0
000 test_loss=0.000011 test_acc=1.0000
step=458000 train_loss=0.000040 train_acc=1.0000 val_loss=0.000041 val_acc=1.0
000 test_loss=0.000041 test_acc=1.0000
step=460000 train_loss=0.000059 train_acc=1.0000 val_loss=0.000061 val_acc=1.0
000 test_loss=0.000061 test_acc=1.0000
step=462000 train_loss=0.000047 train_acc=1.0000 val_loss=0.000049 val_acc=1.0
000 test_loss=0.000049 test_acc=1.0000
step=464000 train_loss=0.013434 train_acc=0.9946 val_loss=0.023890 val_acc=0.9
925 test_loss=0.020509 test_acc=0.9923
n_loss=0.016249 train_acc=0.9957 val_loss=0.010362 val_acc=0.9
979 test_loss=0.018679 test_acc=0.9939
```

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```
step=468000 train_loss=0.000160 train_acc=1.0000 val_loss=0.000156 val_acc=1.0
000 test_loss=0.000241 test_acc=1.0000
step=470000 train_loss=0.000024 train_acc=1.0000 val_loss=0.000025 val_acc=1.0
000 test_loss=0.000025 test_acc=1.0000
step=472000 train_loss=0.000010 train_acc=1.0000 val_loss=0.000010 val_acc=1.0
000 test_loss=0.000011 test_acc=1.0000
step=474000 train_loss=0.000008 train_acc=1.0000 val_loss=0.000009 val_acc=1.0
000 test_loss=0.000009 test_acc=1.0000
step=476000 train_loss=0.000018 train_acc=1.0000 val_loss=0.000018 val_acc=1.0
000 test_loss=0.000019 test_acc=1.0000
step=478000 train_loss=0.000017 train_acc=1.0000 val_loss=0.000018 val_acc=1.0
000 test_loss=0.000018 test_acc=1.0000
step=480000 train_loss=0.000012 train_acc=1.0000 val_loss=0.000012 val_acc=1.0
000 test_loss=0.000012 test_acc=1.0000
step=482000 train_loss=0.000033 train_acc=1.0000 val_loss=0.000037 val_acc=1.0
000 test_loss=0.000036 test_acc=1.0000
step=484000 train_loss=0.000004 train_acc=1.0000 val_loss=0.000004 val_acc=1.0
000 test_loss=0.000004 test_acc=1.0000
step=486000 train_loss=0.000077 train_acc=1.0000 val_loss=0.000078 val_acc=1.0
000 test_loss=0.000080 test_acc=1.0000
step=488000 train_loss=0.000017 train_acc=1.0000 val_loss=0.000018 val_acc=1.0
000 test_loss=0.000018 test_acc=1.0000
step=490000 train_loss=0.000007 train_acc=1.0000 val_loss=0.000007 val_acc=1.0
000 test_loss=0.000007 test_acc=1.0000
step=492000 train_loss=0.000013 train_acc=1.0000 val_loss=0.000013 val_acc=1.0
000 test_loss=0.000013 test_acc=1.0000
step=494000 train_loss=0.000021 train_acc=1.0000 val_loss=0.000021 val_acc=1.0
000 test_loss=0.000021 test_acc=1.0000
step=496000 train_loss=0.000005 train_acc=1.0000 val_loss=0.000006 val_acc=1.0
000 test_loss=0.000006 test_acc=1.0000
step=498000 train_loss=0.000067 train_acc=1.0000 val_loss=0.000087 val_acc=1.0
000 test_loss=0.000097 test_acc=1.0000
step=500000 train_loss=0.000007 train_acc=1.0000 val_loss=0.000008 val_acc=1.0
000 test_loss=0.000007 test_acc=1.0000
best metrics: {
  "step": 484000,
  "lr": 0.00010735037272494874,
  "train_loss": 4.069368765296682e-06,
  "train_acc": 1.0,
  "val_loss": 4.2513188506445885e-06,
  "val_acc": 1.0,
  "test_loss": 4.2687410484672676e-06,
  "test_acc": 1.0,
  "elapsed_sec": 3686.838659763336
}
checkpoint dir: /Users/shobhit/Documents/UW/CSE 493/cse493s-spring26-hw1/out/s
ubmission/part1/ablations/abl_adamw_wd1_trainfrac0p3

$ /opt/homebrew/Caskroom/miniconda/base/envs/cse493/bin/python train.py --mode
modular --op / --p 97 --n_layer 2 --n_head 4 --n_embd 128 --steps 500000 --see
d 42 --batch_size 128 --lr 0.003 --min_lr 0.0001 --lr_schedule cosine --warmup
_steps 1000 --grad_clip 1.0 --beta1 0.9 --beta2 0.95 --weight_decay 1.0 --opti
mizer adamw --train_frac 0.5 --val_frac 0.1 --eval_interval 2000 --log_interva
l 2000 --out_dir '/Users/shobhit/Documents/UW/CSE 493/cse493s-spring26-hw1/ou
t/submission/part1/ablations/abl_adamw_wd1_trainfrac0p5' --bias --resume
number of parameters: 0.41M
[resume] model_last.pt not found in /Users/shobhit/Documents/UW/CSE 493/cse493
s-spring26-hw1/out/submission/part1/ablations/abl_adamw_wd1_trainfrac0p5; star
```

Loading [MathJax]/extensions/Safe.js

```
device=mps train=4656 val=931 test=3725 batch_size=128
```

```
step=1 train_loss=4.689445 train_acc=0.0000 val_loss=4.690156 val_acc=0.0000 t
est_loss=4.681565 test_acc=0.0000
step=2000 train_loss=0.675389 train_acc=0.8232 val_loss=0.695319 val_acc=0.806
7 test_loss=0.684461 test_acc=0.8185
step=4000 train_loss=0.578206 train_acc=0.8741 val_loss=0.603018 val_acc=0.866
8 test_loss=0.589792 test_acc=0.8757
step=6000 train_loss=0.863428 train_acc=0.7270 val_loss=0.884374 val_acc=0.711
1 test_loss=0.942859 test_acc=0.6894
step=8000 train_loss=0.579331 train_acc=0.8926 val_loss=0.588867 val_acc=0.877
6 test_loss=0.603641 test_acc=0.8784
step=10000 train_loss=0.693818 train_acc=0.8082 val_loss=0.680829 val_acc=0.82
49 test_loss=0.712257 test_acc=0.7979
step=12000 train_loss=0.755778 train_acc=0.7661 val_loss=0.766803 val_acc=0.77
12 test_loss=0.759868 test_acc=0.7678
step=14000 train_loss=0.742453 train_acc=0.7762 val_loss=0.737133 val_acc=0.79
48 test_loss=0.780208 test_acc=0.7667
step=16000 train_loss=0.522368 train_acc=0.9171 val_loss=0.540905 val_acc=0.90
12 test_loss=0.548704 test_acc=0.9044
step=18000 train_loss=0.584103 train_acc=0.8907 val_loss=0.622429 val_acc=0.88
51 test_loss=0.610357 test_acc=0.8776
step=20000 train_loss=0.505339 train_acc=0.9044 val_loss=0.501323 val_acc=0.90
87 test_loss=0.526757 test_acc=0.8961
step=22000 train_loss=0.639983 train_acc=0.8370 val_loss=0.653138 val_acc=0.82
49 test_loss=0.665069 test_acc=0.8295
step=24000 train_loss=0.689167 train_acc=0.8018 val_loss=0.653644 val_acc=0.82
60 test_loss=0.704604 test_acc=0.8078
step=26000 train_loss=0.636356 train_acc=0.8668 val_loss=0.631324 val_acc=0.86
36 test_loss=0.671862 test_acc=0.8502
step=28000 train_loss=0.737559 train_acc=0.8091 val_loss=0.743499 val_acc=0.80
45 test_loss=0.767760 test_acc=0.7954
step=30000 train_loss=0.471362 train_acc=0.9223 val_loss=0.486465 val_acc=0.92
37 test_loss=0.491676 test_acc=0.9144
step=32000 train_loss=0.569350 train_acc=0.8890 val_loss=0.569064 val_acc=0.88
08 test_loss=0.595976 test_acc=0.8728
step=34000 train_loss=0.608092 train_acc=0.8529 val_loss=0.610315 val_acc=0.85
50 test_loss=0.633700 test_acc=0.8387
step=36000 train_loss=0.498292 train_acc=0.9182 val_loss=0.540529 val_acc=0.89
15 test_loss=0.546087 test_acc=0.8940
step=38000 train_loss=0.543774 train_acc=0.8763 val_loss=0.566007 val_acc=0.84
64 test_loss=0.555846 test_acc=0.8703
step=40000 train_loss=0.566523 train_acc=0.8741 val_loss=0.589250 val_acc=0.85
07 test_loss=0.590933 test_acc=0.8596
step=42000 train_loss=0.509319 train_acc=0.9119 val_loss=0.535814 val_acc=0.89
58 test_loss=0.530462 test_acc=0.8983
step=44000 train_loss=0.575938 train_acc=0.8866 val_loss=0.623682 val_acc=0.84
00 test_loss=0.597902 test_acc=0.8803
step=46000 train_loss=0.787137 train_acc=0.7781 val_loss=0.857814 val_acc=0.73
79 test_loss=0.817314 test_acc=0.7654
step=48000 train_loss=0.480138 train_acc=0.9122 val_loss=0.508895 val_acc=0.90
87 test_loss=0.498911 test_acc=0.9098
step=50000 train_loss=0.545027 train_acc=0.8666 val_loss=0.570025 val_acc=0.85
28 test_loss=0.554296 test_acc=0.8677
step=52000 train_loss=0.613688 train_acc=0.8634 val_loss=0.604714 val_acc=0.86
57 test_loss=0.643289 test_acc=0.8446
step=54000 train_loss=0.537047 train_acc=0.8907 val_loss=0.563956 val_acc=0.87
65 test_loss=0.561439 test_acc=0.8749
step=56000 train_loss=0.661684 train_acc=0.8630 val_loss=0.676164 val_acc=0.85
39 test_loss=0.674410 test_acc=0.8494
step=58000 train_loss=0.463508 train_acc=0.9379 val_loss=0.465139 val_acc=0.94
20 test_loss=0.497776 test_acc=0.9138
```

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```
step=60000 train_loss=0.533634 train_acc=0.8905 val_loss=0.548090 val_acc=0.88
18 test_loss=0.542798 test_acc=0.8915
step=62000 train_loss=0.539815 train_acc=0.8750 val_loss=0.550979 val_acc=0.86
36 test_loss=0.551694 test_acc=0.8711
step=64000 train_loss=0.658452 train_acc=0.8443 val_loss=0.661387 val_acc=0.84
43 test_loss=0.661755 test_acc=0.8467
step=66000 train_loss=0.537937 train_acc=0.8960 val_loss=0.549452 val_acc=0.88
18 test_loss=0.559044 test_acc=0.8854
step=68000 train_loss=0.588048 train_acc=0.8677 val_loss=0.615829 val_acc=0.85
93 test_loss=0.594328 test_acc=0.8717
step=70000 train_loss=0.491868 train_acc=0.9051 val_loss=0.505796 val_acc=0.89
90 test_loss=0.505230 test_acc=0.9026
step=72000 train_loss=0.476276 train_acc=0.9240 val_loss=0.487574 val_acc=0.91
73 test_loss=0.498859 test_acc=0.9208
step=74000 train_loss=0.591847 train_acc=0.8724 val_loss=0.639429 val_acc=0.84
75 test_loss=0.631714 test_acc=0.8475
step=76000 train_loss=0.470683 train_acc=0.9302 val_loss=0.498452 val_acc=0.92
37 test_loss=0.486782 test_acc=0.9275
step=78000 train_loss=0.545403 train_acc=0.8600 val_loss=0.550902 val_acc=0.87
00 test_loss=0.558674 test_acc=0.8599
step=80000 train_loss=0.501648 train_acc=0.9012 val_loss=0.522327 val_acc=0.88
94 test_loss=0.513449 test_acc=0.8977
step=82000 train_loss=0.484829 train_acc=0.9210 val_loss=0.518405 val_acc=0.90
55 test_loss=0.516444 test_acc=0.9009
step=84000 train_loss=0.505056 train_acc=0.9016 val_loss=0.504808 val_acc=0.90
44 test_loss=0.515557 test_acc=0.8956
step=86000 train_loss=0.512107 train_acc=0.9016 val_loss=0.538630 val_acc=0.88
61 test_loss=0.519879 test_acc=0.9047
step=88000 train_loss=0.537505 train_acc=0.8883 val_loss=0.546894 val_acc=0.89
80 test_loss=0.545871 test_acc=0.8961
step=90000 train_loss=0.538099 train_acc=0.9051 val_loss=0.557120 val_acc=0.89
47 test_loss=0.555504 test_acc=0.8980
step=92000 train_loss=0.523762 train_acc=0.8950 val_loss=0.536482 val_acc=0.89
04 test_loss=0.543554 test_acc=0.8821
step=94000 train_loss=0.498021 train_acc=0.9171 val_loss=0.515247 val_acc=0.91
94 test_loss=0.521784 test_acc=0.9060
step=96000 train_loss=0.578351 train_acc=0.8716 val_loss=0.586274 val_acc=0.87
11 test_loss=0.611358 test_acc=0.8548
step=98000 train_loss=0.429086 train_acc=0.9227 val_loss=0.446863 val_acc=0.92
16 test_loss=0.446121 test_acc=0.9200
step=100000 train_loss=0.532984 train_acc=0.9061 val_loss=0.545580 val_acc=0.9
012 test_loss=0.569670 test_acc=0.8830
step=102000 train_loss=0.498781 train_acc=0.9128 val_loss=0.509184 val_acc=0.9
087 test_loss=0.517863 test_acc=0.9066
step=104000 train_loss=0.578835 train_acc=0.8675 val_loss=0.629163 val_acc=0.8
335 test_loss=0.601350 test_acc=0.8534
step=106000 train_loss=0.668678 train_acc=0.8131 val_loss=0.692791 val_acc=0.8
099 test_loss=0.691504 test_acc=0.8064
step=108000 train_loss=0.536690 train_acc=0.9025 val_loss=0.569638 val_acc=0.8
840 test_loss=0.551385 test_acc=0.8926
step=110000 train_loss=0.533262 train_acc=0.8812 val_loss=0.540302 val_acc=0.8
743 test_loss=0.550661 test_acc=0.8733
step=112000 train_loss=0.427731 train_acc=0.9414 val_loss=0.436155 val_acc=0.9
280 test_loss=0.444289 test_acc=0.9310
step=114000 train_loss=0.456064 train_acc=0.9308 val_loss=0.474159 val_acc=0.9
216 test_loss=0.472960 test_acc=0.9227
step=116000 train_loss=0.570977 train_acc=0.8546 val_loss=0.578491 val_acc=0.8
507 test_loss=0.586355 test_acc=0.8467
n_loss=0.578835 train_acc=0.8731 val_loss=0.592580 val_acc=0.8
593 test_loss=0.602423 test_acc=0.8650
```

Loading [MathJax]/extensions/Safe.js

```
step=120000 train_loss=0.535276 train_acc=0.8842 val_loss=0.567916 val_acc=0.8
604 test_loss=0.555186 test_acc=0.8668
step=122000 train_loss=0.549673 train_acc=0.8475 val_loss=0.551427 val_acc=0.8
636 test_loss=0.571062 test_acc=0.8395
step=124000 train_loss=0.726801 train_acc=0.7734 val_loss=0.750042 val_acc=0.7
787 test_loss=0.742327 test_acc=0.7699
step=126000 train_loss=0.632943 train_acc=0.8359 val_loss=0.639461 val_acc=0.8
195 test_loss=0.655645 test_acc=0.8255
step=128000 train_loss=0.483771 train_acc=0.9192 val_loss=0.489319 val_acc=0.9
205 test_loss=0.491659 test_acc=0.9227
step=130000 train_loss=0.481900 train_acc=0.9261 val_loss=0.483768 val_acc=0.9
216 test_loss=0.497043 test_acc=0.9211
step=132000 train_loss=0.416383 train_acc=0.9448 val_loss=0.427160 val_acc=0.9
441 test_loss=0.431815 test_acc=0.9412
step=134000 train_loss=0.512691 train_acc=0.8892 val_loss=0.539687 val_acc=0.8
690 test_loss=0.540466 test_acc=0.8765
step=136000 train_loss=0.413980 train_acc=0.9259 val_loss=0.419303 val_acc=0.9
259 test_loss=0.426784 test_acc=0.9219
step=138000 train_loss=0.430518 train_acc=0.9201 val_loss=0.453808 val_acc=0.9
076 test_loss=0.447730 test_acc=0.9103
step=140000 train_loss=0.724304 train_acc=0.8078 val_loss=0.754053 val_acc=0.7
852 test_loss=0.734051 test_acc=0.8046
step=142000 train_loss=0.486121 train_acc=0.9145 val_loss=0.507147 val_acc=0.9
033 test_loss=0.509532 test_acc=0.9063
step=144000 train_loss=0.480542 train_acc=0.9250 val_loss=0.504688 val_acc=0.9
087 test_loss=0.493409 test_acc=0.9224
step=146000 train_loss=0.470667 train_acc=0.9216 val_loss=0.474910 val_acc=0.9
227 test_loss=0.480705 test_acc=0.9179
step=148000 train_loss=0.500377 train_acc=0.8969 val_loss=0.522572 val_acc=0.8
980 test_loss=0.522159 test_acc=0.8891
step=150000 train_loss=0.442351 train_acc=0.9414 val_loss=0.462735 val_acc=0.9
216 test_loss=0.461715 test_acc=0.9307
step=152000 train_loss=0.514758 train_acc=0.9102 val_loss=0.531509 val_acc=0.8
990 test_loss=0.539209 test_acc=0.8945
step=154000 train_loss=0.387742 train_acc=0.9706 val_loss=0.401546 val_acc=0.9
635 test_loss=0.399666 test_acc=0.9667
step=156000 train_loss=0.511837 train_acc=0.8868 val_loss=0.537974 val_acc=0.8
808 test_loss=0.534201 test_acc=0.8760
step=158000 train_loss=0.442438 train_acc=0.9139 val_loss=0.432908 val_acc=0.9
141 test_loss=0.455277 test_acc=0.9122
step=160000 train_loss=0.566443 train_acc=0.8679 val_loss=0.570981 val_acc=0.8
668 test_loss=0.593894 test_acc=0.8561
step=162000 train_loss=0.445462 train_acc=0.9330 val_loss=0.451858 val_acc=0.9
291 test_loss=0.457394 test_acc=0.9278
step=164000 train_loss=0.466483 train_acc=0.8963 val_loss=0.462548 val_acc=0.9
076 test_loss=0.488490 test_acc=0.8867
step=166000 train_loss=0.499056 train_acc=0.8797 val_loss=0.509375 val_acc=0.8
754 test_loss=0.507248 test_acc=0.8795
step=168000 train_loss=0.506593 train_acc=0.8763 val_loss=0.518343 val_acc=0.8
690 test_loss=0.526604 test_acc=0.8623
step=170000 train_loss=0.380059 train_acc=0.9381 val_loss=0.385750 val_acc=0.9
345 test_loss=0.394043 test_acc=0.9291
step=172000 train_loss=0.641553 train_acc=0.8243 val_loss=0.623663 val_acc=0.8
453 test_loss=0.650354 test_acc=0.8153
step=174000 train_loss=0.425817 train_acc=0.9347 val_loss=0.428598 val_acc=0.9
323 test_loss=0.452668 test_acc=0.9192
step=176000 train_loss=0.416136 train_acc=0.9487 val_loss=0.426764 val_acc=0.9
452 test_loss=0.425461 test_acc=0.9477
n_loss=0.404222 train_acc=0.9319 val_loss=0.418368 val_acc=0.9
237 test_loss=0.413019 test_acc=0.9345
```

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```
step=180000 train_loss=0.499474 train_acc=0.9074 val_loss=0.486441 val_acc=0.9
141 test_loss=0.521235 test_acc=0.8934
step=182000 train_loss=0.366719 train_acc=0.9708 val_loss=0.380891 val_acc=0.9
560 test_loss=0.381877 test_acc=0.9600
step=184000 train_loss=0.385555 train_acc=0.9454 val_loss=0.380512 val_acc=0.9
549 test_loss=0.406631 test_acc=0.9307
step=186000 train_loss=0.511202 train_acc=0.8894 val_loss=0.531997 val_acc=0.8
872 test_loss=0.524017 test_acc=0.8779
step=188000 train_loss=0.374631 train_acc=0.9555 val_loss=0.393905 val_acc=0.9
388 test_loss=0.378691 test_acc=0.9519
step=190000 train_loss=0.351453 train_acc=0.9643 val_loss=0.367562 val_acc=0.9
581 test_loss=0.368049 test_acc=0.9581
step=192000 train_loss=0.320665 train_acc=0.9689 val_loss=0.333505 val_acc=0.9
624 test_loss=0.325480 test_acc=0.9670
step=194000 train_loss=0.465190 train_acc=0.9167 val_loss=0.491793 val_acc=0.8
990 test_loss=0.499691 test_acc=0.8993
step=196000 train_loss=0.494073 train_acc=0.8984 val_loss=0.507068 val_acc=0.8
883 test_loss=0.515654 test_acc=0.8872
step=198000 train_loss=0.425663 train_acc=0.9418 val_loss=0.453167 val_acc=0.9
141 test_loss=0.445056 test_acc=0.9315
step=200000 train_loss=0.391830 train_acc=0.9448 val_loss=0.397665 val_acc=0.9
334 test_loss=0.397171 test_acc=0.9442
step=202000 train_loss=0.516153 train_acc=0.8860 val_loss=0.535823 val_acc=0.8
808 test_loss=0.527946 test_acc=0.8792
step=204000 train_loss=0.525700 train_acc=0.8726 val_loss=0.512094 val_acc=0.8
861 test_loss=0.523961 test_acc=0.8733
step=206000 train_loss=0.380644 train_acc=0.9465 val_loss=0.387092 val_acc=0.9
356 test_loss=0.385188 test_acc=0.9482
step=208000 train_loss=0.474421 train_acc=0.9109 val_loss=0.498453 val_acc=0.8
969 test_loss=0.496143 test_acc=0.8991
step=210000 train_loss=0.379959 train_acc=0.9416 val_loss=0.376274 val_acc=0.9
495 test_loss=0.378910 test_acc=0.9420
step=212000 train_loss=0.319209 train_acc=0.9716 val_loss=0.311654 val_acc=0.9
742 test_loss=0.330141 test_acc=0.9681
step=214000 train_loss=0.387244 train_acc=0.9409 val_loss=0.392728 val_acc=0.9
366 test_loss=0.399109 test_acc=0.9340
step=216000 train_loss=0.414658 train_acc=0.9184 val_loss=0.411163 val_acc=0.9
194 test_loss=0.433240 test_acc=0.9098
step=218000 train_loss=0.436502 train_acc=0.9197 val_loss=0.443517 val_acc=0.9
141 test_loss=0.450653 test_acc=0.9128
step=220000 train_loss=0.484141 train_acc=0.8823 val_loss=0.494333 val_acc=0.8
872 test_loss=0.513052 test_acc=0.8639
step=222000 train_loss=0.406423 train_acc=0.9338 val_loss=0.411641 val_acc=0.9
388 test_loss=0.415428 test_acc=0.9289
step=224000 train_loss=0.469953 train_acc=0.9072 val_loss=0.482009 val_acc=0.9
001 test_loss=0.505862 test_acc=0.8886
step=226000 train_loss=0.309319 train_acc=0.9766 val_loss=0.318354 val_acc=0.9
764 test_loss=0.321474 test_acc=0.9766
step=228000 train_loss=0.362769 train_acc=0.9482 val_loss=0.368841 val_acc=0.9
441 test_loss=0.387287 test_acc=0.9332
step=230000 train_loss=0.338646 train_acc=0.9474 val_loss=0.335277 val_acc=0.9
538 test_loss=0.339592 test_acc=0.9493
step=232000 train_loss=0.304045 train_acc=0.9815 val_loss=0.320794 val_acc=0.9
646 test_loss=0.319665 test_acc=0.9777
step=234000 train_loss=0.257099 train_acc=0.9895 val_loss=0.259460 val_acc=0.9
882 test_loss=0.270107 test_acc=0.9863
step=236000 train_loss=0.380816 train_acc=0.9467 val_loss=0.385980 val_acc=0.9
398 test_loss=0.393036 test_acc=0.9439
n_loss=0.418375 train_acc=0.9053 val_loss=0.411849 val_acc=0.9
076 test_loss=0.422248 test_acc=0.9103
```

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```
step=240000 train_loss=0.292725 train_acc=0.9764 val_loss=0.298596 val_acc=0.9
785 test_loss=0.294897 test_acc=0.9817
step=242000 train_loss=0.388374 train_acc=0.9347 val_loss=0.392197 val_acc=0.9
366 test_loss=0.389509 test_acc=0.9412
step=244000 train_loss=0.303513 train_acc=0.9708 val_loss=0.320530 val_acc=0.9
689 test_loss=0.315644 test_acc=0.9662
step=246000 train_loss=0.331143 train_acc=0.9768 val_loss=0.334776 val_acc=0.9
753 test_loss=0.348576 test_acc=0.9715
step=248000 train_loss=0.323489 train_acc=0.9635 val_loss=0.356735 val_acc=0.9
398 test_loss=0.332449 test_acc=0.9659
step=250000 train_loss=0.327555 train_acc=0.9624 val_loss=0.341023 val_acc=0.9
603 test_loss=0.338230 test_acc=0.9597
step=252000 train_loss=0.283703 train_acc=0.9800 val_loss=0.293743 val_acc=0.9
796 test_loss=0.304075 test_acc=0.9683
step=254000 train_loss=0.296256 train_acc=0.9843 val_loss=0.308449 val_acc=0.9
764 test_loss=0.307933 test_acc=0.9777
step=256000 train_loss=0.341778 train_acc=0.9708 val_loss=0.357448 val_acc=0.9
678 test_loss=0.359677 test_acc=0.9675
step=258000 train_loss=0.341792 train_acc=0.9622 val_loss=0.369555 val_acc=0.9
463 test_loss=0.352079 test_acc=0.9611
step=260000 train_loss=0.311819 train_acc=0.9607 val_loss=0.306560 val_acc=0.9
764 test_loss=0.328130 test_acc=0.9536
step=262000 train_loss=0.325223 train_acc=0.9787 val_loss=0.331424 val_acc=0.9
817 test_loss=0.334722 test_acc=0.9769
step=264000 train_loss=0.254661 train_acc=0.9925 val_loss=0.264747 val_acc=0.9
903 test_loss=0.260178 test_acc=0.9941
step=266000 train_loss=0.275052 train_acc=0.9882 val_loss=0.295212 val_acc=0.9
828 test_loss=0.285173 test_acc=0.9847
step=268000 train_loss=0.294477 train_acc=0.9759 val_loss=0.307326 val_acc=0.9
678 test_loss=0.302928 test_acc=0.9766
step=270000 train_loss=0.352202 train_acc=0.9669 val_loss=0.360770 val_acc=0.9
646 test_loss=0.366932 test_acc=0.9638
step=272000 train_loss=0.282176 train_acc=0.9871 val_loss=0.292784 val_acc=0.9
817 test_loss=0.291379 test_acc=0.9860
step=274000 train_loss=0.306317 train_acc=0.9796 val_loss=0.340073 val_acc=0.9
603 test_loss=0.323942 test_acc=0.9710
step=276000 train_loss=0.413419 train_acc=0.9240 val_loss=0.406730 val_acc=0.9
323 test_loss=0.444056 test_acc=0.9077
step=278000 train_loss=0.285251 train_acc=0.9854 val_loss=0.289057 val_acc=0.9
850 test_loss=0.294233 test_acc=0.9834
step=280000 train_loss=0.365584 train_acc=0.9485 val_loss=0.391312 val_acc=0.9
356 test_loss=0.385080 test_acc=0.9417
step=282000 train_loss=0.283755 train_acc=0.9783 val_loss=0.299855 val_acc=0.9
710 test_loss=0.289137 test_acc=0.9764
step=284000 train_loss=0.446987 train_acc=0.9259 val_loss=0.476241 val_acc=0.9
162 test_loss=0.471399 test_acc=0.9256
step=286000 train_loss=0.318323 train_acc=0.9699 val_loss=0.326710 val_acc=0.9
613 test_loss=0.328585 test_acc=0.9664
step=288000 train_loss=0.279564 train_acc=0.9843 val_loss=0.282539 val_acc=0.9
785 test_loss=0.295629 test_acc=0.9753
step=290000 train_loss=0.351022 train_acc=0.9725 val_loss=0.372481 val_acc=0.9
635 test_loss=0.369535 test_acc=0.9627
step=292000 train_loss=0.404341 train_acc=0.9233 val_loss=0.428616 val_acc=0.9
012 test_loss=0.427339 test_acc=0.9179
step=294000 train_loss=0.317017 train_acc=0.9626 val_loss=0.321934 val_acc=0.9
581 test_loss=0.330466 test_acc=0.9579
step=296000 train_loss=0.277748 train_acc=0.9757 val_loss=0.292067 val_acc=0.9
678 test_loss=0.296078 test_acc=0.9699
n_loss=0.253333 train_acc=0.9800 val_loss=0.265457 val_acc=0.9
721 test_loss=0.265856 test_acc=0.9723
```

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```
step=300000 train_loss=0.238580 train_acc=0.9888 val_loss=0.248823 val_acc=0.9
839 test_loss=0.252364 test_acc=0.9847
step=302000 train_loss=0.217300 train_acc=0.9927 val_loss=0.220579 val_acc=0.9
903 test_loss=0.220286 test_acc=0.9938
step=304000 train_loss=0.279936 train_acc=0.9759 val_loss=0.294061 val_acc=0.9
721 test_loss=0.293424 test_acc=0.9737
step=306000 train_loss=0.405770 train_acc=0.9298 val_loss=0.397265 val_acc=0.9
323 test_loss=0.419234 test_acc=0.9203
step=308000 train_loss=0.353428 train_acc=0.9495 val_loss=0.368384 val_acc=0.9
495 test_loss=0.374275 test_acc=0.9383
step=310000 train_loss=0.225234 train_acc=0.9972 val_loss=0.228598 val_acc=0.9
957 test_loss=0.235224 test_acc=0.9930
step=312000 train_loss=0.267761 train_acc=0.9770 val_loss=0.269008 val_acc=0.9
774 test_loss=0.282457 test_acc=0.9723
step=314000 train_loss=0.313876 train_acc=0.9650 val_loss=0.324550 val_acc=0.9
656 test_loss=0.333079 test_acc=0.9581
step=316000 train_loss=0.476239 train_acc=0.8761 val_loss=0.484436 val_acc=0.8
700 test_loss=0.508002 test_acc=0.8604
step=318000 train_loss=0.254074 train_acc=0.9807 val_loss=0.274262 val_acc=0.9
699 test_loss=0.265908 test_acc=0.9785
step=320000 train_loss=0.238026 train_acc=0.9783 val_loss=0.248232 val_acc=0.9
764 test_loss=0.249099 test_acc=0.9745
step=322000 train_loss=0.228449 train_acc=0.9856 val_loss=0.239053 val_acc=0.9
828 test_loss=0.237991 test_acc=0.9801
step=324000 train_loss=0.204560 train_acc=0.9910 val_loss=0.200936 val_acc=0.9
957 test_loss=0.210595 test_acc=0.9901
step=326000 train_loss=0.248122 train_acc=0.9777 val_loss=0.256125 val_acc=0.9
774 test_loss=0.255732 test_acc=0.9777
step=328000 train_loss=0.212262 train_acc=0.9916 val_loss=0.221649 val_acc=0.9
925 test_loss=0.227456 test_acc=0.9858
step=330000 train_loss=0.242642 train_acc=0.9869 val_loss=0.245786 val_acc=0.9
839 test_loss=0.253748 test_acc=0.9868
step=332000 train_loss=0.293951 train_acc=0.9583 val_loss=0.306258 val_acc=0.9
603 test_loss=0.314884 test_acc=0.9493
step=334000 train_loss=0.165053 train_acc=0.9987 val_loss=0.168349 val_acc=0.9
979 test_loss=0.170140 test_acc=0.9989
step=336000 train_loss=0.182017 train_acc=0.9959 val_loss=0.191080 val_acc=0.9
914 test_loss=0.189180 test_acc=0.9938
step=338000 train_loss=0.241313 train_acc=0.9667 val_loss=0.244449 val_acc=0.9
635 test_loss=0.251992 test_acc=0.9648
step=340000 train_loss=0.225407 train_acc=0.9817 val_loss=0.231430 val_acc=0.9
807 test_loss=0.238848 test_acc=0.9783
step=342000 train_loss=0.185917 train_acc=0.9931 val_loss=0.192271 val_acc=0.9
925 test_loss=0.196245 test_acc=0.9906
step=344000 train_loss=0.218418 train_acc=0.9848 val_loss=0.244031 val_acc=0.9
678 test_loss=0.232416 test_acc=0.9764
step=346000 train_loss=0.235371 train_acc=0.9835 val_loss=0.242779 val_acc=0.9
774 test_loss=0.248781 test_acc=0.9791
step=348000 train_loss=0.208936 train_acc=0.9886 val_loss=0.218095 val_acc=0.9
882 test_loss=0.218216 test_acc=0.9890
step=350000 train_loss=0.183976 train_acc=0.9888 val_loss=0.190432 val_acc=0.9
882 test_loss=0.192335 test_acc=0.9879
step=352000 train_loss=0.177903 train_acc=0.9921 val_loss=0.189006 val_acc=0.9
936 test_loss=0.183805 test_acc=0.9941
step=354000 train_loss=0.214559 train_acc=0.9811 val_loss=0.229904 val_acc=0.9
764 test_loss=0.228922 test_acc=0.9750
step=356000 train_loss=0.210712 train_acc=0.9805 val_loss=0.221699 val_acc=0.9
753 test_loss=0.223012 test_acc=0.9753
n_loss=0.192365 train_acc=0.9923 val_loss=0.202950 val_acc=0.9
860 test_loss=0.200235 test_acc=0.9901
```

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```
step=360000 train_loss=0.150154 train_acc=0.9966 val_loss=0.153062 val_acc=0.9
989 test_loss=0.159659 test_acc=0.9933
step=362000 train_loss=0.189191 train_acc=0.9931 val_loss=0.194875 val_acc=0.9
925 test_loss=0.197586 test_acc=0.9911
step=364000 train_loss=0.159258 train_acc=0.9903 val_loss=0.168862 val_acc=0.9
893 test_loss=0.163275 test_acc=0.9877
step=366000 train_loss=0.486574 train_acc=0.9104 val_loss=0.467761 val_acc=0.9
162 test_loss=0.510609 test_acc=0.9060
step=368000 train_loss=0.172305 train_acc=0.9869 val_loss=0.180307 val_acc=0.9
828 test_loss=0.182128 test_acc=0.9828
step=370000 train_loss=0.156416 train_acc=0.9957 val_loss=0.166702 val_acc=0.9
914 test_loss=0.164547 test_acc=0.9917
step=372000 train_loss=0.170947 train_acc=0.9890 val_loss=0.182639 val_acc=0.9
860 test_loss=0.177984 test_acc=0.9879
step=374000 train_loss=0.153393 train_acc=0.9910 val_loss=0.155238 val_acc=0.9
903 test_loss=0.166703 test_acc=0.9852
step=376000 train_loss=0.160560 train_acc=0.9914 val_loss=0.168667 val_acc=0.9
893 test_loss=0.167028 test_acc=0.9922
step=378000 train_loss=0.136881 train_acc=0.9970 val_loss=0.141822 val_acc=0.9
968 test_loss=0.141558 test_acc=0.9970
step=380000 train_loss=0.144860 train_acc=0.9953 val_loss=0.146094 val_acc=0.9
979 test_loss=0.152488 test_acc=0.9941
step=382000 train_loss=0.123147 train_acc=0.9994 val_loss=0.132262 val_acc=0.9
946 test_loss=0.132278 test_acc=0.9970
step=384000 train_loss=0.179191 train_acc=0.9961 val_loss=0.194237 val_acc=0.9
946 test_loss=0.185019 test_acc=0.9941
step=386000 train_loss=0.141389 train_acc=0.9987 val_loss=0.142950 val_acc=0.9
989 test_loss=0.144946 test_acc=0.9995
step=388000 train_loss=0.179809 train_acc=0.9837 val_loss=0.177600 val_acc=0.9
903 test_loss=0.192121 test_acc=0.9793
step=390000 train_loss=0.124711 train_acc=0.9989 val_loss=0.131573 val_acc=0.9
968 test_loss=0.132413 test_acc=0.9973
step=392000 train_loss=0.156593 train_acc=0.9951 val_loss=0.165228 val_acc=0.9
925 test_loss=0.165932 test_acc=0.9925
step=394000 train_loss=0.156528 train_acc=0.9942 val_loss=0.164057 val_acc=0.9
893 test_loss=0.165025 test_acc=0.9917
step=396000 train_loss=0.149469 train_acc=0.9972 val_loss=0.159463 val_acc=0.9
946 test_loss=0.163546 test_acc=0.9938
step=398000 train_loss=0.180183 train_acc=0.9828 val_loss=0.201099 val_acc=0.9
731 test_loss=0.199785 test_acc=0.9761
step=400000 train_loss=0.157333 train_acc=0.9976 val_loss=0.164377 val_acc=0.9
968 test_loss=0.168855 test_acc=0.9938
step=402000 train_loss=0.253099 train_acc=0.9467 val_loss=0.265528 val_acc=0.9
474 test_loss=0.266972 test_acc=0.9391
step=404000 train_loss=0.141577 train_acc=0.9985 val_loss=0.151538 val_acc=0.9
968 test_loss=0.147209 test_acc=0.9984
step=406000 train_loss=0.135898 train_acc=0.9994 val_loss=0.144510 val_acc=0.9
989 test_loss=0.145306 test_acc=0.9997
step=408000 train_loss=0.157467 train_acc=0.9869 val_loss=0.164549 val_acc=0.9
860 test_loss=0.166146 test_acc=0.9852
step=410000 train_loss=0.178443 train_acc=0.9966 val_loss=0.183719 val_acc=0.9
989 test_loss=0.187233 test_acc=0.9968
step=412000 train_loss=0.187637 train_acc=0.9921 val_loss=0.199251 val_acc=0.9
914 test_loss=0.201242 test_acc=0.9890
step=414000 train_loss=0.193239 train_acc=0.9938 val_loss=0.195845 val_acc=0.9
936 test_loss=0.203222 test_acc=0.9930
step=416000 train_loss=0.238918 train_acc=0.9875 val_loss=0.248707 val_acc=0.9
871 test_loss=0.247219 test_acc=0.9871
n_loss=0.252937 train_acc=0.9860 val_loss=0.259741 val_acc=0.9
871 test_loss=0.267089 test_acc=0.9847
```

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```
step=420000 train_loss=0.299957 train_acc=0.9757 val_loss=0.303275 val_acc=0.9
785 test_loss=0.314361 test_acc=0.9710
step=422000 train_loss=0.401744 train_acc=0.9549 val_loss=0.419288 val_acc=0.9
452 test_loss=0.418016 test_acc=0.9557
step=424000 train_loss=0.303501 train_acc=0.9878 val_loss=0.327741 val_acc=0.9
753 test_loss=0.324297 test_acc=0.9834
step=426000 train_loss=0.345447 train_acc=0.9807 val_loss=0.354983 val_acc=0.9
796 test_loss=0.359658 test_acc=0.9785
step=428000 train_loss=0.364971 train_acc=0.9785 val_loss=0.372916 val_acc=0.9
742 test_loss=0.390032 test_acc=0.9681
step=430000 train_loss=0.601131 train_acc=0.8799 val_loss=0.619524 val_acc=0.8
808 test_loss=0.632374 test_acc=0.8698
step=432000 train_loss=0.422285 train_acc=0.9779 val_loss=0.441300 val_acc=0.9
689 test_loss=0.444028 test_acc=0.9689
step=434000 train_loss=0.487430 train_acc=0.9622 val_loss=0.514189 val_acc=0.9
441 test_loss=0.518772 test_acc=0.9493
step=436000 train_loss=0.555010 train_acc=0.9268 val_loss=0.589760 val_acc=0.9
033 test_loss=0.581055 test_acc=0.9227
step=438000 train_loss=0.875889 train_acc=0.7629 val_loss=0.917338 val_acc=0.7
347 test_loss=0.894598 test_acc=0.7565
step=440000 train_loss=0.515965 train_acc=0.9661 val_loss=0.541982 val_acc=0.9
517 test_loss=0.545879 test_acc=0.9498
step=442000 train_loss=0.513187 train_acc=0.9671 val_loss=0.532416 val_acc=0.9
592 test_loss=0.547417 test_acc=0.9536
step=444000 train_loss=0.577600 train_acc=0.9463 val_loss=0.625243 val_acc=0.9
087 test_loss=0.603561 test_acc=0.9310
step=446000 train_loss=0.535373 train_acc=0.9553 val_loss=0.556537 val_acc=0.9
484 test_loss=0.565097 test_acc=0.9409
step=448000 train_loss=0.881304 train_acc=0.7762 val_loss=0.934065 val_acc=0.7
562 test_loss=0.940512 test_acc=0.7557
step=450000 train_loss=0.487292 train_acc=0.9714 val_loss=0.508930 val_acc=0.9
603 test_loss=0.520964 test_acc=0.9554
step=452000 train_loss=0.489965 train_acc=0.9648 val_loss=0.514299 val_acc=0.9
441 test_loss=0.527447 test_acc=0.9490
step=454000 train_loss=0.508881 train_acc=0.9568 val_loss=0.535996 val_acc=0.9
441 test_loss=0.543605 test_acc=0.9428
step=456000 train_loss=0.499613 train_acc=0.9592 val_loss=0.530053 val_acc=0.9
484 test_loss=0.532886 test_acc=0.9460
step=458000 train_loss=0.658855 train_acc=0.8754 val_loss=0.709193 val_acc=0.8
464 test_loss=0.735809 test_acc=0.8440
step=460000 train_loss=0.522535 train_acc=0.9790 val_loss=0.556474 val_acc=0.9
613 test_loss=0.558711 test_acc=0.9646
step=462000 train_loss=0.455155 train_acc=0.9860 val_loss=0.483060 val_acc=0.9
839 test_loss=0.486950 test_acc=0.9826
step=464000 train_loss=0.595616 train_acc=0.9274 val_loss=0.636049 val_acc=0.9
184 test_loss=0.635273 test_acc=0.9125
step=466000 train_loss=0.441818 train_acc=0.9663 val_loss=0.465545 val_acc=0.9
517 test_loss=0.475147 test_acc=0.9554
step=468000 train_loss=0.430866 train_acc=0.9880 val_loss=0.454987 val_acc=0.9
753 test_loss=0.461062 test_acc=0.9769
step=470000 train_loss=0.561145 train_acc=0.9577 val_loss=0.590698 val_acc=0.9
495 test_loss=0.600025 test_acc=0.9452
step=472000 train_loss=0.446018 train_acc=0.9616 val_loss=0.455677 val_acc=0.9
592 test_loss=0.474003 test_acc=0.9501
step=474000 train_loss=0.446137 train_acc=0.9639 val_loss=0.471534 val_acc=0.9
560 test_loss=0.478523 test_acc=0.9541
step=476000 train_loss=0.425935 train_acc=0.9948 val_loss=0.443075 val_acc=0.9
914 test_loss=0.449766 test_acc=0.9874
n_loss=0.382314 train_acc=0.9933 val_loss=0.401274 val_acc=0.9
893 test_loss=0.410464 test_acc=0.9842
```

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```
step=480000 train_loss=0.415346 train_acc=0.9828 val_loss=0.448878 val_acc=0.9
656 test_loss=0.449430 test_acc=0.9654
step=482000 train_loss=0.476277 train_acc=0.9654 val_loss=0.518886 val_acc=0.9
398 test_loss=0.512324 test_acc=0.9485
step=484000 train_loss=0.450407 train_acc=0.9751 val_loss=0.481707 val_acc=0.9
506 test_loss=0.482805 test_acc=0.9638
step=486000 train_loss=0.380750 train_acc=0.9942 val_loss=0.402499 val_acc=0.9
914 test_loss=0.406818 test_acc=0.9909
step=488000 train_loss=0.412932 train_acc=0.9860 val_loss=0.423196 val_acc=0.9
850 test_loss=0.445767 test_acc=0.9769
step=490000 train_loss=0.381972 train_acc=0.9942 val_loss=0.405501 val_acc=0.9
850 test_loss=0.408725 test_acc=0.9903
step=492000 train_loss=0.405353 train_acc=0.9929 val_loss=0.434718 val_acc=0.9
850 test_loss=0.434314 test_acc=0.9844
step=494000 train_loss=0.404893 train_acc=0.9777 val_loss=0.425574 val_acc=0.9
742 test_loss=0.433425 test_acc=0.9670
step=496000 train_loss=0.459177 train_acc=0.9618 val_loss=0.499354 val_acc=0.9
452 test_loss=0.495085 test_acc=0.9514
step=498000 train_loss=0.419892 train_acc=0.9860 val_loss=0.441860 val_acc=0.9
817 test_loss=0.449799 test_acc=0.9777
step=500000 train_loss=0.438212 train_acc=0.9755 val_loss=0.457854 val_acc=0.9
646 test_loss=0.472094 test_acc=0.9621
best metrics: {
  "step": 390000,
  "lr": 0.0004340370415046953,
  "train_loss": 0.12471106392411432,
  "train_acc": 0.998926116838488,
  "val_loss": 0.13157309707010087,
  "val_acc": 0.9967776584317938,
  "test_loss": 0.13241256498850432,
  "test_acc": 0.9973154362416108,
  "elapsed_sec": 5740.155249118805
}
checkpoint dir: /Users/shobhit/Documents/UW/CSE 493/cse493s-spring26-hw1/out/s
ubmission/part1/ablations/abl_adamw_wd1_trainfrac0p5
ablation runs configured: 2 executed now: 2
```

In [19]:

```
def first_step_at_or_above(df: pd.DataFrame, col: str, threshold: float):
    s = df.loc[df[col] >= threshold, "step"]
    return int(s.iloc[0]) if not s.empty else None

def grokking_delay(history_df: pd.DataFrame, train_thr: float, test_thr: float
    train_step = first_step_at_or_above(history_df, "train_acc", train_thr)
    if train_step is None:
        return train_step, None, None

    after = history_df[history_df["step"] >= train_step]
    s = after.loc[after["test_acc"] >= test_thr, "step"]
    test_step = int(s.iloc[0]) if not s.empty else None
    delay = None if test_step is None else test_step - train_step
    return train_step, test_step, delay

rows = []
all_runs = [("baseline_div_p97", baseline_div_dir)] + [
    (ab["name"], PART1_DIR / "ablations" / ab["name"]) for ab in ablation_runs
]

for run_name, run_dir in all_runs:
    history(run_dir)
```

Loading [MathJax]/extensions/Safe.js

```
    train_1, test_1, delay_1 = grokking_delay(h, train_thr=1.0, test_thr=1.0)
    train_99, test_99, delay_99 = grokking_delay(h, train_thr=0.99, test_thr=0

    rows.append(
        {
            "run": run_name,
            "train_acc_1_step": train_1,
            "test_acc_1_step": test_1,
            "grokking_delay_steps_strict": delay_1,
            "train_acc_99_step": train_99,
            "test_acc_99_step": test_99,
            "grokking_delay_steps_99": delay_99,
        }
    )

ablation_df = pd.DataFrame(rows)
ablation_df.to_csv(TABLES_DIR / "1_4_ablation_comparison.csv", index=False)
display(ablation_df)
```

	run	train_acc_1_step	test_acc_1_step	grokking_delay_steps_strict	t
0	baseline_div_p97	NaN	NaN	NaN	
1	abl_adamw_wd1_trainfrac0p3	222000.0	270000.0	48000.0	
2	abl_adamw_wd1_trainfrac0p5	NaN	NaN	NaN	

In [ ]:

```
# Overlay baseline + ablations on train/val/test accuracy
baseline_hist = load_history(baseline_div_dir)

fig, axes = plt.subplots(1, 3, figsize=(16, 4), sharex=True, sharey=True)

series = [
    ("train_acc", "Train Accuracy"),
    ("val_acc", "Val Accuracy"),
    ("test_acc", "Test Accuracy"),
]

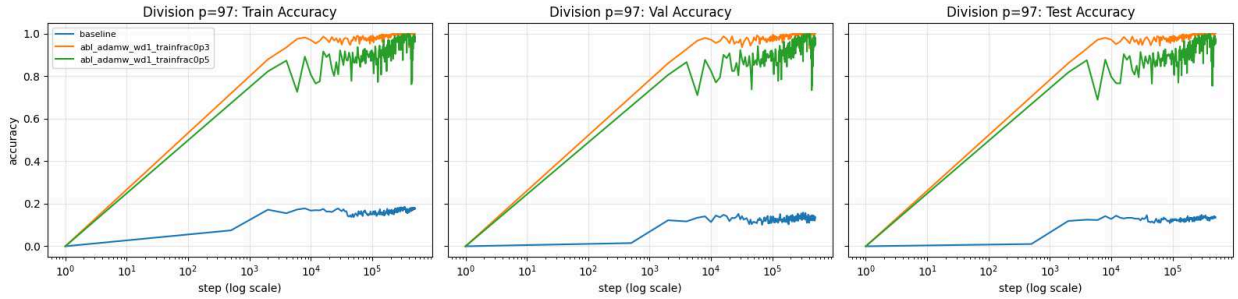
all_runs = [("baseline", baseline_hist)]
for ab in ablation_runs:
    run_dir = PART1_DIR / "ablations" / ab["name"]
    all_runs.append((ab["name"], load_history(run_dir)))

for ax, (col, title) in zip(axes, series):
    for name, h in all_runs:
        ax.plot(h["step"], h[col], label=name)
    ax.set_title(f"Division p=97: {title}")
    ax.set_xscale("log")
    ax.set_xlabel("step (log scale)")
    ax.grid(alpha=0.3)

axes[0].set_ylabel("accuracy")
axes[0].legend(fontsize=8)

fig.tight_layout()
fig.savefig(PLOTS_DIR / "1_4_ablation_overlay.png", dpi=180)
plt.show()
```

Loading [MathJax]/extensions/Safe.js



1.4 Write-Up

We test paper style regularization/optimizer settings for faster and more reliable grokking.

Ablation A: `abl_adamw_wd1_trainfrac0p3` (`AdamW` , `weight_decay=1.0` , `train_frac=0.3`)

- Relaxed criterion (`>=0.99`):
  - `train_acc>=0.99` at step `58,000`
  - `test_acc>=0.99` at step `102,000`
  - delay = `44,000` steps
- Strict criterion (`=1.0`):
  - `train_acc=1.0` at step `222,000`
  - `test_acc=1.0` at step `270,000`
  - delay = `48,000` steps
- This shows grokking behavior

Ablation B: `abl_adamw_wd1_trainfrac0p5` (`AdamW` , `weight_decay=1.0` , `train_frac=0.5`)

- Relaxed criterion (`>=0.99`):
  - `train_acc>=0.99` and `test_acc>=0.99` both at step `264,000`
  - delay = `0` steps
- Strict criterion (`=1.0`) wasn't reached by `500,000` steps
- Peak test accuracy was quite high (`0.9997` at step `406,000`), but final test accuracy at step `500,000` dropped to `0.9621`

Conclusion:

- Compared to the baseline, `AdamW + weight_decay=1.0` dramatically improved generalization dynamics and showed signs of grokking.
- `train_frac=0.3` produced the cleanest delayed generalization signature which is clear train first, test-later transition.
- `train_frac=0.5` achieved strong generalization too but with less clean strict-criterion convergence by the end of the run

In [ ]:

Loading [MathJax]/extensions/Safe.js

Part 2 Test-Time Scaling

some outputs are pasted due to runtime in a separate core for efficiency

In [1]:
```
import re

import pandas as pd
import torch
from datasets import load_dataset
from transformers import AutoModelForCausalLM, AutoTokenizer

# !pip install vllm==0.20.0
from vllm import LLM, SamplingParams

import gc
import torch

gc.collect()
torch.cuda.empty_cache()
```

In [2]:
```
import os

HF_TOKEN = os.environ.get("HF_TOKEN")
```

In [3]:
```
MODEL_NAME = "Qwen/Qwen3-4B"
# or allenai/Olmo-3-7B-Thinking
DATASET_NAME = "OpenRLHF/aime-2024"
MAX_NEW_TOKENS = 34000
```

Loading the model and the data

In [4]:
```
import os
import sys

# Using vLLM instead for faster inference
dtype = "float16" if torch.cuda.is_available() else "float32"
llm = LLM(model=MODEL_NAME, dtype=dtype, max_model_len=MAX_NEW_TOKENS, hf_token=HF_TOKEN)

tokenizer = llm.get_tokenizer()

dataset = load_dataset(DATASET_NAME, split="train", token=HF_TOKEN)
```

```
INFO 05-02 17:11:12 [utils.py:233] non-default args: {'dtype': 'float16', 'max_model_len': 34000, 'disable_log_stats': True,
'hf_token': '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', 'model': 'Qwen/Qwen3-4B'}
INFO 05-02 17:11:12 [model.py:555] Resolved architecture: Qwen3ForCausalLM
WARNING 05-02 17:11:12 [model.py:2018] Casting torch.bfloat16 to torch.float16.
INFO 05-02 17:11:12 [model.py:1680] Using max model len 34000
INFO 05-02 17:11:12 [scheduler.py:239] Chunked prefill is enabled with max_num_batched_tokens=8192.
INFO 05-02 17:11:12 [vllm.py:840] Asynchronous scheduling is enabled.
INFO 05-02 17:11:12 [kernel.py:205] Final IR op priority after setting platform defaults: IrOpPriorityConfig(rms_norm=['nativ
e'])
(EngineCore pid=101416) INFO 05-02 17:11:14 [core.py:109] Initializing a V1 LLM engine (v0.20.0) with config: model='Qwen/Qwe
n3-4B', speculative_config=None, tokenizer='Qwen/Qwen3-4B', skip_tokenizer_init=False, tokenizer_mode=auto, revision=None, to
kenizer_revision=None, trust_remote_code=False, dtype=torch.float16, max_seq_len=34000, download_dir=None, load_format=auto,
tensor_parallel_size=1, pipeline_parallel_size=1, data_parallel_size=1, decode_context_parallel_size=1, dcp_comm_backend=ag_r
s, disable_custom_all_reduce=False, quantization=None, quantization_config=None, enforce_eager=False, enable_return_routed_ex
perts=False, kv_cache_dtype=auto, device_config=cuda, structured_outputs_config=StructuredOutputsConfig(backend='auto', disab
le_any_whitespace=False, disable_additional_properties=False, reasoning_parser='', reasoning_parser_plugin='', enable_in_reas
oning=False), observability_config=ObservabilityConfig(show_hidden_metrics_for_version=None, otlp_traces_endpoint=None, colle
ct_detailed_traces=None, kv_cache_metrics=False, kv_cache_metrics_sample=0.01, cudagraph_metrics=False, enable_layerwise_nvtx
_tracing=False, enable_mfu_metrics=False, enable_mm_processor_stats=False, enable_logging_iteration_details=False), seed=0, s
erved_model_name=Qwen/Qwen3-4B, enable_prefix_caching=True, enable_chunked_prefill=True, pooler_config=None, compilation_conf
ig={'mode': <CompilationMode.VLLM_COMPILE: 3>, 'debug_dump_path': None, 'cache_dir': '', 'compile_cache_save_format': 'binar
y', 'backend': 'inductor', 'custom_ops': ['none'], 'ir_enable_torch_wrap': True, 'splitting_ops': ['vllm::unified_attention_w
ith_output', 'vllm::unified_mla_attention_with_output', 'vllm::mamba_mixer2', 'vllm::mamba_mixer', 'vllm::short_conv', 'vll
m::linear_attention', 'vllm::plamo2_mamba_mixer', 'vllm::gdn_attention_core', 'vllm::gdn_attention_core_xpu', 'vllm::olmo_hyb
rid_gdn_full_forward', 'vllm::kda_attention', 'vllm::sparse_attn_indexer', 'vllm::rocm_aiter_sparse_attn_indexer', 'vllm::dee
pseek_v4_attention', 'vllm::unified_kv_cache_update', 'vllm::unified_mla_kv_cache_update'], 'compile_mm_encoder': False, 'cud
agraph_mm_encoder': False, 'encoder_cudagraph_token_budgets': [], 'encoder_cudagraph_max_vision_items_per_batch': 0, 'encoder
_cudagraph_max_frames_per_batch': None, 'compile_sizes': [], 'compile_ranges_endpoints': [8192], 'inductor_compile_config':
{'enable_auto_functionalized_v2': False, 'size_asserts': False, 'alignment_asserts': False, 'scalar_asserts': False, 'combo_k
ernels': True, 'benchmark_combo_kernel': True}, 'inductor_passes': {}, 'cudagraph_mode': <CUDAGraphMode.FULL_AND_PIECEWISE:
(2, 1)>, 'cudagraph_num_of_warmups': 1, 'cudagraph_capture_sizes': [1, 2, 4, 8, 16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 1
04, 112, 120, 128, 136, 144, 152, 160, 168, 176, 184, 192, 200, 208, 216, 224, 232, 240, 248, 256, 272, 288, 304, 320, 336, 3
52, 368, 384, 400, 416, 432, 448, 464, 480, 496, 512], 'cudagraph_copy_inputs': False, 'cudagraph_specialize_lora': True, 'us
e_inductor_graph_partition': False, 'pass_config': {'fuse_norm_quant': False, 'fuse_act_quant': False, 'fuse_attn_quant': Fal
se, 'enable_sp': False, 'fuse_gemm_comms': False, 'fuse_allreduce_rms': False}, 'max_cudagraph_capture_size': 512, 'dynamic_s
hapes_config': {'type': <DynamicShapesType.BACKED: 'backed'>, 'evaluate_guards': False, 'assume_32_bit_indexing': False}, 'lo
cal_cache_dir': None, 'fast_moe_cold_start': False, 'static_all_moe_layers': []}, kernel_config=KernelConfig(ir_op_priority=I
rOpPriorityConfig(rms_norm=['native']), enable_flashinfer_autotune=True, moe_backend='auto')
(EngineCore pid=101416) INFO 05-02 17:11:14 [nixl_utils.py:20] Setting UCX_RCACHE_MAX_UNRELEASED to '1024' to avoid a rare me
mory leak in UCX when using NIXL.
(EngineCore pid=101416) WARNING 05-02 17:11:14 [nixl_utils.py:34] NIXL is not available
(EngineCore pid=101416) WARNING 05-02 17:11:14 [nixl_utils.py:44] NIXL agent config is not available
(EngineCore pid=101416) INFO 05-02 17:11:15 [parallel_state.py:1402] world_size=1 rank=0 local_rank=0 distributed_init_method
=tcp://10.192.11.34:60289 backend=nccl
(EngineCore pid=101416) INFO 05-02 17:11:15 [parallel_state.py:1715] rank 0 in world size 1 is assigned as DP rank 0, PP rank
0, PCP rank 0, TP rank 0, EP rank N/A, EPLB rank N/A
(EngineCore pid=101416) INFO 05-02 17:11:15 [gpu_model_runner.py:4777] Starting to load model Qwen/Qwen3-4B...
(EngineCore pid=101416) INFO 05-02 17:11:16 [cuda.py:368] Using FLASH_ATTN attention backend out of potential backends: ['FLA
SH_ATTN', 'FLASHINFER', 'TRITON_ATTN', 'FLEX_ATTENTION'].
(EngineCore pid=101416) INFO 05-02 17:11:16 [flash_attn.py:646] Using FlashAttention version 2
(EngineCore pid=101416) INFO 05-02 17:11:17 [weight_utils.py:904] Filesystem type for checkpoints: LIGHTNING. Checkpoint siz
e: 7.49 GiB. Available RAM: 54.35 GiB.
(EngineCore pid=101416) INFO 05-02 17:11:17 [weight_utils.py:927] Auto-prefetch is disabled because the filesystem (LIGHTNIN
G) is not a recognized network FS (NFS/Lustre). If you want to force prefetching, start vLLM with --safetensors-load-strategy
=prefetch.
Loading safetensors checkpoint shards:   0% Completed | 0/3 [00:00<?, ?it/s]
(EngineCore pid=101416) INFO 05-02 17:11:19 [default_loader.py:384] Loading weights took 2.17 seconds
(EngineCore pid=101416) INFO 05-02 17:11:20 [gpu_model_runner.py:4879] Model loading took 7.56 GiB memory and 3.170874 second
s
(EngineCore pid=101416) INFO 05-02 17:11:24 [backends.py:1069] Using cache directory: /teamspace/studios/this_studio/.cache/v
llm/torch_compile_cache/3b93b3280e/rank_0_0/backbone for vLLM's torch.compile
(EngineCore pid=101416) INFO 05-02 17:11:24 [backends.py:1128] Dynamo bytecode transform time: 3.93 s
(EngineCore pid=101416) INFO 05-02 17:11:26 [backends.py:290] Directly load the compiled graph(s) for compile range (1, 8192)
from the cache, took 2.122 s
(EngineCore pid=101416) INFO 05-02 17:11:26 [decorators.py:305] Directly load AOT compilation from path /teamspace/studios/th
is_studio/.cache/vllm/torch_compile_cache/torch_aot_compile/ff66badfe79830dcc826a314d37b2c1f3f7f42f20edd3f21444ae8aeccada857/
rank_0_0/model
(EngineCore pid=101416) INFO 05-02 17:11:26 [monitor.py:53] torch.compile took 6.36 s in total
(EngineCore pid=101416) INFO 05-02 17:11:26 [monitor.py:81] Initial profiling/warmup run took 0.15 s
(EngineCore pid=101416) INFO 05-02 17:11:28 [gpu_model_runner.py:5963] Profiling CUDA graph memory: PIECEWISE=51 (largest=51
2), FULL=35 (largest=256)
(EngineCore pid=101416) INFO 05-02 17:11:30 [gpu_model_runner.py:6042] Estimated CUDA graph memory: 0.51 GiB total
(EngineCore pid=101416) INFO 05-02 17:11:30 [gpu_worker.py:440] Available KV cache memory: 11.53 GiB
(EngineCore pid=101416) INFO 05-02 17:11:30 [gpu_worker.py:455] CUDA graph memory profiling is enabled (default since v0.21.
0). The current --gpu-memory-utilization=0.9200 is equivalent to --gpu-memory-utilization=0.8968 without CUDA graph memory pr
ofiling. To maintain the same effective KV cache size as before, increase --gpu-memory-utilization to 0.9432. To disable, set
VLLM_MEMORY_PROFILER_ESTIMATE_CUDAGRAPHS=0.
(EngineCore pid=101416) INFO 05-02 17:11:30 [kv_cache_utils.py:1711] GPU KV cache size: 83,936 tokens
(EngineCore pid=101416) INFO 05-02 17:11:30 [kv_cache_utils.py:1716] Maximum concurrency for 34,000 tokens per request: 2.47x
Capturing CUDA graphs (mixed prefill-decode, PIECEWISE): 100%|██████████| 51/51 [00:03<00:00, 15.04it/s]
Capturing CUDA graphs (decode, FULL): 100%|██████████| 35/35 [00:01<00:00, 17.77it/s]
(EngineCore pid=101416) INFO 05-02 17:11:36 [gpu_model_runner.py:6133] Graph capturing finished in 6 secs, took 0.50 GiB
(EngineCore pid=101416) INFO 05-02 17:11:36 [gpu_worker.py:599] CUDA graph pool memory: 0.5 GiB (actual), 0.51 GiB (estimate
d), difference: 0.01 GiB (1.6%).
(EngineCore pid=101416) INFO 05-02 17:11:36 [core.py:299] init engine (profile, create kv cache, warmup model) took 16.70 s
(compilation: 6.36 s)
(EngineCore pid=101416) INFO 05-02 17:11:37 [kernel.py:205] Final IR op priority after setting platform defaults: IrOpPriorit
yConfig(rms_norm=['native'])
```

Evaluation helpers

In [5]:
```
from collections import Counter

def majority(answers):
    valid = [a for a in answers]
    return Counter(valid).most_common(1)[0][0] if valid else None

def extract_thinking_trace(text):
    # complete thinking
    match = re.search(r'<think>(.*?)</think>', text, re.DOTALL)
    if match:
        return match.group(1).strip()
    # truncated due to max tokens
    match = re.search(r'<think>(.*)', text, re.DOTALL)
    return match.group(1).strip() if match else ""

def think_end_ids(tokenizer):
    for tag in ["<|/thinking|>", "</think>"]:
        ids = tokenizer.encode(tag, add_special_tokens=False)
        if ids:
            return ids, tag

def make_prompt_think(text, tokenizer):
    msgs = [
        {"role": "system", "content": "You are a careful competition math assistant. Always output your final answer in \\boxe
        {"role": "user",   "content": text},
    ]
    return tokenizer.apply_chat_template(msgs, tokenize=False, add_generation_prompt=True, enable_thinking=True)

def strip_thinking_trace(text):
    text = re.sub(r"<think>.*?</think>", "", text, flags=re.DOTALL)
    text = re.sub(r"<\|begin_of_thought\|>.*?<\|end_of_thought\|>", "", text, flags=re.DOTALL)
    return text.strip()


def extract_answer(text: str, mode="exact_match") -> int | None:
    """Extract an AIME-style integer answer from a model completion."""
    answer_text = strip_thinking_trace(text)
    if not answer_text:
        if mode == "exact_match":
            return None
        else:
            answer_text = text  # fall back to full text


    # 1. Boxed LaTeX answer: \boxed{123}
    if mode == "exact_match":
        boxed = re.findall(r"\\boxed\{(\d+)\}", answer_text)
        if boxed:
            val = int(boxed[-1])
            return val
        else:
            return None

    elif mode == "flexible_extract":
        # 2. "The answer is N" or "answer: N" patterns
        patterns = [
            r"(?:the\s+)?answer\s+is\s+[:\s]*(\d+)",
            r"answer[:\s]+(\d+)",
            r"=\s*(\d+)\s*$",
            r"(?:therefore|thus|so),?\s+(\d+)\s*(?:\.|$)",
        ]
        for pattern in patterns:
            matches = re.findall(pattern, answer_text, re.IGNORECASE)
            if matches:
                val = int(matches[-1])
                return val

        # 3. Last integer in [0, 999] in the answer portion
        integers = re.findall(r"\b(\d{1,3})\b", answer_text)
        for candidate in reversed(integers):
            val = int(candidate)
            return val
        return None
```

2.1 Warm-Up

It was run separately with MAX_NEW_TOKENS = 8192

In [ ]:
```
#vLLM instead!
ANSWER_MODE = 'flexible_extract'
prompts, gold_answers, problems = [], [], []
for example in dataset:
    problem = example["prompt"][0]["content"]
```

```
    problems.append(problem)
    gold_answers.append(int(example["label"]))
    prompts.append(tokenizer.apply_chat_template(
        [{"role": "system", "content": "You are a careful competition math assistant. Always output your final answer in \\box
         {"role": "user",   "content": problem}],
        tokenize=False, add_generation_prompt=True, enable_thinking=True,
    ))

sampling = SamplingParams(temperature=0.0, max_tokens=MAX_NEW_TOKENS, detokenize=True, skip_special_tokens=False)
outputs  = llm.generate(prompts, sampling_params=sampling)

print("RAW OUTPUT SNIPPET:", repr(outputs[0].outputs[0].text[:300]))

records = []
for i, (out, gold, problem) in enumerate(zip(outputs, gold_answers, problems)):
    model_output = out.outputs[0].text
    extracted    = extract_answer(model_output, mode=ANSWER_MODE)
    correct      = extracted == gold if extracted is not None else False
    thinking     = extract_thinking_trace(model_output)
    think_len    = len(tokenizer.encode(thinking, add_special_tokens=False))

    records.append(dict(problem=problem, gold_answer=gold, model_output=model_output,
                        extracted_answer=extracted, correct=correct, thinking_length=think_len))
    print(f"[{i+1}/{len(dataset)}] gold={gold} pred={extracted} correct={correct} thinking_len={think_len}")

results_df = pd.DataFrame(records)
print("\nDistribution of Thinking Lengths (in tokens):")
print(results_df["thinking_length"].describe())

prompts_no_thinking = []
for i, example in enumerate(dataset):
    prompts_no_thinking.append(tokenizer.apply_chat_template(
        [{"role": "system", "content": "You are a careful competition math assistant. Always output your final answer in \\box
         {"role": "user",   "content": problems[i]}],
        tokenize=False, add_generation_prompt=True, enable_thinking=False,
    ))

sampling = SamplingParams(temperature=0.0, max_tokens=MAX_NEW_TOKENS, detokenize=True, skip_special_tokens=False)
outputs_no_thinking  = llm.generate(prompts_no_thinking, sampling_params=sampling)

print("RAW OUTPUT SNIPPET:", repr(outputs_no_thinking[0].outputs[0].text[:300]))

records_no_thinking = []
for i, (out, gold, problem) in enumerate(zip(outputs_no_thinking, gold_answers, problems)):
    model_output = out.outputs[0].text
    extracted    = extract_answer(model_output, mode=ANSWER_MODE)
    correct      = extracted == gold if extracted is not None else False
    thinking     = extract_thinking_trace(model_output)
    think_len    = len(tokenizer.encode(thinking, add_special_tokens=False))

    records_no_thinking.append(dict(problem=problem, gold_answer=gold, model_output=model_output,
                        extracted_answer=extracted, correct=correct, thinking_length=think_len))
    print(f"[{i+1}/{len(dataset)}] gold={gold} pred={extracted} correct={correct} thinking_len={think_len}")

results_df_no_thinking = pd.DataFrame(records_no_thinking)
print("\nDistribution of Thinking Lengths (in tokens):")
print(results_df_no_thinking["thinking_length"].describe())
```

```
RAW OUTPUT SNIPPET: '<think>\nOkay, so I need to find the largest possible real part of the expression $(75 + 117i)z + \\frac
{96 + 144i}{z}$ where $z$ is a complex number with $|z| = 4$. Hmm, let me think about how to approach this.\n\nFirst, since
$z$ is a complex number with modulus 4, I can represent $z$ in polar form. T'
[1/30] gold=540 pred=540 correct=True thinking_len=4269
[2/30] gold=204 pred=204 correct=True thinking_len=2474
[3/30] gold=721 pred=2 correct=False thinking_len=7688
[4/30] gold=236 pred=236 correct=True thinking_len=8054
[5/30] gold=809 pred=809 correct=True thinking_len=8032
[6/30] gold=45 pred=45 correct=True thinking_len=7990
[7/30] gold=385 pred=1 correct=False thinking_len=8049
[8/30] gold=104 pred=16 correct=False thinking_len=8053
[9/30] gold=601 pred=100 correct=False thinking_len=8044
[10/30] gold=23 pred=16 correct=False thinking_len=7933
[11/30] gold=321 pred=8321 correct=False thinking_len=8070
[12/30] gold=104 pred=104 correct=True thinking_len=7978
[13/30] gold=902 pred=2 correct=False thinking_len=8029
[14/30] gold=211 pred=0 correct=False thinking_len=7983
[15/30] gold=113 pred=484 correct=False thinking_len=8023
[16/30] gold=33 pred=33 correct=True thinking_len=5051
[17/30] gold=110 pred=155 correct=False thinking_len=8065
[18/30] gold=25 pred=1 correct=False thinking_len=8007
[19/30] gold=73 pred=73 correct=True thinking_len=8018
[20/30] gold=55 pred=55 correct=True thinking_len=4047
[21/30] gold=127 pred=9 correct=False thinking_len=7990
[22/30] gold=294 pred=294 correct=True thinking_len=5329
[23/30] gold=468 pred=468 correct=True thinking_len=8077
[24/30] gold=197 pred=34 correct=False thinking_len=8036
[25/30] gold=480 pred=240 correct=False thinking_len=8046
[26/30] gold=699 pred=5 correct=False thinking_len=8059
[27/30] gold=315 pred=2 correct=False thinking_len=7951
[28/30] gold=371 pred=7 correct=False thinking_len=8054
[29/30] gold=80 pred=200 correct=False thinking_len=8068
[30/30] gold=116 pred=116 correct=True thinking_len=2216

Distribution of Thinking Lengths (in tokens):
count      30.000000
mean     7189.433333
std      1759.758186
min      2216.000000
25%      7937.500000
50%      8020.500000
75%      8052.000000
max      8077.000000
Name: thinking_length, dtype: float64


RAW OUTPUT SNIPPET: 'We are given the expression:\n\n$$\n(75 + 117i)z + \\frac{96 + 144i}{z}\n$$\n\nand we are told that $ |z
| = 4 $, where $ z $ is a complex number.\n\nWe are to **find the largest possible real part** of this expression.\n\n---\n\n
### Step 1: Let $ z = re^{i\\theta} $\n\nSince $ |z| = 4 $, we can write:\n\n$$\nz = 4e^{i'
[1/30] gold=540 pred=540 correct=True thinking_len=0
[2/30] gold=204 pred=204 correct=True thinking_len=0
[3/30] gold=721 pred=37 correct=False thinking_len=0
[4/30] gold=236 pred=9 correct=False thinking_len=0
[5/30] gold=809 pred=404 correct=False thinking_len=0
[6/30] gold=45 pred=0 correct=False thinking_len=0
[7/30] gold=385 pred=2 correct=False thinking_len=0
[8/30] gold=104 pred=180 correct=False thinking_len=0
[9/30] gold=601 pred=100 correct=False thinking_len=0
[10/30] gold=23 pred=2 correct=False thinking_len=0
[11/30] gold=321 pred=2 correct=False thinking_len=0
[12/30] gold=104 pred=169 correct=False thinking_len=0
[13/30] gold=902 pred=1 correct=False thinking_len=0
[14/30] gold=211 pred=89 correct=False thinking_len=0
[15/30] gold=113 pred=665 correct=False thinking_len=0
[16/30] gold=33 pred=33 correct=True thinking_len=0
[17/30] gold=110 pred=0 correct=False thinking_len=0
[18/30] gold=25 pred=25 correct=True thinking_len=0
[19/30] gold=73 pred=234 correct=False thinking_len=0
[20/30] gold=55 pred=255 correct=False thinking_len=0
[21/30] gold=127 pred=152 correct=False thinking_len=0
[22/30] gold=294 pred=42 correct=False thinking_len=0
[23/30] gold=468 pred=384 correct=False thinking_len=0
[24/30] gold=197 pred=1 correct=False thinking_len=0
[25/30] gold=480 pred=0 correct=False thinking_len=0
[26/30] gold=699 pred=995 correct=False thinking_len=0
[27/30] gold=315 pred=132 correct=False thinking_len=0
[28/30] gold=371 pred=4 correct=False thinking_len=0
[29/30] gold=80 pred=66 correct=False thinking_len=0
[30/30] gold=116 pred=116 correct=True thinking_len=0

Distribution of Thinking Lengths (in tokens):
count    30.0
mean      0.0
std       0.0
min       0.0
25%       0.0
50%       0.0
75%       0.0
max       0.0
Name: thinking_length, dtype: float64
```

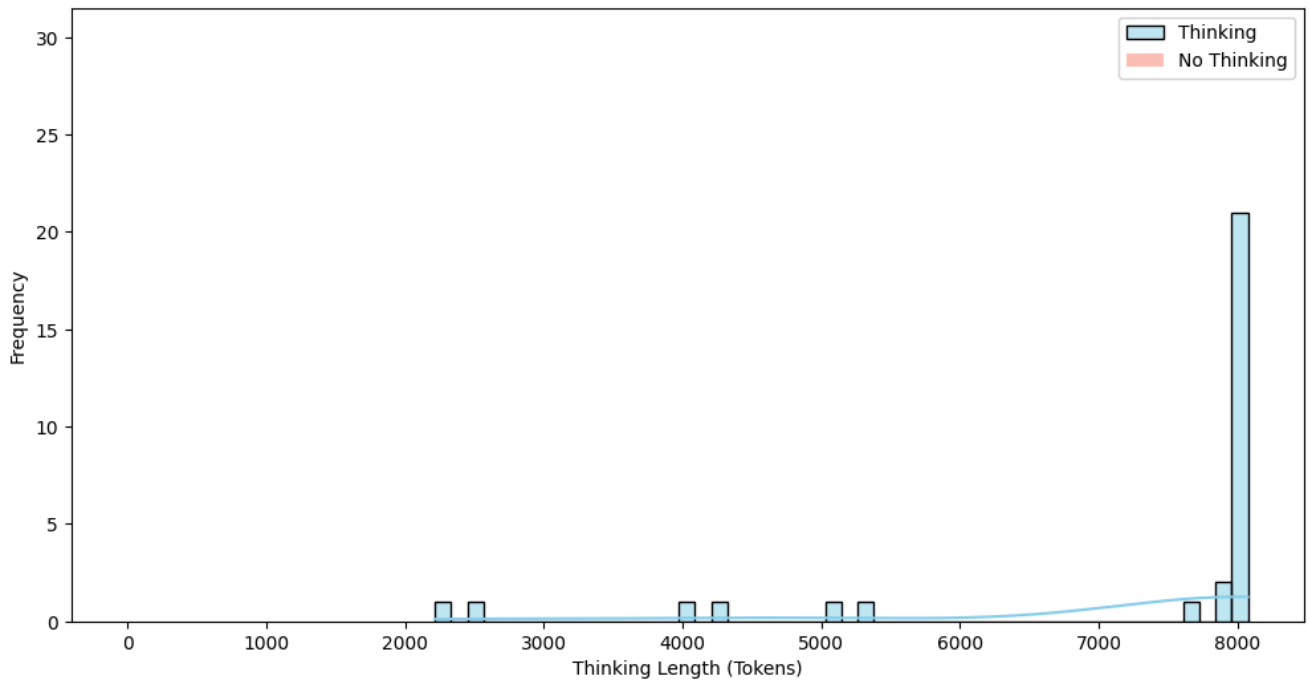
In [ ]:
```
print(f'Thinking-enabled accuracy: {results_df["correct"].mean()}')
print(f'No-thinking accuracy: {results_df_no_thinking["correct"].mean()}')

results_df.to_csv("results_df.csv", index=False)
results_df_no_thinking.to_csv("results_df_no_thinking.csv", index=False)
```

```
Thinking-enabled accuracy: 0.4
No-thinking accuracy: 0.16666666666666666
```

In [ ]:
```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(12, 6))
sns.histplot(results_df['thinking_length'], bins=50, color='skyblue', label='Thinking', kde=True)
sns.histplot(results_df_no_thinking['thinking_length'], bins=50, color='salmon', label='No Thinking', kde=True)
plt.xlabel('Thinking Length (Tokens)')
plt.ylabel('Frequency')
plt.legend()
plt.tight_layout(); plt.savefig("hist_lthink.png", dpi=150); plt.show()
```



2.2 Scaling Experiments

In [6]:
```
SEQ_BUDGETS = [1024, 4096, 8192, 32000]
PAR_M_VALUES = [1, 2, 4, 8]
PAR_BUDGET = 4000
MAX_ANS_TOKENS = 1024
N_RUNS = 1          # re-runs for parallel error bars
T, TOP_K, TOP_P = 0.6, 50, 0.95
```

In [7]:
```
def run_one(llm, tokenizer, problem, budget, strategy="wait",
            temperature=0.6, top_k=50, top_p=0.95):
    end_ids, end_tag = think_end_ids(tokenizer)
    prompt   = make_prompt_think(problem, tokenizer)
    sampling = SamplingParams(
        temperature      = temperature,
        top_k            = top_k,
        top_p            = top_p,
        max_tokens       = 1,
        detokenize       = False,
    )

    token_ids = tokenizer.encode(prompt)
    prompt_len = len(token_ids)

    # ── thinking loop ──────────────────────────────────────────────
    while True:
        think_len = len(token_ids) - prompt_len
        out = llm.generate({"prompt_token_ids": token_ids},
                   sampling_params=sampling,
                   use_tqdm=False)
        next_tok  = out[0].outputs[0].token_ids[0]

        if think_len >= budget:
            token_ids += end_ids
            break
```

```
        elif next_tok in end_ids and strategy == "wait":
            wait_id = tokenizer.encode("Wait", add_special_tokens=False)[0]
            token_ids.append(wait_id)
        else:
            token_ids.append(next_tok)
            if next_tok in end_ids:
                break

    # ── generate final answer ──────────────────────────────────────────
    ans_sampling = SamplingParams(
        temperature = temperature,
        top_k       = top_k,
        top_p       = top_p,
        max_tokens = MAX_ANS_TOKENS, detokenize=True, skip_special_tokens=False
    )
    out = llm.generate({"prompt_token_ids": token_ids},
                       sampling_params=ans_sampling,
                       use_tqdm=False)
    decoded  = out[0].outputs[0].text
    full_out = tokenizer.decode(token_ids) + decoded

    thinking  = extract_thinking_trace(full_out)
    think_tok = len(tokenizer.encode(thinking, add_special_tokens=False))
    total_tok = len(token_ids) - len(tokenizer.encode(prompt)) + len(out[0].outputs[0].token_ids)
    return full_out, think_tok, total_tok
```

In [10]:

```
def run_sequential(budgets=SEQ_BUDGETS, save_dir="seq_results"):
    os.makedirs(save_dir, exist_ok=True)
    all_rows = []

    for budget in budgets:
        save_path = os.path.join(save_dir, f"budget{budget}.csv")

        # Skip if already completed
        if os.path.exists(save_path):
            print(f"\n── Budget {budget} ── [SKIPPING, already saved]")
            all_rows.append(pd.read_csv(save_path))
            continue

        print(f"\n── Budget {budget} ──")
        rows = []

        for i, ex in enumerate(dataset):
            problem = ex["prompt"][0]["content"]
            gold    = int(ex["label"])
            decoded, tlen, total = run_one(llm, tokenizer, problem, budget)
            rows.append(dict(
                budget        = budget,
                gold          = gold,
                correct_flex  = extract_answer(decoded, mode="flexible_extract") == gold,
                correct_exact = extract_answer(decoded, mode="exact_match")      == gold,
                think_len     = tlen,
                total_len     = total,
            ))
            print(f"  [{i+1}/30] think_len={tlen} total={total} flex={rows[-1]['correct_flex']}")

        # Save this budget's results
        budget_df = pd.DataFrame(rows)
        budget_df.to_csv(save_path, index=False)
        print(f"  ✓ Saved → {save_path}")
        all_rows.append(budget_df)

    return pd.concat(all_rows, ignore_index=True)


seq_df = run_sequential()
seq_df.to_csv("seq_results.csv", index=False)
```

```
── Budget 1024 ── [SKIPPING, already saved]

── Budget 4096 ── [SKIPPING, already saved]

── Budget 8192 ── [SKIPPING, already saved]

── Budget 32000 ── [SKIPPING, already saved]
```

I ran this cell in a separate notebook, but here's the output —— Budget 1024 —— [1/30] think_len=2052 total=2054 flex=False [2/30] think_len=2052 total=2054 flex=False [3/30] think_len=2052 total=2054 flex=False [4/30] think_len=1028 total=2054 flex=False [5/30] think_len=2052 total=2054 flex=False [6/30] think_len=1028 total=2054 flex=False [7/30] think_len=2051 total=2054 flex=False [8/30] think_len=2052 total=2054 flex=False [9/30] think_len=2052 total=2054 flex=False [10/30] think_len=1028 total=2054 flex=False [11/30] think_len=2052 total=2054 flex=False [12/30] think_len=2052 total=2054 flex=False [13/30] think_len=2052 total=2054 flex=False [14/30] think_len=2052 total=2054 flex=False [15/30] think_len=2052 total=2054 flex=False [16/30] think_len=2052 total=2054 flex=False [17/30] think_len=2052 total=2054 flex=False [18/30] think_len=2052 total=2054 flex=True [19/30] think_len=1028 total=2054 flex=False [20/30] think_len=2052 total=2054 flex=False [21/30] think_len=2052 total=2054 flex=False [22/30] think_len=2052 total=2054 flex=False [23/30] think_len=2051 total=2054 flex=False [24/30] think_len=2052 total=2054 flex=False [25/30] think_len=1028 total=2054 flex=False [26/30] think_len=2051 total=2054 flex=False [27/30] think_len=2052 total=2054 flex=False [28/30] think_len=2052 total=2054 flex=False [29/30] think_len=2052 total=2054 flex=False [30/30] think_len=2051 total=2054 flex=False ✓ Saved → /content/drive/My Drive/Colab Notebooks/CSE_ML_HW1/seq_results/budget1024.csv —— Budget 4096 —— [1/30] think_len=4100 total=5126 flex=False [2/30] think_len=4099 total=5126 flex=False [3/30] think_len=5124 total=5126 flex=False [4/30] think_len=5124 total=5126 flex=False [5/30] think_len=5124 total=5126 flex=False [6/30] think_len=4100 total=5126 flex=False [7/30] think_len=4100 total=5126 flex=False [8/30] think_len=5123 total=5126 flex=False [9/30] think_len=5124 total=5126 flex=False [10/30] think_len=4100 total=5126 flex=False [11/30] think_len=5124 total=5126 flex=True [12/30]

think_len=5124 total=5126 flex=False [13/30] think_len=4100 total=5126 flex=False [14/30] think_len=5124 total=5126 flex=False [15/30] think_len=5124 total=5126 flex=False [16/30] think_len=4100 total=5126 flex=False [17/30] think_len=5124 total=5126 flex=False [18/30] think_len=5124 total=5126 flex=True [19/30] think_len=4100 total=5126 flex=False [20/30] think_len=4100 total=4717 flex=True [21/30] think_len=4100 total=5126 flex=False [22/30] think_len=4100 total=4782 flex=True [23/30] think_len=5124 total=5126 flex=False [24/30] think_len=4100 total=5126 flex=False [25/30] think_len=4100 total=5126 flex=False [26/30] think_len=5123 total=5126 flex=False [27/30] think_len=4100 total=4772 flex=False [28/30] think_len=5124 total=5126 flex=False [29/30] think_len=4100 total=4103 flex=False [30/30] think_len=4100 total=4890 flex=True ✓ Saved → /content/drive/My Drive/Colab Notebooks/CSE_ML_HW1/seq_results/budget4096.csv —— Budget 8192 —— [1/30] think_len=8195 total=9073 flex=False [2/30] think_len=8196 total=9222 flex=False [3/30] think_len=8195 total=9222 flex=False [4/30] think_len=9219 total=9222 flex=False [5/30] think_len=8195 total=9176 flex=True [6/30] think_len=8196 total=9222 flex=False [7/30] think_len=8195 total=9222 flex=False [8/30] think_len=8195 total=9222 flex=False [9/30] think_len=9220 total=9222 flex=False [10/30] think_len=8195 total=9222 flex=False [11/30] think_len=8196 total=9222 flex=False [12/30] think_len=8196 total=9222 flex=False [13/30] think_len=9218 total=9222 flex=False [14/30] think_len=9218 total=9222 flex=False [15/30] think_len=8194 total=9222 flex=False [16/30] think_len=8196 total=9222 flex=False [17/30] think_len=7513 total=9222 flex=False [18/30] think_len=8196 total=8788 flex=True [19/30] think_len=3581 total=8199 flex=True [20/30] think_len=5575 total=8445 flex=True [21/30] think_len=8196 total=9222 flex=False [22/30] think_len=4913 total=8199 flex=True [23/30] think_len=9220 total=9222 flex=False [24/30] think_len=8196 total=9222 flex=False [25/30] think_len=8196 total=9222 flex=False [26/30] think_len=9219 total=9222 flex=False [27/30] think_len=9220 total=9222 flex=False [28/30] think_len=9220 total=9222 flex=False [29/30] think_len=9218 total=9222 flex=False [30/30] think_len=8194 total=8921 flex=False ✓ Saved → /content/drive/My Drive/Colab Notebooks/CSE_ML_HW1/seq_results/budget8192.csv —— Budget 32000 —— [1/30] think_len=32004 total=33030 flex=False [2/30] think_len=32004 total=33030 flex=False [3/30] think_len=32004 total=33030 flex=False [4/30] think_len=9646 total=32131 flex=True [5/30] think_len=9230 total=32007 flex=False [6/30] think_len=10645 total=33029 flex=True [7/30] think_len=32003 total=33030 flex=False [8/30] think_len=32004 total=33030 flex=False [9/30] think_len=15074 total=32010 flex=False [10/30] think_len=32004 total=33030 flex=False [11/30] think_len=32003 total=33030 flex=False [12/30] think_len=32004 total=33030 flex=False [13/30] think_len=6930 total=32053 flex=False [14/30] think_len=25970 total=32319 flex=True [15/30] think_len=32004 total=33030 flex=False [16/30] think_len=32004 total=33030 flex=False [17/30] think_len=8978 total=32007 flex=True [18/30] think_len=32002 total=32431 flex=True [19/30] think_len=6257 total=32007 flex=False

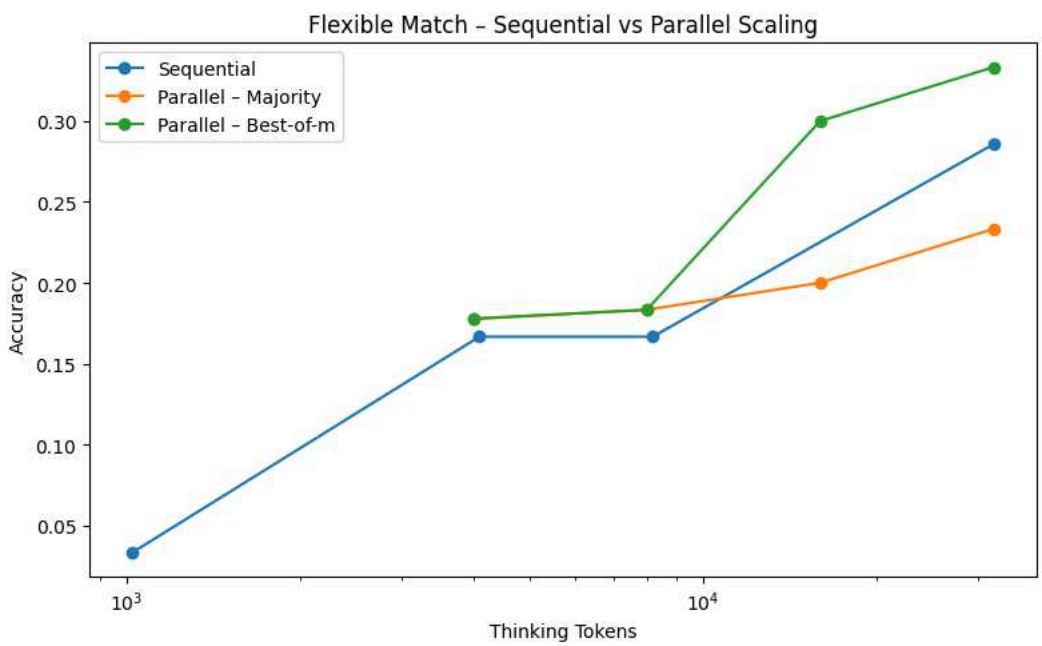
In [ ]:
```python
# 1. majority -- majority voting; 2. bom -- best-of-m
def run_parallel(m_values=PAR_M_VALUES, budget=PAR_BUDGET, n_runs=N_RUNS, save_dir="par_results"):
    os.makedirs(save_dir, exist_ok=True)
    all_rows = []
    print('Start running parallel scaling')
    for m in m_values:
        for run in range(n_runs):
            save_path = os.path.join(save_dir, f"m{m}_run{run}.csv")

            # Skip if already completed
            if os.path.exists(save_path):
                print(f"\n-- m={m}  run={run+1}/{n_runs} -- [SKIPPING, already saved]")
                all_rows.append(pd.read_csv(save_path))
                continue

            print(f"\n-- m={m}  run={run+1}/{n_runs} --")
            rows = []

            for i, ex in enumerate(dataset):
                problem = ex["prompt"][0]["content"]
                gold    = int(ex["label"])
                results = [run_one(llm, tokenizer, problem, budget, strategy="wait",
                                   temperature=T, top_k=TOP_K, top_p=TOP_P)
                           for _ in range(m)]
        # Extract answers and keep track of the decoded text for flexible extraction
                preds_flex_with_decoded = [(extract_answer(d, "flexible_extract"), d) for d, *_ in all_individual_results]
                preds_exact_with_decoded = [(extract_answer(d, "exact_match"), d) for d, *_ in all_individual_results]

                preds_flex  = [p for p, _ in preds_flex_with_decoded]
                preds_exact = [p for p, _ in preds_exact_with_decoded]
                maj_flex_ans = majority(preds_flex)
                for extracted_ans, decoded_text in preds_flex_with_decoded:
                    if extracted_ans == maj_flex_ans:
                        majority_flex_full_output = decoded_text

                rows.append(dict(
                    m=m, run=run, gold=gold,
                    majority_flex  = maj_flex_ans  == gold,
                    majority_exact = majority(preds_exact) == gold,
                    bom_flex       = any(p == gold for p in preds_flex),
                    bom_exact      = any(p == gold for p in preds_exact),
                    total_think    = sum(t   for _, t, _   in results),
                    total_toks     = sum(tot for _, _, tot in results),
                    majority_flex_full_output = majority_flex_full_output # Add the full output to the DataFrame
                ))
                print(f"  [{i+1}/30] maj_flex={rows[-1]['majority_flex']} bom_flex={rows[-1]['bom_flex']}")

            # Save this run's results
            run_df = pd.DataFrame(rows)
            run_df.to_csv(save_path, index=False)
            print(f"  ✓ Saved → {save_path}")
            all_rows.append(run_df)

    return pd.concat(all_rows, ignore_index=True)


par_df = run_parallel()
par_df.to_csv("par_results.csv", index=False)
```

Start running parallel scaling —— m=8 run=1/1 —— [1/30] maj_flex=True bom_flex=True [2/30] maj_flex=False bom_flex=True [3/30] maj_flex=False bom_flex=False [4/30] maj_flex=False bom_flex=True [5/30] maj_flex=True bom_flex=True [6/30] maj_flex=True bom_flex=True [7/30] maj_flex=False bom_flex=False [8/30] maj_flex=False bom_flex=False [9/30] maj_flex=False bom_flex=False [10/30] maj_flex=False bom_flex=False [11/30] maj_flex=False bom_flex=False [12/30] maj_flex=False bom_flex=False [13/30] maj_flex=False bom_flex=False [14/30] maj_flex=False bom_flex=False [15/30] maj_flex=False bom_flex=False [16/30] maj_flex=False bom_flex=False [17/30] maj_flex=False bom_flex=False [18/30] maj_flex=True bom_flex=True [19/30] maj_flex=False bom_flex=False [20/30] maj_flex=True bom_flex=True [21/30] maj_flex=False bom_flex=False [22/30] maj_flex=True bom_flex=True [23/30] maj_flex=False bom_flex=False [24/30]

```
maj_flex=False bom_flex=False [25/30] maj_flex=False bom_flex=False [26/30] maj_flex=False bom_flex=False [27/30] maj_flex=False bom_flex=False [28/30]
maj_flex=False bom_flex=False [29/30] maj_flex=False bom_flex=True [30/30] maj_flex=True bom_flex=True ✓ Saved → par_results/m8_run0.csv
```

In [6]:
```
import glob
par_df = pd.concat(
    [pd.read_csv(f) for f in sorted(glob.glob("par_results/m*_run*.csv"))],
    ignore_index=True
)
par_df.to_csv("par_results.csv", index=False)

seq_df = pd.concat(
    [pd.read_csv(f) for f in sorted(glob.glob("seq_results/*.csv"))],
    ignore_index=True
)
seq_df.to_csv("seq_results.csv", index=False)
```

In [16]:
```
import matplotlib.pyplot as plt

seq_df = pd.read_csv("seq_results.csv")
par_df = pd.read_csv("par_results.csv")

fig, ax = plt.subplots(figsize=(8, 5))

g = seq_df.groupby("budget")["correct_flex"].mean()
ax.plot(g.index, g.values, marker="o", label="Sequential")

g_maj = par_df.groupby("m")["majority_flex"].mean()
ax.plot(g_maj.index * PAR_BUDGET, g_maj.values, marker="o", label="Parallel — Majority")

g_bom = par_df.groupby("m")["bom_flex"].mean()
ax.plot(g_bom.index * PAR_BUDGET, g_bom.values, marker="o", label="Parallel — Best-of-m")

ax.set_xscale("log")
ax.set_xlabel("Thinking Tokens")
ax.set_ylabel("Accuracy")
ax.set_title("Flexible Match — Sequential vs Parallel Scaling")
ax.legend()

plt.tight_layout()
plt.savefig("fig_scaling_seq_par_flex.png", dpi=150)
plt.show()
```



2.3 Qualitative Analysis

• Find 2 questions which are unsolved by sequential scaling at low budgets but solved at higher budgets. Do you see any change in strategy?

High budget strategy tend to restating the problem and straight diving into the numbers and equations. At low budget=1024 the model runs out of tokens mid-reasoning — it's still setting up variables (a, b, c) and never reaches a numerical answer, so ans= is empty.

In [8]:
```
for problem_idx in range(18, 22):
  ex = dataset[problem_idx]
  problem = ex["prompt"][0]["content"]
  gold    = int(ex["label"])
  print(f"####Problem[{problem_idx}] gold={gold}, problem={problem}####")
```

```
decoded, tlen, total = run_one(llm, tokenizer, problem, 1024)
print(f"* Low budget 1024: correct_flex={extract_answer(decoded, mode="flexible_extract") == gold},ans={decoded[-200:]}")
decoded, tlen, total = run_one(llm, tokenizer, problem, 8192)
print(f"* High budget 8192: correct_flex={extract_answer(decoded, mode="flexible_extract") == gold},ans={decoded[-200:]}")
```

```
####Problem[18] gold=73, problem=Solve the following math problem step by step. Please put your final answer within \boxed{}.

Among the 900 residents of Aimeville, there are 195 who own a diamond ring, 367 who own a set of golf clubs, and 562 who own
a garden spade. In addition, each of the 900 residents owns a bag of candy hearts. There are 437 residents who own exactly tw
o of these things, and 234 residents who own exactly three of these things. Find the number of residents of Aimeville who own
all four of these things.

Remember to put your final answer within \boxed{}.####
* Low budget 1024: correct_flex=False,ans= 437
- Total with exactly three = 234

So, total with at least two is:

$$
437 + 234 = 671
$$

Thus, total with **at least one** is:

$$
x + y + z = 900 - w
$$

But we can also use the formula:

$$
|A
* High budget 8192: correct_flex=True,ans= = 890 - 2c
$$

But $ |R \cup G \cup S| $ is also equal to:
$$
a + b + c = 437 + 234 + c = 671 + c
$$

So:
$$
671 + c = 890 - 2c
$$
$$
3c = 219
$$
$$
c = 73
$$

---

### Final Answer

$$
\boxed{73}
$$
####Problem[19] gold=55, problem=Solve the following math problem step by step. Please put your final answer within \boxed{}.

Alice chooses a set $A$ of positive integers. Then Bob lists all finite nonempty sets $B$ of positive integers with the prope
rty that the maximum element of $B$ belongs to $A$. Bob's list has 2024 sets. Find the sum of the elements of A.

Remember to put your final answer within \boxed{}.####
* Low budget 1024: correct_flex=False,ans=).

So the binary representation is 11111101000.

Let me check how many bits that is. Let's count:

Starting from the first remainder (the last division):

Position 10: 1

Position 9: 1

Position 8: 1
* High budget 8192: correct_flex=False,ans=responding a values are k - 1 + 1 = k + 1, so a = 4, 6, 7, 8, 9, 10, 11.

Summing these: 4 + 6 + 7 + 8 + 9 + 10 + 11. Let me compute that:

4 + 6 = 10
10 + 7 = 17
17 + 8 = 25
25 + 9<|/thinking|>
####Problem[20] gold=127, problem=Solve the following math problem step by step. Please put your final answer within \boxed
{}.

Torus $\mathcal T$ is the surface produced by revolving a circle with radius $3$ around an axis in the plane a distance $6$ f
rom the center of the circle. When a sphere of radius $11$ rests inside $\mathcal T$, it is internally tangent to $\mathcal T
$ along a circle with radius $r_{i}$, and when it rests outside $\mathcal T$, it is externally tangent along a circle with ra
dius $r_{o}$. The difference $r_{i}-r_{o}=\tfrac{m}{n}$, where $m$ and $n$ are relatively prime positive integers. Find $m+n
$.

Remember to put your final answer within \boxed{}.####
```

```
* Low budget 1024: correct_flex=False,ans=the cross-section through the axis, then the torus is actually a circle with radius
R + r, but with a hole of radius R - r. Wait, but that would mean that if R - r is positive, which it is here (6 - 3
* High budget 8192: correct_flex=False,ans=, y, z) * [sqrt(x^2 + y^2) - R] + (0, 0, z).

Wait, no. Let me compute it properly.

Let me denote sqrt(x^2 + y^2) as ρ. Then, F = (ρ - R)^2 + z^2 - r^2 = 0.

The partial derivatives:

dF/dx = 2(ρ - R
####Problem[21] gold=294, problem=Solve the following math problem step by step. Please put your final answer within \boxed
{}.

Consider the paths of length $16$ that follow the lines from the lower left corner to the upper right corner on an $8\times 8
$ grid. Find the number of such paths that change direction exactly four times, as in the examples shown below.

Remember to put your final answer within \boxed{}.####
* Low budget 1024: correct_flex=False,ans=

Therefore, for splitting 8 R moves into three segments: C(8-1, 3-1) = C(7,2) = 21. Similarly, splitting 8 U moves into two se
gments: C(8-1, 2-1) = C(7,1) = 7. Therefore, the number of paths starting
* High budget 8192: correct_flex=False,ans=of ways to split 8 into 3 positive integers is:
    $$
    \binom{8 - 1}{3 - 1} = \binom{7}{2} = 21
    $$

- **U segments**: We need to split 8 U moves into 2 segments, each at least 1.
<|/thinking|>
```

• Find 2 questions which are solved correctly at a lower budget but not at a higher budget for sequential scaling. What do you observe?

We only found one question in this case. For simple (short) questions, lower budget leads to faster convergence to true answer. We observed that the low budget stumbles onto the right algebraic path early — factoring via roots of unity products — and commits to it, reaching the correct answer before running out of tokens. The high budget has more room to second-guess itself. It starts exploring a polar/trigonometric approach (arctan, e^{iπ/4}, α^13), which is a valid but much longer detour. With more budget it keeps going down this rabbit hole and never recovers to box the answer correctly. This is overthinking — more compute actively hurts because the model explores a harder path when it didn't need to. It's a known failure mode of sequential scaling.

In [ ]:
```
for problem_idx in [10]:
  ex = dataset[problem_idx]
  problem = ex["prompt"][0]["content"]
  gold    = int(ex["label"])
  print(f"####Problem[{problem_idx + 1}] gold={gold}, problem={problem}####")
  decoded, tlen, total = run_one(llm, tokenizer, problem, 4096)
  print(f"* Low budget 4096: correct_flex={extract_answer(decoded, mode="flexible_extract") == gold},ans={decoded[-200:]}")
  decoded, tlen, total = run_one(llm, tokenizer, problem, 8192)
  print(f"* High budget 8192: correct_flex={extract_answer(decoded, mode="flexible_extract") == gold},ans={decoded[-200:]}")
```

####Problem[11] gold=321, problem=Solve the following math problem step by step. Please put your final answer within \boxed{}. Let $\omega \neq 1$ be a 13th root of unity. Find the remainder when

$$\prod_{k=0}^{12}(2-2\omega^k+\omega^{2k})$$

is divided by 1000. Remember to put your final answer within \boxed{}.#### * Low budget 4096: correct_flex=True,ans=α^13 - 1)(β^13 - 1). Is that correct? Yes, because if you have a polynomial P(x) = x^13 - 1, then the product over (x - ζ) is P(x). But here, we have product over (ζ^2 - 2ζ + 2) = product over (ζ - α * High budget 8192: correct_flex=False,ans=\tan^{-1}(1) = \frac{\pi}{4}$So, \alpha = \sqrt{2} e^{i\pi/4}$Then: \alpha^{13} = (\sqrt{2})^{13} e^{i \cdot 13 \cdot \pi/4} = 2^{13/2} e^{i \cdot \frac{13\pi}{4}}$ Now, simplify the angle

• Are there any questions that the parallel strategy solves correctly but the sequential does not at a particular budget? Why might this be happening?

By printing out the trace of each independent inference in parallel (m=8), we found the core advantage of parallel: diversity of reasoning paths. With 8 independent attempts, at least one finds the right route even if the majority don't. Sequential is stuck on one path with no escape despite of matching thinking token budget (32,000).

In [ ]:
```
for problem_idx in [0, 1]:
    ex = dataset[problem_idx]
    problem = ex["prompt"][0]["content"]
    gold    = int(ex["label"])
    print(f"####Problem[{problem_idx + 1}] gold={gold}, problem={problem}####")

    all_individual_results = []
    for _ in range(8):
    decoded, tlen, total = run_one(llm, tokenizer, problem, PAR_BUDGET, strategy="wait",
                                   do_sample=True, temperature=T, top_k=TOP_K, top_p=TOP_P)
    all_individual_results.append((decoded, tlen, total))

    preds_flex_with_decoded = [(extract_answer(d, "flexible_extract"), d) for d, *_ in all_individual_results]
    preds_exact_with_decoded = [(extract_answer(d, "exact_match"), d) for d, *_ in all_individual_results]
```

```
    preds_flex  = [p for p, _ in preds_flex_with_decoded]
    preds_exact = [p for p, _ in preds_exact_with_decoded]

    maj_flex_ans = majority(preds_flex)

    for extracted_ans, decoded_text in preds_flex_with_decoded:
        print(f'ans:{extracted_ans}, trace:{decoded_text}')
      if extracted_ans == maj_flex_ans:
        majority_flex_full_output = decoded_text
    print(f"* Parallel m=8 (32000 tokens): correct_flex={maj_flex_ans == gold},ans={majority_flex_full_output}")

    decoded, tlen, total = run_one(llm, tokenizer, problem, 32000)
    print(f"* Sequence budget 32000: correct_flex={extract_answer(decoded, mode="flexible_extract") == gold},ans={decoded}")
```

####Problem[1] gold=540, problem=Solve the following math problem step by step. Please put your final answer within \boxed{}. Find the largest possible real part of

$$(75+117i)z+\frac{96+144i}{z}$$

where z is a complex number with $|z|=4$. Remember to put your final answer within \boxed{}.#### m0 -- [ans:1, trace:s √(a² + b²). So if x² + y² = r², then maximum is √(a² + b²) * r. Wait, let me check with Cauchy-Sch] m1 -- [ans:540, trace:ssion: (75 + 117i)*4e^{iθ} + (96 + 144i)/4 * e^{-iθ} Which is 4*(75 + 117i)e^{iθ} + (24 + 36i)e^{-] m2 -- [ans:2, trace:right)

$$Compute\$R\$:$$

R = \sqrt{324^2 + 432^2} = \sqrt{(324)^2 + (432)^2}

= \sqrt{10497] m3 -- [ans:50, trace:ts: [300 sinθ + 468 cosθ] + [-24 sinθ + 36 cosθ] = (300 - 24) sinθ + (468 + 36) cosθ = 276 sinθ + 50] m4 -- [ans:11, trace:m Az + Imaginary part from B/z]. So total real part is (75x - 117y) + (6x + 9y) = (75x + 6x) + (-11] m5 -- [ans:540, trace:he maximum of A cosθ + B sinθ is achieved when θ is such that tanθ = B/A or something? Wait, no. Let] m6 -- [ans:540, trace:al derivatives: ∂L/∂x = 81 - 2λx = 0 ⇒ 81 = 2λx ⇒ x = 81/(2λ). ∂L/∂y = -108 - 2λy = 0 ⇒ -108 = 2λy] m7 -- [ans:36, trace:m + Real part from second term:

$$4(75\cos\theta-117\sin\theta)+24\cos\theta+36\sin\theta$$

] * Parallel m=8 (32000 tokens): bom_flex=True * Sequence budget 32000: correct_flex=False,ans=## Step 5: Maximize $396\cos\theta$ The maximum value of $\cos\theta$ is 1, so the maximum value of the real part is:

$$396\cdot 1=396$$

--- ### ✅ Final Answer:

$$396$$

####Problem[2] gold=204, problem=Solve the following math problem step by step. Please put your final answer within \boxed{}. Every morning Aya goes for a 9-kilometer-long walk and stops at a coffee shop afterwards. When she walks at a constant speed of s kilometers per hour, the walk takes her 4 hours, including t minutes spent in the coffee shop. When she walks $s+2$ kilometers per hour, the walk takes her 2 hours and 24 minutes, including t minutes spent in the coffee shop. Suppose Aya walks at $s+\frac{1}{2}$ kilometers per hour. Find the number of minutes the walk takes her, including the t minutes spent in the coffee shop. Remember to put your final answer within \boxed{}.#### m0 -- [ans:60, trace:{ hours}

$$Totaltimeincludingcoffeeshop:$$

3 + \frac{t}{60} \text{ hours}

We need to find]
m1 -- [ans:45, trace:ion:

s = [-2 ± sqrt(4 + 45)] / 2

Wait, discriminant is b² - 4ac = (2)^2 - 4*1*(-11.25) = 4 + 45 =]
m2 -- [ans:2, trace: \pm \sqrt{40.96 - 9.6}}{2}

$$x=\frac{6.4\pm\sqrt{31.36}}{2}=\frac{6.4\pm 5.6}{2}$$

So,] m3 -- [ans:204, trace:rt to minutes:

$$3.4\times 60=204\text{ minutes}$$

--- ### Final Answer:

$$204$$

] m4 -- [ans:144, trace:

$$\frac{s(144-48s)}{60}=\frac{144s-48s^2}{60}$$

So the equation becomes:

$$4s-\frac{144]m5--[ans:24,trace:enwalkingat3km/h:-Timewalking:\$\frac{9}{3}=3\$hours-Timeincoffeeshop:24minutes=]m6--[ans:5,trace:dor\quad s=\frac{-36}{8}=-4.5\$\$Sincespeedcannotbenega}{2}$$

Time spent walking:

\frac{9}{3} \times 60 = 3 \times 60 =]
* Parallel m=8 (32000 tokens): bom_flex=True
* Sequence budget 32000: correct_flex=False,ans=ve)

Thus, $ s = 2.5 $ km/h

Step 4: Find $ t $

Use equation (1):

\frac{9}{2.5} + \frac{t}{60} = 4

3.6 + \frac{t}{60} = 4

\frac{t}{60} = 0.4

t = 24 \text{ minutes} $$

- Do the parallel traces lead to diverse answers or solving strategies?

Based on the traces, parallel scaling does lead to diverse answers per m. All independent completions start with completely different token for the same question. However, for some questions, the majority of them still agree with the right answer despite of diverse initialization and reasoning. For example, Problem 1 above, 3 out of 8 samples get 540 correct (m1, m5, m6); each takes a different approach — Lagrange multipliers, polar form, direct parametrization. Multiple valid solution strategies lead to the correct answer. In some more complicated problem, at least one finds the right route even if the majority don't (for example, Problem 2). This leads to the superior performance of parallel test-time scaling with the best-of-m strategy.

2.4 Improving Parallel Scaling

We chose using different prompting strategies and changing the temperature. Here, we tested for a total thinking token budget of 8,000 (m=2) and 16,000 (m=4) tokens because of the lack of runtime -- m=8 took 13 hours. This still gives us clear comparison with the sequential scaling and baseline parallel.

In [ ]:
```
# rewrte prompt strategies
def make_prompt(text, tokenizer, enable_thinking=True, prompt_strategy="default"):
    system_message = "You are a careful competition math assistant. Always output your final answer in \\boxed{}."
    user_content = text

    if prompt_strategy == "structured":
        user_content = f"Here is a math problem: {text}\n\nTo ensure accuracy, please follow these steps:\n1. Understand the p
    elif prompt_strategy == "cot": # New 'chain of thought' strategy
        user_content = f"Here is a math problem: {text}\n\nLet's think step by step to solve this problem."
    elif prompt_strategy == "default":
        user_content = text
    else:
        raise ValueError(f"Unknown prompt strategy: {prompt_strategy}")

    msgs = [
        {"role": "system", "content": system_message},
        {"role": "user",   "content": user_content},
    ]
    return tokenizer.apply_chat_template(msgs, tokenize=False, add_generation_prompt=True, enable_thinking=enable_thinking)

def run_one(llm, tokenizer, problem, budget, strategy="wait",
            do_sample=False, temperature=0.6, top_k=50, top_p=0.95, enable_thinking=True, prompt_strategy="default"):
    end_ids, end_tag = think_end_ids(tokenizer)
    # Changed call to use the new make_prompt function
    prompt   = make_prompt(problem, tokenizer, enable_thinking=enable_thinking, prompt_strategy=prompt_strategy)
    sampling = SamplingParams(
        temperature      = temperature if do_sample else 0.0,
        top_k            = top_k  if do_sample else -1,
        top_p            = top_p  if do_sample else 1.0,
        max_tokens       = 1,
        detokenize       = False,
    )

    token_ids = tokenizer.encode(prompt)

    # —— thinking loop ————————————————————————————————————————————
    while True:
        think_len = len(token_ids) - len(tokenizer.encode(prompt))
        out       = llm.generate({"prompt_token_ids": token_ids},
                    sampling_params=sampling,
                    use_tqdm=False)
        next_tok  = out[0].outputs[0].token_ids[0]

        if think_len >= budget:
            token_ids += end_ids
            break
        elif next_tok in end_ids and strategy == "wait":
```

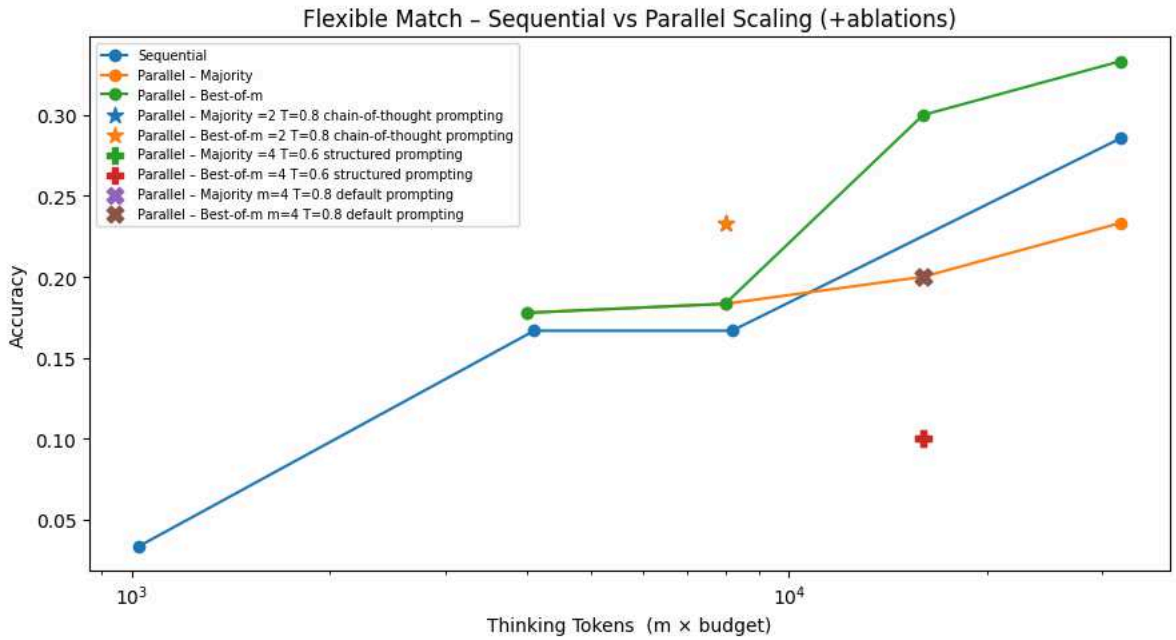
```
            wait_id = tokenizer.encode("Wait", add_special_tokens=False)[0]
            token_ids.append(wait_id)
        else:
            token_ids.append(next_tok)
            if next_tok in end_ids:
                break

    # ── generate final answer ──────────────────────────────────────────
    ans_sampling = SamplingParams(
        temperature = temperature if do_sample else 0.0,
        top_k       = top_k  if do_sample else -1,
        top_p       = top_p  if do_sample else 1.0,
        max_tokens  = MAX_ANS_TOKENS,
    )
    out      = llm.generate({"prompt_token_ids": token_ids},
                   sampling_params=ans_sampling,
                   use_tqdm=False)
    decoded  = out[0].outputs[0].text
    full_out = tokenizer.decode(token_ids) + decoded

    thinking  = extract_thinking_trace(full_out)
    think_tok = len(tokenizer.encode(thinking, add_special_tokens=False))
    total_tok = len(token_ids) - len(tokenizer.encode(prompt)) + len(out[0].outputs[0].token_ids)
    return full_out, think_tok, total_tok

def run_parallel(m_values=PAR_M_VALUES, budget=PAR_BUDGET, n_runs=N_RUNS, save_dir="par_results",
                 temperature=T, top_k=TOP_K, top_p=TOP_P, enable_thinking=True, prompt_strategy="default"):
    os.makedirs(save_dir, exist_ok=True)
    all_rows = []
    print('Start running parallel scaling')
    for m in m_values:
        for run in range(n_runs):
            # Updated save_path to reflect sampling and thinking parameters
            save_path = os.path.join(save_dir, f"m{m}_b{budget}_run{run}_temp{temperature}_topk{top_k}_topp{top_p}_think{enabl

            # Skip if already completed
            if os.path.exists(save_path):
                print(f"\n── m={m}  run={run+1}/{n_runs} ── [SKIPPING, already saved]")
                all_rows.append(pd.read_csv(save_path))
                continue

            print(f"\n── m={m}  run={run+1}/{n_runs} ──")
            rows = []

            for i, ex in enumerate(dataset):
                problem = ex["prompt"][0]["content"]
                gold    = int(ex["label"])
                results = [run_one(llm, tokenizer, problem, budget, strategy="wait",
                                   temperature=temperature, top_k=top_k, top_p=top_p, enable_thinking=enable_thinking, prompt
                           for _ in range(m)]
                preds_flex  = [extract_answer(d, "flexible_extract") for d, *_ in results]
                preds_exact = [extract_answer(d, "exact_match")      for d, *_ in results]
                rows.append(dict(
                    m=m, run=run, gold=gold,
                    majority_flex  = majority(preds_flex)  == gold,
                    majority_exact = majority(preds_exact) == gold,
                    bom_flex       = any(p == gold for p in preds_flex),
                    bom_exact      = any(p == gold for p in preds_exact),
                    total_think    = sum(t   for _, t, _   in results),
                    total_toks     = sum(tot for _, _, tot in results),
                ))
                print(f"  [{i+1}/30] maj_flex={rows[-1]['majority_flex']} bom_flex={rows[-1]['bom_flex']}")

            # Save this run's results
            run_df = pd.DataFrame(rows)
            run_df.to_csv(save_path, index=False)
            print(f"  ✓ Saved → {save_path}")
            all_rows.append(run_df)

    return pd.concat(all_rows, ignore_index=True)

SAVE_DIR = "par_results"

par_df_m2_8k_think_cot = run_parallel(m_values=[2], budget=4000, n_runs=1, temperature=0.8, top_k=TOP_K, top_p=TOP_P, enable_t

par_df_m2_8k_think_structured = run_parallel(m_values=[2], budget=4000, n_runs=1, temperature=T, top_k=TOP_K, top_p=TOP_P, ena

par_df_m4_16k_think_structured = run_parallel(m_values=[4], budget=4000, n_runs=1, temperature=T, top_k=TOP_K, top_p=TOP_P, er

par_df_m4_16k_think_default = run_parallel(m_values=[4], budget=4000, n_runs=1, temperature=0.8, top_k=TOP_K, top_p=TOP_P, ena
```

```
Start running parallel scaling

── m=2  run=1/1 ──
  [1/30] maj_flex=True bom_flex=True
  [2/30] maj_flex=False bom_flex=False
  [3/30] maj_flex=False bom_flex=False
  [4/30] maj_flex=True bom_flex=True
  [5/30] maj_flex=False bom_flex=False
  [6/30] maj_flex=False bom_flex=False
  [7/30] maj_flex=False bom_flex=False
  [8/30] maj_flex=False bom_flex=False
  [9/30] maj_flex=False bom_flex=False
  [10/30] maj_flex=False bom_flex=False
  [11/30] maj_flex=False bom_flex=False
  [12/30] maj_flex=False bom_flex=False
  [13/30] maj_flex=False bom_flex=False
  [14/30] maj_flex=False bom_flex=False
  [15/30] maj_flex=False bom_flex=False
  [16/30] maj_flex=False bom_flex=False
  [17/30] maj_flex=False bom_flex=False
  [18/30] maj_flex=True bom_flex=True
  [19/30] maj_flex=True bom_flex=True
  [20/30] maj_flex=True bom_flex=True
  [21/30] maj_flex=False bom_flex=False
  [22/30] maj_flex=True bom_flex=True
  [23/30] maj_flex=False bom_flex=False
  [24/30] maj_flex=False bom_flex=False
  [25/30] maj_flex=False bom_flex=False
  [26/30] maj_flex=False bom_flex=False
  [27/30] maj_flex=False bom_flex=False
  [28/30] maj_flex=False bom_flex=False
  [29/30] maj_flex=False bom_flex=False
  [30/30] maj_flex=True bom_flex=True
  ✓ Saved → par_results/m2_b4000_run0_temp0.8_topk50_topp0.95_thinkTrue_promptcot.csv
Start running parallel scaling

── m=2  run=1/1 ──
  [1/30] maj_flex=False bom_flex=False
  [2/30] maj_flex=False bom_flex=False
  [3/30] maj_flex=False bom_flex=False
  [4/30] maj_flex=False bom_flex=False
  [5/30] maj_flex=False bom_flex=False
  [6/30] maj_flex=False bom_flex=False
  [7/30] maj_flex=False bom_flex=False
  [8/30] maj_flex=False bom_flex=False
  [9/30] maj_flex=False bom_flex=False
  [10/30] maj_flex=False bom_flex=False
  [11/30] maj_flex=False bom_flex=False
  [12/30] maj_flex=False bom_flex=False
  [13/30] maj_flex=False bom_flex=False
  [14/30] maj_flex=False bom_flex=False
  [15/30] maj_flex=False bom_flex=False
  [16/30] maj_flex=False bom_flex=False
  [17/30] maj_flex=False bom_flex=False
  [18/30] maj_flex=False bom_flex=False
  [19/30] maj_flex=False bom_flex=False
  [20/30] maj_flex=True bom_flex=True
  [21/30] maj_flex=False bom_flex=False
  [22/30] maj_flex=True bom_flex=True
  [23/30] maj_flex=False bom_flex=False
  [24/30] maj_flex=False bom_flex=False
```

In [9]:

```python
import matplotlib.pyplot as plt
import pandas as pd
SAVE_DIR = "par_results"

seq_df = pd.read_csv("seq_results.csv")
par_df = pd.read_csv("par_results.csv")

extra_configs = [
    ("m2_b4000_run0_temp0.8_topk50_topp0.95_thinkTrue_promptcot.csv",          "=2 T=0.8 chain-of-thought prompting"),
    ("m4_b4000_run0_temp0.6_topk50_topp0.95_thinkTrue_promptstructured.csv",   "=4 T=0.6 structured prompting"),
    ("m4_b4000_run0_temp0.8_topk50_topp0.95_thinkTrue_promptdefault.csv",      "m=4 T=0.8 default prompting"),
]
extra_dfs = [(pd.read_csv(SAVE_DIR + "/" + f), label) for f, label in extra_configs]

fig, ax = plt.subplots(figsize=(9, 5))

g = seq_df.groupby("budget")["correct_flex"].mean()
ax.plot(g.index, g.values, marker="o", label="Sequential")

g_maj = par_df.groupby("m")["majority_flex"].mean()
ax.plot(g_maj.index * PAR_BUDGET, g_maj.values, marker="o", label="Parallel — Majority")

g_bom = par_df.groupby("m")["bom_flex"].mean()
ax.plot(g_bom.index * PAR_BUDGET, g_bom.values, marker="o", label="Parallel — Best-of-m")

markers_extra = ["*", "P", "X"]
for (df, label), mk in zip(extra_dfs, markers_extra):
    m_val   = df["m"].iloc[0]
```

```
    x       = m_val * PAR_BUDGET
    maj_acc = df["majority_flex"].mean()
    ax.scatter(x, maj_acc, marker=mk, s=80, zorder=5, label="Parallel – Majority "+f"{label}")
    maj_acc = df["bom_flex"].mean()
    ax.scatter(x, maj_acc, marker=mk, s=80, zorder=5, label="Parallel – Best-of-m "+f"{label}")

ax.set_xscale("log")
ax.set_xlabel("Thinking Tokens  (m × budget)")
ax.set_ylabel("Accuracy")
ax.set_title("Flexible Match – Sequential vs Parallel Scaling (+ablations)")
ax.legend(fontsize=7, loc="upper left")
plt.tight_layout()
plt.savefig("fig_parallel_improv_flex.png", dpi=150)
plt.show()
```



We explored two strategies to improve parallel scaling within a total thinking-token budget of **8,000 and 16,000 tokens**. Note that only half the problem set was evaluated due to runtime constraints, though this did not noticeably undermine performance.

Strategy 1 — Sampling Temperature

We compared higher-temperature ($T = 0.8$) inference (cross marks) against the baseline ($T = 0.6$) at $m = 4$ (16,000 token budget) (green line).

> **Result:** Higher temperature *decreased* best-of-m accuracy.

This suggests that parallel sampling already introduces sufficient solution diversity through independent rollouts. What the strategy actually needs is **convergence** — i.e., multiple samples agreeing on the correct answer — which benefits from a *lower* temperature. Increasing temperature pushes samples further apart, making majority agreement harder to achieve. This is also reflected in the closing gap between the performance of best-of-m and majority strategies (crosses overlaid on each other compared to the gap between orange and green lines).

Strategy 2 — Prompting Strategy

We evaluated three system prompts:

Label	Prompt
Baseline	`"You are a careful competition math assistant. Always output your final answer in \boxed{}."`
Structured	`"Here is a math problem: {text}\n\nTo ensure accuracy, please follow these steps:\n1. Understand the problem and identify key information.\n2. Outline a detailed plan or strategy for solving the problem.\n3. Execute your plan, showing all necessary calculations and reasoning.\n4. Critically review your solution for any errors or alternative approaches.\nFinally, provide your answer."`
Chain-of-Thought (CoT)	`"Let's think step by step to solve this problem."`

> **Result:** CoT prompting yielded the largest accuracy gain; structured prompting *underperformed* the baseline.

The CoT improvement (star marks) is notable given that it also used the suboptimal $T = 0.8$ temperature, suggesting the prompt itself is doing meaningful work.

The underperformance of structured prompting (plus marks) is perhaps the most surprising finding. Despite providing detailed, explicit guidance, it hurt accuracy relative to the baseline. We hypothesize that **overly prescriptive prompts reduce answer diversity** in parallel sampling — by forcing all m samples through the same rigid reasoning template, they are more likely to commit to the same early mistake and converge on a shared wrong answer rather than independently exploring different solution paths.

Part 3: Synthesis and Analysis

1. Characterize when each strategy helps

Our main takeaway is that train time and test time compute both help but for different reasons and at different stages.

For train time compute, the baseline modular division run (`p=97`) showed that just running longer isn't enough. After a long run, test accuracy still stayed low (around 0.13), so extra steps alone didn't produce grokking. Once we changed the setup (AdamW + stronger weight decay) then we were able to observe grokking.

Train time Summary

Intervention	Effect	Interpretation
AdamW + `weight_decay=1.0` (tuned-up), `train_frac=0.3` (tuned-down)	Train >= 0.99 at step 58k, test >= 0.99 at step 102k	Evident grokking signature; clear train-first, test-later transition
AdamW + `weight_decay=1.0` (tuned-up), `train_frac=0.5` (tuned-down)	Train and test >= 0.99 at the same step 264k	Generalization is strong but simultaneous, so the delayed grokking phase is less visible

For test time compute, our preliminary scaling results looked roughly linear across the tested compute range for both sequential and parallel methods (with parallel generally giving a better return per extra token). The intervention study also made it clear that not all extra inference compute is equal.

Test time Summary

Intervention	Effect	Interpretation
`T=0.8` (higher temperature)	Lower accuracy	Too much divergence; harder to reach consensus
Structured prompting	Lower accuracy	Reduced diversity; more correlated failures across samples
Chain-of-thought prompting	Higher accuracy	Better reasoning guidance without over constraining exploration

Overall, train time compute helps most when optimization pushes the model toward a generalizable algorithm while test time compute helps when it improves reasoning quality and sample diversity.

2. Failure modes

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The common theme across failures is that more compute does not automatically mean better generalization.

At train time, long runs can still fail if the optimizer/data setup keeps the model in a memorization heavy regime. At test time, larger budgets can be wasted if the model keeps extending weak reasoning from early mistakes. The temperature and prompting results highlight this directly, that too much randomness hurts consistency, but too much structure (more input tokens) can also hurt by collapsing diversity.

What this suggests is that compute only helps when it improves search over useful solution paths. If it only amplifies the same bad trajectory, in weights during training or in reasoning traces during inference, performance saturates or drops.

3. Propose a strategy

With a limited budget and mixed difficulty, we would use a complexity allocation policy. We would estimate the complexity of each problem by doing a preliminary training (part 1.2) and inference (part 2.0) with the lowest budget; then we would assign each problem a level of complexity for different scaling strategies.

For easy problems, we keep inference cheap (low thinking budget) and only scale on test-time inference with a small sample number and spend almost no extra train time compute.

For medium problems, we use moderate inference scaling first (small parallel samples with CoT prompting), then if confidence is low we escalate.

For hard problems, we combine using finetuned regularization settings with partial training set that showed better generalization in Part 1, and a larger test time scaling especially parallel sampling with more samples and higher thinking budget (eg. raise form 4000 per m to 8000).

In other words, we would route compute adaptively, starting with low cost and then scale up only when needed. That matches what we observed in this homework as well which is better than applying a uniformly large budget to every problem in order to isolate the problem and save compute.

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